

FRM Part I Exam

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Questions with Answers - Financial Markets and Products

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Reading 27: Banks

Q.1093 Banks face the following main risks:

Scenario I: A bank has invested a substantial amount in commercial loans to real estate development projects in a region prone to natural disasters. If a disaster strikes, it may result in the default of these loans.

Scenario II: A bank maintains a considerable portfolio of various financial instruments. A sudden change in interest rates, exchange rates, or commodity prices may significantly alter the value of this portfolio.

Scenario III: A bank has an extensive, automated IT system to manage its transactions and customer data. However, the risk of cyberattacks or significant system failures persists, which may lead to extensive monetary and reputational losses.

The types of risks described here are:

- A. Credit risk, operational risk, and market risk, respectively.
- B. Operational risk, market risk, and credit risk, respectively.
- C. Market risk, operational risk, and credit risk, respectively.
- D. Credit risk, market risk, and operational risk, respectively.

The correct answer is **D**.

Scenario I: This describes credit risk - the risk that counterparties in loan transactions will default, in this case due to natural disasters. Credit risk arises when a borrower is not able to repay a debt.

Scenario II: This scenario refers to market risk - the possibility that the bank's assets and financial instruments will decline in value due to movement in key market variables like interest rates, exchange rates, or commodity prices.

Scenario III: This is operational risk - the risk of loss resulting from inadequate or failed internal processes, people and systems, or from external events. Here it refers to losses from cyberattacks or significant system failures.

Things to Remember

- Credit risk management in banks often involves assessing the potential impact of external factors, like natural disasters, on the ability of borrowers to repay loans.
- Market risk is quantified using models such as Value-at-Risk (VaR), which estimates the potential losses in an investment portfolio due to movements in market prices over a set period at a given confidence interval.

- Operational risk in financial institutions includes cybersecurity threats, which are mitigated through continuous monitoring, robust IT security protocols, and regular system audits.
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Q.1095 Regulatory capital and economic capital are two critical concepts in risk management and financial regulation, serving distinct purposes in the financial industry. Distinguish between regulatory capital and economic capital.

A. Regulatory capital is the amount of capital a bank is required to hold in accordance with regulatory guidance to sufficiently mitigate the risk of failure, whereas economic capital is the amount of capital a bank needs as prescribed by its own (risk) models.

B. Regulatory capital is the amount of capital a bank needs as prescribed by its own (risk) models, whereas economic capital is the amount of capital a bank is required to hold to sufficiently mitigate the risk of failure.

C. Regulatory capital is the amount of capital a bank needs to hold in accordance with stipulated rules and regulations while economic capital is the amount of capital every bank needs to deposit at the Federal Reserve Bank.

D. Regulatory capital is the amount of capital a bank needs to hold in cash at any given time. In contrast, economic is the total amount of capital held, including cash deposits and tangible/intangible financial assets.

The correct answer is **A**.

Regulatory capital and economic capital are two different concepts in banking. Regulatory capital refers to the minimum amount of capital that a bank is required to hold by regulatory authorities. This is to ensure that the bank has enough capital to absorb a reasonable amount of loss and mitigates the risk of failure. The amount is determined based on the bank's financial condition and compliance with regulatory standards. On the other hand, economic capital is the amount of money that a bank chooses to hold based on its own risk models. This is determined by the bank's internal assessment of potential risks and the capital required to cover those risks. Therefore, the statement 'Regulatory capital is the amount of capital a bank is required to hold in accordance with regulatory guidance to sufficiently mitigate the risk of failure, whereas economic capital is the amount of capital a bank needs as prescribed by its own (risk) models' accurately differentiates between the two concepts.

B, C, and D are incorrect as per the explanation for A above.

Things to Remember

- Both economic and regulatory capital are crucial for maintaining financial stability in banking institutions. While regulatory capital ensures compliance with legal

requirements, economic capital helps banks manage their risks effectively.

- The Basel Accords provide international standards for calculating regulatory capital for banks. These accords aim at ensuring that a bank holds enough reserve in order to safeguard its solvency and overall economic stability.
 - Economic Capital is often calculated using Value-at-Risk (VaR) models which estimate the potential losses a firm might suffer due to market risk or credit risk within a certain confidence level over a specific time horizon.
-

Q.1096 Following several high-profile bank failures, the Central Bank of a certain Asian country is advocating the creation of a deposit insurance corporation to protect depositors in the event that banks fail in the future. How might the establishment of the corporation create a **moral hazard**?

- A. Banks might refuse to make premium payments to the corporation, crippling it financially in the process.
- B. Depositors might channel more of their savings to banks, reducing investments in other sectors of the economy.
- C. Banks may increasingly venture into risky businesses that would not otherwise be feasible.
- D. The corporation may encourage banks to manipulate accounts so as to appear healthier than they actually are, hence be eligible for lower premium payments.

The correct answer is **C**.

The establishment of a deposit insurance corporation can indeed create a moral hazard by encouraging banks to engage in riskier business ventures. The term 'moral hazard' refers to the risk that a party insulated from risk may behave differently than it would if it were fully exposed to the risk. In this case, banks, knowing that their depositors are protected by the insurance corporation, may feel emboldened to venture into riskier businesses that they would otherwise avoid due to the potential for high losses. This is because the insurance corporation essentially provides a safety net for the banks, reducing the consequences of their potential failure. As a result, banks may take on excessive risk, leading to instability in the banking sector. This phenomenon is a classic example of moral hazard, where the provision of insurance leads to changes in the behavior of the insured party.

Choice A is incorrect. While banks might refuse to make premium payments, this does not constitute a moral hazard. Moral hazard refers to the risk that a party insulated from risk may behave differently than it would if it were fully exposed to the risk. In this case, refusing to pay

premiums would expose banks to more risks rather than insulate them.

Choice B is incorrect. Depositors channeling more of their savings into banks due to the establishment of a deposit insurance corporation does not represent a moral hazard for the banks themselves. This scenario could potentially lead to an imbalance in the economy, but it doesn't change the behavior of banks in terms of taking on more risks because they are insured.

Choice D is incorrect. Although manipulation of accounts by banks could be an unintended consequence of establishing a deposit insurance corporation, this action represents fraudulent behavior rather than moral hazard. The moral hazard would occur if banks take on higher risks knowing they have insurance coverage, not through manipulation of their financial health status.

Things to Remember

- Moral hazard refers to the risk that one party has an incentive to engage in risky behavior because it knows that it is protected against the consequences of that behavior.
- In the context of banking, moral hazard can occur when banks take on excessive risk knowing they are insured by a deposit insurance corporation. This is because they know any losses incurred will be covered by the insurance.
- Deposit insurance corporations are typically funded by premiums paid by member banks. The level of these premiums may be linked to the risk profile of each bank.
- The establishment of a deposit insurance corporation can lead to increased confidence among depositors, potentially leading to an increase in deposits at banks. However, this could also lead to distortions in capital allocation across different sectors of the economy.
- While deposit insurance can protect individual depositors from bank failures, it does not eliminate systemic risk in the banking sector. Systemic risk refers to the possibility that a triggering event such as a major bank failure could cause a chain reaction leading to broader financial instability or crisis.

Q.1097 As the chief officer in charge of bank risk monitoring at the Federal Reserve Bank, Peter Musk is asked to advise the regulator on the best strategy to curb moral hazards among banks after the establishment of a deposit insurance agency. Mr. Musk could most likely advise the regulator to:

- A. Implement stringent measures against banks awarding their senior executives disproportionately high compensations.
- B. Instruct the insurance agency to routinely request comprehensive information on each bank's financial transactions.
- C. Advise the insurance agency to adjust the premiums required, reflecting the individual risk profile of each bank.
- D. Encourage the deregulation of banks to stimulate competition and self-regulation, despite the increased risks.

The correct answer is **C**.

Directing the agency to tailor premiums payable according to each bank's risk level is the most effective strategy to curb moral hazards among banks. This approach would discourage banks from engaging in excessively risky strategies that could potentially put depositors' funds at risk. Each bank would be required to pay a premium that is proportional to the level of risk it presents. This would incentivize banks to manage their risks more effectively, as higher risks would result in higher premiums. Furthermore, this strategy aligns with the principle of risk-based pricing, which is a common practice in the insurance industry. Risk-based pricing ensures that the price of an insurance policy reflects the risk profile of the insured entity. In this case, banks with higher risk profiles would pay higher premiums, thereby discouraging risky behavior. This approach also promotes fairness, as banks that pose less risk would not be required to subsidize those with higher risk profiles.

Choice A is incorrect. While cracking down on banks for paying their senior management excessive sums may discourage reckless behavior, it does not directly address the issue of moral hazard associated with deposit insurance. The moral hazard problem arises when banks take on excessive risk knowing that they are insured, and this is not necessarily linked to the compensation of senior management.

Choice B is incorrect. Demanding all details of each bank's transactions on a regular basis could be an overwhelming task for the agency and may not effectively mitigate moral hazards. It would be more efficient to focus on key risk indicators rather than every single transaction.

Choice D is incorrect. While competition and self-regulation can indeed play a positive role in a functioning financial system, this approach might not be effective in curbing moral hazards, particularly in the context of a deposit insurance agency. Deregulation might actually exacerbate moral hazards, as there would be fewer regulatory checks and balances to prevent risk-taking behavior.

Things to Remember

- Moral hazard refers to the risk that one party has an incentive to engage in risky behavior because the cost of that behavior will be borne, at least in part, by another party.

- Deposit insurance is a measure implemented by many countries to protect bank depositors, in full or in part, from losses caused by a bank's inability to pay its debts when due. However, it can also lead to moral hazard if banks take on excessive risk knowing they are insured.
 - Regulators often use various strategies such as capital requirements, supervision and regulation of banks' activities and risk management practices to mitigate moral hazards among banks.
 - Tailoring premiums payable according to each bank's risk level is one way of mitigating moral hazard. This means that banks with higher levels of risk would have higher insurance premiums which could incentivize them to reduce their risks.
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Q.1098 In the context of financial risk management, different types of risks are associated with different aspects of a business's operations. These risks include market risk, liquidity risk, credit risk, and operational risk. Under which category of risk would losses due to fraudulent activities carried out by employees be classified?

- A. Market risk
- B. Liquidity risk
- C. Credit risk
- D. Operational risk

The correct answer is **D**.

Operational risk refers to the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events. This definition includes legal risk, but excludes strategic and reputational risk. In the context of this question, losses resulting from employee fraud fall under operational risk. This is because employee fraud is a result of internal flaws or failures, such as inadequate internal controls, lack of oversight, or failure to enforce compliance with company policies. These are all aspects of operational risk. Therefore, any losses that a company incurs due to fraudulent activities by its employees would be classified as operational risk. It's important to note that operational risk is inherent in all business activities, regardless of the industry or sector. Therefore, effective management of operational risk is crucial for the overall risk management strategy of any organization.

Choice A is incorrect. Market risk refers to the potential for financial loss due to changes in the market conditions such as interest rates, exchange rates, commodity prices, and equity

prices. It does not cover losses due to fraudulent activities carried out by employees.

Choice B is incorrect. Liquidity risk pertains to a company's inability to meet its short-term financial obligations because it cannot convert its assets into cash quickly enough or at a reasonable cost. This type of risk does not encompass losses resulting from employee fraud.

Choice C is incorrect. Credit risk involves the possibility that a borrower or counterparty will fail to fulfill their obligations under a contract, leading to financial loss for the lender or other party. It does not include losses caused by fraudulent activities of employees.

Things to Remember

- Operational risk includes both direct and indirect losses. Direct losses include financial loss due to fraud, while indirect losses may cover additional investigation costs and reputational damage.
 - Legal risk, often considered a subset of operational risk, involves losses due to lawsuits or adverse judgments. Frauds leading to legal repercussions would also fall under this domain.
 - Operational risk management often involves the creation of a Risk and Control Self-Assessment (RCSA) framework, where each department evaluates their risk and control environment.
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Q.1100 Investment banks play a crucial role in the financial sector, performing a variety of functions. What is the primary role of investment banks in the financial market?

- A. Acting as a broker for institutional investors and executing large-volume trades.
- B. Providing insurance policies against financial and operational risks to corporations.
- C. Facilitating mergers and acquisitions, and raising capital through issuing debt or equity for corporations.
- D. Providing short-term loans and accepting deposits from corporations and individuals.

The correct answer is **C**.

The primary role of an investment bank like Fairchild is to facilitate mergers and acquisitions (M&A) and assist corporations in raising capital by issuing debt or equity. Investment banks serve as intermediaries between investors and those in need of capital. They provide the expertise and knowledge required to structure and execute complex financial transactions,

including M&As and public offerings.

A is incorrect. While some investment banks do have brokerage divisions that cater to large institutional investors, acting as a broker is not the primary role of an investment bank. Brokerage is just one of the many services provided by investment banks.

B is incorrect. Providing insurance policies against financial and operational risks is typically the role of insurance companies, not investment banks. While some investment banks do engage in risk management consulting, they do not directly provide insurance policies.

D is incorrect. Offering short-term loans and accepting deposits are typically functions associated with commercial banks, not investment banks.

Things to Remember

- Investment banks are key players in primary markets where they help in underwriting new issues of securities. Underwriting involves the investment bank purchasing securities from the issuer and selling them to the public, thus absorbing a lot of the risk.
 - In mergers and acquisitions, investment banks act as advisors, helping with valuation, negotiation, and structuring the deal, ensuring compliance with regulatory requirements.
 - During an Initial Public Offering (IPO), investment banks prepare the prospectus, gauge market conditions, and often guarantee a minimum sale price of the shares by purchasing them first.
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Q.1101 A manufacturing company has an ambitious plan to expand its factory operations and has approached an investment bank for financial assistance. The bank is considering raising the necessary funds through a private placement. This means that:

- A. The bank will provide the funds itself without enlisting any third-party investor.
- B. The bank will sell the desired security to a few large investors such as insurance companies.
- C. The manufacturing company will only accept proposals/bids from privately owned limited liability companies.
- D. The bank must sell the securities to one and only one investor.

The correct answer is **B**.

In the context of investment banking, a private placement refers to the process where securities are sold to a select group of large institutional investors. These investors can include insurance companies, pension funds, or mutual funds. The investment bank that underwrites the arrangement receives a fee that is negotiated with the issuer; in this case, the manufacturing company. The bank does not need to find just one investor; the issuer may prefer more than one financier. This method of raising funds is often preferred when the issuer wants to avoid the regulatory requirements and costs associated with a public offering. It also allows the issuer to raise funds more quickly as the securities do not have to be registered with the Securities and Exchange Commission.

Choice A is incorrect. In a private placement, the investment bank does not provide the funds itself. Instead, it acts as an intermediary to facilitate the transaction between the company seeking funds and potential investors.

Choice C is incorrect. The type of companies that can participate in a private placement (publicly traded or privately owned) is not determined by the manufacturing company's preferences but by securities laws and regulations.

Choice D is incorrect. While a private placement involves selling securities to a limited number of investors, there's no requirement that it must be only one investor. It could be multiple institutional investors such as insurance companies, pension funds or mutual funds.

Things to Remember

- Private placements provide a streamlined fundraising avenue without the need for SEC registration, which is mandatory for public offerings. This exemption from registration saves time and reduces disclosure requirements.
- Institutional investors participating in private placements often benefit from purchasing securities at a discount compared to their market value, compensating for the lack of liquidity associated with privately placed securities.
- The negotiated fee structure between the issuer and the investment bank in a private placement can significantly affect the net proceeds received by the issuer, highlighting the importance of negotiation skills in such transactions.

Q.1102 The management of XYX Inc. wishes to raise some \$50 million via a public offering. Which of the following methods would be most appropriate, given that the total amount **MUST** be raised?

A. Private placement.

- B. Best efforts.
- C. Firm commitment.
- D. Dutch auction.

The correct answer is **C**.

In a firm commitment public offering, an underwriting investment bank guarantees the sale of the entire issue of securities. The bank agrees to purchase the entire issue from the issuer, and then resells the securities to the public. The issuer is assured of raising the full amount of capital, as the bank bears the risk of not being able to sell all the securities. This method is most appropriate for XYZ Corporation as it ensures that the entire \$50 million will be raised. The bank's commitment to buy the entire issue provides the issuer with certainty of funding, which is the primary requirement of XYZ Corporation in this scenario.

Choice A is incorrect. Private placement is not a method of public offering. It involves selling securities to a small group of investors, typically institutional, and not to the general public.

Choice B is incorrect. In a best efforts offering, the underwriters agree to do their best to sell all the securities being offered by the issuer but do not guarantee that all will be sold. This does not ensure that XYZ Corporation will raise its desired capital of \$50 million.

Choice D is incorrect. A Dutch auction also does not guarantee that all shares will be sold or that XYZ Corporation will raise its desired capital of \$50 million as it depends on bidding process where price can fluctuate based on demand.

Things to Remember

- Firm commitment underwriting mitigates the issuer's financial risk by transferring it to the underwriter, who commits to purchasing all shares at a specified price.
- This method is particularly advantageous for issuers needing guaranteed funds by a specific deadline, as it eliminates uncertainties related to market demand.
- While firm commitment underwriting provides capital certainty, issuers must carefully consider the terms, as they might receive a lower per-share price compared to more market-driven methods like Dutch auctions due to the underwriter's risk assumption.
- Issuers opting for this type of offering typically undergo a rigorous due diligence process by the underwriter to protect their financial interests, reinforcing the need for thorough preparation and transparency from the issuer.

Q.1103 In financial markets, a public offering is a critical event that involves the sale of securities. This process can be executed in several ways, one of which is a 'best efforts' basis. Which of the following is the most accurate description of a public offering on a best efforts basis?

- A. The issuer works with multiple investment banks as underwriters.
- B. The underwriter does as well as they can to place securities with investors and gets paid a fee commensurate with the extent of its success.
- C. The underwriter buys the securities and then sells them to investors at a premium.
- D. The issuer does not enlist the services of an underwriter but instead offers securities to investors directly.

The correct answer is **B**.

A public offering on a 'best efforts' basis is a type of securities sale where the underwriter, typically an investment bank, strives to place the securities with investors to the best of its ability. The underwriter's compensation is commensurate with the degree of its success in selling the securities at the predetermined price. This arrangement is beneficial for both the issuer and the underwriter. The issuer does not have to worry about unsold securities, as the underwriter does not guarantee the sale of all securities. On the other hand, the underwriter does not bear the risk of buying the securities and not being able to sell them. This type of offering is commonly used for riskier investments, where it is uncertain how well the securities will be received by the market.

Choice A is incorrect. While it's true that an issuer can work with multiple investment banks as underwriters, this is not a defining characteristic of a 'best efforts' basis public offering. In such an offering, the underwriter does their best to sell the securities but doesn't guarantee the sale of all securities.

Choice C is incorrect. This choice describes a firm commitment offering, not a 'best efforts' basis public offering. In a firm commitment offering, the underwriter buys all the securities from the issuer and then sells them to investors at a premium.

Choice D is incorrect. This choice describes direct listing or direct public offerings where no underwriters are involved in selling securities to investors directly from issuers. It does not describe 'best efforts' basis public offerings where an underwriter attempts to sell as many shares as possible without guaranteeing full sale.

Things to Remember

- In a 'best efforts' offering, the underwriter acts more like a broker than a buyer, attempting to sell the securities to the public without assuming the risk of purchasing the securities outright.
- This arrangement is particularly favorable for smaller or less-known issuers seeking to

access capital markets without the additional burden of unsold shares, which could occur in a firm commitment offering.

- The fee structure in a best efforts arrangement incentivizes underwriters to maximize their sales efforts, as their compensation is directly tied to the amount of capital they successfully raise for the issuer.
 - Regulatory requirements for best efforts offerings, including disclosures and due diligence, remain stringent to protect investors, despite the underwriter not taking on inventory risk.
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Q.1104 ABC Company Limited, a privately held firm, has been considering expansion. However, internal funding is insufficient for the intended scale of growth. To raise the required funds, the company is planning to issue 50 million shares. Notably, the firm wants to retain some level of control over the ownership distribution and minimize public disclosure about its business due to competitive reasons. Moreover, the company is unsure about the suitable price per share. In this scenario, what is the most likely method for the company to issue these shares?

- A. Best efforts IPO
- B. Rights issue
- C. Firm commitment
- D. Private placement

The correct answer is **D**.

In a private placement, a company sells securities to a select group of investors (typically institutional or high-net-worth investors) rather than to the public. This method allows ABC Company Limited to have more control over who buys its shares and to reduce public disclosure, thus preserving its competitive advantage. The uncertainty over the share price could also be mitigated through private negotiations with the selected investors.

A is incorrect. In this scenario, an investment bank would make its best effort to sell the securities to the public, but there is no guarantee of sale for all the shares. Moreover, the company's desire to limit disclosure and maintain control over ownership is not compatible with a public offering.

B is incorrect. A rights issue involves offering new shares to existing shareholders, usually at a discount. However, as a private company, ABC Company Limited might not have a broad shareholder base to raise the needed capital.

C is incorrect. In this case, an investment bank would buy all the shares and resell them to the public. While this would ensure raising the needed funds, it would involve wide public disclosure and does not allow the company to control its ownership distribution.

Things to Remember

- Private placement is often chosen by companies seeking to minimize regulatory requirements and maintain confidentiality about their financial and strategic operations.
 - This funding method allows companies to tailor agreements directly with investors, such as setting bespoke terms for the share price and investment duration, which can be more flexible compared to public offerings.
 - Companies opting for private placements often engage with venture capital firms, private equity firms, or other institutional investors who can provide not just capital but also strategic value such as management expertise or industry connections.
 - Regulatory oversight for private placements in many jurisdictions is generally less stringent than for public offerings, making it a quicker and less costly option for raising capital.
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Q.1105 A company intends to employ the Dutch Auction Approach (DAA) to sell 1 million shares. It receives the following bids:

Bidder	Number of shares	Price per share
A	200,000	\$40.50
B	150,000	\$39.50
C	400,000	\$41.00
D	100,000	39.40
E	300,000	39.00
F	50,000	38.75
G	120,000	37.00

Which of the following is closest to the price all successful buyers will pay per share?

- A. 37
- B. 41
- C. 40.5
- D. 39

The correct answer is **D**.

Under the DAA, potential investors enter their bids quoting the number of shares they intend to purchase and the price they are willing to pay per share. Once the bids have been submitted, the allotment is done starting from the highest bid down, until all the allotted shares have been assigned. However, the final price paid per share is that which has been quoted by the last successful bidder – the buyer whose bid coincides with the end of the intended allotment. Following the methodology outlined above, here’s what would happen in this scenario: First, 400,000 shares would be allocated to C, 200,000 shares to A, 150,000 shares to B, and 100,000 shares to D. At this point, only 150,000 shares remain out of the planned 1 million shares allotment. This means the next highest bid of 300,000 shares at \$39 each can only be half-filled. As such, \$39 is the price that would be paid by all the other successful bidders.

Q.4883 Which of the following best defines a broker's discretionary account?

- A. An account where the broker can trade the investor's funds without the investor's explicit consent.
- B. An account where a trader can buy and sell privately traded securities only
- C. An account that holds securities traded solely for the benefit of the broker, not investors.

D. An account that doesn't have to comply with the rules set by the Securities and Exchange Commission.

The correct answer is **A**.

A broker's discretionary account is an investment account that allows an approved broker to buy and sell securities without obtaining the client's permission for each transaction. But to be able to do so, the client must sign a discretionary disclosure with the broker.

A discretionary account is also called a managed account.

Option B is incorrect. There are no limitations as to which securities can be traded in a discretionary account.

Option C is incorrect. All the proceeds of a discretionary account flow to the investor, less the agreed-upon broker fee.

Option D is incorrect. All trading accounts must comply with the rules and regulations set by the Securities and Exchange Commission, including discretionary accounts.

Things to Remember

- Broker discretionary accounts require a formal discretionary agreement, which sets out the terms under which the broker may act and typically includes a statement of investment objectives and limitations.
- Such accounts are subject to regular oversight and compliance checks to ensure that all trades align with the client's stated investment goals and risk tolerance.
- In the U.S., discretionary trading is regulated under the Investment Advisers Act of 1940, ensuring brokers manage these accounts with a fiduciary duty of utmost good faith and full and fair disclosure of all material facts.
- The use of discretionary accounts can be advantageous for clients who may not have the time or expertise to manage their investments actively.

Q.4884 In the context of banks, the banking book and the trading book represent two distinct portfolios used to manage different types of assets and risks. What is the difference between a banking book and a trading book as used in banks?

A. The banking book consists of assets on the bank's balance sheet expected to be held until maturity while the trading book consists of assets that are available for sale.

B. The banking book consists of assets held on the bank's balance sheet while the trading book consists of assets held off the balance sheet.

C. The banking book reports assets whose value is fixed (e.g. fixed income bonds) while the trading book reports assets whose value fluctuates in response to market variables.

D. The banking book only shows primary financial instruments such as cash while the trading book shows secondary financial derivative instruments such as interest rate futures and options.

The correct answer is **A**.

The banking book and trading book are two different portfolios within a bank. The banking book consists of assets on the bank's balance sheet that are expected to be held until maturity. These assets are not marked to market, meaning they are usually held at historical cost. The banking book typically includes loans, mortgages, and bonds. The purpose of the banking book is to earn interest over a long period, and the risks associated with the banking book are credit risk and interest rate risk.

On the other hand, the trading book consists of assets that are available for sale, meaning that they are eligible for day-to-day trading. Under Basel II and III, the trading book has to be marked to market on a daily basis. The trading book typically includes financial instruments like government and corporate bonds, derivatives, and equities. The purpose of the trading book is to earn profits from short-term price fluctuations, and the risks associated with the trading book are market risk and liquidity risk.

Choice B is incorrect. Both the banking book and trading book consist of assets that are held on the bank's balance sheet. The distinction between them lies in their purpose and how they are managed, not where they are held.

Choice C is incorrect. While it's true that the value of assets in the trading book can fluctuate due to market variables, it's not accurate to say that all assets in the banking book have fixed values. The banking book can include loans and bonds whose values may change due to interest rate fluctuations or credit risk changes.

Choice D is incorrect. The banking book does not only show primary financial instruments such as cash; it also includes other long-term investments like loans and bonds which are expected to be held until maturity. Similarly, while the trading book may include derivative instruments, it also includes other securities intended for active trading.

Things to Remember

- The existence of separate banking and trading books allows banks to manage and assess risks appropriately based on the differing nature and purpose of the assets within each book.
- Maintaining distinct books aids in regulatory compliance, as different rules apply to the

valuation and risk management of each book under international standards like Basel III.

- The segregation helps in optimizing capital allocation and maximizing returns, as the banking book focuses on long-term stability and interest income, while the trading book targets short-term gains through active trading.
-

Q.4885 In corporate finance, companies often employ various defensive strategies to protect against hostile takeovers. Which of the following is an example of a poison pill?

- A. Granting a bonus to executives if the company remains independent after a takeover attempt.
- B. Implementing a staggered board system where only a few directors are elected each year.
- C. Issuing preferred shares that immediately convert to common shares in the case of a takeover.
- D. Creating a subsidiary that holds all the intellectual property, which can be sold in the event of a takeover.

The correct answer is **C**.

Issuing preferred shares that automatically convert to common shares upon a takeover attempt dilutes the voting power and ownership percentage of the acquirer, directly impacting the feasibility and attractiveness of the takeover. This is a classic example of a poison pill tactic.

A is incorrect. While granting a bonus to executives if the company remains independent might seem like a deterrent, it primarily serves as a retention tool rather than a structural mechanism that complicates or deters a hostile takeover directly. It does not affect the company's ownership structure or operational control during a takeover.

B is incorrect. A staggered board system, where only a fraction of directors are elected in any single year, can indeed make hostile takeovers more difficult by prolonging the process required to gain control of the board. However, this is generally considered a separate governance mechanism rather than a classic poison pill, which typically involves changes to share distribution or ownership rights.

D is incorrect. Creating a subsidiary to hold critical assets like intellectual property might make the main company less attractive to take over, but it does not directly interfere with the mechanics of a takeover or change the conditions during a takeover attempt. This strategy is more about asset protection and less about the immediate defensive reaction characteristic of poison pills.

Things to Remember

- A poison pill, or shareholder rights plan, is a defensive strategy used by corporations to deter hostile takeovers by diluting the shares held by potential acquirers, making a takeover less attractive or more costly.
 - Examples of poison pills include issuing new shares to existing shareholders at a discount, allowing shareholders except the acquirer to buy more shares at a discount, or creating new classes of shares that increase voting rights for existing shareholders upon a trigger event.
 - In 2012, Netflix adopted a poison pill after activist investor Carl Icahn acquired a significant stake. This plan was designed to be triggered if an individual or group acquired 10% (or 20% for institutional investors) of Netflix's shares, effectively preventing a hostile takeover without board approval.
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Q.4887 The following table shows information about the bid and bidders in a Dutch Auction to sell 10,000 shares.

Bidder	Number of Shares	Price (USD)
A	1, 500	40
B	1, 000	36
C	2, 500	45
D	3, 000	43
E	4, 500	35
F	2, 000	42
G	4, 000	44

At what price are the shares sold?

- A. 44
- B. 42
- C. 36
- D. 40

The correct answer is **B**.

In a Dutch auction, the auctioneer starts with the highest asking price and lowers it until it reaches a price level where the bids received will cover the entire offer quantity.

First, we sort the bids from the highest to the lowest (according to bid price)

Bidder	Number of Shares Requested	Cumulative Number of Shares Requested	Price (USD)
C	2, 500	2, 500	45
G	4, 000	6, 500	44
D	3, 000	9, 500	43
F	2, 000	11, 500	42
A	1, 500	13, 000	40
B	1, 000	14, 000	36
E	2, 000	16, 000	35

From the table above, it can be noted that a cumulative total of

9,500 shares can be sold for 43 or more. 11,500 shares can be sold for 42 or more. This is the price that coincides with a cumulative total of 10,000 shares as targeted by the auctioneer. Shares therefore sell for 42 each.

Q.5041 Which of the following best describes the liquidity coverage ratio?

- A. A requirement to ensure that banks have enough funding to survive a 30-day period of acute stress such as downgrading, losing deposits or drawdowns on its lines of credit.
- B. A requirement that limits the size of mismatches between the maturity of assets and the maturity of the assets.
- C. A capital requirement designed to cover losses arising from defaults on loans and derivatives contracts.
- D. A standardized ratio for determining operational risk capital.

The correct answer is **A**.

The liquidity coverage ratio (LCR) is a regulatory requirement introduced by the Basel Committee on Banking Supervision. It is designed to ensure that banks have sufficient high-quality liquid assets to survive a 30-day period of acute stress. This stress could be caused by a variety of factors, such as a downgrade in the bank's credit rating, a loss of deposits, or drawdowns on its lines of credit. The LCR is calculated by dividing a bank's high-quality liquid assets by its total net cash outflows over a 30-day stress period. The aim is to promote short-term resilience of a bank's liquidity risk profile by ensuring it has sufficient high-quality liquid assets to survive a significant stress scenario lasting for one month.

Choice B is incorrect. While it is true that managing maturity mismatches between assets and liabilities is crucial for a bank's liquidity management, this does not specifically describe the purpose of the Liquidity Coverage Ratio (LCR). The LCR is designed to ensure that banks have sufficient high-quality liquid assets to survive a 30-day period of acute stress.

Choice C is incorrect. This choice describes a capital requirement, which aims to cover potential losses from loan defaults or derivative contracts. However, this does not accurately represent the function of the LCR. The LCR focuses on liquidity risk rather than credit risk.

Choice D is incorrect. Operational risk capital determination involves assessing risks associated with failures in systems, processes or personnel within an organization. This concept differs from the Liquidity Coverage Ratio which primarily addresses liquidity risk during periods of financial stress.

Things to Remember

- High-quality liquid assets (HQLA) are assets that can be easily and quickly converted into cash with little to no loss in value.
- The Liquidity Coverage Ratio (LCR) is part of the Basel III regulatory framework aimed at strengthening the resilience of the banking sector.
- Net cash outflows are calculated by considering both expected cash outflows and

expected cash inflows under stressed conditions.

- Banks are required to maintain a minimum LCR of 100%, meaning they must hold enough high-quality liquid assets to cover their net cash outflows over a 30-day stress period.
 - The LCR helps to ensure that banks can meet their short-term obligations even under severe liquidity stress scenarios.
-

Q.5334 Which of the following is a potential drawback of the originate-to-distribute banking model?

- A. It allows banks to hold onto assets that are difficult to sell in the market.
- B. It introduces flexibility into the banks' financial statements.
- C. It can lead to a misalignment of incentives between banks and investors.
- D. It increases the amount of capital that banks need to hold.

The correct answer is **C**.

The originate-to-distribute banking model can indeed lead to a misalignment of incentives between banks and investors. This is because banks, under this model, may prioritize the quantity of loans originated over their quality. This is due to the fact that banks do not retain the loans on their balance sheets and, therefore, do not bear the risk of default. As a result, they may be incentivized to originate loans without adequately assessing the creditworthiness of the borrowers, leading to increased risk for the investors who purchase these loans. This misalignment of incentives was one of the factors that contributed to the subprime mortgage crisis in 2007-2008.

Choice A is incorrect. The originate-to-distribute banking model does not allow banks to hold onto assets that are difficult to sell in the market. In fact, it's quite the opposite. This model allows banks to offload loans from their balance sheets by selling them to investors, thereby reducing their exposure to these potentially risky assets.

Choice B is incorrect. While this model may introduce some level of flexibility into a bank's financial statements by allowing it to manage its risk exposure and capital requirements more effectively, this is generally considered an advantage rather than a disadvantage of the originate-to-distribute banking model.

Choice D is incorrect. The originate-to-distribute banking model does not increase the amount of capital that banks need to hold. Instead, it can actually reduce capital requirements as loans are sold off and removed from the bank's balance sheet.

Things to Remember

- Originate-to-distribute banking model involves banks originating loans and then selling them off to investors, rather than holding them on their balance sheets.
 - Securitization is a key process in the originate-to-distribute model, where loans are pooled together and sold as securities to investors.
 - Misalignment of incentives can occur when banks prioritize loan quantity over quality in this model, leading to increased risk for investors.
 - The subprime mortgage crisis of 2007-2008 was partly attributed to issues related to the originate-to-distribute banking model.
 - Regulatory oversight and risk management practices are crucial in mitigating the risks associated with the originate-to-distribute model.
-

Q.5335 A risk manager at ABC Bank is discussing the moral hazard problem in risk management with newly hired employees. Which of the following **correctly** describes the moral hazard problem associated with deposit insurance?

- A. Depositors are incentivized to take on more risk due to the protection of deposit insurance.
- B. Banks are incentivized to take on riskier investments due to the protection of deposit insurance.
- C. Regulators are incentivized to reduce oversight of banks due to the protection of deposit insurance.
- D. Bank employees are incentivized to engage in fraudulent activities due to the protection of deposit insurance.

The correct answer is **B**.

Deposit insurance provides protection to depositors in case their bank fails, which reduces the risk for depositors. However, it also creates a moral hazard problem for banks, as they are incentivized to take on riskier investments since the cost of default is borne by the insurance fund, not the bank itself.

A is incorrect because depositors are not the ones who are taking on the risk. Banks are the ones who are taking on the risk by making investments. Depositors are simply the ones who are

entrusting their money to banks. If a bank fails, depositors may lose their money, but it is the bank that is taking on the risk of failure. Instead, insurance makes depositors less keen on monitoring the bank's finances and decisions.

C is incorrect. Deposit insurance does not incentivize regulators to reduce oversight of banks, as regulators are responsible for ensuring that banks meet certain standards of safety and soundness at all times.

D is incorrect. Deposit insurance does not incentivize bank employees to engage in fraudulent activities, as fraudulent activities can lead to regulatory sanctions, lawsuits, and other adverse consequences.

Things to Remember

- Deposit insurance is a protection scheme that guarantees depositor funds up to a certain amount, even if a bank fails. For example, the FDIC in the U.S. insures deposits up to \$250,000 per depositor per insured bank.
 - An example of deposit insurance at work is when a bank enters receivership; insured depositors are reimbursed up to the insured limit, typically without significant delays.
 - A real-world example of moral hazard due to deposit insurance is seen in the 2008 financial crisis, where some banks engaged in risky mortgage lending because they knew deposits were insured, reducing their risk of depositor withdrawals.
 - The presence of deposit insurance can lead banks to assume greater risks in investment strategies, under the assumption that deposit protection will buffer against depositor panic and withdrawal.
-

Q.5336 A bank executive is attending a seminar on the major risks faced by banks and the ways in which these risks can arise. The seminar is meant to provide an overview of the different types of risks and their implications for the banking industry. Which of the following is an example of operational risk?

- A. The risk of losses in positions arising from movements in market variables.
- B. The risk of default on a debt that may arise from a borrower failing to make required payments.
- C. The risk of losses due to inadequate or failed internal processes, people, and systems.
- D. The risk of losses due to events outside of the bank's control, such as natural disasters or political events.

The correct answer is **C**.

Operational risk is defined as the risk of losses resulting from inadequate or failed internal processes, people, and systems. This can encompass a wide range of potential issues, including:

- **Inadequate or failed internal processes.** These can range from errors in transaction processing to system failures, or even insufficient controls that fail to prevent fraud or misuse of the bank's resources. For example, a bank might have a flawed process for approving loans, leading to the approval of loans that should have been denied. This could result in significant losses for the bank.
- **People.** Human errors or misconduct can also lead to operational risk. Employees might make mistakes in data entry, fail to follow established procedures, or engage in fraudulent activities. For instance, an employee might accidentally transfer funds to the wrong account, or intentionally embezzle funds from the bank.
- **Systems.** Technical issues, such as software bugs, hardware failures, or cyberattacks, can lead to operational risk by disrupting normal banking operations and potentially causing losses. For example, a cyberattack might result in a data breach, leading to financial losses and damage to the bank's reputation.

Choice A is incorrect. This refers to market risk, not operational risk. Market risk arises from fluctuations in values of, or income from, assets or in interest or exchange rates.

Choice B is incorrect. This describes credit risk which occurs when a borrower fails to repay a loan and the lender loses the principal of the loan or the interest associated with it.

Choice D is incorrect. While this could be considered a form of operational risk under certain circumstances, it's more accurately described as systemic or external risk. These risks are outside of an organization's control and can impact an entire industry rather than just one institution.

Things to Remember

- Operational risk is one of the key risks faced by banks and financial institutions, alongside credit risk, market risk, liquidity risk, and strategic risk.
- Operational risk can also include legal risks, such as fines or penalties resulting from non-compliance with regulations or laws.

- Risk management frameworks, such as Basel III, require banks to have robust processes in place to identify, assess, monitor, and mitigate operational risks.
 - Operational risk can have both direct and indirect impacts on a bank's financial performance and reputation.
-

Q.5337 A compliance officer at a commercial bank is concerned about potential conflicts of interest with the investment banking division of the same bank. The investment banking division has recently taken on a new client that competes directly with an existing client of the commercial banking division. What is the primary concern of the compliance officer in this case?

- A. The investment banking division pressuring the bank's brokers to buy certain securities for clients.
- B. The investment banking division pressuring researchers to generate buy recommendations for the new client.
- C. The investment banking division receiving material nonpublic information from the commercial banking division about the existing client.
- D. The investment banking division pressuring commercial bankers to offer preferential rates to the new client.

The correct answer is **C**.

The primary concern for the compliance officer in this scenario would be the possibility of the investment banking division receiving material nonpublic information from the commercial banking division about the existing client. This is because such information could potentially be used against the existing client, leading to a conflict of interest for the bank. Moreover, this could also result in a violation of securities laws and regulations, which prohibit the use of material nonpublic information for trading purposes. This is a serious concern as it could lead to legal repercussions for the bank and damage its reputation. Therefore, the compliance officer would need to ensure that appropriate measures are in place to prevent such a situation from occurring.

Choice A is incorrect. While it's possible that the investment banking division could pressure the bank's brokers to buy certain securities for clients, this isn't directly related to the conflict of interest between serving a new client who is a direct competitor of an existing client in the commercial banking division. The compliance officer's primary concern would be about information flow and potential misuse, not about influencing investment decisions.

Choice B is incorrect. Pressuring researchers to generate buy recommendations for the new client might be a concern, but it doesn't represent the primary conflict of interest in this scenario. The main issue here revolves around access and potential misuse of material nonpublic

information from one division (commercial banking) by another (investment banking).

Choice D is incorrect. Although offering preferential rates to the new client could potentially create conflicts within different divisions of the bank, it does not address directly the compliance officer's main concern which would be about maintaining confidentiality and preventing misuse of sensitive information across divisions.

Things to Remember

- Material nonpublic information (MNPI) refers to confidential information that has not been made available to the general public and could impact the price of a security if disclosed.
 - Conflicts of interest can arise when individuals or entities have competing interests that could make it difficult to fulfill their duties impartially.
 - Compliance officers are responsible for ensuring that organizations adhere to relevant laws, regulations, and internal policies to prevent misconduct and maintain ethical standards.
 - Securities laws and regulations, such as insider trading laws, prohibit the use of material nonpublic information for trading or other purposes that could give an unfair advantage.
 - Reputational risk is a significant concern for financial institutions, as any misconduct or unethical behavior can damage their reputation and lead to loss of trust from clients and stakeholders.
-

Reading 28: Insurance Companies and Pension Plans

Q.1109 Term life insurance is a type of life insurance policy that provides coverage for a specified period. Under term life insurance, the sum assured is payable only if:

- A. The insured dies within the specified time period.
- B. The beneficiary is alive at the end of the specified term.
- C. The insured lives beyond the specified term period.
- D. The beneficiary dies within the specified term period.

The correct answer is **A**.

Term life insurance is a type of life insurance that provides coverage for a certain period of time, or a 'term'. If the insured dies during this term, the death benefit is paid out to the beneficiary. This is the fundamental principle of term life insurance. The purpose of this type of insurance is to provide financial protection to the beneficiary in the event of the insured's death during the term of the policy. The sum assured is not payable if the insured survives beyond the term of the policy. Therefore, the sum assured is payable only if the insured dies within the specified time period.

Choice B is incorrect. The sum assured in a term life insurance policy is not dependent on whether the beneficiary is alive at the end of the specified term. The policy pays out if the insured dies within the specified time period, regardless of whether or not the beneficiary survives this period.

Choice C is incorrect. If the insured lives beyond the specified term period, no death benefit will be paid out under a term life insurance policy. This type of insurance only provides coverage for a specific 'term' or duration; if this duration expires and no claim has been made (i.e., if the insured does not die), then no benefit will be paid.

Choice D is incorrect. The death of a beneficiary within a specified term does not trigger payment in a term life insurance policy. The payout condition for such policies revolves around whether or not insured dies within that specific timeframe, irrespective of when beneficiaries pass away.

Things to Remember

- Term life insurance is typically more affordable than permanent life insurance because it only provides coverage for a specific term.
- Term life insurance does not have a cash value component like some permanent life insurance policies.
- Term life insurance is often used to cover specific financial obligations that will

decrease over time, such as a mortgage or a child's education expenses.

- Term life insurance policies can usually be renewed at the end of the term, but the premiums may increase based on the insured's age and health at that time.
 - Some term life insurance policies offer the option to convert to a permanent life insurance policy without the need for a medical exam.
-

Q.1112 The following data gives the mortality experience among males in Europe in 1931.

Age in years	Probability of death within one year	Survival probability	Life expectancy
30	0.001419	0.97372	47.52
31	0.001445	0.97234	46.59
32	0.001478	0.97093	45.65
33	0.001519	0.97093	44.73

Calculate the probability of a new-born male dying between his 30th and 31st birthday.

- A. 0.97372
- B. 0.99862
- C. 0.001418
- D. 0.00138

The correct answer is **D**.

From the table, the probability of a man surviving to age 30 is 0.97372.

The probability that a new-born male dies between age 30 and 31 equals the probability that the newborn lives until age 30 (0.97372) and dies in the 30th year (0.001419)

Thus, the probability of a new-born male dying between his 30th and 31st birthday is:
 $0.97372 \times 0.001419 = 0.00138$

Q.1113 The following data gives the mortality experience among males in Europe in 1931.

Age in years	Probability of death within one year	Survival probability	Life expectancy
30	0.001419	0.97372	47.52
31	0.001445	0.97234	46.59
32	0.001478	0.97093	45.65
33	0.001519	0.97093	44.73

Calculate the probability of a man aged 30 dying in the **second** year?

- A. 0.001407
- B. 0.001443
- C. 0.998557
- D. 0.972338

The correct answer is **B**.

The probability of a man who is aged 30 dying between ages 31 and 32 means the man survives for one year but dies in the second year. This is equal to the probability that the man survives in the first year (between ages 30 and 31) multiplied by the probability that he dies in the second year (between ages 31 and 32). This is:

$$(1 - 0.001419) * 0.001445 \approx 0.001443$$

Q.1114 The following data gives the mortality experience among males in Europe in 1931.

Age in years	Probability of death within one year	Survival probability	Life expectancy
30	0.001419	0.97372	47.52
31	0.001445	0.97234	46.59
32	0.001478	0.97093	45.65
33	0.001519	0.97093	44.73

Assuming that: I. Interest rate = 4% II. Premiums are paid annually in advance (at the beginning of the year) III. Compounding is semi-annual. A 30-year-old man takes up a term insurance policy that expires in two years. Calculate the present value of the expected payout given that the policy has a sum assured of \$100,000. (Assume that deaths occur mid-way through the year).

- A. 144.62
- B. 275.09
- C. 283.42
- D. 125

The correct answer is **B**.

First, let's calculate the present value of the expected payout:

If the man dies during the first year, the expected payout is the probability of a 30-year-old dying within one year multiplied by the sum assured.

$$= 0.001419 * 100,000 = \$141,90$$

And because we assume that the payout is made approximately half-way through the year, we should discount the expected payout for 6 months.

$$= 141.90 * 1.02^{-1} = \$139.12$$

Similarly, if the man dies during the second year, the expected payout is the probability of a 30-year-old surviving for the first year and then dying in the second year multiplied by the sum assured.

$$[(1 - 0.001419) * 0.001445 * 100,000] * 1.02^{-3} = \$135.97$$

Hence, total expected payout = $139.12 + 135.97 = \$275.09$

Q.1115 The following data gives the mortality experience among males in Europe in 1931.

Age in years	Probability of death within one year	Survival probability	Life expectancy
30	0.001419	0.97372	47.52
31	0.001445	0.97234	46.59
32	0.001478	0.97093	45.65
33	0.001519	0.97093	44.73

Assuming that:

I. Interest rate = 4%

II. Premiums are paid annually in advance (at the beginning of the year)

III. Compounding is semi-annual

IV. Assume that payouts are made at at the mid of the year.

If the policy has a sum assured of \$100,000, and a 30-year-old man takes up a term insurance policy that expires in two years, then which of the following is closest to the break-even premium payable by the policyholder?

A. 140.37

B. 123.62

C. 150

D. 80

The correct answer is **A**.

First, let's calculate the present value of the expected payout:

If the man dies during the first year, the expected payout is the probability of a 30-year-old dying within one year multiplied by the sum assured.

$$= 0.001419 * 100,000 = \$141.90$$

And because we assume that the payout is made approximately half-way through the year, we should discount the expected payout for 6 months.

$$= 141.90 * 1.02^{-1} = \$139.12$$

Similarly, if the man dies during the second year, the expected payout is the probability of a 30-year-old surviving for the first year and then dying in the second year multiplied by the sum assured.

$$[(1-0.001419) * 0.001445 * 100,000] * 1.02^{-3} = \$135.97$$

Hence, total expected payout = $139.12 + 135.97 = \$275.09$

If we let P be the annual premium payable, then the first premium is received at time zero with a probability of 1. The second premium is received at the beginning of the second year. This particular premium is subject to the probability that the man does not die in the first year. It also has to be discounted for 12 months.

Thus, present value of premiums = $p + \frac{(1-0.001419)p}{1.02^2} = 1.9598p$

To calculate the value of p , we then equate the present value of the expected payout to the present value of premiums:

$$\begin{aligned} 275.09 &= 1.9598p \\ p &= \$140.37 \end{aligned}$$

This is the break-even premium.

Q.1116 How does an increase in longevity risk affect the profitability of lifelong annuity contracts?

- A. It increases profits made by the insurer.
- B. It decreases the profits made by the insurer.
- C. It reduces the return on investment to the policyholder.
- D. It has no effect on profitability.

The correct answer is **B**.

Longevity risk refers to the risk that policyholders may live longer than initially estimated. In the context of lifelong annuity contracts, this risk directly impacts the profitability of the insurance company. Lifelong annuity contracts are structured such that the insurer agrees to make regular payments to the policyholder from a certain age until their death. If the policyholder lives longer than initially estimated, the insurer is obligated to continue making these payments for a longer period. This results in higher costs for the insurer, thereby reducing their profitability. Therefore, an increase in longevity risk decreases the profits made by the insurer.

Choice A is incorrect. Longevity risk, or the potential increase in the average lifespan of policyholders, does not increase profits made by the insurer. In fact, it's quite the opposite. If policyholders live longer than expected, insurers will have to make more payments than initially planned which can lead to a decrease in profitability.

Choice C is incorrect. While longevity risk may impact the return on investment for an insurance company due to increased payouts over time, it does not directly reduce the return on investment to the policyholder. The return on investment for a policyholder is typically determined by factors such as premium rates and terms of contract rather than longevity risk.

Choice D is incorrect. Longevity risk certainly has an effect on profitability of an insurance company offering lifelong annuity contracts. As mentioned earlier, if policyholders live longer than expected, this means that insurers will have to make more payments over time which can negatively impact their profitability.

Things to Remember

- Longevity risk is a key consideration for insurance companies offering annuity products, as it directly impacts their profitability.
 - Lifelong annuity contracts involve regular payments to policyholders until their death, making longevity risk a significant factor in determining costs for insurers.
 - An increase in longevity risk can lead to higher costs for insurers, reducing their profitability as they may have to make payments for a longer period than initially estimated.
 - Insurance companies use actuarial science and mortality tables to estimate longevity risk and set appropriate premium rates for annuity products.
 - Longevity risk management strategies include reinsurance, hedging, and product design adjustments to mitigate the financial impact of policyholders living longer than expected.
-

Q.1117 In the field of life insurance, understanding the impact of various risk factors on profitability is crucial for effective policy management. How does increased mortality risk affect the profitability of life insurance contracts?

- A. It increases profits made by the insurer.
- B. It decreases the profits made by the insurer.
- C. It reduces the return on investment to the policyholder.
- D. It has no effect on profitability.

The correct answer is **B**.

An increase in mortality risk decreases the profits made by the insurer. This is because mortality risk is the risk that policyholders will die sooner than expected due to factors such as epidemics,

pandemics, and wars. When mortality risk increases, policyholders live for shorter periods of time than expected and therefore make fewer premium payments to the insurance company before the insurance company needs to make payments for the sum assured. The insurance company will receive less in payments but still be required to pay the policy assured amount to beneficiaries. This reduces the profitability of the life insurance business.

Choice A is incorrect. An increase in mortality risk does not increase the profits made by the insurer. In fact, it's quite the opposite. Higher mortality rates mean that insurers have to pay out more death benefits, which reduces their profitability.

Choice C is incorrect. Mortality risk has no direct impact on the return on investment to the policyholder. The return on investment for a life insurance policyholder is determined by factors such as premium payments, cash value accumulation and dividends, not mortality rates.

Choice D is incorrect. Mortality risk certainly affects profitability of life insurance contracts as it directly influences claim payouts from insurers. Therefore, stating that an increase in mortality risk has no effect on profitability is inaccurate.

Things to Remember

- Life insurance contracts involve the insurer agreeing to pay a specified sum of money upon the death of the insured individual.
- Mortality risk is a key factor in determining the pricing and profitability of life insurance contracts.
- Insurers use actuarial tables and mortality rates to assess the risk of policyholders dying and adjust premiums accordingly.
- Factors such as age, health status, occupation, and lifestyle habits can impact mortality risk.
- Reinsurance is a common practice in the life insurance industry to help insurers manage their exposure to mortality risk.

Q.1119 Richard Brad, FRM, owns a high-rise mixed-use building located in the heart of London. Although he has complied with all quality and safety standards, he fears that a major accident, such as a fire, might result in injuries to residents and third parties and he might be forced to pay for such damages. To protect his building and avoid losses resulting from large-scale compensations, Mr. Brad could most likely:

- A. Make life insurance and disability insurance mandatory requirements for every tenant as well as visitors.

- B. Reinsure the building against large-scale loss.
- C. Seek property insurance.
- D. Seek property-casualty insurance.

The correct answer is **D**.

Seeking property-casualty insurance is the most appropriate measure for Mr. Brad to protect his building and avoid losses resulting from large-scale compensations. Property insurance would provide protection against damage to the building resulting from fire, theft, water damage, etc. Casualty insurance, on the other hand, would provide protection against legal liability exposures, such as injuries to visitors in case of an inferno or the occurrence of any risk. This combination of property and casualty insurance would ensure that Mr. Brad is covered for both physical damages to his property and any liabilities arising from injuries or damages sustained by others. Therefore, this choice is the most comprehensive and suitable option for Mr. Brad to safeguard his interests.

Choice A is incorrect. Making life insurance and disability insurance mandatory for every tenant and visitor would not provide Mr. Brad with protection against potential losses. These types of insurances are personal coverages that protect the insured individuals, not the property owner, in case of death or disability.

Choice B is incorrect. Reinsuring the building against large-scale loss could potentially help to mitigate some financial risk associated with a major accident, but it does not directly protect Mr. Brad from liability claims arising from injuries to residents and third parties.

Choice C is incorrect. While property insurance would cover damages to the building itself due to accidents like fire, it does not typically include liability coverage for injuries or damages suffered by third parties on the premises.

Things to Remember

- Property insurance provides coverage for damage to the building itself resulting from fire, theft, water damage, etc.
- Casualty insurance provides protection against legal liability exposures, such as injuries to visitors or third parties on the premises.
- Reinsurance is a risk management strategy where an insurance company transfers a portion of its risk to another insurer to reduce the potential for large losses.
- Liability insurance covers legal liabilities arising from injuries or damages to third parties and is an important component of property-casualty insurance.
- Insurance policies can be tailored to cover specific risks and liabilities based on the

needs of the insured party.

Q.1120 Suppose you are the chief risk officer of a growing insurance company, "SafeGuard Insurances Ltd." Recently, you noticed an increase in the number of claims. Digging deeper, you found that the claims primarily originated from policies bought in the last 6 months. You suspect adverse selection may be causing this increase in claims, and thus, you are conducting a thorough review to identify potential examples. Which of the following is most likely an example of adverse selection risk?

- A. The company has seen a surge in life insurance policies sold to elderly customers with pre-existing health conditions.
- B. There is an increase in car insurance claims due to severe weather conditions in areas where the company has a significant market presence.
- C. The company's stock value has recently decreased, leading to a decline in market confidence.
- D. The IT system of the company has been hacked, causing significant losses in digital assets.

The correct answer is **A**.

Adverse selection occurs when the party with more information takes advantage of the party with less information. In this scenario, the elderly customers with pre-existing health conditions are more likely to claim insurance than the average person. This is because they know they have health issues, which is information not initially disclosed or factored in during the underwriting process, which leads to a higher-than-expected loss rate for the insurance company. This is a classic example of adverse selection.

Options B, C, and D are incorrect. Adverse selection specifically pertains to risks that arise due to asymmetrical information between the insurer and the insured before the agreement. Severe weather conditions, stock market performance, and IT breaches are all external factors not influenced by the policyholder's actions or choices and do not involve information asymmetry.

Things to Remember

- Adverse selection is a phenomenon where individuals with higher risk levels are more likely to purchase insurance compared to those with lower risk levels.
- Insurance companies use underwriting processes to assess the risk of potential policyholders before issuing a policy.

- One way to mitigate adverse selection is through risk-based pricing, where individuals with higher risk profiles are charged higher premiums.
 - Adverse selection can lead to financial losses for insurance companies if not properly managed.
 - Insurers often use data analytics and actuarial models to predict and manage the impact of adverse selection on their portfolios.
-

Q.1121 If an insurance company offers the same premium to both smokers and non-smokers, it is likely to attract high-risk policyholders and might contend with more payouts than initially expected. This problem is called:

- A. Moral hazard.
- B. Poor selection.
- C. Adverse selection.
- D. Adverse risk modeling.

The correct answer is **C**.

Adverse selection refers to a situation where an insurer faces the challenge of distinguishing between good and bad risks when processing policy applications. If the insurer does not gather sufficient information about the applicant, they risk overlooking certain 'grey areas' that could make the policy too risky to underwrite. For instance, if the insurer does not inquire about the applicant's smoking status, they might end up charging the same premium for both smokers and non-smokers. This could result in higher-than-expected claims and payouts, as a large number of smokers, including those rejected by other insurers, might seek to purchase policies from this insurer. There have been instances where insurers have suffered significant losses due to such practices.

Choice A is incorrect. Moral hazard refers to the risk that a party insulated from risk behaves differently than it would if it were fully exposed to the risk. In this case, charging the same premium for both smokers and non-smokers does not change their behavior, hence moral hazard is not applicable here.

Choice B is incorrect. Poor selection refers to a situation where an insurer fails to select suitable policyholders based on their individual risk profiles. While this scenario does involve poor decision-making by the insurance company, it doesn't accurately describe the problem of attracting high-risk policyholders due to uniform premium rates.

Choice D is incorrect. Adverse risk modeling refers to a situation where an insurer's model for

assessing and pricing risks proves inaccurate or ineffective in practice. Although this could potentially be part of the problem in this scenario, it doesn't capture the specific issue of attracting more high-risk policyholders due to equal premiums for smokers and non-smokers.

Things to Remember

- Insurance underwriting: The process of evaluating the risk of insuring a person or asset and determining the premium to be charged.
 - Actuarial science: The discipline that applies mathematical and statistical methods to assess risk in the insurance and finance industries.
 - Risk assessment: The process of identifying, analyzing, and evaluating potential risks to determine their impact on an organization or individual.
 - Underwriting risk: The risk that an insurance company faces due to inaccurate assessment of the risk associated with a policyholder.
 - Policyholder behavior: How the actions and choices of policyholders can impact the risk profile and profitability of an insurance company.
-

Q.1122 The main difference between a defined benefit scheme and a defined contribution scheme is that:

- A. While defined benefit schemes are employer-sponsored, defined contribution schemes are run by employees who contribute funds for investment with no input from employers.
- B. Defined contribution schemes last for a maximum of 40 work-years while there's no time limit for defined benefit schemes.
- C. Defined benefit schemes are tax-deductible but defined contribution schemes are not.
- D. A defined benefit scheme promises a specific income upon retirement whereas, with a defined contribution scheme, the income depends on factors such as the size of monthly contributions and investment performance.

The correct answer is **D**.

A defined benefit scheme promises a specific income upon retirement whereas, with a defined contribution scheme, the income depends on factors such as the size of monthly contributions and investment performance. Defined benefit schemes, as the name suggests, offer a 'defined' or 'fixed' benefit upon retirement. This benefit is usually calculated based on the employee's salary

and years of service. The employer bears the investment risk in this type of scheme, as they are obligated to provide the promised benefit regardless of the performance of the underlying investments. On the other hand, in a defined contribution scheme, both the employee and the employer contribute to the employee's individual account. The final benefit upon retirement depends on the total contributions made and the performance of the investments. Therefore, the investment risk in this type of scheme is borne by the employee. The key difference between the two schemes lies in who bears the investment risk and how the retirement benefit is determined.

Choice A is incorrect. Both defined benefit schemes and defined contribution schemes can be employer-sponsored. The key difference lies in the structure and benefits, not in who runs the scheme.

Choice B is incorrect. The duration of both defined contribution and defined benefit schemes can vary depending on the specific terms set by employers or plan administrators, not necessarily limited to 40 work-years for defined contribution schemes.

Choice C is incorrect. Both types of pension plans have tax implications, but their tax-deductibility does not distinguish them from each other. In many jurisdictions, contributions to both types of plans are tax-deductible up to certain limits.

Things to Remember

- Defined Benefit Scheme:
 - Promises a specific income upon retirement
 - Employer bears the investment risk
 - Benefit calculated based on salary and years of service
- Defined Contribution Scheme:
 - Income depends on contributions and investment performance
 - Employee bears the investment risk
 - Final benefit determined by total contributions and investment performance
- Employer
- Sponsored Plans:
 - Both types of schemes can be employer-sponsored
 - Employer may contribute to the plan
- Tax Implications:
 - Both types of pension plans have tax implications

- Contributions may be tax-deductible up to certain limits
-

Q.4888 Which of the following factors poses the **greatest** risk of loss to whole life insurance business?

- A. Low rates of interest.
- B. Mortality risk.
- C. Longevity risk.
- D. Currency risk.

The correct answer is **B**.

Mortality risk is the most significant risk faced by life insurance companies. This risk arises when policyholders die earlier than expected, which could be due to various reasons such as illness, disease, or fatal accidents. When this happens, the insurance company is forced to make life insurance payouts earlier than anticipated. This can lead to significant financial losses for the company, especially if the number of early deaths is higher than what was initially projected when pricing the insurance policies. Therefore, managing mortality risk is a critical aspect of the operations of a life insurance company.

Choice A is incorrect. While low rates of interest can impact the profitability of a life insurance company, they do not pose the most significant threat to a whole life insurance business. This is because these companies have strategies in place to manage interest rate risk, such as asset-liability management.

Choice C is incorrect. Longevity risk, which refers to the risk that policyholders live longer than expected, can indeed affect profitability. However, it does not pose the most significant threat for whole life insurance businesses as these policies are designed with this risk in mind and premiums are set accordingly.

Choice D is incorrect. Currency risk could potentially impact an international life insurance company's profitability if it has operations in different countries with varying currencies. However, this risk isn't typically considered as significant for a whole life insurance business compared to mortality risk.

Things to Remember

- Interest rate risk: Life insurance companies are exposed to interest rate risk as they invest policyholder premiums in various assets to generate returns. Fluctuations in interest rates can impact the profitability of these investments.
- Asset-liability management: Life insurance companies use asset-liability management

strategies to match the duration and cash flows of their assets with their liabilities, such as policyholder payouts.

- Longevity risk: This risk refers to the possibility that policyholders may live longer than expected, leading to higher payouts for the insurance company.
 - Policy pricing: Life insurance companies carefully price their policies taking into account factors such as mortality risk, interest rates, and longevity risk to ensure profitability.
-

Q.4889 In property insurance, different types of coverage present varying levels of risk to insurers. Which of the following types of property insurance poses the greatest risk of loss for an insurer?

- A. Fire insurance.
- B. Theft insurance.
- C. Insurance against automobile accidents.
- D. Insurance against natural disasters.

The correct answer is **D**.

This type of insurance falls under Category B risks, where a single event can lead to many claims at the same time. These risks are particularly challenging to model due to the scarcity of data and the unpredictable nature of catastrophic events. For instance, a hurricane can either occur, leading to most policyholders filing a claim, or not occur, resulting in no claims. This all-or-nothing nature of natural disasters makes them the greatest risk of loss for insurers.

Choice A is incorrect. Fire insurance does pose a risk to insurers, but it is not the highest risk. This is because fire incidents can be reasonably predicted based on historical data and the independence of claims. Therefore, insurers can model and manage this risk effectively.

Choice B is incorrect. Theft insurance also poses a risk to insurers, but it's not the highest one either. Similar to fire insurance, theft incidents can be reasonably predicted based on historical data and the independence of claims which allows for effective modeling and management of this type of risk.

Choice C is incorrect. Insurance against automobile accidents does pose a significant risk due to its frequency; however, these risks are still manageable as they are independent events that can be modeled using historical data.

Things to Remember

- Property insurance is a type of insurance that provides financial reimbursement to the owner or renter of a structure and its contents in the event of damage or theft.
 - Insurers use risk assessment to determine the likelihood of a loss occurring and the potential magnitude of that loss.
 - Underwriting is the process of evaluating the risk of insuring a particular property and determining the appropriate premium to charge.
 - Reinsurance is a way for insurers to protect themselves against large losses by transferring some of their risk to other insurance companies.
 - Catastrophe modeling is a tool used by insurers to estimate potential losses from natural disasters and other catastrophic events.
-

Q.4891 The following data gives the mortality experience among males in Europe in 1931.

Age in years	Probability of death within one year	Survival probability	Life expectancy
30	0.001419	0.97372	47.52
31	0.001445	0.97234	46.59
32	0.001478	0.97093	45.65
33	0.001519	0.97093	44.73

What is the probability that a man aged 30 will live to age 33?

- A. 0.9454
- B. 0.9956
- C. 0.0255
- D. 0.0029

The correct answer is **B**.

The probability that a man aged 30 lives to age 33 is calculated by multiplying the survival probabilities for each year from age 30 through age 32:

$$\begin{aligned} P(\text{live from 30 to 33}) &= (1 - 0.001419) \times (1 - 0.001445) \times (1 - 0.001478) \\ &= 0.998581 \times 0.998555 \times 0.998522 = 0.9956642804164275 \end{aligned}$$

Q.4892 Which of the following is the best definition of variable life insurance?

- A. Whole life assurance with an investment component.
- B. The policyholder makes a lump sum payment to the insurer in exchange for a stream of regular payments for a specified period of time.
- C. Term life insurance that is renewable on expiry.
- D. Whole life assurance with an investment component gives the policyholder a lot more flexibility in terms of the premium payable.

The correct answer is **A**.

Variable life insurance is indeed a type of whole life assurance that includes an investment component. This means that a portion of the premium paid by the policyholder is invested in

various sub-accounts available within the policy. For instance, if a policyholder pays an annual premium of \$10,000, half of this amount could be allocated towards the sum assured (death benefit), say \$1 million, while the remaining half is invested in different financial instruments. Consequently, the total benefit received upon the death of the policyholder would be the sum assured plus a variable amount generated from the investment account. This feature of variable life insurance makes it a unique and attractive option for policyholders who are not only interested in life coverage but also in investment opportunities.

Choice B is incorrect. This description refers to an annuity, not a variable life insurance policy. In an annuity, the policyholder makes a lump sum payment and in return receives regular payments for a specified period of time or until death.

Choice C is incorrect. This describes term life insurance that can be renewed at the end of the term, not variable life insurance. Variable life insurance has an investment component and provides lifelong coverage.

Choice D is incorrect. While it does mention whole life assurance with an investment component which aligns with variable life insurance, it incorrectly emphasizes on flexibility in terms of premium payable which is not a defining feature of variable life insurance.

Things to Remember

- Variable life insurance is a type of permanent life insurance that combines a death benefit with an investment component.
 - Policyholders have the option to allocate a portion of their premium payments to different investment sub-accounts within the policy.
 - The cash value of the policy can fluctuate based on the performance of the chosen investments.
 - Policyholders can potentially benefit from the growth of their investments in addition to the guaranteed death benefit.
 - Variable life insurance offers flexibility and the potential for higher returns compared to traditional whole life insurance policies.
-

Q.5042 Given the information below, the operating ratio for a property-casualty insurance company is closest to?

Loss Ratio	65%
Expense Ratio	15%
Combined Ratio	80%
Dividends	3%
Investment Income	5%

- A. 78%
- B. 88%
- C. 72%
- D. 85%

The correct answer is **A**.

The insurance company's operating ratio is a gross profitability measure. It is calculated by adding the loss ratio, expense ratio, and dividends and deducting the investment income earned if the investment earned a positive return. If the investment had a negative return, it should be added to the equation.

$$\text{Operating Ratio} = 65\% + 15\% + 3\% - 5\% = 78\%$$

Q.5338 A group of insurance agents is discussing the risks facing life insurance companies. The agents are particularly concerned about the potential for adverse selection. Which of the following scenarios may result in adverse selection in the life insurance market?

- A. A life insurance applicant not being asked to disclose a pre-existing medical condition.
- B. A life insurance policyholder lying about their age to get a lower premium.
- C. A life insurance policyholder dying before the policy matures.
- D. A life insurance company failing to pay out a valid claim.

The correct answer is **A**.

Adverse selection occurs when policyholders with higher risk are more likely to seek insurance coverage than those with lower risk. In the case of life insurance, adverse selection can occur when the insurance company fails to ask the applicants to disclose pre-existing medical conditions, resulting in a higher-risk pool of policyholders.

B is incorrect. This is an example of fraud, not adverse selection.

C is incorrect. This is an example of the insured event occurring, which is not related to adverse selection.

D is incorrect. This is an example of a failure of the insurance company to fulfill its obligations, which is not related to adverse selection

Things to Remember

- Adverse selection occurs when individuals with higher risk are more likely to purchase insurance compared to those with lower risk.
 - Insurance companies use underwriting processes to assess the risk profile of applicants and determine appropriate premiums.
 - Pre-existing medical conditions are important factors in assessing the risk of an applicant for life insurance.
 - Fraud in insurance involves intentional deception for financial gain, which is different from adverse selection.
 - Insurance companies have a legal obligation to pay out valid claims to policyholders.
-

Q.5339 An insurance company has made payouts of \$15 million and \$5 million in expenses over the past year. It received premiums of \$25 million over the same period. Calculate the loss ratio, expense ratio, and combined ratio for this insurance company.

- A. Loss ratio = 60%, expense ratio = 20%, combined ratio = 80%
- B. Loss ratio = 50%, expense ratio = 25%, combined ratio = 75%
- C. Loss ratio = 60%, expense ratio = 10%, combined ratio = 80%
- D. Loss ratio = 40%, expense ratio = 15%, combined ratio = 55%

The correct answer is **A**.

To calculate the loss ratio, expense ratio, and combined ratio for the insurance company, we'll use the following formulas:

- Loss Ratio = $\left(\frac{\text{Losses Paid}}{\text{Premiums Earned}}\right) \times 100$
- Expense Ratio = $\left(\frac{\text{Expenses}}{\text{Premiums Earned}}\right) \times 100$
- Combined Ratio = Loss Ratio + Expense Ratio

Given the information provided:

- Losses Paid = \$15 million
- Expenses = \$5 million
- Premiums Earned = \$25 million

Thus,

$$\begin{aligned}\text{Loss ratio} &= \frac{\$15 \text{ million}}{\$25 \text{ million}} = 60\% \\ \text{Expense ratio} &= \frac{\$5 \text{ million}}{\$25 \text{ million}} = 20\% \\ \text{Combined ratio} &= 60\% + 20\% = 80\%\end{aligned}$$

Q.5341 Under Solvency II regulations, how is the relationship between the Solvency Capital Requirement (SCR) and the Minimum Capital Requirement (MCR) best characterized for insurance companies?

- A. The MCR is always equal to the SCR to ensure strict capital adequacy standards.
- B. The SCR is a higher capital requirement, while the MCR serves as a lower threshold that triggers supervisory intervention if breached.
- C. The MCR is a higher capital requirement that focuses on risk management, while the SCR serves as a lower threshold for capital adequacy.
- D. The SCR and MCR are unrelated concepts that are independently used to assess an insurance company's capital adequacy.

The correct answer is **B**.

The Solvency Capital Requirement (SCR) and the Minimum Capital Requirement (MCR) under Solvency II regulations are two different capital requirements that insurance companies must meet. The SCR is a higher capital requirement that is designed to ensure that insurance companies have sufficient capital to absorb significant losses with a 99.5% confidence level over a one-year horizon. This requirement takes into account various risk factors, including underwriting, market, credit, operational, and other risks. On the other hand, the MCR serves as a lower threshold for capital adequacy. If an insurance company's capital falls below this level, it triggers immediate supervisory intervention. The insurance company may be prevented from taking on new business, and existing policies might be transferred to another insurance company. Therefore, the SCR and MCR work together to ensure the solvency and financial stability of insurance companies.

Choice A is incorrect. The MCR is not always equal to the SCR. While both are capital requirements under Solvency II, they serve different purposes and are calculated differently. The SCR reflects the capital that an insurance company needs to absorb significant losses and continue its operations, while the MCR represents a lower limit below which policyholders and beneficiaries would be exposed to an unacceptable level of risk.

Choice C is incorrect. Contrary to this statement, it's actually the SCR that is a higher capital requirement focusing on risk management, while the MCR serves as a lower threshold for capital adequacy.

Choice D is incorrect. The SCR and MCR are not unrelated concepts; they are interconnected components of Solvency II regulations designed to assess an insurance company's capital adequacy from different perspectives.

Things to Remember

- SCR (Solvency Capital Requirement) is a higher capital requirement that ensures insurance companies have enough capital to absorb significant losses with a 99.5% confidence level over a one-year horizon.
- MCR (Minimum Capital Requirement) is a lower threshold for capital adequacy, triggering supervisory intervention if breached.

- SCR considers various risk factors like underwriting, market, credit, and operational risks.
 - MCR is a regulatory safety net to protect policyholders and beneficiaries.
 - SCR and MCR are both essential components of Solvency II regulations for assessing capital adequacy in insurance companies.
-

Q.5364 An insurance company has been facing financial difficulties in recent years. The management is trying to identify the underlying issues by analyzing various financial ratios. Which of the following scenarios is most likely to result in the company facing financial difficulties in the long run?

- A. High loss ratio, low expense ratio, and low combined ratio.
- B. Low loss ratio, high expense ratio, and low combined ratio.
- C. High loss ratio, high expense ratio, and high combined ratio.
- D. Low loss ratio, low expense ratio, and high combined ratio.

The correct answer is **C**.

The loss ratio measures the proportion of claims paid (losses) to the premiums earned by the insurance company. The expense ratio shows the proportion of an insurance company's operating expenses (such as commissions, salaries, and administrative costs) relative to the premiums earned. The combined ratio is the sum of the loss ratio and the expense ratio.

A high loss ratio indicates that the company is paying out a significant proportion of its premiums as claims, suggesting that the company's underwriting standards may be inadequate or that it is not pricing its policies effectively. A high expense ratio shows that the company's operating expenses are not being managed efficiently, which can erode profitability. A high combined ratio signifies that the company is experiencing an underwriting loss.

A company with high loss and expense ratios, as well as a high combined ratio, is likely to face financial difficulties in the long run, as it is not generating underwriting profits and may struggle to maintain its operations. This situation can lead to reduced financial stability and potentially harm the company's reputation and competitiveness in the market.

In contrast, the other scenarios (A, B, and D) depict a combination of low loss ratios, low expense ratios, or low combined ratios, which generally indicate better financial health, profitability, and operational efficiency.

Things to Remember

- Loss Ratio: Measures the proportion of claims paid to premiums earned by the insurance company.
 - Expense Ratio: Shows the proportion of operating expenses relative to premiums earned.
 - Combined Ratio: Sum of the loss ratio and the expense ratio, indicating overall underwriting performance.
 - Underwriting Standards: Criteria used by insurance companies to evaluate and accept risks.
 - Pricing Policies: Strategies used by insurance companies to determine the premiums charged for policies.
 - Operational Efficiency: Refers to how well a company utilizes its resources to generate profits.
 - Financial Stability: Ability of a company to maintain its financial position over time.
 - Market Competitiveness: Company's ability to compete effectively in the insurance market.
-

Reading 29: Fund Management

Q.1124 An investor joins a mutual fund and buys shares at \$200 each. In the trading course, the fund leads to a capital gain of \$15 per share in the first year and a capital loss of \$20 per share in the second year. If the investor decided to sell the shares during the second year, what would be the purchase price to calculate the capital gain/loss on the transaction during the second year?

- A. \$200
- B. \$215
- C. \$195
- D. \$205

The correct answer is **B**.

The investor has to declare a capital gain of \$15 in the first year and a capital loss of \$20 in the second year. To avoid double counting, the purchase price must be adjusted to take into account the capital gains or losses that have already accrued to the investor. By selling the shares in the second year, only the \$15 capital gain has accrued, and thus the purchase price would be $(200 + 15) = \$215$.

Note: If the investor were to sell during the third year, both the capital gain of \$15 and the capital loss of \$20 would have accrued, giving an adjusted purchase price of $(200 + 15 - 20) = \$195$

Things to Remember

- **Capital Gain:** The profit that results from a sale of a capital asset, such as stocks, bonds, or real estate, where the sale price exceeds the purchase price.
- **Capital Loss:** The loss that results from the sale of a capital asset for less than the purchase price.
- **Adjusted Purchase Price:** The purchase price of an asset adjusted for any capital gains or losses that have already accrued to the investor.
- **Realized Gain/Loss:** The actual gain or loss experienced by an investor when an investment is sold.

Q.1125 In an investment, there are various types of funds that investors can choose from, each with its unique characteristics and investment strategies. One such type of fund is designed specifically to mirror the performance of a particular equity index, such as the S&P 500. What are these types of funds called?

- A. Open-end funds.
- B. Closed-end funds.
- C. Equity index funds.
- D. Hedge funds.

The correct answer is **C**.

Equity index funds are used to track the performance of particular equity indexes, such as the S&P 500 or the FTSE 100. To achieve this, all the shares in the chosen index are bought in amounts reflective of their weight. That means if XYZ stock has a 2% weight in a particular index, 2% of the tracking portfolio for the index would be invested in XYZ stock.

Option A is incorrect. In an open mutual fund, shares are traded at their net asset value, (NAV), which is calculated as:

Where,

$$\text{NAV} = \frac{\text{Market value of assets at the close of the day} - \text{Liabilities}}{\text{Number of outstanding shares}}$$

The number of shares in an open mutual fund increases and decreases as investors buy and sell their shares.

Option B is incorrect. In a closed-end mutual fund, shares are traded at a discount/premium to their net asset value (NAV), and the number of shares remains constant throughout the fund's life. The changes in the price of its shares are determined by supply and demand.

Option D is incorrect. Hedge funds have fewer regulations than mutual funds, follow a diverse approach of trading strategies, and are not required to disclose their holdings daily. They, however, have additional restrictions on how to solicit funds from investors.

Q.1126 Michael Bauer wishes to buy shares in a front-end loaded mutual fund. He is likely to:

- A. Pay a front-end purchase fee at the time of purchase.
- B. Pay a back-end purchase fee at the time of purchase.
- C. Pay a front-end fee whenever he decides to sell his shares.

D. Pay a back-end purchase fee when he decides to sell the shares.

The correct answer is **A**.

A front-end load is a charge that investors pay when they purchase shares in a mutual fund, and it is typically deducted from the initial investment. This fee is used to cover various costs associated with the management and operation of the fund, including sales commissions, administrative expenses, and other related costs. In the case of Michael Bauer, if he decides to buy shares in a front-end loaded mutual fund, he will be required to pay this fee at the time of purchase. This fee is a one-time charge and does not recur when the investor decides to sell his shares. The front-end load is usually expressed as a percentage of the total investment, and it can significantly impact the net return on the investment, especially in the short term. Therefore, investors should carefully consider this fee when choosing a mutual fund.

Choice B is incorrect. A back-end purchase fee, also known as a deferred sales charge or exit fee, is not paid at the time of purchase. Instead, it is charged when the investor sells his shares in the mutual fund.

Choice C is incorrect. A front-end fee is not charged when an investor decides to sell his shares. It's a type of commission that's paid upfront at the time of purchase.

Choice D is incorrect. While this statement correctly describes a back-end load (it's paid when shares are sold), it does not accurately represent what would happen if Michael Bauer invests in a mutual fund with a front-end load as stated in the question.

Things to Remember

- Front-end load: A fee charged to investors at the time of purchasing shares in a mutual fund.
- Back-end load: A fee charged to investors when they sell their shares in a mutual fund.
- Deferred sales charge: Another term for a back-end load, which is paid when shares are sold.
- Net return: The actual return on an investment after accounting for all fees and expenses.
- Investors should carefully consider fees, such as front-end loads, as they can impact the overall return on their investment.

Q.1127 In hedge fund management, performance incentives are often structured to align the interests of fund managers with those of investors. Describe the meaning of hurdle rate in regard

to hedge funds.

- A. The minimum rate of return necessary for fund managers to be paid.
- B. The minimum rate of return necessary for the incentive fee to be applicable.
- C. The highest peak in value that an investment fund or account has reached.
- D. The lowest value that an investment fund or account has reached.

The correct answer is **B**.

The hurdle rate in the context of hedge funds refers to the minimum rate of return that must be achieved before the incentive fee can be applied. This is a critical aspect of the compensation structure for hedge fund managers. The incentive fee is a performance-based fee that is typically a percentage of the profits generated by the fund. However, this fee is only applicable if the fund's returns exceed a specified minimum level, known as the hurdle rate. This mechanism is designed to align the interests of the fund managers with those of the investors. It ensures that the managers are rewarded for generating high returns, but only if they exceed a certain threshold, thereby protecting the interests of the investors.

Choice A is incorrect. While it might seem logical that fund managers would require a minimum rate of return to be paid, this is not what the term 'hurdle rate' refers to in the context of hedge funds. The hurdle rate does not directly determine the payment of fund managers but rather sets a benchmark for performance-based incentive fees.

Choice C is incorrect. This choice describes the concept of 'high watermark', not 'hurdle rate'. A high watermark refers to the highest peak in value that an investment fund or account has reached, which is used as a reference point for future performance and incentive fee calculations.

Choice D is incorrect. This choice describes a situation where an investment fund or account has reached its lowest value, which does not relate to the concept of 'hurdle rate'. The hurdle rate pertains to setting a minimum level of return necessary for applying incentive fees, and it doesn't concern with lowest values achieved by an investment.

Things to Remember

- **High Watermark:** Refers to the highest peak in value that an investment fund or account has reached, used as a reference point for future performance and incentive fee calculations.
- **Performance Incentives:** Designed to align the interests of fund managers with those of investors by rewarding managers for generating high returns.
- **Incentive Fee:** A performance-based fee that is typically a percentage of the profits

generated by the fund, applicable only if returns exceed the hurdle rate.

- **Alignment of Interests:** Important concept in hedge fund management where structures like hurdle rates ensure that fund managers are rewarded for exceeding performance benchmarks.
-

Q.1129 Sarah wants to invest in the financial markets but first wants to understand the differences among open-end mutual funds, closed-end mutual funds, and exchange-traded funds (ETFs). Which of the following statements accurately describes one of the distinguishing features of closed-end mutual funds?

- A. Closed-end mutual funds are continuously offered to investors and allow for daily redemptions at the net asset value (NAV).
- B. Closed-end mutual funds are actively managed and trade on an exchange like individual stocks.
- C. Closed-end mutual funds issue a fixed number of shares through an initial public offering (IPO) and trade at a premium or discount to their net asset value (NAV).
- D. Closed-end mutual funds are passively managed and seek to replicate the performance of a specific market index.

The correct answer is **C**.

Closed-end mutual funds differ from open-end mutual funds and ETFs in several ways. Closed-end funds issue a fixed number of shares through an initial public offering (IPO), meaning that once the shares are sold, new shares are not continuously offered to investors. This is in contrast to open-end mutual funds and ETFs, which can create or redeem shares on a daily basis. Furthermore, closed-end funds trade on exchanges like stocks, and their share prices can deviate from the net asset value (NAV), leading to them trading at a premium or discount. This price deviation is a key distinguishing feature of closed-end funds.

Option A is incorrect. Closed-end mutual funds do not allow for daily redemptions at the NAV. As mentioned earlier, closed-end funds issue a fixed number of shares through an IPO and do not continuously offer new shares to investors. Therefore, daily redemptions at the NAV are not possible.

Option B is incorrect. The level of management, whether active or passive, is not a defining characteristic of closed-end funds. Both actively managed and passively managed funds can be structured as closed-end funds. The key distinguishing feature of closed-end funds is their fixed number of shares and trading on exchanges, not the management style.

Option D is incorrect. Closed-end funds can be either actively managed or passively managed. While some closed-end funds may indeed seek to replicate the performance of a specific market

index, this is not a universal characteristic. The management style of a closed-end fund can vary, and it is not solely limited to passive management.

Things to Remember

- Open-end mutual funds continuously offer shares to investors and allow for daily redemptions at the NAV.
 - ETFs are similar to closed-end funds in that they trade on exchanges, but they typically have mechanisms in place to keep their share prices close to their NAV.
 - Closed-end funds can trade at a premium or discount to their NAV due to supply and demand dynamics in the market.
 - Closed-end funds may use leverage to enhance returns, which can introduce additional risks compared to open-end mutual funds.
-

Q.1130 Proud Mutual Fund had year-end assets worth \$335 million and liabilities of \$68 million. Given that there were a total of 100,000 shares outstanding, compute the NAV.

- A. 120
- B. 2,600
- C. 1,500
- D. 2,670

The correct answer is **D**.

The Net Asset Value (NAV) is the market value of the assets of the fund minus its liabilities divided by the number of shares in the fund.

$$\text{NAV} = \frac{(335,000,000 - 68,000,000)}{100,000} = \$2,670$$

Q.1131 Brighter Market Portfolio had end-year liabilities amounting to \$43 million and assets worth \$279 million. Given that the fund's NAV was \$20, how many shares must have been held in the fund?

- A. 5000 shares.
- B. 11,000,000 shares.
- C. 11,800,000 shares.
- D. 10,000,000 shares.

The correct answer is **C**.

Using the formula used to calculate a fund's Net Asset Value, the number of a fund's shares can be calculated as:

$$\begin{aligned}\text{NAV} &= \frac{\text{Market value of assets} - \text{Liabilities}}{\text{Number of outstanding shares}} \\ &= \frac{(279,000,000 - 43,000,000)}{\text{Number of outstanding shares}} \\ \Rightarrow 20 &= \frac{(279,000,000 - 43,000,000)}{\text{Number of outstanding shares}} \\ \Rightarrow \text{Number of outstanding shares} &= \frac{(279,000,000 - 43,000,000)}{20} = 11,800,000\end{aligned}$$

Q.1132 When most actively managed mutual funds are compared to index funds such as the S&P 500, they:

- A. Beat the market return in most years.
- B. Underperform the market.
- C. Generally do not outperform the market.
- D. Exceed the return earned on index funds.

The correct answer is **C**.

The statement that actively managed mutual funds generally do not outperform the market is accurate. This is a well-documented phenomenon in the world of finance. The reason behind this is multifaceted. Firstly, actively managed funds incur higher transaction costs due to frequent buying and selling of securities, which can eat into the returns. Secondly, predicting the market consistently over the long term is extremely difficult, even for seasoned fund managers. Lastly, the fees associated with actively managed funds are typically higher than those of index funds, further reducing the net return for investors. Therefore, while some actively managed funds do

outperform the market in certain years, on average and over the long term, they do not tend to outperform the market.

Choice A is incorrect. Most actively managed mutual funds do not beat the market return in most years. This is due to a variety of factors, including high fees and expenses associated with active management, as well as the difficulty in consistently predicting market movements.

Choice B is incorrect. While it's true that many actively managed funds underperform the market, this statement is too broad and doesn't accurately reflect the performance of all such funds. Some actively managed funds may indeed underperform, but others may perform on par with or even outperform their benchmark index in certain years.

Choice D is incorrect. Actively managed mutual funds generally do not exceed the return earned on index funds like S&P 500 over long periods of time due to reasons mentioned above such as high fees and difficulty in consistently predicting market movements.

Things to Remember

- Actively managed mutual funds are managed by professional portfolio managers who make decisions on which securities to buy, hold, and sell in an attempt to outperform the market.
 - Index funds, such as those tracking the S&P 500, aim to replicate the performance of a specific market index rather than outperform it.
 - Expense ratios and fees associated with mutual funds can significantly impact investor returns over time.
 - The efficient market hypothesis suggests that it is difficult to consistently outperform the market due to the speed at which information is incorporated into security prices.
 - Diversification, risk management, and investment strategy are key factors to consider when evaluating mutual funds.
-

Q.1134 Which of the following statements is correct regarding hedge funds?

- A. Hedge funds are subject to the Investment Company Act of 1940 and the Securities Act of 1933.
- B. Hedge funds outline their investment agenda in their prospectus.
- C. Hedge funds must be set up as partnerships and have to provide detailed investment strategies to investors.

D. A majority of hedge funds commit to the use of leverage and short-selling and have a wide investment latitude, including land, derivatives, stocks, currencies, real estate, etc.

The correct answer is **D**.

Hedge funds are known for their wide investment latitude and the use of complex strategies such as leverage and short-selling. Unlike mutual funds, they are not restricted to investing in certain types of assets. They can invest in a wide range of assets, including land, derivatives, stocks, currencies, real estate, and more. This wide investment latitude allows them to potentially generate high returns, but it also exposes them to a higher level of risk. The use of leverage and short-selling is a common strategy among hedge funds. Leverage involves borrowing money to invest, which can amplify both gains and losses. Short-selling involves selling borrowed securities with the expectation that their price will fall, allowing the fund to buy them back at a lower price and profit from the difference. However, these strategies can also increase the risk of losses. Therefore, investing in hedge funds is generally considered suitable for sophisticated investors who can tolerate a high level of risk.

Choice A is incorrect. Hedge funds are not subject to the Investment Company Act of 1940 and the Securities Act of 1933. These regulations apply to mutual funds, but hedge funds are exempt from these due to their limited availability to accredited investors.

Choice B is incorrect. While it's true that hedge funds do outline their investment strategies, they do not typically provide a prospectus like mutual funds do. Instead, they provide a private placement memorandum which contains information about the fund's strategies, risks and fees among other things.

Choice C is incorrect. Hedge funds can be set up in various structures including limited partnerships, limited liability companies or offshore corporations. They don't necessarily have to provide detailed investment strategies as this information might be proprietary and confidential.

Things to Remember

- Hedge funds are typically only available to accredited investors, who are individuals or entities that meet certain income or net worth requirements.
- Hedge funds often charge both management fees (a percentage of assets under management) and performance fees (a percentage of profits).
- Due to their complex strategies and higher risk profile, hedge funds are not as heavily regulated as mutual funds, allowing them more flexibility in their investment approach.
- Hedge funds often have lock-up periods, which are specific time frames during which investors cannot redeem their investments.
- Investing in hedge funds requires a thorough understanding of the fund's strategies,

risks, and fees, as well as a willingness to accept the potential for significant losses.

Q.1135 Which of the following financial institutions **must** provide to all investors the information regarding portfolio composition?

- A. Mutual funds.
- B. Hedge funds.
- C. Both mutual funds and hedge funds.
- D. None of the above.

The correct answer is **A**.

Mutual funds are investment vehicles that pool together funds from many investors to purchase a diversified portfolio of stocks, bonds, or other securities. Due to their nature as investment companies, mutual funds are subject to strict regulatory oversight. This includes the requirement to disclose detailed information about their portfolio composition to all investors. The rationale behind this requirement is to ensure transparency and protect the interests of investors. By knowing the composition of the mutual fund's portfolio, investors can make informed decisions about whether the fund aligns with their investment goals and risk tolerance. This transparency also allows investors to monitor the fund's performance and the fund manager's adherence to the stated investment strategy.

Choice B is incorrect. Hedge funds are not legally obligated to disclose their portfolio composition to all investors. They typically only provide this information to their clients or potential investors, and even then, the level of detail can vary significantly.

Choice C is incorrect. As explained above, hedge funds are not required by law to disclose their portfolio composition to all investors. Therefore, it cannot be both mutual funds and hedge funds.

Choice D is incorrect. This option suggests that no financial institution has a legal obligation to provide details about their portfolio composition which contradicts the fact that mutual funds are indeed required by law to do so.

Things to Remember

- Transparency in financial markets is crucial for investor protection and market efficiency.
- Regulatory oversight helps ensure that financial institutions adhere to laws and regulations designed to protect investors.

- Investors should always carefully review the disclosure documents provided by financial institutions before making investment decisions.
 - Understanding the composition of a portfolio can help investors assess risk, diversification, and alignment with their investment objectives.
 - Different types of financial institutions may have varying levels of disclosure requirements based on their structure and regulatory framework.
-

Q.1136 Which of the following best describes the long/short equity hedge fund strategy?

- A. Taking a long position in undervalued stocks and a short position in overvalued stocks.
- B. Taking a long position in overvalued stocks and a long position in undervalued stocks.
- C. Taking a long position in both overvalued and undervalued stocks.
- D. Taking a long position in overvalued stocks.

The correct answer is **A**.

The long/short equity strategy involves taking a long position in stocks that are undervalued and a short position in overvalued stocks. This strategy is based on the fund manager's ability to accurately assess the value of stocks. In a long position, the manager buys stocks that are undervalued with the expectation that their price will increase. On the other hand, in a short position, the manager borrows and sells overvalued stocks with the expectation that their price will decrease, allowing them to be bought back at a lower price for a profit. The success of this strategy depends on the manager's ability to accurately identify overvalued and undervalued stocks and to balance the long and short positions so that the value of the shares shorted equals the value of those bought and both portfolios have the same sensitivity to market movements. If executed well, this strategy can yield good returns in both bull and bear markets.

Choice B is incorrect. This choice suggests taking a long position in both overvalued and undervalued stocks, which contradicts the basic principle of the long/short equity strategy. The strategy involves buying undervalued stocks (long position) and selling overvalued ones (short position), not buying both.

Choice C is incorrect. Similar to Choice B, this option also proposes taking a long position in both overvalued and undervalued stocks. This does not align with the concept of the long/short equity strategy as it does not involve shorting any stock.

Choice D is incorrect. Taking a long position only in overvalued stocks would be against the principles of value investing which forms part of the basis for a long/short equity strategy. In such strategies, investors typically take short positions in overvalued stocks while going long on

undervalued ones.

Things to Remember

- Long/short equity hedge fund strategy involves taking both long and short positions in stocks.
 - Managers aim to buy undervalued stocks (long position) and sell overvalued stocks (short position).
 - The success of this strategy depends on the manager's ability to accurately assess the value of stocks.
 - Balance between long and short positions is crucial to manage risk and returns.
 - Long/short equity strategies can provide returns in both bullish and bearish market conditions.
-

Q.1137 Washington Mutual Fund had year-end assets worth \$240 million and liabilities of \$12 million. Given that there were a total of 1,000,000 shares outstanding, which of the following is closest to the net asset value (NAV) of the fund?

- A. 240
- B. 220
- C. 20
- D. 228

The correct answer is **D**.

The Net Asset Value (NAV) is the market value of the assets of the fund minus its liabilities divided by the number of shares in the fund.

$$\text{No. of shares} = \frac{(\text{Market value of assets} - \text{Liabilities})}{\text{NAV}}$$

$$\text{NAV} = \frac{(240,000,000 - 12,000,000)}{1,000,000} = 228$$

Q.3493 An analyst gathered the following information regarding a mutual fund:

- Total shares outstanding: 500,000 shares.
- Assets: \$1,000,000.
- Liabilities: \$300,000.

What is the fund's net asset value (NAV)?

- A. 2
- B. 1.4
- C. 0.6
- D. 0.7

The correct answer is **B**.

The Net Asset Value (NAV) is the market value of the assets of the fund minus its liabilities divided by the number of shares in the fund.

$$\begin{aligned}\text{NAV} &= \frac{\text{Market value of assets} - \text{Liabilities}}{\text{No. of shares outstanding}} \\ &= \frac{1,000,000 - 300,000}{500,000} = 1.4\end{aligned}$$

Q.3496 Hedge funds are known for their unique investment strategies and operational structures compared to traditional investment vehicles. Which of the following is a correct characteristic of hedge funds?

- A. Hedge funds are usually listed on exchanges.
- B. Management fees are a fixed percentage of the funds under management, but managers also collect fees based on performance.
- C. Hedge funds are readily available to all investors.
- D. Hedge funds are free to advertise to the public.

The correct answer is **B**.

Hedge funds typically charge a management fee and a performance fee. The management fee is a fixed percentage of the total assets under management. This fee is used to cover the operational costs of the fund, including salaries, office space, and administrative costs. The performance fee, on the other hand, is a percentage of the fund's profits. This fee structure aligns the interests of the fund managers with those of the investors. If the fund performs well, the managers earn a higher fee. This incentivizes the managers to strive for high returns. However, it's important to note that this fee structure can also encourage excessive risk-taking. If the fund performs poorly, the managers do not share in the losses but simply earn a lower fee.

Choice A is incorrect. Hedge funds are not usually listed on exchanges. They are typically private investment vehicles and their shares are not traded publicly.

Choice C is incorrect. Hedge funds are not readily available to all investors. They are typically only accessible to accredited investors or those with a high net worth due to the risk and complexity associated with these types of investments.

Choice D is incorrect. Hedge funds are not free to advertise to the public. In many jurisdictions, they face restrictions on advertising and solicitation activities due to regulatory requirements aimed at protecting less sophisticated investors.

Things to Remember

- Hedge funds are alternative investment vehicles that typically use a variety of strategies to generate returns for their investors.
- They are known for their flexibility in investment strategies, which can include long and short positions, leverage, derivatives, and other complex instruments.
- Hedge funds often have high minimum investment requirements, making them inaccessible to retail investors.
- Investors in hedge funds are usually required to lock up their capital for a certain period, known as the lock-up period, to align the interests of the investors and the fund managers.
- Hedge funds are subject to less regulatory oversight compared to traditional investment vehicles like mutual funds, which can lead to higher risks but also potentially higher returns.

Q.3497 Restrictions on redemptions of funds invested in hedge fund until the specific time during which withdrawals are not allowed is called:

- A. Redemption restriction.
- B. Lock-up period.
- C. Non-withdrawal period.
- D. High-water mark.

The correct answer is **B**.

A lock-up period in hedge funds is a predetermined time span during which investors are not allowed to redeem or sell shares. The lock-up period is typically set at the inception of the fund and can range from a few months to a few years. This period is implemented to give the hedge fund manager the freedom to implement their investment strategy without the concern of withdrawals. It also protects the fund and its investors from the potential negative effects of large, unexpected redemptions. During the lock-up period, the fund manager can invest in less liquid assets, which might have higher returns, without worrying about liquidity for redemptions. After the lock-up period ends, investors can typically redeem their shares at regular intervals, such as quarterly or annually.

Choice A is incorrect. Redemption restriction is a broad term that refers to any limitation on the investor's ability to withdraw their funds from an investment. While it could technically include a lock-up period, it also includes other types of restrictions such as notice periods and redemption fees. Therefore, it does not specifically describe the restriction mentioned in the question.

Choice C is incorrect. Non-withdrawal period is not a term commonly used in the hedge fund industry to describe a specific time during which investors are not allowed to withdraw their funds.

Choice D is incorrect. High-water mark refers to a provision in hedge fund agreements that protects investors from paying performance fees on reinvested money that has previously been lost and recovered. It does not refer to any kind of withdrawal restriction.

Things to Remember

- Hedge funds are alternative investment vehicles that pool capital from accredited investors and invest in a variety of assets and strategies.
- Lock-up periods are common in hedge funds to provide fund managers with stability and flexibility in implementing their investment strategies.
- Investors in hedge funds should carefully consider the terms and conditions, including lock-up periods, before committing capital.
- Redemption restrictions can also include notice periods, gates, and other limitations on

withdrawals from investment funds.

- High-water mark is a concept related to performance fees in hedge funds, ensuring that investors only pay fees on new profits earned after previous losses have been recovered.
-

Q.3498 Anna Smith is a hedge fund manager who tries to exploit price discrepancies between convertible bonds and common stocks of companies. The strategy that Smith uses is known as:

- A. Long/short equity.
- B. Fixed income arbitrage.
- C. Distressed debt.
- D. Convertible arbitrage.

The correct answer is **D**.

Convertible arbitrage is a market-neutral investment strategy often employed by hedge funds. It involves the simultaneous buying of convertible securities and the short selling of the same issuer's common stock. The premise of the strategy is that the convertible is sometimes priced inefficiently relative to the stock, and this inefficiency can be exploited to generate profits. This strategy is typically used by hedge funds and involves sophisticated risk management to handle market movements and other risks. The strategy is known for its complexity and requires a deep understanding of both the bond and equity markets. It is not a strategy for the average investor due to its complexity and the level of risk involved.

Choice A is incorrect. Long/short equity strategy involves buying undervalued stocks (going long) and selling overvalued stocks (going short). This strategy does not specifically involve exploiting price discrepancies between convertible bonds and common stocks.

Choice B is incorrect. Fixed income arbitrage involves capitalizing on price differentials between related fixed income securities. While convertible bonds are a type of fixed income security, this strategy does not specifically focus on the relationship between these bonds and the common stock of the same company.

Choice C is incorrect. Distressed debt investing involves purchasing the debt of companies that are in or near bankruptcy at a significant discount to face value, with the hope that a turnaround or restructuring will increase its value. This strategy does not involve exploiting pricing discrepancies between convertible bonds and common stocks.

Things to Remember

- Convertible arbitrage involves buying convertible securities and short selling the same

issuer's common stock simultaneously.

- This strategy aims to exploit pricing inefficiencies between the convertible bond and the common stock of a company.
 - Convertible arbitrage is a market-neutral strategy, meaning it is designed to be unaffected by overall market movements.
 - Risk management is crucial in convertible arbitrage due to the complexity of the strategy and the potential for market fluctuations.
 - Convertible arbitrage requires a deep understanding of both the bond and equity markets to be successful.
-

Q.3499 A hedge fund has a beginning year value of \$200 million, 2% management fee, and 20% incentive fee with a hurdle rate of 10%. The management fee is applied to the end-of-year assets under management, and the incentive fee is calculated net of the management fee. If the fund's ending value is \$300 million, then what is the total fee of the hedge fund?

- A. \$18.8 million.
- B. \$20.8 million.
- C. \$14.8 million.
- D. \$6 million.

The correct answer is **B**.

Management fee = \$300 million * 2% = \$6 million

Incentive fee = (\$300 million - \$6 million - \$200 million - (\$200 million * 10%)) * 20% = \$14.8 million

Total fees = Management fee + Incentive fee = \$6 million + \$14.8 million = \$20.8 million

Things to Remember

- Management fee is a fixed percentage of the assets under management charged by the hedge fund regardless of the fund's performance.
- Incentive fee is a performance-based fee calculated as a percentage of the profits generated by the fund above a certain threshold known as the hurdle rate.
- Hurdle rate is the minimum rate of return that a fund must achieve before the incentive fee is charged.
- Net of fee means that the fee is deducted from the fund's profits before calculating the incentive fee.
- Assets under management (AUM) is the total market value of the investments that a hedge fund manages on behalf of its clients.

Q.3500 A hedge fund has a beginning year value of \$370 million and a 2 plus 20 fee structure with no hurdle rate or watermark. The hedge fund structure is set up such that the management fee is calculated on the assets at the beginning of the year and that the incentive fee is calculated net of the management fee. If the fund's ending value is \$400 million, then what are the total fees paid to the hedge fund for the period?

- A. \$12.4 million.
- B. \$11.92 million.
- C. \$16 million.
- D. \$4.4 million.

The correct answer is **B**.

Management fee = \$370 million * 2% = \$7.4 million

Incentive fee = (\$400 million - \$7.4 million - \$370 million) * 20% = \$4.52 million

Total fees = Management fee + Incentive fee = \$7.4 million + \$4.52 million = \$11.92 million

Things to Remember

- Hedge funds typically charge management fees and incentive fees to compensate the fund managers for their services and performance.
- The management fee is usually a percentage of the assets under management and is charged annually.
- The incentive fee is typically a percentage of the profits earned by the fund and is charged only if the fund achieves a certain level of return, known as the hurdle rate.
- A watermark is a high-water mark provision that ensures that the manager only receives incentive fees on new profits earned by the fund, not on previously earned profits that have been lost.
- The 2 plus 20 fee structure is a common fee arrangement in the hedge fund industry, where the management fee is 2% of assets under management and the incentive fee is 20% of profits.

Q.3502 Mega Star Investment is a hedge fund with \$550 million initial capital and a '2 and 20' fee structure. The 2% management fee is based on year-end assets under management and the 20% incentive fee is not independent of the management fee. The value of the fund at the end of year one is \$652 million. What is the investor's net return?

A. 0.1247

B. 0.1294

C. 0.1531

D. 0.1779

The correct answer is **B**.

Year end fund value = \$652 million

Management fee = $2\% \times 652 = \$13.04$ million

Incentive fee = $(652 - 550 - 13.04) \times 20\% = \17.79 million

Investor's net return = $(652 - 550 - 13.04 - 17.79)/550 = 71.17/550 = 12.94\%$

Things to Remember

- Management fee is a fixed percentage of the assets under management charged by the hedge fund to cover its operating costs.
- Incentive fee is a performance-based fee calculated as a percentage of the profits generated by the hedge fund.
- '2 and 20' fee structure is a common fee arrangement in the hedge fund industry where the fund charges a 2% management fee and a 20% incentive fee.
- Investor's net return is the return earned by the investor after deducting all fees and expenses from the fund's performance.

Q.3503 Rosy Garcia is considering investing in a hedge fund or a fund of funds.

Garcia invests \$50 million in the hedge fund and receives a yearly gross return of 10%. The fund has a '2 and 20' fee structure with no hurdle rate, and management fees are calculated on an annual basis on assets under management at the beginning of the year. Incentive fees are calculated independently of management fees.

Garcia also invests \$60 million in a fund of funds (FOF) and earns a 5% yearly gross return. Assuming that the fund of funds fee structure is '1 and 10' and that all other fee structures in the FOF are similar to that of the hedge fund, the return to the investor of investing directly in the hedge fund will be:

- A. 2.5% greater than the return generated by investing in the FOF.
- B. 2.3% greater than the return generated by investing in the FOF.
- C. 3.1% greater than the return generated by investing in the FOF.
- D. Lower than the return generated by investing in the FOF.

The correct answer is **A**.

For investing directly \$50 million in the hedge fund: \$50 million (10%) = \$5 million profit

Management fee: \$50 million (2%) = \$1 million gross profit

Incentive fee: \$5 million (0.20) = \$1 million

Total fees = \$1 million + \$1 million = \$2 million

Return: (\$5 million - \$2 million)/\$50 million = 6%

For investing \$60 million in the FOF: \$60 million (5%) = \$3 million gross profit

The FOF charges a fee of 1%: 60 million (1%) = \$0.6 million and an incentive fee of \$3 million (0.10) = \$0.3 million

Return: (\$3 million - \$0.6 million - \$0.3 million)/\$60 million = 3.5%

So 6% - 3.5% = 2.5%

Note that in this chapter, it is usually assumed that the management fee is calculated on the assets at the beginning of the year and that the incentive fee is calculated **after** subtracting management fees.

So, in a special case where the incentive fee is calculated independent of the management fee, then this simply means that we will now calculate the incentive fee "independently". In other words, the incentive fee is calculated **before** subtracting management fees.

Things to Remember

- Management fee: A fee paid to the investment manager for managing the assets, usually calculated as a percentage of the total assets under management.
- Incentive fee: A performance fee paid to the investment manager based on the fund's performance, often calculated as a percentage of the profits generated.
- '2 and 20' fee structure: A common fee structure in hedge funds where the management fee is 2% of assets under management and the incentive fee is 20% of profits.

- '1 and 10' fee structure: A fee structure in fund of funds where the management fee is 1% of assets under management and the incentive fee is 10% of profits.
 - Hurdle rate: A minimum rate of return that a fund must achieve before incentive fees are charged.
-

Q.3504 Right-Lance Capital is a hedge fund with \$250 million as initial investment capital. A 2% management fee based on assets under management is charged at the beginning of the year, and a 20% incentive fee is charged on the performance net of management fees. In the first year of operations, the fund earned a return of 16%.

What is the investor's effective return given this fee structure?

- A. 0.1094
- B. 0.112
- C. 0.125
- D. 0.0943

The correct answer is **B**.

Management fee = \$250 million * 2% = \$5 million

Assets under management at end of period = \$250 million * 1.16=\$290 million

Incentive fee = (\$290 million - \$250 million - \$5 million) * 20% = \$ 7. million

Total fees to Right-Lance Capital = \$5. million + \$7 million = \$12 million

Investor's return = (\$290 million - \$250 million - \$12 million)/\$250 million = 11.20%

Things to Remember

- Management fee is charged based on the assets under management at the beginning of the year.
 - Incentive fee is charged on the performance net of management fees.
 - Effective return for the investor is the return after deducting all fees from the total return.
 - Hedge funds often have complex fee structures that can impact the investor's overall return.
 - Understanding fee structures is crucial for investors to accurately assess the performance of a hedge fund.
-

Q.3505 Clock Limited is a hedge fund with a total asset base of \$10 million. The fund charges a 2% management fee based on assets under management at year-end and a 20% incentive fee in excess of a 0.5% hurdle rate. During the first year, the fund appreciates by 15%. If incentive fees are calculated independently and management fees are calculated at year-end, what is the investor's return net of performance fees?

A. 0.068

B. 0.081

C. 0.098

D. 0.0852

The correct answer is **C**.

Fund value at year-end = \$10 million * 1.15 = \$11.5 million

Management fees = \$11.5 million * 0.02 = \$0.23 million

Hurdle amount = \$10 million * 0.005 = \$0.05 million

Incentive fees = (\$11.5 million - \$10 million - \$0.05 million) * 0.20 = \$0.29 million

Total fees paid to Clock Limited = \$0.23 million + \$0.29 million = \$0.52 million

Investor's net return = (\$11.5 million - \$10 million - \$0.52 million)/\$10 million = 9.8%

Things to Remember

- Management fees are charged based on the total assets under management at year-end, typically calculated as a percentage of the fund's total value.
- Incentive fees are performance-based fees charged on the fund's profits, usually calculated as a percentage of the returns exceeding a specified hurdle rate.
- Hurdle rate is the minimum rate of return that must be achieved before incentive fees are charged to investors.
- Investor's net return is the return earned by the investor after deducting all fees and expenses from the gross return.
- It's important to understand how fees impact the overall return for investors in hedge funds and other investment vehicles.

Q.3506 Which of the following statements correctly describes a fixed income arbitrage hedge fund strategy?

- A. This strategy seeks beta-positive investment strategies.
- B. This strategy seeks to employ a pricing discrepancy between related securities.
- C. This strategy involves buying a convertible bond of one issuer while selling another issuer's common stock.
- D. This strategy seeks to make investment decisions guided by the economic/political outlook of a country.

The correct answer is **B**.

A fixed income arbitrage hedge fund strategy typically involves exploiting pricing discrepancies between related fixed income securities. The goal is to profit from these inefficiencies by buying underpriced securities and simultaneously selling overpriced ones. These related securities might include bonds of different maturities, issuers, or structures that have mispriced relationships based on market factors.

Option A is incorrect. A fixed income convertible arbitrage strategy is a market-neutral (zero beta portfolio) strategy.

Option C is incorrect. The fixed income convertible arbitrage strategy involves buying a convertible bond of one issue and simultaneously selling the issuer's common stock.

Option D is incorrect. Global macro is a general investment strategy that involves making investment decisions guided by the economic/political outlook of a country.

Things to Remember

- Fixed income arbitrage involves exploiting pricing discrepancies in fixed income securities.
- Convertible arbitrage is a specific type of fixed income arbitrage strategy that involves buying convertible bonds and selling the underlying stock.
- Relative value strategies aim to profit from mispricings between related securities.
- Market-neutral strategies seek to eliminate systematic risk (beta) from the portfolio.
- Global macro strategies involve making investment decisions based on macroeconomic and geopolitical factors.

Q.3509 MZJ Income Fund is a mutual fund that does not issue new shares, and its shares can only be bought or sold like equity on exchange markets. Identify this type of fund.

- A. Closed-end fund.
- B. Open-end fund.
- C. Exchange-traded fund.
- D. Hedge fund.

The correct answer is **A**.

Closed-end funds are a type of investment fund and exchange-traded product (ETP). They are structured as publicly traded investment companies by the Securities and Exchange Commission (SEC). Closed-end funds raise a prescribed amount of capital only once, through an initial public offering (IPO), and list shares for trade on a stock exchange. Unlike open-end funds (mutual funds), closed-end funds do not stand ready to issue and redeem shares on a continuous basis. Instead, the shares can be purchased and sold only in the market. This is the reason why MZJ Income Fund, which does not issue new shares and its shares can only be bought or sold like equity on exchange markets, is a closed-end fund.

Choice B is incorrect. An open-end fund continuously issues and redeems units or shares based on the net asset value (NAV) of the fund. This is not consistent with MZJ Income Fund's structure, which does not issue new shares.

Choice C is incorrect. While exchange-traded funds (ETFs) are traded on exchanges like equities, they also create and redeem shares based on demand, unlike MZJ Income Fund which does not issue new shares.

Choice D is incorrect. Hedge funds are typically private investment vehicles that have limited liquidity and do not trade their shares publicly on exchange markets as described for MZJ Income Fund.

Things to Remember

- Closed-end funds raise a prescribed amount of capital only once, through an initial public offering (IPO).
- Closed-end funds do not stand ready to issue and redeem shares on a continuous basis.
- Open-end funds continuously issue and redeem units or shares based on the net asset value (NAV) of the fund.
- Exchange-traded funds (ETFs) are traded on exchanges like equities, but they create

and redeem shares based on demand.

- Hedge funds are typically private investment vehicles with limited liquidity and do not trade their shares publicly on exchange markets.
-

Q.3510 The Belta fund trades on the Chicago Stocks Exchange. Its most recent price is \$1850, but its NAV is \$1,600. We know then that:

- A. the fund is closed-end, selling at a premium.
- B. the fund is closed-end, selling at a discount.
- C. the fund is open-end, selling at a premium.
- D. the fund is open-end, selling at a discount.

The correct answer is **A**.

The Belta fund is a closed-end fund selling at a premium. Closed-end funds are a type of investment company whose shares are traded on the open market, like stocks or ETFs. The price of these shares fluctuates based on market demand and can deviate from the fund's Net Asset Value (NAV). When the trading price of a closed-end fund is higher than its NAV, it is said to be trading at a 'premium'. In this case, the Belta fund's trading price (\$1850) is higher than its NAV (\$1600), indicating that it is indeed a closed-end fund selling at a premium.

Choice B is incorrect. A closed-end fund selling at a discount would mean that the trading price of the fund is less than its Net Asset Value (NAV). In this case, however, the trading price (\$1850) is higher than the NAV (\$1600), indicating that it's not selling at a discount.

Choice C is incorrect. Open-end funds do not trade on exchanges and their prices are not determined by market forces of supply and demand but are instead directly related to their NAV which is calculated at the end of each trading day. Therefore, an open-end fund cannot sell at a premium or discount.

Choice D is incorrect. Similar to Choice C, open-end funds do not trade on exchanges and their prices are directly related to their NAV calculated daily. Hence, they cannot sell at a premium or discount.

Things to Remember

- Net Asset Value (NAV) is the value of a fund's assets minus its liabilities, divided by the number of shares outstanding. It represents the per-share value of the fund.
- Closed-end funds have a fixed number of shares outstanding and their prices are

determined by market forces, leading to potential deviations from NAV.

- Open-end funds continuously issue and redeem shares at their NAV, which is calculated at the end of each trading day.
 - Premium and discount refer to the relationship between the trading price of a fund and its NAV. A premium means the trading price is higher than NAV, while a discount means the trading price is lower than NAV.
 - Investors should consider the premium or discount when investing in closed-end funds, as it can impact the potential returns and risks associated with the investment.
-

Q.4932 In the hedge fund industry, compensation structures are designed to incentivize managers based on performance and fund management. Hedge fund managers are compensated by

- A. deducting management fees from fund assets and receiving incentive bonuses for beating a specified benchmark.
- B. deducting a percentage of any gains in asset value.
- C. Buying shares in the fund at a discount.
- D. charging portfolio turnover fees.

The correct answer is **A**.

Hedge fund managers are typically compensated by deducting management fees from fund assets and receiving incentive bonuses for beating a specified benchmark. This is often referred to as the '2 and 20' structure, where the management fee is usually 2% of the fund's net asset value, and the incentive fee is 20% of the fund's profits above a certain benchmark. The management fee is meant to cover the operational costs of the fund, while the incentive fee aligns the interests of the fund managers with those of the investors, as the managers stand to gain significantly if the fund performs well. This structure is designed to motivate the managers to generate high returns, but it also means that they stand to earn substantial amounts even if the fund's performance is mediocre, as the management fee is charged regardless of the fund's performance.

Choice B is incorrect. While hedge fund managers do often receive a percentage of the gains in asset value, this is typically in addition to a management fee and not deducted from the gains themselves. This compensation structure is known as "two and twenty", where the manager receives a 2% management fee and 20% of any profits.

Choice C is incorrect. Hedge fund managers do not typically buy shares in their own funds at a discount as part of their compensation. While they may invest their own money into the fund to align their interests with those of the investors, this is not considered part of their compensation structure.

Choice D is incorrect. Charging portfolio turnover fees isn't typical for hedge funds' compensation structures. Portfolio turnover fees are more commonly associated with mutual funds, where they are used to cover transaction costs associated with buying and selling securities within the fund's portfolio.

Things to Remember

- Hedge fund compensation structures are designed to align the interests of fund managers with those of the investors.
 - The '2 and 20' structure is a common compensation model in the hedge fund industry, with a 2% management fee and a 20% incentive fee.
 - Hedge fund managers may also invest their own money in the fund to show commitment and align their interests with investors.
 - Performance fees are a key component of hedge fund compensation, incentivizing managers to generate high returns.
 - Management fees are meant to cover operational costs of the fund and are typically deducted from fund assets.
-

Q.4933 On January 1, 2020, a hedge fund began with \$100 million in assets from investors. The 10-year Treasury, yielding 1.25% at the time, was chosen as the hurdle rate for the next five years. In addition, the fund operates on a 2 plus 20% fee structure and is bound by a high-water mark clause. In the first year of operation, a combination of a challenging macroeconomic environment and some bad decisions culminated in end-of-the-year assets under management dropping to \$90 million after payment of the management fee. In 2021, the fund bounced back, with its total assets coming at \$110 million by the end of the year. Calculate the total fees earned by the management in 2021. (Assume that the management fee is calculated on the assets at the beginning of the year and that the incentive fee is calculated after subtracting management fees)

- A. \$5.8m
- B. \$1.8m
- C. \$3.5m

D. \$2.94m

The correct answer is **D**.

A high-water mark clause states that previous losses must first be recouped and hurdle rates surpassed before incentive fees once again apply.

In the first year, the fund managers earned a management fee of \$2m ($= 100 * 0.02$). However, it earned \$0 in incentive fees since it made a loss. Incentive fees would only have applied to any profits earned above a 1.25% return, meaning that only an ending balance higher than \$101.25 million would have triggered the 20% incentive fee.

The management pocketed \$1.8m in management fees in the second year ($= 90 * 0.02$). As per the fund's high-water mark clause, investors expected its total worth to be above \$102.5 at the end of year 2 before any incentive fees can be earned (i.e., $100 + 1.25 + 1.25$). In other words, the high-water mark for year 1 is \$101.25m and \$102.5m for year 2. At \$110m, at the end of the second year, therefore, the fund has outperformed by a margin of \$7.5m (i.e., $110 - 102.5$). As such, 20% of this less the management fee [i.e., $0.2 * (7.5 - 1.8) = \$1.14m$] is earned as incentive fees.

Total fees earned in year 2 = $\$1.8m + \$1.14m = \$2.94m$

Things to Remember

- Management fee: A fee charged by the fund manager for managing the investments, typically calculated as a percentage of the total assets under management.
- Incentive fee: A performance-based fee charged by the fund manager for generating profits above a certain threshold, usually calculated as a percentage of the profits earned.
- High-water mark clause: A provision in a hedge fund's agreement that ensures the fund manager only earns incentive fees on new profits generated after previous losses have been recovered and hurdle rates surpassed.
- Hurdle rate: A minimum rate of return that must be achieved before incentive fees can be earned by the fund manager.
- Assets under management (AUM): The total market value of the investments managed by a financial institution or fund manager on behalf of clients.

Q.5043 The following are returns for a fund and the index it is tracking. Given the information below, the tracking error is closest to?

Period	Fund Return	Index Return
Year 1	5%	4.5%
Year 2	7%	6%
Year 3	-2.5%	-1.5%
Year 4	1%	0.5%

A. 0.68

B. 0.79

C. 0.62

D. 0.59

The correct answer is **B**.

The difference between the funds' returns and the index return are:

Year 1: 5%-4.5%=0.5%

Year 2: 7%-6%=1%

Year 3: -2.5%-(-1.5%)=-1.0%

Year 4: 1%-0.5%=0.5%

$$\sqrt{\frac{(0.5)^2 + (1.0)^2 + (-1.0)^2 + (0.5)^2}{4}} = \sqrt{\frac{2.5}{4}} = 0.79$$

Q.5342 In the context of hedge funds, which of the following statements best describes the purpose and characteristics of incentive fees?

A. Incentive fees are a fixed percentage of the fund's net asset value and are used to align the interests of fund managers and investors.

B. Incentive fees are based on a fund's performance and serve to penalize fund managers for poor performance.

C. Incentive fees are performance-based fees that reward fund managers for generating returns above a specified benchmark or hurdle rate.

D. Incentive fees are always equal to the management fees and are charged regardless of

the fund's performance.

The correct answer is **C**.

Incentive fees in hedge funds are indeed performance-based fees that reward fund managers for generating returns above a specified benchmark or hurdle rate. This fee structure is designed to align the interests of the fund managers with those of the investors. By tying a portion of the fund managers' compensation to the fund's performance, it incentivizes them to strive for superior returns. If the fund's returns exceed the specified benchmark or hurdle rate, the fund managers receive a percentage of those excess returns as incentive fees. This not only rewards the fund managers for their skill and effort but also ensures that they share in the upside when the investors do well. However, if the fund's returns do not exceed the benchmark or hurdle rate, the fund managers do not receive any incentive fees. This ensures that the fund managers are only rewarded when they generate value for the investors.

Choice A is incorrect. While it is true that incentive fees are used to align the interests of fund managers and investors, they are not a fixed percentage of the fund's net asset value. Instead, they are typically based on a percentage of the returns generated by the fund that exceed a specified benchmark or hurdle rate.

Choice B is incorrect. Incentive fees do not serve to penalize fund managers for poor performance. Rather, they serve as a reward mechanism for generating superior returns above a certain benchmark or hurdle rate.

Choice D is incorrect. Incentive fees are not always equal to management fees and their charge depends on the performance of the fund rather than being charged regardless of it. They are specifically designed to provide additional compensation if the manager outperforms certain benchmarks.

Things to Remember

- Hedge funds are alternative investment vehicles that typically use a variety of strategies to generate returns for their investors.
- Management fees are a separate fee charged by hedge funds for managing the fund's investments, while incentive fees are performance-based.
- Hedge funds often have high minimum investment requirements and are typically only open to accredited investors due to their complex and risky nature.
- Investors in hedge funds are often looking for diversification, risk management, and potentially higher returns compared to traditional investments.
- Hedge funds are known for their flexibility in investment strategies, which can include long and short positions, leverage, derivatives, and other complex instruments.

Q.5343 In the context of hedge fund strategies, which of the following statements best describes the primary objective and approach of a long-short equity hedge fund?

- A. A long-short equity hedge fund primarily invests in undervalued equities and shorts overvalued equities with the objective of generating consistent returns regardless of market conditions.
- B. A long-short equity hedge fund takes long positions in high-yield bonds and short positions in lower-yielding bonds, aiming to profit from the yield spread.
- C. A long-short equity hedge fund invests solely in long equity positions, while using short positions in index futures to hedge against market risk.
- D. A long-short equity hedge fund focuses exclusively on short-selling equities with poor fundamentals and high levels of debt, seeking to profit from their price declines.

The correct answer is **A**.

A long-short equity hedge fund primarily invests in undervalued equities and shorts overvalued equities with the objective of generating consistent returns regardless of market conditions. This strategy involves taking long positions in stocks that are expected to increase in value and short positions in stocks that are expected to decrease in value. The primary objective of this strategy is to generate consistent returns by capitalizing on both the appreciation of undervalued equities and the depreciation of overvalued equities, while mitigating market risk. This approach allows the fund to profit from both rising and falling markets, providing a level of protection against market volatility. The fund manager uses their expertise to identify undervalued and overvalued stocks based on a variety of factors, including financial analysis, market trends, and economic indicators. The long-short strategy is a common approach used by hedge funds and is considered a relatively aggressive investment strategy due to the use of short selling and leverage.

Choice B is incorrect. This choice incorrectly suggests that a long-short equity hedge fund primarily invests in high-yield bonds and shorts lower-yielding bonds. While this could be a strategy employed by some hedge funds, it does not accurately represent the primary objective and methodology of a long-short equity hedge fund, which focuses on equities rather than bonds.

Choice C is incorrect. This option inaccurately states that a long-short equity hedge fund only invests in long equity positions and uses short positions in index futures to hedge against market risk. Although hedging against market risk can be part of the strategy, the main characteristic of a long-short equity hedge fund is its ability to take both long and short positions in individual stocks based on their expected performance.

Choice D is incorrect. While short-selling equities with poor fundamentals and high levels of debt can be part of the strategy, it's not accurate to say that a long-short equity hedge fund focuses exclusively on this approach. The primary goal of such funds is to generate consistent returns regardless of market conditions by taking both long positions in stocks expected to increase in value and short positions in stocks expected to decrease.

Things to Remember

- Long-short equity hedge funds aim to generate consistent returns by taking both long and short positions in individual stocks.
 - These funds typically focus on investing in undervalued equities and shorting overvalued equities to capitalize on market inefficiencies.
 - Managers of long-short equity hedge funds use a variety of factors such as financial analysis, market trends, and economic indicators to identify potential investment opportunities.
 - Short selling and leverage are common tools used in long-short equity strategies to enhance returns and manage risk.
 - Long-short equity hedge funds provide a level of protection against market volatility by profiting from both rising and falling markets.
-

Q.5344 Which of the following correctly explains the concept of net asset value (NAV) of an open-end mutual fund and how it relates to the share price?

- A. NAV represents the total value of all assets held by the mutual fund, minus any liabilities, divided by the total number of shares outstanding.
- B. NAV represents the price at which the mutual fund's shares are bought and sold on the open market.
- C. NAV represents the total return of the mutual fund over a specific time period.
- D. NAV represents the total fees and expenses charged by the mutual fund, divided by the total number of shares outstanding.

The correct answer is **A**.

The Net Asset Value (NAV) of an open-end mutual fund is indeed the total value of all assets held by the mutual fund, minus any liabilities, divided by the total number of shares outstanding. This value is calculated at the end of each trading day based on the closing market prices of the securities held by the fund. The NAV is used to determine the price at which shares of the mutual fund are bought and sold. If demand for the mutual fund is high, the share price will be above NAV, and if demand is low, the share price will be below NAV. The inverse relationship between share price and NAV is due to the fact that new shares are issued or redeemed at NAV, but

shares are bought and sold on the open market at the prevailing market price.

Choice B is incorrect. The Net Asset Value (NAV) does not represent the price at which the mutual fund's shares are bought and sold on the open market. Instead, it represents the per-share value of the fund's assets, after subtracting liabilities. The buying and selling price of mutual fund shares in an open-end fund is determined by this NAV, not by market forces as in case of stocks.

Choice C is incorrect. NAV does not represent the total return of a mutual fund over a specific time period. Total return would include all changes in value during that period including capital gains or losses and dividends or interest received, whereas NAV only reflects current value of assets minus liabilities divided by number of outstanding shares.

Choice D is incorrect. While fees and expenses do impact a mutual fund's net asset value to some extent because they reduce its total assets, stating that NAV represents these costs divided by total number of shares outstanding oversimplifies and misrepresents what NAV actually stands for - which is essentially a per-share valuation based on current worth of all assets after accounting for liabilities.

Things to Remember

- Open-end mutual funds continuously issue and redeem shares based on investor demand, unlike closed-end funds which have a fixed number of shares.
- NAV is calculated by taking the current market value of the fund's assets, subtracting any liabilities, and dividing by the number of shares outstanding.
- NAV per share is used to determine the buying and selling price of shares in an open-end mutual fund.
- Investors can buy or sell shares of an open-end mutual fund at the NAV price, which ensures that all investors receive a fair price based on the fund's underlying assets.
- NAV is a key metric for investors to evaluate the performance and value of a mutual fund.

Q.5345 Which measurement bias is primarily associated with voluntary reporting of performance results by hedge funds and mutual funds?

- A. Survivorship bias
- B. Self-reporting bias
- C. Look-ahead bias

D. Confirmation bias

The correct answer is **B**.

Self-reporting bias is a type of measurement bias that is primarily associated with the voluntary reporting of performance results by hedge funds and mutual funds. This bias occurs when funds with strong performance are more likely to voluntarily report their results, while funds with poor performance may not report or may selectively report their results. This leads to an overestimation of performance for both types of funds, as the reported performance data may be skewed towards better-performing funds. This bias can significantly distort the perceived performance of these funds and can mislead investors who rely on this data to make investment decisions. Therefore, it is crucial for investors to be aware of this bias when evaluating the performance of hedge funds and mutual funds based on voluntarily reported data.

Choice A is incorrect. Survivorship bias refers to the tendency of failed companies being left out of performance studies because they no longer exist. It does not directly relate to the voluntary reporting of performance results by funds.

Choice C is incorrect. Look-ahead bias occurs when a study or simulation incorporates data that would not have been known or available during the period being analyzed, thus skewing results. This type of bias is not specifically associated with voluntary reporting of performance results.

Choice D is incorrect. Confirmation bias refers to a type of selective thinking whereby one tends to notice and look for what confirms one's beliefs, and ignore, not look for, or undervalue the relevance of what contradicts one's beliefs. This type of cognitive bias does not pertain directly to the voluntary reporting practices by funds.

Things to Remember

- Survivorship bias: Refers to the tendency of failed companies being left out of performance studies because they no longer exist.
- Look-ahead bias: Occurs when a study or simulation incorporates data that would not have been known or available during the period being analyzed, thus skewing results.
- Confirmation bias: Refers to a type of selective thinking whereby one tends to notice and look for what confirms one's beliefs, and ignore, not look for, or undervalue the relevance of what contradicts one's beliefs.

Q.5346 An investment advisor is discussing the performance of mutual funds and hedge funds with a client. Which of the following best describes the impact of survivorship bias on performance measurement?

- A. Survivorship bias tends to overstate the performance of active funds relative to passive funds.
- B. Survivorship bias tends to understate the performance of surviving funds by including the results of closed or liquidated funds.
- C. Survivorship bias tends to have minimal or no impact on performance measurement because it affects all types of funds equally.
- D. Survivorship bias tends to create an accurate representation of fund performance because it only includes successful and surviving funds.

The correct answer is **A**.

Survivorship bias is a form of bias that can significantly distort the performance measurement of funds. It occurs when the performance measurements only consider funds that have survived over a specific time period, while excluding those that have been closed, liquidated, or merged. This exclusion can lead to an overestimation of the average performance, as the failed or underperforming funds are not included in the analysis. Therefore, survivorship bias tends to overstate the performance of active funds, which include both mutual funds and hedge funds, relative to passive funds. This is because the active funds that have performed poorly and did not survive are not taken into account in the performance measurement.

Choice B is incorrect. Survivorship bias does not understate the performance of surviving funds by including the results of closed or liquidated funds. On the contrary, it overstates the performance because it only considers those funds that have survived and excludes those that have failed or been liquidated, which generally have lower returns.

Choice C is incorrect. The statement that survivorship bias has minimal or no impact on performance measurement because it affects all types of funds equally is false. Survivorship bias can significantly distort performance measurements as it ignores those funds that have been closed down due to poor performance, thereby overstating average fund returns.

Choice D is incorrect. Survivorship bias does not create an accurate representation of fund performance because it only includes successful and surviving funds. This leads to an overestimation of average fund returns as poorly performing, closed or liquidated funds are excluded from calculations.

Things to Remember

- Survivorship bias occurs when only successful and surviving funds are considered in performance measurement, leading to an overestimation of average performance.
- It excludes failed, underperforming, closed, or liquidated funds, skewing the results in favor of the surviving funds.
- Survivorship bias is a common issue in fund performance analysis and needs to be

accounted for in order to have a more accurate assessment.

- It is important to be aware of survivorship bias when comparing the performance of different types of funds, such as active versus passive funds.
-

Q.5348 An investment advisor is educating clients about potential undesirable trading behaviors in mutual funds. Which of the following is a type of undesirable trading behavior in mutual funds that involves traders using acquired information to trade for themselves before trading for their clients or firm?

- A. Late trading.
- B. Market Timing.
- C. Front running.
- D. Directed brokerage.

The correct answer is **C**.

Front running is an unethical and illegal trading practice where a trader uses acquired information to execute trades for themselves before trading for their clients or firm. This behavior can be particularly harmful to clients, as it can result in the execution of their trades at less favorable prices, ultimately reducing their potential returns. It is essential for investment advisors to educate clients about this type of undesirable trading behavior to ensure they understand the risks involved and can make informed decisions when choosing a mutual fund or investment manager.

Choice A is incorrect. Late trading refers to the practice of placing orders after market close but reporting them at the closing net asset value. This does not involve using acquired information for personal gain before executing trades for clients or the firm.

Choice B is incorrect. Market Timing involves frequent buying and selling of shares in a mutual fund to take advantage of inefficiencies in mutual fund pricing, not using acquired information for personal gain before executing trades for clients or the firm.

Choice D is incorrect. Directed brokerage refers to an arrangement where a mutual fund directs its portfolio transactions to a particular broker in return for that broker's agreement to promote the sale of the fund's shares. It does not involve traders using acquired information to trade for themselves before trading for their clients or firm.

Things to Remember

- Late trading is a prohibited practice where investors place orders after the market closes but execute them at the closing price, which is illegal and unethical.

- Market Timing involves frequent buying and selling of shares in a mutual fund to exploit pricing inefficiencies, which can disrupt the fund's investment strategy and harm long-term investors.
 - Directed brokerage is an arrangement where a mutual fund directs its trades to a specific broker in exchange for the broker promoting the sale of the fund's shares, which can create conflicts of interest.
 - Front running is a serious violation of trust where traders use non-public information to trade for their own benefit before executing trades for clients or their firm, leading to unfair advantages and potential harm to clients.
-

Reading 30: Introduction to Derivatives

Q.27 Allan enters into a derivative contract with one of his clients. The client is expected to sell the underlying asset to Allan at the expiration date at price P. Allan wishes to fully hedge his position using derivatives. Which of the following can help him achieve his goal?

- A. Sell a p-strike call.
- B. Purchase a p-strike call and sell a p-strike put.
- C. Sell a p-strike call and buy a p-strike put.
- D. Subscribe to a long forward contract with a forward price P.

The correct answer is **C**.

Selling a p-strike call and buying a p-strike put is the correct strategy for Allan to hedge his position. This is because the client's obligation to sell the underlying asset implies that Allan is in a long-forward position. To hedge this position, Allan needs to create a synthetic short futures contract. This can be achieved by selling a call option and buying a put option at the same strike price. This strategy effectively replicates a short forward position, thus hedging Allan's long-forward position.

For instance, if an investor has entered into a long futures contract to buy crude oil at \$60 a barrel on June 30, 2022, he can create a synthetic short futures contract on oil for the same date by buying a put with a \$60 strike price and selling a call with a \$60 strike price. If the asset price is above the strike price on the expiration date, the investor will be obligated to buy at \$60 under the futures contract and sell at \$60 to the short position in the call, resulting in zero loss. If the asset price at expiration is below the strike price, the investor will be obligated to buy at \$60 under the futures contract and will want to exercise the put option and sell at \$60, again resulting in no loss. Thus, the long futures position combined with the synthetic short forward will hedge the contract and result in no loss for the investor, regardless of the direction the price of the asset takes.

Choice A is incorrect. Selling a p-strike call would not hedge Allan's position completely. This strategy would only limit his potential profit if the price of the underlying asset rises above P, but it does not protect him from losses if the price falls below P.

Choice B is incorrect. Purchasing a p-strike call and selling a p-strike put would create a synthetic long position in the underlying asset, which is not what Allan wants. He already has an obligation to buy the asset at price P, so this strategy would double his exposure instead of hedging it.

Choice D is incorrect. Subscribing to a long forward contract with a forward price P means that Allan agrees to buy the underlying asset at price P in future, which again increases his exposure rather than hedging it as he already has an obligation to buy under his existing contract.

Things to Remember

- **Long Forward Position:** Represents an obligation to buy an asset at a predetermined price in the future. It indicates a commitment to acquire the underlying asset.
 - **Synthetic Short Futures Contract:** Created by selling a call option and buying a put option at the same strike price. This setup mirrors the payoff of a short position in a futures contract.
 - **Short Forward Position:** Involves a commitment to sell an asset at a specified price in the future. This position is beneficial if the price of the asset is expected to decrease.
 - **Call Option:** A financial derivative that gives the buyer the right, but not the obligation, to buy an asset at a specified price within a specific time frame.
 - **Put Option:** A financial derivative that gives the buyer the right, but not the obligation, to sell an asset at a specified price within a specific time frame.
 - **Long Futures Contract:** An agreement to buy an asset at a future date at a price agreed upon today. This contract obligates the holder to purchase the asset.
-

Q.28 Matthew enters into a derivative position with one of his real estate customers. Under the terms of the contract, the customer is obligated to sell the underlying asset to Matthew if the spot price at the expiration is more than P. Matthew, on the other hand, has the right to sell the underlying asset to the customer if the spot price at expiration is less than P. Which of the following describes Matthew's position?

- A. Matthew enters into a short forward contract.
- B. Matthew enters into a long forward contract.
- C. Matthew purchases a P-strike call and a P-strike put.
- D. Matthew purchases a P-strike call and sells a P-strike put.

The correct answer is **C**.

Matthew purchases a P-strike call and a P-strike put. In the context of derivative contracts, a call option gives the holder the right, but not the obligation, to buy an asset at a specified price within a specific time period. On the other hand, a put option gives the holder the right, but not the obligation, to sell an asset at a specified price within a specific time period. In this scenario, Matthew has the right to buy the underlying asset from the client if the spot price at expiration is

more than P , which is characteristic of a call option. Similarly, Matthew has the right to sell the underlying asset to the client if the spot price at expiration is less than P , which is characteristic of a put option. Therefore, Matthew's position can be described as purchasing a P -strike call and a P -strike put.

Choice A is incorrect. A short forward contract would obligate Matthew to sell the underlying asset at a predetermined price, P , regardless of the spot price at expiration. This does not match the conditions of the derivative contract described in the question where Matthew has rights but not obligations based on different scenarios.

Choice B is incorrect. In a long forward contract, Matthew would be obligated to buy the underlying asset at a predetermined price, P , irrespective of its spot price at expiration. This contradicts with the scenario given in which Matthew has options depending on whether or not the spot price exceeds or falls below P .

Choice D is incorrect. If Matthew purchases a P -strike call and sells a P -strike put, he would have an obligation to buy if prices fall below P (due to selling put) and right to buy if prices rise above P (due to buying call). This does not align with conditions mentioned in question where he has right but no obligation based on different scenarios.

Things to Remember

- Call Option: Gives the holder the right, but not the obligation, to buy an asset at a specified price within a specific time period.
 - Put Option: Gives the holder the right, but not the obligation, to sell an asset at a specified price within a specific time period.
 - Long Forward Contract: Obligates the holder to buy the underlying asset at a predetermined price, regardless of its spot price at expiration.
 - Short Forward Contract: Obligates the holder to sell the underlying asset at a predetermined price, regardless of the spot price at expiration.
-

Q.589 David Dillion, head of the treasury department of Dutch Monks Corp., entered into a derivative contract to purchase ₺350 million (Turkish lira) 3-month forwards from a Lirika Bank 3-month forward exchange rate of ₺3.9 per euro. Which of the following correctly describes Lirika Bank's position on the euro?

- A. Short forward contract.
- B. Long forward contract.
- C. Short futures contract.

D. Long futures contract.

The correct answer is **B**.

Dutch Monks Corp. entered into a derivative contract with Lirika Bank to purchase Turkish Lira (TRY) 350 million in 3-month forwards from Lirika Bank at a rate of TRY 3.9 per Euro (EUR). This means that Dutch Monks Corp. will exchange EUR for TRY in the future, while Lirika Bank will exchange TRY for EUR. Therefore, Dutch Monks Corp. (buyer) is in a long position on the Turkish Lira (TRY) and a short position on the Euro (EUR) because they will buy TRY and sell EUR in the future. On the other hand, Lirika Bank (seller) is in a short position on the Turkish Lira (TRY) and a long position on the Euro (EUR) because they will sell TRY and buy EUR in the future.

Therefore,

- Dutch Monks Corp. is in a short position on the Euro (EUR) and a long position on the Turkish Lira (TRY).
- Lirika Bank is in a long position on the Euro (EUR) and a short position on the Turkish Lira (TRY).

Choice A is incorrect. Lirika Bank does not hold a short forward contract on the euro. In a short forward contract, the holder agrees to sell an asset at a specified future date for a price agreed upon today. However, in this scenario, Lirika Bank has agreed to sell Turkish lira and buy euros, which means it holds a long position on the euro.

Choice C is incorrect. The bank does not hold a short futures contract either. A short futures contract would mean that the bank agrees to sell an asset (in this case euros) at a future date for an agreed-upon price today. But as explained above, Lirika Bank has entered into a forward contract where it will be buying euros and not selling them.

Choice D is incorrect. The bank does not have any futures contracts in this scenario; it only has forwards contracts with Dutch Monks Corp., so any option involving futures contracts can be ruled out.

Things to Remember

- Forward contracts are agreements between two parties to buy or sell an asset at a specified future date for a price agreed upon today.
- In a forward contract, the party agreeing to buy the asset in the future assumes a long position, while the party agreeing to sell the asset in the future assumes a short position.
- Long forward contracts benefit from an increase in the price of the underlying asset,

while short forward contracts benefit from a decrease in the price of the underlying asset.

- Forward contracts are customized agreements and are traded over-the-counter (OTC), unlike futures contracts which are standardized and traded on exchanges.
 - Financial institutions often use forward contracts to hedge against currency risk, interest rate risk, or commodity price risk.
-

Q.590 A trader at Prime Investments entered into a standardized derivatives contract to purchase one lot (or 100 troy ounces) of gold at the price of \$1,200/ounce and take delivery 3 months from now. Determine the appropriate position of the trader in the derivatives contract.

- A. A long gold futures contract.
- B. A long dollar futures contract.
- C. A long gold forward contract.
- D. A short dollar forward contract.

The correct answer is **A**.

The trader is in a long gold futures contract. A futures contract is a legal agreement to buy or sell a particular commodity or asset at a predetermined price at a specified time in the future. The trader is said to be 'long' on the futures contract because they have agreed to buy the underlying asset in the future. In this case, the underlying asset is gold. The trader's position is a long position because they expect the price of gold to rise in the future. If the price of gold rises above the agreed price of \$1,200 per ounce, the trader will make a profit. If the price of gold falls below the agreed price, the trader will make a loss. The futures contract is standardized in terms of quantity and delivery date, which is why it is a futures contract and not a forward contract. The trader does not know the counterparty of the derivatives contract, which is another characteristic of futures contracts. In contrast, in forward contracts, both parties know each other.

Choice B is incorrect. A long dollar futures contract would mean the trader expects the value of the dollar to increase. However, in this scenario, the trader is buying gold, not dollars.

Choice C is incorrect. While a long gold forward contract also involves an agreement to buy gold at a future date, it differs from a futures contract in terms of standardization and trading location. Futures contracts are standardized and traded on an exchange while forward contracts are private agreements between two parties and can be customized according to their needs.

Choice D is incorrect. A short dollar forward contract would imply that the trader expects the

value of the dollar to decrease in future. This does not align with our scenario where he has agreed to buy gold at a predetermined price.

Things to Remember

- Forward Contracts: Private agreements between two parties to buy or sell an asset at a specified price on a future date. They are customizable and not traded on exchanges.
 - Futures Contracts: Standardized agreements to buy or sell an asset at a predetermined price on a specified future date. Traded on exchanges with margin requirements.
 - Long Position: A position where an investor expects the price of an asset to rise in the future.
 - Short Position: A position where an investor expects the price of an asset to fall in the future.
 - Delivery Date: The date on which the underlying asset must be delivered in a futures or forward contract.
-

Q.591 Which of the following equations accurately demonstrates the payoff of the short position holder in a forwards contract?

- A. $K - S_T$
- B. $S_T - K$
- C. $\max(0, X - S_T)$
- D. $\max(0, S_T - X)$

The correct answer is **A**.

The payoff for the holder of a short position in a forward contract is accurately represented by the equation $K - S_T$. In this equation, K represents the delivery price, which is the price agreed upon at the inception of the contract for the delivery of the underlying asset in the future. S_T represents the spot price of the underlying asset at the time of delivery. The difference between these two prices ($K - S_T$) represents the payoff for the short position holder. If the delivery price is higher than the spot price at the time of delivery ($K > S_T$), the short position holder will make a profit. Conversely, if the spot price at the time of delivery is higher than the delivery price ($S_T > K$), the short position holder will incur a loss. This is because the short position holder is obligated to sell the underlying asset at the delivery price, which could be lower than the current

market price.

Choice B is incorrect. The expression $S_T - K$ represents the payoff for a long position in a forward contract, not a short position. In this case, the holder of the contract agrees to buy an asset at a predetermined price (K) and would benefit if the spot price at maturity (S_T) is higher than this agreed price.

Choice C is incorrect. The expression $\max(0, X - S_T)$ represents the payoff for a long position in a put option, not a short position in a forward contract. Here, X denotes the strike price of the option and S_T denotes the spot price at maturity. If $X > S_T$ (i.e., if it's more profitable to exercise than to sell on market), then profit equals $X - S_T$; otherwise, it's more profitable not to exercise and profit equals 0.

Choice D is incorrect. The expression $\max(0, S_T - X)$ represents the payoff for a long position in call option rather than that of short forward contract holder. In this case, if $S_T > X$, then profit equals $S_T - X$; otherwise, it's more profitable not to exercise and profit equals 0.

Things to Remember

- Forwards contracts are agreements between two parties to buy or sell an asset at a specified price on a future date.
- Short position holder in a forwards contract is obligated to sell the underlying asset at the delivery price.
- Payoff for the short position holder in a forwards contract is calculated as the difference between the delivery price and the spot price at the time of delivery.
- Short position holders in forwards contracts are exposed to the risk of price movements in the underlying asset.
- Forwards contracts are traded over-the-counter (OTC) and are customizable in terms of contract size, expiration date, and delivery terms.

Q.592 A number of derivatives are used to hedge the risk or earn a profit with speculation and arbitrage strategies. Forwards, futures and options are different from each other in terms of their properties. Which of the following statement correctly differentiates forward, futures, and options?

- A. Forward contracts and options are different from futures, as it takes a certain cost to enter into a forward contract.
- B. Options and futures are different from forwards contracts as they give an option or

futures contract holder the right, but not the obligation, to exercise the contract.

C. Forwards and futures are different from options because the holder of the forwards and futures are obligated to buy or sell the underlying.

D. Forward contracts and options are different from futures because forwards and options trade on OTC markets.

The correct answer is **C**.

Forwards and futures are different from options because the holder of the forwards and futures are obligated to buy or sell the underlying. This statement accurately captures the fundamental difference between these types of derivatives. In a forward or futures contract, the buyer and seller are obligated to buy or sell the underlying asset at a predetermined price on a specified future date. This obligation exists regardless of the market price of the underlying asset at the time of contract execution. This characteristic of forwards and futures contracts exposes the contract holders to a significant amount of risk, especially if the market price of the underlying asset moves unfavorably.

On the other hand, options contracts provide the holder with the right, but not the obligation, to buy (in the case of a call option) or sell (in the case of a put option) the underlying asset at a predetermined price before or on the expiration date. This means that the holder of an options contract can choose to exercise the option if it is profitable to do so, or let the option expire worthless if it is not. This flexibility is a key feature of options contracts and is what differentiates them from forwards and futures contracts.

Choice A is incorrect. It is not accurate to say that it takes a certain cost to enter into a forward contract. Forward contracts do not require an upfront payment or premium like options do. They are agreements between two parties to buy or sell an asset at a specified future date for a price agreed upon today.

Choice B is incorrect. While it's true that options give the holder the right, but not the obligation, to exercise the contract, futures contracts also obligate the holder to buy or sell the underlying asset at expiration, similar to forwards. Therefore, this statement does not accurately distinguish between these derivatives.

Choice D is incorrect. Although many forward contracts and options do trade on over-the-counter (OTC) markets, this isn't exclusively true for all such instruments and hence doesn't serve as a distinguishing factor among them. For instance, some types of futures can also be traded OTC.

Things to Remember

- Derivatives are financial instruments whose value is derived from an underlying asset, index, or rate.
- Forwards are customized contracts between two parties to buy or sell an asset at a specified future date for a price agreed upon today.

- Futures are standardized contracts traded on exchanges that obligate the buyer to purchase and the seller to sell the underlying asset at a specified price on a future date.
 - Options give the holder the right, but not the obligation, to buy or sell the underlying asset at a predetermined price before or on the expiration date.
 - OTC markets refer to decentralized markets where trading occurs directly between two parties without a centralized exchange.
 - Speculation involves taking on risk in the hope of making a profit, while hedging involves using derivatives to offset or reduce existing risks.
-

Q.593 Kapil Kumar is an individual investor who invests a portion of his salary in stocks and derivatives at the beginning of every month. Kumar is interested in the stocks of Geneva Computers Inc., which are currently trading at the price of \$14. However, he believes the stock will trade above \$17 at the beginning of next month. If Kapil is interested in entering into an options contract that gives him the right to take exposure in the stock at \$17, then suggest the most appropriate option position for Kumar.

- A. A long call option with a strike price of \$14.
- B. A short call option with a strike price of \$17.
- C. A long put option with a strike price of \$17.
- D. A Short put option with a strike price of \$14.

The correct answer is **A**.

A long call option with a strike price of \$14 is the most suitable position for Kumar. This is because a long call option gives the buyer (in this case, Kumar) the right, but not the obligation, to buy the underlying asset (in this case, the stock of Geneva Computers Inc.) at the strike price before the option expires. If Kumar's prediction is correct and the stock price rises above \$17, he can exercise his option to buy the stock at \$14 and then sell it at the higher market price, making a profit. The payoff in this scenario would be the difference between the market price and the strike price, which is ($\$17 - \14). The profit would be this payoff minus the premium paid for the option, which is ($\$17 - \$14 - \text{premium paid}$). Therefore, this option position aligns with Kumar's investment strategy and market prediction.

Choice B is incorrect. A short call option would obligate Kumar to sell the stock at \$17 if the option is exercised. This would not be beneficial for him if he anticipates that the stock price will

rise above \$17, as he would miss out on potential profits from selling at a higher market price.

Choice C is incorrect. A long put option gives the holder the right to sell a stock at a specified price within a certain period of time. Since Kumar expects the stock price to rise, not fall, this strategy does not align with his expectations and hence it's not suitable for him.

Choice D is incorrect. A short put option obligates Kumar to buy more of Geneva Computers Inc.'s stocks if they fall below \$14, which contradicts his bullish outlook on the company's shares.

Things to Remember

- Long call option: Gives the buyer the right to buy the underlying asset at a specified price within a certain time frame.
 - Short call option: Obligates the seller to sell the underlying asset at a specified price if the option is exercised.
 - Long put option: Gives the holder the right to sell the underlying asset at a specified price within a certain time frame.
 - Short put option: Obligates the seller to buy the underlying asset at a specified price if the option is exercised.
 - Strike price: The price at which the option holder can buy or sell the underlying asset.
 - Premium: The price paid by the option buyer to the seller for the right to buy or sell the underlying asset.
-

Q.594 Consider a European call option for 100 shares of Tesla, Inc., whose strike price is \$870 per share and which matures 12 months from now. What does this option entitle you to do?

- A. Between now and 12 months from now, you are entitled to make a phone call to the European headquarters of Tesla, Inc., to inquire about the value of 100 shares.
- B. Between now and 12 months from now, you have the right, but not the obligation to purchase 100 shares of Tesla, Inc., for \$870 per share.
- C. At the maturity date, that is 12 months from now, you have the right, but not the obligation to sell 100 shares of Tesla, Inc., for \$870 per share
- D. At the maturity date, that is 12 months from now, you have the right, but not the obligation to purchase 100 shares of Tesla, Inc., for \$870 per share.

The correct answer is **D**.

A European call option gives the holder the right, but not the obligation, to purchase a specified amount of an underlying asset at a specified price (the strike price) within a specified time period. In this case, the underlying asset is 100 shares of Tesla, Inc., and the strike price is \$870 per share. The specified time period is 12 months from now, which is when the option matures. Therefore, at the maturity date, the holder of this European call option has the right, but not the obligation, to purchase 100 shares of Tesla, Inc., for \$870 per share. This is the fundamental principle of a call option. The holder can decide to exercise the option and purchase the shares if the market price of the shares is higher than the strike price at the time of maturity. If the market price is lower, the holder can choose not to exercise the option, thereby limiting their loss to the premium paid for the option.

Choice A is incorrect. This choice misinterprets the concept of a call option. A call option does not entitle the holder to make a phone call to inquire about the value of shares. It provides the right, but not an obligation, to buy an underlying asset at a specified price within a specific time period.

Choice B is incorrect. This choice incorrectly suggests that European options can be exercised at any time before expiration. However, unlike American options, European options can only be exercised at expiration.

Choice C is incorrect. This choice describes a put option rather than a call option. A put option gives the holder the right but not an obligation to sell an underlying asset at a specified price within a specific time period.

Things to Remember

- European call option gives the holder the right, but not the obligation, to purchase a specified amount of an underlying asset at a specified price within a specified time period.
- Call options provide the opportunity for potential profit if the market price of the underlying asset rises above the strike price.
- European options can only be exercised at expiration, unlike American options which can be exercised at any time before expiration.
- Call options are used by investors to speculate on the price movement of an underlying asset or to hedge existing positions.
- Understanding the difference between call options and put options is crucial in options trading. Put options give the holder the right, but not the obligation, to sell an

underlying asset at a specified price within a specific time period.

Q.595 In derivatives trading, different financial instruments impose various obligations and rights on the parties involved. Which of the following options requires a party to purchase the underlying asset at a specific date?

- A. European short call option.
- B. American short call option.
- C. American short put option.
- D. European short put option.

The correct answer is **D**.

A European short put option is a type of derivative contract that obligates the seller (writer) of the option to buy the underlying asset at a predetermined price (strike price) on a specific date (expiration date). This obligation arises if the buyer (holder) of the option decides to exercise the option. The term 'short' refers to the act of selling the option, and 'European' denotes that the option can only be exercised on the expiration date, not before. This is in contrast to 'American' options, which can be exercised at any time up to the expiration date. The seller of a European short put option is essentially betting that the price of the underlying asset will be higher than the strike price on the expiration date. If this is the case, the option will not be exercised, and the seller will profit from the premium received for selling the option. However, if the price of the underlying asset is lower than the strike price on the expiration date, the buyer will exercise the option, and the seller will be obligated to buy the asset at the higher strike price, resulting in a loss.

Choice A is incorrect. A European short call option does not obligate the party to buy the underlying asset. Instead, it gives the holder of the option (the buyer) the right, but not obligation, to buy an underlying asset at a specified price on a specific date.

Choice B is incorrect. An American short call option also does not obligate a party to buy an underlying asset. Similar to a European short call option, it provides the holder with the right but not obligation to purchase an underlying asset at any time up until expiration.

Choice C is incorrect. An American short put option obligates a party (the seller of put) to buy an underlying asset if exercised by buyer but it can be exercised anytime before expiration and not necessarily on specific date.

Things to Remember

- European options can only be exercised on the expiration date, while American options can be exercised at any time up to the expiration date.

- A short call option gives the holder the right, but not the obligation, to buy the underlying asset at a specified price on a specific date.
 - A short put option obligates the seller to buy the underlying asset if exercised by the buyer, but it can be exercised at any time before expiration.
 - The strike price is the price at which the underlying asset will be bought or sold if the option is exercised.
 - The expiration date is the date on which the option contract expires and the rights and obligations of the parties are settled.
-

Q.597 A trader sold five July put options, each at \$7.60, pledging to buy 500 shares of Galaxy Carpets Co. at a price of \$13.50 per share. If at maturity of the contract, Galaxy Carpet's stock is trading at \$9.30 per share, then which of the following statements accurately describes the net cash flow of the transaction?

- A. The trader profited \$3,800 from the transaction.
- B. The trader lost \$2,100 from the transaction.
- C. The trader profited \$2,100 from the transaction.
- D. The trader profited \$1,700 from the transaction.

The correct answer is **D**.

At initiation, the trader gets $5 \times 7.6 \times 100 = \$3,800$ from the sale of the put options. This is because for stock options, the premium is quoted as a dollar amount per share, and most contracts represent the commitment of 100 shares.

At maturity, the trader made a loss of $(\$9.3 - \$13.5) \times 500 = \$2,100$

Therefore, the net cash flow for the trader is $+\$1,700$ ($3,800 - 2,100$).

Note: On the derivatives market, options are quoted in per-share prices but only sold in 100 share lots. In other words, each put has 100 shares. In this case, for example, the put option is quoted at \$7.6, but the buyer pays $\$7.6 \times 100 = \760 per put option. For 5 puts, that's $\$760 \times 5 = \$3,800$

Things to Remember

- Put options give the holder the right, but not the obligation, to sell a specified amount

of an underlying asset at a set price within a specific time frame.

- When selling put options, the seller receives a premium from the buyer and is obligated to buy the underlying asset at the strike price if the buyer chooses to exercise the option.
 - The net cash flow of an options transaction is calculated by considering the premium received at initiation and any gains or losses at maturity.
 - Options contracts typically represent the commitment of 100 shares of the underlying asset.
 - It's important to understand the impact of stock price movements on options positions to assess potential profits or losses accurately.
-

Q.598 Steve Hellmuth, a former derivatives trader, runs an online derivatives investment and trading tutorial portal. Every weekend he educates hundreds of subscribers through weekly webinars. In his last webinar, he presented the following properties of each trader type:

I. Hedgers use derivatives to guard against the risks related to future movements in market prices of underlying variables.

II. Arbitrageurs use derivatives to bet on the direction of the market of underlying variables.

III. Speculators use derivatives to take offsetting positions in two or more instruments and markets to earn a profit.

Which type of derivatives trader did Hellmuth define inappropriately?

- A. Speculators only.
- B. Hedgers and speculators.
- C. Arbitrageurs and speculator.
- D. Hedgers and arbitrageurs.

The correct answer is **C**.

Steve Hellmuth incorrectly defined the characteristics of Arbitrageurs and Speculators.

I. Hedgers use derivatives to protect against the risks associated with future fluctuations in the market prices of underlying variables. This definition is correct as hedgers use derivatives as a risk management tool to hedge against potential losses that could result from price changes in the underlying asset.

II. Speculators, not arbitrageurs, use derivatives to predict the direction of the market of underlying variables. Speculators are market participants who try to profit from market volatility. They take on risk, betting on future price movements in the hope of making gains. They do not use derivatives to hedge risk but to speculate on the direction of prices.

III. Arbitrageurs, not speculators, use derivatives to establish offsetting positions in multiple instruments and markets to generate profits. Arbitrageurs seek to exploit price discrepancies between related financial instruments. They take offsetting positions in two or more instruments or markets to lock in a risk-free profit from the price difference. Therefore, the definitions of Arbitrageurs and Speculators were swapped in Hellmuth's presentation.

Choice A is incorrect. Speculators were correctly defined by Hellmuth in his presentation. Speculators do use derivatives to establish positions in the market, but not necessarily offsetting ones, with the aim of profiting from future price changes.

Choice B is incorrect. Hedgers were also correctly defined by Hellmuth. They use derivatives to protect against risks associated with future fluctuations in market prices of underlying variables.

Choice D is incorrect. Hedgers were accurately described by Hellmuth as they utilize derivatives to hedge against potential risks due to price fluctuations in the market.

Things to Remember

- Derivatives are financial instruments whose value is derived from an underlying asset or group of assets.
- Hedging is a risk management strategy used to offset potential losses in one investment by taking an opposite position in a related security, such as using derivatives.
- Arbitrage is the practice of simultaneously buying and selling an asset to profit from a price difference in different markets or instruments.
- Speculation involves taking on risk in the hope of making a profit from future price movements in the market.
- Derivatives traders can be categorized into different types based on their objectives and strategies.

Q.599 Donald Brown, an investment manager at a pension fund, manages a portfolio of twenty stocks. Brown is long the stocks of Blue Moon Inc., and he believes that the value of the stock of Blue Motors Inc., which forms part of the portfolio, can decrease due to an increase in oil prices. After analyzing the fundamentals of the stock, Brown decides to take a long position in

put options on Blue Motors Inc. stocks. The given transaction appropriately categorizes Donald Brown as a:

- A. Speculator.
- B. Hedger.
- C. Market maker.
- D. Arbitrageur.

The correct answer is **B**.

In the context of financial markets, a hedger is an individual or institution that uses derivative instruments like futures and options to offset the risk of price movements in an underlying asset. In this case, Donald Brown is using put options to hedge against the risk of a decrease in the value of Blue Motors Inc. stocks. Put options give the holder the right, but not the obligation, to sell the underlying asset at a predetermined price within a specified period. By taking a long position in put options, Brown is securing the right to sell Blue Motors Inc. stocks at a fixed price, thereby protecting himself from potential losses if the stock price decreases as he anticipates. This strategy is a classic example of hedging, where the goal is not to make a profit but to reduce the risk of adverse price movements in an underlying asset.

Choice A is incorrect. A speculator is someone who takes on significant risks in the hopes of making substantial profits. In this case, Donald Brown is not taking a speculative position but rather trying to protect his portfolio from potential losses due to anticipated market conditions. Therefore, he cannot be classified as a speculator.

Choice C is incorrect. A market maker provides liquidity to the market by buying and selling securities at publicly quoted prices. Donald Brown's actions do not involve providing liquidity or setting prices for securities; instead, he's managing risk within his portfolio by using options as a hedging tool.

Choice D is incorrect. An arbitrageur exploits price discrepancies in different markets or different forms of the same financial instrument to make risk-free profits. In this scenario, Donald Brown isn't exploiting any price discrepancies but rather protecting his portfolio from potential losses due to rising oil prices affecting Blue Motors Inc.'s stock value.

Things to Remember

- Derivative instruments like futures and options are commonly used for hedging purposes in financial markets.
- A put option gives the holder the right to sell the underlying asset at a predetermined price within a specified period.
- Hedging is a risk management strategy aimed at reducing the impact of adverse price

movements on a portfolio.

- Speculators take on significant risks in the hopes of making substantial profits in the financial markets.
 - Market makers provide liquidity to the market by buying and selling securities at publicly quoted prices.
 - Arbitrageurs exploit price discrepancies in different markets to make risk-free profits.
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Q.601 Samuel Simpson is a commodities trader at one of the largest asset management firm in Abu Dhabi. He believes that due to a resolution passed by all members of the OPEC committee to cut the supply of oil, the prices of oil are expected to increase. In order to capitalize on his vision, Simpson purchased 2,000 lots of crude oil futures for the price of \$45.6 per barrel. Which type of derivatives trader is Samuel Simpson?

- A. Speculator.
- B. Hedger.
- C. Option trader.
- D. Arbitrageur.

The correct answer is **A**.

Samuel Simpson is indeed a speculator. Speculators are individuals or entities who attempt to profit from market inefficiencies or price fluctuations. In this case, Simpson is speculating that the price of oil will increase due to the decision by the OPEC committee to cut oil supply. By purchasing crude oil futures, he is betting on the future price of oil. If his prediction is correct and the price of oil increases, he stands to make a profit. Speculators play a crucial role in the financial markets by providing liquidity and bearing the risk that hedgers seek to avoid. However, speculation involves a high level of risk as the markets may not move in the direction anticipated by the speculator.

Choice B is incorrect. A hedger is someone who enters into a derivative contract to reduce or eliminate the risk associated with price fluctuations in an underlying asset. In this case, Samuel Simpson is not trying to mitigate risk but rather capitalize on anticipated price movements, which aligns more with the activities of a speculator than a hedger.

Choice C is incorrect. An option trader buys and sells options contracts that give them the right, but not obligation, to buy or sell an underlying asset at a predetermined price within a specific time period. Samuel Simpson's activity involves futures contracts and not options; hence he cannot be categorized as an option trader.

Choice D is incorrect. An arbitrageur takes advantage of price discrepancies in different markets for the same asset to make risk-free profits. Samuel Simpson's actions do not involve exploiting such market inefficiencies; instead, he's making speculative trades based on expected future oil prices.

Things to Remember

- Speculators aim to profit from market inefficiencies or price fluctuations by taking on risk in the hopes of making a profit.
 - Hedgers use derivative contracts to reduce or eliminate the risk associated with price fluctuations in an underlying asset.
 - Option traders buy and sell options contracts that give them the right, but not obligation, to buy or sell an underlying asset at a predetermined price within a specific time period.
 - Arbitrageurs take advantage of price discrepancies in different markets for the same asset to make risk-free profits.
 - Speculation involves a high level of risk as market movements may not align with the speculator's predictions.
-

Q.602 Which of the following positions offers the most effective protective strategy against declines in the underlying asset's price while allowing for some potential upside profit if the price remains stable or increases?

- A. Writing (shorting) a naked call option on the asset.
- B. Entering into a collar strategy by purchasing a put option and writing a call option on the asset.
- C. Purchasing a futures contract on the asset.
- D. Writing (shorting) a covered call option on the asset.

The correct answer is **B**.

A collar strategy provides protection against declines in the price of the underlying asset. By purchasing a put option, the investor has a right to sell the asset at a pre-determined price, offering protection against price declines. Writing a call option provides premium income and

caps the potential profit if the asset's price rises significantly. This strategy provides both downside protection and potential for some upside gain.

A is incorrect. Writing a naked call option exposes the writer to potentially unlimited losses if the underlying asset's price rises significantly. However, the writer benefits if the price remains stable or declines, as they can keep the option premium.

C is incorrect. Purchasing a futures contract exposes the holder to potential losses if the price of the underlying asset declines. The holder benefits if the price increases.

D is incorrect. Writing a covered call option means holding the underlying asset and writing a call option on it. If the price of the asset increases beyond the strike price of the call option, the profit is capped at that strike price, minus the premium received. If the price declines, losses on the asset are partially offset by the option premium.

Things to Remember

- A collar strategy involves buying a put option to protect against downside risk and writing a call option to generate income and limit upside potential.
 - Writing a naked call option exposes the writer to unlimited losses if the underlying asset's price rises significantly.
 - Purchasing a futures contract exposes the holder to potential losses if the price of the underlying asset declines.
 - Writing a covered call option involves holding the underlying asset and writing a call option on it to generate income and limit upside potential.
-

Q.603 Assume stock K trades on the New York Stock Exchange (NYSE) and the London Stock Exchange (LSE). The stock currently trades on the NYSE for \$50 and on the LSE for £39. Given the current exchange rate is 1.2658 \$/£, determine the amount of arbitrage profit that could possibly be earned.

- A. \$0.82
- B. \$1.25
- C. \$0.63
- D. Zero: there's no opportunity for arbitrage

The correct answer is C.

Value in dollars of K on LSE = £39 x 1.2658 \$/£ = \$49.37

Arbitrageur could purchase K on LSE for \$49.37 and sell on NYSE for \$50.

Profit per share = \$50 - \$49.37 = \$0.63

Things to Remember

- Arbitrage involves taking advantage of price differences of the same asset in different markets to make a profit.
- Exchange rates play a crucial role in determining arbitrage opportunities between different markets.
- Efficient markets should theoretically eliminate arbitrage opportunities quickly due to the actions of arbitrageurs.
- Arbitrage opportunities can arise due to market inefficiencies, delays in information dissemination, or regulatory differences.
- Arbitrageurs need to consider transaction costs, currency risks, and other factors when engaging in arbitrage strategies.

Q.3512 In the derivatives market, the structure and trading venues for financial instruments can significantly affect their characteristics and risks. Relative to over-the-counter (OTC) derivatives, exchange-traded derivatives are:

- A. Traded in larger lot sizes.

- B. Transacted through a process that is verified by a central exchange.
- C. Traded in markets where there is zero potential to earn arbitrage profits.
- D. Flexible and easily customizable.

The correct answer is **B**.

Exchange-traded derivatives are indeed transacted through a process that is verified by a central exchange. This is one of the key distinguishing features of exchange-traded derivatives. The central exchange acts as an intermediary between the buyer and the seller, ensuring the integrity of the transaction. This process involves verifying the execution of the transaction and recording the identities of the participants. The central exchange also facilitates the creation of clearing and settlement operations, which further enhances the transparency and security of the transaction. This is in contrast to OTC derivatives, which are traded directly between two parties without the involvement of an intermediary. As a result, OTC derivatives are typically associated with a higher level of counterparty risk compared to exchange-traded derivatives.

Choice A is incorrect. While it's true that exchange-traded derivatives are often traded in larger lot sizes than OTC derivatives, this is not a defining characteristic. The size of the lots can vary greatly depending on the specific derivative and market conditions.

Choice C is incorrect. It's not accurate to say that there is zero potential to earn arbitrage profits in markets where exchange-traded derivatives are traded. While these markets are typically more efficient and thus offer fewer arbitrage opportunities than less regulated OTC markets, it's still possible for savvy traders to identify and exploit pricing inefficiencies.

Choice D is incorrect. Exchange-traded derivatives are actually less flexible and customizable than OTC derivatives. This is because they're standardized contracts with fixed terms and conditions, whereas OTC derivatives can be tailored to meet the specific needs of the parties involved.

Things to Remember

- OTC derivatives are traded directly between two parties without the involvement of an intermediary, leading to higher counterparty risk compared to exchange-traded derivatives.
- Exchange-traded derivatives are standardized contracts with fixed terms and conditions, making them less flexible and customizable than OTC derivatives.
- Central exchanges in exchange-traded derivatives verify transactions, record participant identities, and facilitate clearing and settlement operations, enhancing transparency and security.
- Efficiency in exchange-traded derivative markets reduces but does not eliminate the

potential for earning arbitrage profits compared to OTC markets.

Q.3515 If Michael Emery takes a long position in copper futures, which of the following parties will take the opposite position to the futures contract?

- A. Another investor/trader.
- B. The clearinghouse.
- C. A large commercial bank.
- D. The commodity exchange.

The correct answer is **B**.

The clearinghouse acts as the counterparty to every transaction in the futures markets. This means that when an investor like Michael Emery takes a long position in a futures contract, the clearinghouse takes the short position, effectively becoming the opposite party to the contract. The role of the clearinghouse is crucial in providing stability and reducing risk in the futures markets. It ensures that the terms of the contracts are fulfilled by both parties, thereby eliminating counterparty risk. The clearinghouse achieves this by stepping in between the two parties involved in the contract, agreeing to buy from the seller and sell to the buyer. This mechanism ensures that even if one party defaults, the other party's interests are protected. The clearinghouse, therefore, plays a pivotal role in maintaining the integrity and smooth functioning of the futures markets.

Choice A is incorrect. While another investor or trader could theoretically take the counter position in a futures contract, it is not typically the case in practice. Futures contracts are standardized and traded on an exchange, where the clearinghouse acts as a counterparty to all trades. This ensures that the risk of default is minimized.

Choice C is incorrect. Large commercial banks may participate in futures markets, but they do not typically act as counterparties to individual investors or traders like Michael Emery. They might use futures for hedging their own risks or for proprietary trading, but they do not usually take on the role of a counterparty in such transactions.

Choice D is incorrect. Commodity exchange provides the platform for trading futures contracts but does not take positions in the contracts. The exchange facilitates the trades and ensures transparency and efficiency in the market.

Things to Remember

- Counterparty risk: The risk that one party in a contract will default on their obligations.
- Clearinghouse: An entity that acts as an intermediary in futures markets, ensuring the fulfillment of contracts and reducing counterparty risk.

- Long position: A position in which an investor expects the price of an asset to increase.
 - Short position: A position in which an investor expects the price of an asset to decrease.
 - Standardization of futures contracts: Futures contracts have standardized terms and are traded on exchanges to ensure liquidity and transparency.
-

Q.3516 The everyday process of adjusting the margin for the gains and losses on the value of futures contracts is known as:

- A. Marking to market.
- B. Value adjusting.
- C. Clearing.
- D. Initial margining.

The correct answer is **A**.

Marking to market is the daily process of settling gains and losses on futures contracts. This process involves adjusting the margin account to reflect the daily fluctuations in the value of the futures contracts. The margin account is a type of collateral that is used to cover potential losses on futures contracts. When the value of the futures contracts changes due to changes in the prices of the underlying assets, the margin account is adjusted accordingly. This process ensures that the margin account always reflects the current value of the futures contracts, thereby protecting both the buyer and the seller from potential losses. Marking to market is a critical aspect of risk management in futures trading.

Choice B is incorrect. Value adjusting is not a term used in futures trading to describe the daily adjustment process of margin accounts. It may be confused with 'marking to market', but it does not accurately represent the procedure of adjusting for gains and losses due to fluctuations in contract values.

Choice C is incorrect. Clearing refers to the process of settling trades, which includes transferring funds from buyer's account to seller's account and vice versa, but it does not specifically refer to the daily adjustment of margins based on contract value fluctuations.

Choice D is incorrect. Initial margining refers to the initial deposit required by a broker when a futures contract is first entered into. This term does not describe the daily adjustment process that accounts for gains and losses due to changes in futures contracts' values.

Things to Remember

- Marking to market is a crucial risk management tool in futures trading to ensure that margin accounts reflect the current value of futures contracts.
 - Margin accounts act as collateral to cover potential losses on futures contracts.
 - Initial margining is the initial deposit required by a broker when entering into a futures contract.
 - Clearing involves the settlement of trades by transferring funds between buyer and seller accounts.
 - Value adjusting is not a term used in futures trading for the daily adjustment process of margin accounts.
-

Q.3517 Which of the following best describes a forward commitment?

- A. A forward commitment is a legally binding promise to perform some action in the future.
- B. A forward commitment is a claim (to a payoff) that depends on a particular event.
- C. A forward commitment is a contingent claim that depends on a stock price at some future date.
- D. A forward commitment is a derivative contract through which two parties exchange the cash flows or liabilities from two different financial instruments.

The correct answer is **A**.

A forward commitment is a legally binding promise to perform some action in the future. Forward commitments include forward contracts, futures contracts, and swaps.

A forward commitment can also be defined as a contract entered into between two parties that require both parties to transact in the future at a pre-specified price known as the forward price. The parties and the identity and quantity of the underlying are specified as well as the date of the future transaction (expiration) and the nature of the settlement. The parties have to transact; they are obligated to do so. In the event of non-performance, because of the obligation of the forward contract, a legal remedy is possible to enforce the obligation.

The payoff profiles of forward commitments are linear in nature and move upwards or downwards in direct relation to the price of the underlying asset. Forward commitments include futures contracts and forwards contracts.

Options B and C are incorrect. A forward commitment and a contingent claim are two different things. A **forward commitment** creates an obligation between the transacting parties whereas a contingent **claim** creates the right but not the obligation to transact at a future date.

Option D is the definition of a swap.

Things to Remember

- Forward commitments are commonly used in risk management to hedge against future price fluctuations.
 - Forward contracts are traded over-the-counter (OTC) and are customized agreements between two parties.
 - Forward commitments are different from options as they involve an obligation to transact in the future, whereas options provide the right but not the obligation to transact.
 - Forward commitments are widely used in commodities markets to lock in prices for future delivery.
 - Counterparty risk is an important consideration when entering into forward commitments, as there is a risk of default by the other party.
-

Q.3520 Which of the following statements is correct regarding the value of a forward contract to a short party at expiration?

The value of the forward contract is:

- A. Valueless.
- B. Equal to the value to the long party multiplied by -1.
- C. Positive if the spot price of the underlying exceeds the forward price.
- D. Equal to 1 divided by the value of the long party.

The correct answer is **B**.

The value of a forward contract to a party holding a short position at expiration is indeed equal to the value to the long party multiplied by -1. This is because the short party has agreed to sell

the underlying asset at a predetermined price, known as the forward price. If the spot price of the asset at expiration is higher than the forward price, the short party incurs a loss, while the long party makes a gain. Conversely, if the spot price is lower than the forward price, the short party makes a gain, while the long party incurs a loss. Therefore, the value of the contract to the short party is the negative of the value to the long party.

Choice A is incorrect. The value of a forward contract to the short party at expiration is not valueless. It depends on the difference between the spot price and the forward price at expiration.

Choice C is incorrect. This statement contradicts how a short position in a forward contract works. The value to the short party would be negative if the spot price of the underlying exceeds the forward price, not positive.

Choice D is incorrect. The value of a forward contract to a short party at expiration does not equal 1 divided by the value of long party. This choice seems to confuse some concepts related with financial derivatives and does not accurately describe how this calculation should be made.

Things to Remember

- Forward contracts are agreements between two parties to buy or sell an asset at a specified future date for a price agreed upon today.
 - The party taking a short position in a forward contract agrees to sell the underlying asset at a predetermined price in the future.
 - The value of a forward contract to the short party at expiration is the negative of the value to the long party.
 - Forward contracts are often used for hedging against price fluctuations in the underlying asset.
 - The value of a forward contract is determined by the difference between the spot price of the underlying asset and the forward price at expiration.
-

Q.3522 Which of the following factors differentiates futures contracts from forward contracts?

- A. Futures contracts are cash-settled contracts.
- B. The value of a futures contract is derived from its underlying asset.
- C. Forward contracts require physical assets for settlement, not cash.
- D. Futures contracts trade on regulated markets.

The correct answer is **D**.

Futures contracts are indeed traded on regulated markets. This is one of the key differences between futures and forward contracts. Futures contracts are standardized contracts that are traded on organized exchanges. These exchanges act as intermediaries between the buyer and the seller, and they provide a regulated and transparent environment for trading. The exchanges also provide clearing and settlement services, reducing the risk of default by either party. The standardization of futures contracts includes details such as the quantity and quality of the underlying asset, and the date and location of delivery. This standardization facilitates liquidity and allows the contracts to be traded on the exchange. On the other hand, forward contracts are private agreements between two parties and are not traded on an exchange. They are customized to the needs of the parties involved, and as such, they are not as liquid as futures contracts. The lack of a regulated market for forward contracts also means that they carry a higher counterparty risk compared to futures contracts.

Choice A is incorrect. While it's true that futures contracts can be cash-settled, this is not a distinguishing factor between futures and forwards. Both types of contracts can be settled either in cash or by physical delivery of the underlying asset.

Choice B is incorrect. The statement that the value of a futures contract is derived from its underlying asset applies to both futures and forward contracts, not just futures. Therefore, this does not serve as a distinguishing factor between the two.

Choice C is incorrect. Forward contracts do not necessarily require physical assets for settlement; they can also be cash-settled. Hence, this statement does not accurately distinguish forward contracts from futures contracts.

Things to Remember

- Forward contracts are customizable agreements between two parties, while futures contracts are standardized and traded on exchanges.
- Futures contracts require daily marking-to-market, where gains and losses are settled daily, while forward contracts settle gains and losses at the end of the contract term.
- Futures contracts have margin requirements to ensure performance, while forward contracts do not have such requirements.
- Futures contracts have a high degree of liquidity due to trading on exchanges, while forward contracts may lack liquidity and are more tailored to specific needs.
- Regulated markets for futures contracts provide transparency, price discovery, and risk mitigation mechanisms, which are not present in the over-the-counter market for forward contracts.

Q.3523 In the context of stock market investments, managing and mitigating downside risk is crucial for protecting portfolio value. In order to protect from the downside risk of stock prices, investors should:

- A. Buy put options.
- B. Sell put options.
- C. Buy call options.
- D. Sell call options.

The correct answer is **A**.

Put options are financial contracts that give the option buyer the right, but not the obligation, to sell a specified amount of an underlying security at a specified price within a specified time frame. This is often used as a protective strategy by investors who are concerned about potential losses in the underlying stock. When an investor buys a put option, they are essentially securing the right to sell their stock at a set price, regardless of how far the market price falls. This means that if the stock price does fall, the investor can exercise their put option and sell their stock at the higher, predetermined price, thereby limiting their losses. The cost of this protection is the premium paid for the put option. Therefore, buying put options is a strategy that can effectively protect investors from the downside risk of stock prices.

Choice B is incorrect. Selling put options would not be an appropriate strategy to protect against a decrease in stock prices. When you sell a put option, you are obligated to buy the underlying asset at the strike price if the buyer decides to exercise their right. If stock prices fall, this could result in significant losses as you would be forced to buy the asset at a higher price than its current market value.

Choice C is incorrect. Buying call options would also not provide protection against falling stock prices. A call option gives the holder the right but not obligation to buy an asset at a specified price within a certain period of time. If stock prices fall, this option becomes worthless as it would be more cost-effective for investors to purchase shares directly from the market at their lower price.

Choice D is incorrect. Selling call options does not protect against downside risk either; instead, it limits upside potential while providing some income through premiums received from selling these options. In case of falling stock prices, while there may be no direct loss due to this strategy (as calls will likely expire worthless), it does nothing substantial in terms of protecting your portfolio from declining values.

Things to Remember

- Put options provide downside protection by allowing investors to sell a stock at a predetermined price, regardless of how far the market price falls.

- Buying put options is a common strategy for hedging against potential losses in a stock portfolio.
 - Selling put options can expose investors to significant losses if the stock price falls below the strike price.
 - Call options give the holder the right to buy an asset at a specified price within a certain period of time, which does not protect against falling stock prices.
 - Selling call options can limit upside potential and provide income through premiums, but it does not protect against downside risk.
-

Q.5349 A risk manager at a large energy company is presenting at a seminar on derivative contracts to a group of newly hired energy traders. The manager focuses on the advantages and disadvantages of using options contracts and futures contracts for hedging in the energy markets. Which of the following statements is correct?

- A. Options contracts provide the holder more flexibility than futures contracts.
- B. Options contracts are traded exclusively in the OTC (over-the-counter) market.
- C. Options contracts have a linear payoff, while futures contracts have a non-linear payoff.
- D. The value of an option is independent of the volatility of the underlying.

The correct answer is **A**.

Options contracts indeed offer more flexibility than futures contracts. The primary reason for this is that options contracts provide the holder with the right, but not the obligation, to buy or sell the underlying asset at a predetermined price and date in the future. This means that the holder can choose whether or not to exercise the option, depending on market conditions. This gives the holder more flexibility than futures contracts, which obligate both parties to fulfill the terms of the contract. This flexibility is particularly beneficial in volatile markets, where the ability to choose whether or not to exercise the option can help limit potential losses and maximize potential gains.

Choice B is incorrect. Options contracts are not traded exclusively in the OTC market. They can also be traded on organized exchanges.

Choice C is incorrect. The statement about the payoff profiles of options and futures contracts is reversed. Options contracts have a non-linear payoff, while futures contracts have a linear payoff.

Choice D is incorrect. The value of an option is not independent of the volatility of the underlying asset; rather, it's highly dependent on it. Higher volatility generally increases the value of an option because it increases the likelihood that the option will end up in-the-money at expiration.

Things to Remember

- Options contracts provide the holder with the right, but not the obligation, to buy or sell the underlying asset at a predetermined price and date in the future.
 - Options contracts can be traded on organized exchanges as well as in the OTC market.
 - Options contracts have a non-linear payoff, while futures contracts have a linear payoff.
 - The value of an option is highly dependent on the volatility of the underlying asset.
 - Higher volatility generally increases the value of an option because it increases the likelihood that the option will end up in-the-money at expiration.
-

Q.5350 A Canadian-based bank signs a 3-month forward contract with a manufacturer to sell USD 50 million in 3 months at a rate of CAD 1.25 per USD. What is the payoff for the bank from the forward contract if the exchange rate is CAD 1.13 per USD 1 in 3 months?

- A. -6,500,000
- B. 6,000,000
- C. 2,500,000
- D. -6,000,000

The correct answer is **B**.

The value of the contract for the bank at expiration is given by:

$$\text{USD } 50,000,000 \times 1.25 \text{ CAD/USD} = \text{CAD } 62,500,000.$$

At expiration, to close out the contract, it will cost the bank:

$$\text{USD } 50,000,000 \times 1.13 \text{ CAD/USD} = \text{CAD } 56,500,000.$$

Therefore, the final payoff to the bank is:

$$\text{CAD } 62,500,000 - \text{CAD } 56,500,000 = \text{CAD } 6,000,000.$$

Q.5351 A risk manager at a bank is trying to decide whether to use a forward or an option contract to hedge against currency risk. The manager expects to receive a foreign currency payment of €500,000 in three months. Which of the following statements is correct regarding the manager's decision?

- A. The manager will choose a forward contract because it allows for flexibility in exercising the contract.
- B. The manager will choose an option contract because it provides a guaranteed exchange rate for the foreign currency payment.
- C. The manager will choose a forward contract because it has no upfront costs.
- D. The manager will choose an option contract because it limits the downside risk if the exchange rate moves unfavorably.

The correct answer is **D**.

Option contracts are indeed a viable choice for the manager as they limit the downside risk if the exchange rate moves unfavorably. By purchasing an option, specifically a put option, the manager can secure a maximum exchange rate and limit potential losses if the exchange rate moves in the wrong direction. This is because a put option gives the holder the right, but not the obligation, to sell a currency at a specified rate within a certain period. If the exchange rate moves unfavorably, the manager can exercise the option and sell the currency at the agreed rate, thereby limiting the downside risk. If the exchange rate moves favorably, the manager can let the option expire and exchange the currency at the prevailing market rate, thereby benefiting from the favorable movement in the exchange rate. This flexibility to adapt to market conditions is a key advantage of options over forwards.

Choice A is incorrect. Forward contracts do not provide flexibility in exercising the contract. They are binding agreements that obligate the buyer to purchase an asset at a predetermined price at a specific future date, regardless of the prevailing market price.

Choice B is incorrect. While it's true that option contracts can provide a guaranteed exchange rate, this isn't exclusive to options alone. Both forward and option contracts can lock in an exchange rate for future transactions.

Choice C is incorrect. Although forward contracts typically do not have upfront costs, they expose the holder to potential losses if the market moves unfavorably before maturity of the contract, unlike options which limit downside risk.

Things to Remember

- Forward contracts are binding agreements that obligate the buyer to purchase an asset at a predetermined price at a specific future date.
- Option contracts provide the holder with the right, but not the obligation, to buy or sell an asset at a specified price within a certain period.
- Put options specifically give the holder the right to sell a currency at a specified rate within a certain period, providing downside protection.
- Options offer flexibility as the holder can choose whether or not to exercise the contract based on market conditions.
- Forward contracts do not provide this flexibility and lock in the buyer to the agreed-upon terms.

Reading 31: Exchanges and OTC Markets

Q.826 Muhammad Amir recently completed his Ph.D. in finance and economics. After his graduation, he started his career as a college professor. In his first lecture, he said: "Exchanges are more efficient and more liquid than OTC markets as they minimize the risk and promote customization." Which of the following options is correct?

- A. Muhammad Amir is incorrect regarding the liquidity feature of exchanges.
- B. Muhammad Amir is incorrect regarding the enhanced efficiency of exchanges.
- C. Muhammad Amir is incorrect regarding the promoted customization of exchanges.
- D. Muhammad Amir is incorrect regarding the risk reduction of exchanges.

The correct answer is **C**.

Dr. Muhammad Amir's claim that exchanges promote customization is incorrect. In fact, exchanges are known for promoting standardization rather than customization. This is primarily because exchanges operate on a set of predefined rules and regulations that ensure uniformity in the trading process. This standardization is crucial as it facilitates the smooth functioning of the exchange, reduces complexity, and enhances transparency. Customization, on the other hand, implies tailoring the trading process to meet the specific needs of individual traders, which is not typically associated with exchanges. Therefore, Dr. Amir's assertion regarding the promotion of customization by exchanges is incorrect.

Choice A is incorrect. Exchanges are indeed known for their liquidity feature. They have a large number of buyers and sellers, which ensures that transactions can be executed quickly and without significantly affecting the price.

Choice B is incorrect. Exchanges are also recognized for their efficiency. The presence of many participants in exchanges allows for competitive pricing, which leads to market efficiency.

Choice D is incorrect. Dr. Amir's claim about risk reduction in exchanges is correct as well. Exchanges provide a centralized clearing house that guarantees the performance of contracts, thereby reducing counterparty risk.

Exchanges do not typically foster customization as they deal with standardized contracts to ensure liquidity and transparency in trading operations.

Things to Remember

- **OTC Markets:** Over-the-counter (OTC) markets are decentralized markets where trading of financial instruments occurs directly between parties without the supervision of an exchange. OTC markets are known for their flexibility and ability to facilitate customized transactions.

- **Liquidity:** Liquidity refers to the ease with which an asset can be bought or sold in the market without significantly affecting its price. High liquidity is desirable as it allows for quick and efficient trading.
 - **Standardization:** Exchanges often standardize contracts and trading processes to ensure uniformity and transparency in transactions. This standardization helps in reducing risks and promoting market efficiency.
 - **Centralized Clearing House:** Exchanges typically have a centralized clearing house that acts as a counterparty to all trades, guaranteeing the performance of contracts and reducing counterparty risk.
-

Q.837 Which of the following is the accurate difference between the clearing process and the settlement process of over-the-counter derivatives?

- A. The settlement of OTC derivatives is the process by which payment obligations between two or more parties are computed and netted, and clearing is the process by which the contract obligations are fulfilled.
- B. The clearing process of OTC derivatives is the process by which payment obligations between two or more parties are computed and netted, and the settlement is the process by which the contract obligations are fulfilled.
- C. The clearing process of OTC derivatives is the process by which members are required to post cash and assets against their open positions, and the settlement is the process by which the contract obligations are fulfilled.
- D. The clearing process of OTC derivatives is the process by which payment obligations between two or more parties are computed and netted, and the settlement is the process by which the contract cleared through the central clearinghouse.

The correct answer is **B**.

The clearing process of OTC derivatives is the process by which payment obligations between two or more parties are computed and netted, and the settlement is the process by which the contract obligations are fulfilled. Clearing is the first step in the process, where the payment obligations of each party are calculated and netted against each other. This process involves the calculation of the net amount that each party owes to the other, taking into account all the transactions that have taken place between them. Once the net obligations have been determined, the next step is settlement. Settlement is the process by which these obligations are fulfilled. This involves the actual exchange of cash or assets between the parties, in accordance

with the net obligations that were determined during the clearing process. Therefore, the statement in choice B accurately describes the difference between the clearing and settlement processes of OTC derivatives.

Choice A is incorrect. The statement has reversed the definitions of clearing and settlement. Clearing is the process by which payment obligations between two or more parties are computed and netted, not settlement.

Choice C is incorrect. While it's true that members are required to post cash and assets against their open positions during the clearing process, this statement does not accurately define what clearing entails in its entirety. Clearing involves computing and netting payment obligations between parties, not just posting cash or assets.

Choice D is incorrect. This choice incorrectly suggests that settlement only occurs after a contract has been cleared through a central clearinghouse. In reality, settlement refers to the fulfillment of contract obligations regardless of whether they have been cleared through a central clearinghouse or not.

Things to Remember

- Clearing process involves the calculation and netting of payment obligations between parties in OTC derivatives transactions.
- Settlement process is the fulfillment of contract obligations after the clearing process has determined the net obligations.
- Central clearinghouses play a key role in the clearing process by acting as intermediaries between parties and ensuring the integrity of the transactions.
- OTC derivatives are traded directly between parties and are not traded on centralized exchanges, hence the importance of clearing and settlement processes.
- Counterparty risk is a significant concern in OTC derivatives transactions, making the clearing and settlement processes crucial for risk mitigation.

Q.840 Clearing houses play a crucial role in the financial markets by ensuring the smooth execution and settlement of trades. Clearing houses are:

- A. never used in futures markets but are sometimes used in OTC markets.
- B. always used in futures markets and sometimes used in OTC markets.
- C. always used in both futures markets and OTC markets.

D. always used in OTC markets but never used in futures markets.

The correct answer is **B**.

Clearing houses are always used in futures markets and sometimes used in OTC markets. A clearing house is an intermediary between buyers and sellers of financial instruments. It is always used in regulated (exchange) markets to settle trading accounts, clear trades, collect and maintain margin monies, regulate delivery, and report trading data. Futures contracts are exclusively traded in organized exchanges, hence the constant use of clearing houses. Although OTC markets (decentralized markets in which market participants trade stocks, commodities, currencies, or other instruments directly) traditionally do not involve a clearing house, they are increasingly embracing the idea of central clearing through central counterparties (CCPs) which are a type of clearing house. Therefore, clearing houses are sometimes used in OTC markets.

Choice A is incorrect. Clearing houses are always used in futures markets, not never. They act as the counterparty to both sides of a futures contract - buyer and seller - reducing the risk of default by either party. However, they are sometimes used in OTC markets, which is correctly stated.

Choice C is incorrect. While clearing houses are always used in futures markets, they are not always used in OTC markets. The use of clearing houses in OTC markets depends on various factors such as the type of financial instrument being traded and the regulatory environment.

Choice D is incorrect. This choice incorrectly states that clearing houses are never used in futures markets when they actually play a crucial role there by acting as intermediaries between buyers and sellers to reduce default risk. Also, it's wrong to say that clearing houses are always used in OTC markets because their usage varies depending on several factors.

Things to Remember

- Clearing houses are financial intermediaries that facilitate the smooth execution and settlement of trades in financial markets.
- They are crucial in reducing counterparty risk by acting as a buyer to every seller and a seller to every buyer in the market.
- Clearing houses also collect and maintain margin monies, regulate delivery, and report trading data in regulated markets.
- Futures markets exclusively use clearing houses to settle trading accounts and clear trades due to the standardized nature of futures contracts.
- OTC markets, on the other hand, traditionally do not involve clearing houses, but are increasingly adopting central clearing through central counterparties (CCPs).

- The use of clearing houses in OTC markets depends on factors such as the type of financial instrument being traded and the regulatory environment.
-

Q.842 In the insurance industry, certain companies specialize in providing guarantees for specific types of financial products. Which of the following correctly defines monolines?

- A. Monolines are legal entities created to isolate the default risk of the counterparty in a derivatives transaction, so the firm can receive the full settlement of its other transactions.
- B. Monolines are triple-A rated independently capitalized entities created by one or more banks as a bankruptcy-remote subsidiary.
- C. Monolines are dependent central parties in the derivatives market that act as the counterparty in every derivative transaction.
- D. Monolines are types of insurance companies with strong credit ratings that provide credit wraps and credit default swaps to achieve diversification and better returns.

The correct answer is **D**.

Monolines, also known as Credit Derivatives Product Companies (CPDCs), are akin to insurance companies with strong credit ratings. They provide credit wraps (financial guarantees) and credit default swaps (CDC) to achieve diversification and better returns. They are structured as an extension of a Derivative Product Company (DPC) that focuses solely on credit default swaps. This definition accurately encapsulates the role and function of monolines in the financial market.

Choice A is incorrect. Monolines are not created to isolate the default risk of the counterparty in a derivatives transaction. They primarily provide credit enhancement services, such as credit wraps and credit default swaps, to improve the creditworthiness of debt securities.

Choice B is incorrect. While monolines are often highly rated and may be created by banks, they are not necessarily bankruptcy-remote subsidiaries. The primary function of monolines is to provide insurance for financial transactions, particularly in the bond market.

Choice C is incorrect. Monolines do not act as central parties in every derivative transaction. Their role is more specific - they insure financial products against default risk which helps issuers achieve better ratings on their debt issues.

Things to Remember

- Monolines are also known as Credit Derivatives Product Companies (CPDCs) in the financial industry.

- They provide credit wraps and credit default swaps to enhance the creditworthiness of debt securities.
 - Monolines are structured as an extension of Derivative Product Companies (DPCs) focusing on credit default swaps.
 - Monolines play a crucial role in the bond market by providing insurance for financial transactions.
 - Monolines are not central parties in every derivative transaction; their main focus is on providing credit enhancement services.
-

Q.843 Which of the following is a method of risk mitigation in over-the-counter markets where a firm creates a legal entity to transfer its assets and to isolate the firm's financial risk?

- A. Central counterparty
- B. Initial Margins
- C. Derivative Product Company
- D. Special purpose vehicles

The correct answer is **D**.

A Special Purpose Vehicle (SPV), also known as a Special Purpose Entity (SPE), is a separate legal entity created by a firm to isolate itself from financial risk. The firm transfers its assets to the SPV, thereby protecting itself from the financial risk associated with those assets. In the context of a derivatives transaction, if a specific counterparty defaults, the firm can still receive full settlement on its other transactions without having to net the losses on the defaulted transactions. This is because the assets are held by the SPV, which is legally separate from the firm. Therefore, the firm's financial risk is effectively isolated.

Choice A is incorrect. A Central Counterparty (CCP) is a financial institution that takes on counterparty credit risk between parties to a transaction and provides clearing and settlement services for trades in foreign exchange, securities, options, and derivative contracts. CCPs are not separate legal entities created by firms to transfer their assets for risk mitigation.

Choice B is incorrect. Initial Margins are collateral that the holder of a financial instrument has to deposit with a counterparty (most often their broker or an exchange) to cover some or all of the credit risk posed by the holder to the counterparty. This does not involve creating a separate legal entity for transferring assets.

Choice C is incorrect. A Derivative Product Company (DPC) is typically set up by large financial institutions in order to isolate certain types of operational risks associated with derivatives transactions, but it does not involve transferring assets into this entity as part of its risk mitigation strategy.

Things to Remember

- Legal entity creation for risk mitigation
 - Asset transfer to isolate financial risk
 - Separate legal entity -Special Purpose Vehicle (SPV)
 - Protection from financial risk associated with assets
 - SPV holding assets independently from the firm
 - Effectively isolating financial risk
 - Central Counterparty (CCP) in OTC markets
 - Initial Margins as collateral for credit risk
 - Derivative Product Company (DPC) for operational risks
-

Q.3572 What is the definition of a Special Purpose Vehicle (SPV)?

- A. A subsidiary company with an asset/liability structure and legal status that makes its obligations secure even if the parent company goes bankrupt
- B. A type of asset-backed security that is secured by a mortgage or collection of mortgages
- C. A structured financial product that pools together cash flow-generating assets and repackages this asset pool into discrete tranches that can be sold to investors
- D. An institution providing a wide variety of deposit, lending and investment products to individuals, businesses or both.

The correct answer is **A**.

A Special Purpose Vehicle (SPV) is a subsidiary company with a distinct asset/liability structure and legal status that ensures its obligations remain secure, even if the parent company goes

bankrupt. SPVs are typically used in securitization transactions, allowing companies to isolate or pool assets, or to allow investors to invest in a specific set of risks. The SPV's assets are typically held off-balance sheet, which can help to protect them from the parent company's creditors. This structure also allows the parent company to maintain a cleaner balance sheet, while also potentially reducing the amount of capital required to be held against the assets. SPVs are often used in complex financing transactions, such as mortgage-backed securities, collateralized debt obligations, and lease financing.

Choice B is incorrect. While it is true that an SPV can be used to create asset-backed securities, the definition provided in this choice refers specifically to a mortgage-backed security (MBS), which is just one type of asset-backed security. An SPV's structure and purpose are not limited to the creation of MBSs.

Choice C is incorrect. This choice describes a Collateralized Debt Obligation (CDO), which is a type of structured financial product that can indeed be created using an SPV. However, this definition does not encompass all the possible uses and structures of an SPV.

Choice D is incorrect. This option describes a financial institution such as a bank or credit union, not an SPV. An SPV does not provide deposit, lending or investment products directly to individuals or businesses; instead, it serves as a separate legal entity created for specific financial purposes by another company.

Things to Remember

- SPVs are commonly used in structured finance transactions to isolate risk and protect assets.
 - SPVs are often used in securitization transactions, such as mortgage-backed securities and collateralized debt obligations.
 - SPVs have a distinct legal status and asset/liability structure that separates them from their parent companies.
 - SPVs can help companies manage risk, raise capital, and maintain a cleaner balance sheet.
 - SPVs are typically used to pool assets and create investment opportunities for investors.
-

Q.4878 Assume that, before the existence of CCPs, Trader X agreed to buy 10,000 bushels of wheat for 400 cents per bushel from Trader Y for delivery in June. Three weeks later, Trader X sold 5,000 bushels of wheat to Trader Z for 420 cents per bushel for delivery in June and another 5,000 bushels to Trader Q for 440 cents per bushel of wheat for delivery in June. What is the expected profit or loss for Trader X?

- A. A profit of USD 4,000
- B. A loss of USD 3,000
- C. A profit of USD 3,000
- D. A loss of USD 2,000

The correct answer is **C**.

The cost of delivery of 10,000 bushels is:

$$10,000 \times 400 \text{ cents} = \text{USD } 40,000$$

Therefore,

Trader X sold:

- i. 5,000 bushels of wheat to Trader Z for 420 cents per bushel for delivery in June
- ii. 5,000 bushels to Trader Q for 440 cents per bushels of wheat for delivery in June.

Thus, gross earnings will be:

$$5000 \times 420 \text{ cents} + 5000 \times 440 \text{ cents} = \text{USD } 43,000$$

Hence, Trader A makes a net profit of:

$$\text{USD } 43,000 - \text{USD } 40,000 = \text{USD } 3,000$$

Q.4879 A trader agrees with a broker to enter into a futures contract to sell 4,000 bushels of wheat for 500 cents per bushel. The initial margin is USD 20,000, and the maintenance margin is USD 10,000. Which circumstances will lead to withdrawal of USD 400 from the margin account?

- A. If price rise by 10 cents.
- B. If price drops by 10 cents.
- C. If price drops by 13 cents.
- D. If price rises by 13 cents.

The correct answer is **A**.

$$4000F = (\$20,000 + \$400)$$
$$F = \frac{20,400}{4,000} = 510 \text{ cents}$$

So, USD 400 can be withdrawn from the margin account if the price of wheat rises by 10 cents (510-500) or more.

Alternative (More Direct) Approach

The trader has a short position. Thus, they will gain if the price of wheat falls and lose if the price rises. But how much do they lose for each one-cent price increase? That would be $0.01 * 4,000 = \$40$. Therefore, how far would the price need to rise in order for the trader to lose \$400? That would be $400/40 = 10$ cents.

Additional explanation:

In the case of a short futures position, the trader sells the futures contract with the expectation that the price of the underlying asset will decrease. If the price of the asset increases, the value of the short futures position will decrease, resulting in a loss for the trader. This loss will be deducted from the margin account, reducing the amount of available margin and potentially leading to a margin call if the margin account balance falls below the maintenance margin level.

Q.4880 All of the following parties are required to post margin by the Chicago Board Options Exchange EXCEPT:

- A. A trader with a net long position in a call option.
- B. A trader with a net short position in a put option.
- C. A trader with a net short position in a call option.
- D. A trader shorting a stock.

The correct answer is **A**.

A trader with a net long position in a call option is not required to post margin by the Chicago Board Options Exchange (CBOE). This is because when a trader holds a net long position in an exchange-traded stock option, they have no potential future liability. These positions are often purchased upfront, and the option may or may not be exercised. Consequently, there is no reason for the exchange to require margin from a trader who holds a long position in a call option or a long position in a put option. The trader has already paid the premium for the option, which is the maximum amount they can lose. Therefore, the exchange does not require any additional margin from these traders.

Choice B is incorrect. A trader with a net short position in a put option is required to post margin according to the rules of the CBOE. This is because when you sell (or "write") a put option, you are obligated to buy the underlying asset at the strike price if the option holder decides to exercise their right. This exposes you to potential losses if the price of the underlying asset falls below the strike price, hence, margin is required as collateral for this risk.

Choice C is incorrect. Similarly, a trader with a net short position in a call option must also post margin according to CBOE rules. When selling call options, traders are obligated to sell an underlying asset at an agreed-upon price (the strike price) if called upon by an options holder. If market prices rise above this level, sellers face potential losses and therefore need to provide margin as security against these risks.

Choice D is incorrect. A trader who shorts stocks also needs to post margin according to CBOE regulations. Shorting involves borrowing shares and selling them with hopes that they can be bought back later at lower prices for profit; however, if share prices increase instead of falling as anticipated by short sellers, they would incur losses and thus need margins as protection against such adverse movements.

Things to Remember

- Margin requirements are set by exchanges to ensure traders have enough funds to cover potential losses.
- Long positions in options do not require margin as the maximum loss is limited to the premium paid.
- Short positions in options and shorting stocks require margin as they expose traders to potentially unlimited losses.
- Margin is a form of collateral that traders must deposit with their brokers to cover potential losses.
- Margin calls can be issued if the value of the account falls below a certain threshold, requiring additional funds to be deposited.

Q.4882 1,000 shares are sold by a trader at a price of \$45 per share. The initial margin and maintenance margin are 140% and 115%. What is the initial margin required?

- A. \$63,000
- B. \$51,750
- C. \$45,000
- D. \$18,000

The correct answer is **A**.

The initial margin required is 140%,

$$\text{Initial margin} = 1.4 \times 1000 \times 45 = 63,000$$

That is, initially, in addition to the \$45,000 obtained from the selling of the shares, the trader is required to contribute another \$18,000 ($=\$63,000 - \$45,000$)

Q.5039 You sell one December Brent Crude Oil futures contract when the futures price is \$108 per gallon. Each contract is on 1,000 gallons, and the initial margin per contract that you provide is \$6,000. The maintenance margin per contract is \$3,000. During the next day, the futures price rises to \$109.5 per gallon. What is the balance of your margin account at the end of the day?

- A. \$1,500
- B. \$7,500
- C. \$4,500
- D. \$9,000

The correct answer is **C**.

The price has increased by \$1.5. Because you have a short position, you lose $1.5 \times 1,000$ or \$1,500.

The balance in the margin account, therefore, goes down from \$6,000 to \$4,500.

However, there is no margin call since the margin account is still above the maintenance margin level.

Things to Remember

- **Initial Margin:** The amount of money that must be deposited by an investor with their broker when entering into a futures contract.
- **Maintenance Margin:** The minimum amount of equity that must be maintained in a margin account to avoid a margin call.
- **Margin Call:** When the balance in a margin account falls below the maintenance margin level, the broker will issue a margin call requiring the investor to deposit more funds.
- **Short Position:** When an investor sells a futures contract with the expectation that the price will fall, they are said to have a short position.
- **Long Position:** When an investor buys a futures contract with the expectation that the price will rise, they are said to have a long position.

Q.5353 An investor is considering buying stock on margin without using a central counterparty

(CCP) to facilitate the transaction. Which of the following factors is the most relevant when evaluating the risks associated with this investment decision?

- A. The stock's historical dividend yield.
- B. The investor's creditworthiness.
- C. The stock's beta coefficient.
- D. The stock's price-to-earnings (P/E) ratio.

The correct answer is **B**.

The creditworthiness of an investor is the most pertinent factor when buying stock on margin without the involvement of a CCP. This is because the investor is essentially borrowing money from a broker to buy the shares. The investor's creditworthiness will determine the interest rate they will be charged on the borrowed funds, the amount they can borrow, and the terms of the margin agreement. Therefore, an investor with a high credit score may be able to borrow more money at a lower interest rate, while an investor with a low credit score may face higher interest rates or stricter terms. This can significantly impact the profitability of the investment and the risk of a margin call, where the broker demands the investor deposit more money or securities into their account to cover potential losses.

Choice A is incorrect. The stock's historical dividend yield, while important in assessing the potential return on investment, does not directly relate to the risks associated with purchasing shares on margin without a central counterparty. It is more relevant when considering the income generation potential of an investment rather than its risk profile.

Choice C is incorrect. The stock's beta coefficient measures its volatility in relation to the market as a whole and can be useful in assessing market risk. However, it does not address the specific risks tied to purchasing shares on margin without using a CCP such as counterparty risk or credit risk which are more pertinent in this context.

Choice D is incorrect. The stock's price-to-earnings (P/E) ratio can provide insight into whether a stock may be overvalued or undervalued but it doesn't directly address the risks associated with buying shares on margin without using a CCP. This metric primarily helps investors evaluate if they are paying too much for what would essentially be future earnings of that company.

Things to Remember

- Central Counterparty (CCP): A CCP acts as an intermediary between buyers and sellers in financial markets, reducing counterparty risk and ensuring the completion of transactions.
- Margin Trading: Buying stocks on margin involves borrowing funds from a broker to purchase securities, using the investor's existing securities as collateral.

- Creditworthiness: The creditworthiness of an investor determines the terms of the margin agreement, including the amount that can be borrowed and the interest rate charged.
 - Margin Call: A margin call occurs when the value of the investor's account falls below a certain level, prompting the broker to demand additional funds or securities to cover potential losses.
 - Counterparty Risk: The risk that the other party in a financial transaction will not fulfill their obligations, leading to financial losses.
-

Q.5354 An investor buys 100 shares of a stock on margin, with an initial margin requirement of 60% and a maintenance margin requirement of 40%. The purchase price is \$80 per share. After a few days, the stock price drops to \$50 per share. What is the minimum amount of cash the investor must deposit to meet a margin call?

- A. \$200
- B. \$900
- C. \$250
- D. \$1,800

The correct answer is **A**.

The initial margin requirement is 60%, which means the investor must put up 60% of the purchase price as initial margin. The purchase price is \$80 per share, so the total cost of the purchase is 100 shares x \$80 = \$8,000. Therefore, the initial margin required is 60% x \$8,000 = \$4,800.

When the stock price drops to \$50 per share, the market value of the shares is 100 shares x \$50 = \$5,000. The current equity in the investment is equal to the market value of the shares minus the amount borrowed, which is \$5,000 - \$0.40*8,000 = \$1,800.

The maintenance margin requirement is 40%, which means the investor must maintain equity of at least 40% of the market value of the shares. The minimum equity required to avoid a margin call is 40% x \$5,000 = \$2,000.

Since the current equity is only \$1,800, the investor must deposit additional cash to meet the minimum equity requirement. The minimum amount of cash the investor must deposit is the difference between the minimum equity requirement and the current equity, which is \$2,000 - \$1,800 = \$200.

Things to Remember

- Initial Margin Requirement: The percentage of the purchase price that an investor must put up as initial margin.
 - Maintenance Margin Requirement: The minimum amount of equity that must be maintained in a margin account.
 - Market Value of Shares: The total value of the shares owned by an investor based on the current market price.
 - Equity in Investment: The difference between the market value of the shares and the amount borrowed in a margin account.
 - Margin Call: A demand by a broker for an investor to deposit additional cash or securities to meet the minimum equity requirements.
-

Q.5355 A company is considering using collateralization to mitigate risk associated with an over-the-counter (OTC) derivatives transaction. Which of the following statements regarding the role of collateralization in the OTC market is **most accurate**?

- A. Collateralization is not used in the OTC market, as counterparties rely on their creditworthiness to ensure performance.
- B. Collateralization is used to mitigate counterparty credit risk exposure in the OTC market by providing security for the performance of a trade.
- C. Collateralization is used to protect against economic risk, but not counterparty credit risk, in the OTC market.
- D. Collateralization is less effective in the OTC market than in exchange-traded markets due to the lack of transparency in pricing and valuation.

The correct answer is **B**.

Collateralization is used to mitigate counterparty credit risk exposure in the OTC market by providing security for the performance of a trade. In the OTC market, collateralization is a common practice to reduce the exposure to counterparty credit risk. Counterparty credit risk is the risk that the counterparty to a financial contract will not live up to its contractual obligations. In this context, collateralization involves the use of assets, such as cash or securities, pledged by one party to a trade to provide security for the performance of the trade. If one party defaults,

the collateral can be used to cover potential losses. This practice can help to improve market stability and reduce the likelihood of systemic risk. Systemic risk refers to the risk that the failure of one participant in a financial market can cause a cascading failure of other participants due to their interlinkages and interdependencies. By using collateralization, firms can protect themselves against the potential default of their counterparties, thereby reducing their exposure to counterparty credit risk.

Choice A is incorrect. Collateralization is indeed used in the OTC market. While creditworthiness of counterparties can play a role in ensuring performance, it does not provide the same level of security as collateralization. Collateral provides an additional layer of protection by offering assets that can be seized or sold in case of non-performance.

Choice C is incorrect. This statement misrepresents the function of collateralization in the OTC market. Collateralization serves to mitigate counterparty credit risk, not economic risk. Economic risks are broader and include factors such as changes in interest rates, exchange rates, or other macroeconomic conditions which are not directly addressed through collateralization.

Choice D is incorrect. The effectiveness of collateralization does not depend on whether a market is exchange-traded or over-the-counter but rather on how well it's managed and enforced by parties involved. While transparency issues may pose challenges for valuation and pricing in OTC markets, these do not inherently make collateralization less effective as a risk mitigation strategy.

Things to Remember

- Counterparty Credit Risk: The risk that the counterparty to a financial contract will not fulfill their obligations.
- Collateral: Assets such as cash or securities pledged to secure the performance of a trade.
- Systemic Risk: The risk that the failure of one participant in a financial market can lead to a chain reaction of failures.
- Economic Risk: Risks related to changes in interest rates, exchange rates, or other macroeconomic conditions.
- Transparency: The degree to which information is available and accessible in a market.

Q.5356 A risk manager at a bank is considering using a Special Purpose Vehicle (SPV) to mitigate counterparty and credit risk exposure in an over-the-counter (OTC) derivatives transaction. Which of the following statements regarding the use of an SPV is **correct**?

- A. An SPV is a type of bank that is structured as a trust, company, or partnership.
- B. The use of an SPV eliminates all counterparty and credit risk exposure in OTC derivatives transactions.
- C. The default of one party will not cause the SPV to default.
- D. When using an SPV, assets are transferred to the SPV in exchange for shares of the company.

The correct answer is **C**.

SPVs are designed to be bankruptcy remote, which means that the default of one party will not cause the SPV to default. This feature helps to mitigate counterparty and credit risk exposure in OTC derivatives transactions, as it helps to protect the assets held by the SPV in the event of a default.

A is incorrect. An SPV is not a type of bank. It is a non-bank financial institution that is typically structured as a trust, company, or partnership.

B is incorrect. The use of an SPV does not eliminate all counterparty and credit risk exposure in OTC derivatives transactions. While it can help to mitigate this risk, there is still the potential for default by either counterparty.

D is incorrect. When using an SPV, assets are transferred to the SPV in exchange for cash, not shares of the company. This provides the transferring party with cash liquidity while also isolating the risk associated with the OTC derivative transaction.

Things to Remember

- Special Purpose Vehicles (SPVs) are commonly used in structured finance transactions to isolate risk and protect assets.
- SPVs are typically bankruptcy remote entities, meaning they are designed to remain solvent even if other parties involved in the transaction default.
- SPVs are often used in securitization transactions, where assets are transferred to the SPV to back the issuance of securities.
- The use of an SPV can help to achieve off-balance sheet treatment for certain assets and liabilities.
- SPVs are subject to regulatory oversight and must comply with legal and accounting requirements to maintain their status.

Q.5357 An investor purchases \$20,000 worth of stock using a margin account with an initial margin requirement of 50% and a maintenance margin requirement of 25%. What is the minimum amount that the investor must deposit into the margin account to meet the maintenance margin requirement if the stock's value falls to \$11,100?

- A. \$1,400
- B. \$1,675
- C. \$1,100
- D. \$7,500

The correct answer is **B**.

The initial margin is 50%, so the broker-dealer will lend the investor \$10,000, and the investor must put up \$10,000 of their own money. The maintenance margin is 25%, so the minimum value of the account must be \$5,000 (25% of \$20,000).

If the stock's value falls to \$11,100, the amount of equity in the account is \$1,100 (\$11,100 value of the stock - \$10,000 loan from the broker).

The margin as a percentage of the current value of the stock is 9.99% ($\$1100 / \$11,100$).

Since this is less than the 25% maintenance margin requirement, there is a margin call to bring the margin balance up to 25% of the value of the shares, which is \$1,675 ($\$11,100 \times 25\% - \$1,100$).

Therefore, the investor must add \$1,675 to the margin account to meet the maintenance margin requirement. If this is not provided, the broker may sell some or all of the securities without notification to bring the account back up to minimum margins.

Things to Remember

- **Initial Margin Requirement:** The minimum percentage of the total value of the investment that must be paid for with the investor's own money.
- **Maintenance Margin Requirement:** The minimum percentage of the total value of the investment that must be maintained in the margin account.
- **Margin Call:** When the value of the securities in the margin account falls below the maintenance margin requirement, the broker may issue a margin call requiring the investor to deposit more funds.
- **Leverage:** Margin accounts allow investors to leverage their investments by borrowing

funds from the broker-dealer.

- **Margin Account:** An account in which an investor can borrow funds from a broker-dealer to purchase securities.
-

Reading 32: Central Clearing

Q.845 Unlike traditional investments where the transaction takes place in two stages, derivatives transactions are carried out in multiple stages. In which of the following stages of a derivatives trade is the central counterparty most likely involved?

- A. Execution stage
- B. Clearing stage
- C. Settlement stage
- D. Valuation stage

The correct answer is **B**.

The clearing stage of a derivatives transaction is where the central counterparty (CCP) comes into play. The CCP acts as an intermediary between the buyer and seller, reducing the risk of default by either party. During the clearing process, the CCP calculates and collects margin from both parties to cover potential future exposure. This process is known as margining. Additionally, the CCP also performs netting, which involves offsetting the total amount of buy orders against the total amount of sell orders, thereby reducing the overall number of transactions and the associated transaction costs. The involvement of the CCP in the clearing stage ensures the smooth execution of the trade and mitigates the counterparty risk.

Choice A is incorrect. The execution stage of a derivatives transaction typically involves the buyer and seller agreeing on the terms of the contract, such as price, quantity, and maturity date. This stage does not usually involve a central counterparty as it primarily focuses on negotiation between two parties.

Choice C is incorrect. The settlement stage refers to the actual exchange of payment and delivery of the underlying asset in accordance with the terms agreed upon during execution. While this process may involve various intermediaries for facilitating payment or delivery, it does not typically require a central counterparty.

Choice D is incorrect. Valuation stage involves determining the value of the derivatives positions, but it is not a stage where the central counterparty is primarily involved. The central counterparty's role is mainly in clearing and settling trades, not in the valuation of positions.

Things to Remember

- Central Counterparty (CCP): An entity that interposes itself between the buyer and seller of a financial instrument to reduce counterparty risk.
- Margining: The process of calculating and collecting margin from both parties in a derivatives transaction to cover potential future exposure.

- Netting: Offsetting the total amount of buy orders against the total amount of sell orders to reduce the number of transactions and transaction costs.
 - Counterparty Risk: The risk that one party in a transaction may default on its obligations, leading to financial losses for the other party.
 - Derivatives: Financial instruments whose value is derived from an underlying asset, index, or rate.
-

Q.849 Margins are usually of two types – initial margins and variation margins. Both required margins are calculated based on different variables. Which of the following is the determinant of the initial margin?

- A. Ratings of the borrower.
- B. Risk of the transaction.
- C. The creditworthiness of the borrower.
- D. Discretion of the parties involved.

The correct answer is **B**.

Initial margins are primarily designed to cover the potential future exposure of a transaction over a specified period of time. The main determinant of the initial margin is, therefore risk of the transaction. In other words, initial margins serve as a financial buffer in the derivatives and securities trading world. Their primary purpose is to safeguard against potential adverse price movements that could occur over a certain time frame. In essence, they act as a form of security deposit that is taken to ensure that parties to the transaction can meet their obligations.

Things to Remember

- Variation margins are calculated based on the current market value of the position.
- Initial margins are set by clearinghouses or counterparties to cover potential future exposure.
- Ratings of the borrower and creditworthiness are more relevant for determining the cost of borrowing or credit terms, rather than initial margin requirements.
- Discretion of the parties involved is not a determinant of the initial margin, as it is based on objective risk factors.

Q.851 Infrastructure Bank of Congo has a long exposure of \$350 million in a derivatives contract on the Frankfurt futures exchange. Since elections recently took place in Congo and the newly elected government canceled the projects of its predecessor, the nation's bank is likely to default on its obligations. Which of the following is the first alternative a central counterparty (CCP) will apply after default?

- A. Increasing the variation margin.
- B. Auctioning the contract of the defaulting party.

C. Requiring additional initial margin.

D. Loss mutualizing.

The correct answer is **B**.

In the event of a default, the Central Counterparty (CCP) steps in to ensure the obligations of the contract are met. This is achieved not by the CCP directly paying the losses on behalf of the defaulter, but by replacing the defaulting counterparty of the contract with a new counterparty through an auction process. This is the first step taken by the CCP to manage the default situation. The auction process allows the CCP to transfer the risk to another party willing to take on the contract, thereby ensuring the continuity of the contract and minimizing the impact of the default.

Choice A is incorrect. Increasing the variation margin is not the initial step taken by a central counterparty (CCP) to manage a default situation. Variation margin refers to payments made or received due to changes in the price of the underlying asset, and it's typically adjusted daily. However, in case of an expected default, simply increasing this margin would not be sufficient as it does not directly address the risk of non-fulfillment of obligations.

Choice C is incorrect. Requiring additional initial margin may help mitigate future risks but it's not an immediate action taken by CCPs when a party defaults on its obligations. The initial margin serves as collateral to cover potential future exposure arising from price changes, but once a default has occurred, requiring more upfront collateral won't rectify the current situation.

Choice D is incorrect. Loss mutualizing involves spreading out losses among all members of CCP which can be considered as one way to manage defaults but it's usually seen as a last resort measure rather than an initial step in managing defaults.

Things to Remember

- Central Counterparty (CCP) acts as an intermediary in derivatives and securities markets, becoming the buyer to every seller and the seller to every buyer, thereby reducing counterparty risk.
- Default risk is the risk that a party to a transaction will not fulfill its obligations, leading to financial losses for the other party.
- Auctioning the contract of the defaulting party is a common practice in financial markets to transfer the risk to a willing party and ensure continuity of contracts.
- Variation margin is the margin that is paid or received based on the daily changes in the price of the underlying asset.
- Initial margin is the collateral required by CCPs to cover potential future exposure

arising from price changes.

- Loss mutualizing involves spreading out losses among all members of a CCP to manage defaults collectively.
-

Q.852 The central counterparty (CCP) is the center point in the exchanges. It is the counterparty of all the parties or members having exposure in the exchange. Suppose that one of the members is unable to fulfill its obligation and defaults. The CCP will terminate all financial contracts and relations with the defaulting party. The CCP has to manage ways to go about such defaults. Which of the following has the least adverse consequences on the other members if a member defaults?

- A. Let the defaulting member absorb the losses.
- B. Let the CCP pay for the losses.
- C. Let the CCP replace the defaulting member through auction.
- D. Let the CCP distribute loss through loss mutualisation.

The correct answer is **C**.

The strategy of replacing the defaulting member through an auction process is the most effective way to minimize the adverse impact on other members. In this scenario, the CCP immediately terminates all financial contracts with the defaulting member, thereby preventing any further losses. The positions of the defaulting member are then auctioned off to other members, usually by sub-portfolio, such as interest rate swaps. This allows other members to take over the positions of the defaulting member, often at favorable terms. This strategy is generally well-received by other members, as it allows them to potentially profit from the default without incurring any negative consequences. Furthermore, it ensures that the losses from the default are not distributed among all members through loss mutualization, which would have a more significant adverse impact.

Choice A is incorrect. Letting the defaulting member absorb the losses may not be feasible if the member is insolvent or lacks sufficient assets to cover the losses. This could potentially lead to a domino effect, where other members are affected by the default, thereby increasing systemic risk.

Choice B is incorrect. If CCP pays for all losses, it might deplete its own resources and compromise its ability to function as an intermediary in future transactions. This could destabilize the financial market and increase systemic risk.

Choice D is incorrect. Loss mutualisation involves spreading out a loss among all members of CCP which can negatively impact even those members who were not directly involved with the defaulting party. This can create moral hazard issues and discourage prudent risk management

practices among members.

Things to Remember

- Central Counterparty (CCP) is an entity that interposes itself between the counterparties to contracts traded in one or more financial markets, becoming the buyer to every seller and the seller to every buyer.
 - Default management procedures are crucial for CCPs to handle member defaults effectively and minimize systemic risk.
 - Auctioning off the positions of a defaulting member is a common practice in CCPs to transfer the risk and positions to other solvent members in an orderly manner.
 - Loss mutualisation involves spreading the losses from a default across all members of the CCP, which can have implications for risk-sharing and moral hazard.
 - Systemic risk refers to the risk of a breakdown in the financial system where problems in one part of the system can spread and cause disruptions throughout the system.
-

Q.854 Mitigating the counterparty risk is essential to maintain liquidity and reduce systematic risk in the financial system. As an insurance against the counterparty default, all the central counterparty (CCP) members contribute specific resources to a pool that is used if the resources of the defaulting counterparty are insufficient to pay off the losses. In which of the following processes do members contribute to this type of insurance to absorb the adverse consequences of a defaulting counterparty?

- A. Novation
- B. Auction
- C. Loss mutualization
- D. Multilateral netting

The correct answer is **C**.

Loss mutualization is the process where all the central counterparty (CCP) members contribute a specific amount of resources to a pool. This pool is then used to absorb the losses if the resources included in the initial margin, variation margin, and default fund contribution of the defaulting counterparty are insufficient to pay off the losses. This process is a form of insurance

against the default of a counterparty and is essential in maintaining liquidity and reducing systematic risk in the financial system. It ensures that the losses from a defaulting counterparty are shared among all members, thereby reducing the impact on any single member and the system as a whole.

Choice A is incorrect. Novation is a process in which the central counterparty (CCP) becomes the buyer for every seller and the seller for every buyer, thereby eliminating counterparty risk. However, it does not involve CCP members contributing to a pool of resources to cover losses from a defaulting counterparty.

Choice B is incorrect. An auction is a method used by CCPs to sell off the positions of a defaulting member to other non-defaulting members. While this can help mitigate losses, it does not involve contributions from CCP members into an insurance mechanism.

Choice D is incorrect. Multilateral netting involves offsetting multiple positions or payments due between more than two parties, reducing overall exposure and potentially mitigating some risk. However, it does not refer to the process where CCP members contribute to an insurance mechanism designed specifically for absorbing adverse effects of a defaulting counterparty.

Things to Remember

- Central Counterparty (CCP): An entity that interposes itself between the counterparties to contracts traded in one or more financial markets, becoming the buyer to every seller and the seller to every buyer. This helps in reducing counterparty risk.
- Default Fund: A pool of financial resources contributed by CCP members to cover losses that may arise from the default of another member.
- Initial Margin: The collateral that market participants must deposit with a CCP to cover potential future exposure to losses.
- Variation Margin: Additional collateral that may be required to be posted by market participants to cover changes in the value of their positions.
- Systematic Risk: The risk of a widespread adverse effect on the financial system and markets, caused by an external event.

Q.858 Lindy Sago is a project manager at Toronto Fast Brokers Inc. The firm acts as a non-clearing member in the derivatives exchange market it trades in, but the firm's management recently decided to become a clearing member of the exchange. The firm has assigned Sago the task to evaluate the revenue model of the central counterparty (CCP). Which of the following is the appropriate combination of a CCP's revenue?

- A. Trading spreads and initial margins.
- B. Trading spreads and interest on the margins.
- C. Fees on clearing trades and interest on the margins.
- D. Fees on clearing trades and initial margins.

The correct answer is **C**.

Central Counterparties (CCPs) play a crucial role in the financial markets by reducing counterparty risk and ensuring the smooth functioning of transactions. They act as intermediaries between buyers and sellers in a derivatives exchange, assuming the counterparty risk involved in the transactions. To support their operations and manage the risks, CCPs have two primary sources of revenue. The first is the fees they charge for clearing trades. These fees are levied on each trade cleared by the CCP and are a significant source of income. The second source of revenue is the interest earned on the margins posted by the members. These margins are essentially collateral that members are required to deposit with the CCP to cover potential losses in case of a default. The CCP invests these margins in low-risk, liquid assets and earns interest on them. Therefore, the correct combination of a CCP's revenue is the fees on clearing trades and interest on the margins.

Choice A is incorrect. Trading spreads are not a revenue stream for a central counterparty (CCP). CCPs do not participate in trading activities, hence they do not earn from trading spreads. Their primary role is to facilitate trades between parties and manage risk, not to trade themselves.

Choice B is incorrect. Similar to choice A, trading spreads are not a source of revenue for CCPs as they are not involved in the actual trading process. While interest on margins can be a source of income for CCPs, it cannot be paired with trading spreads as a correct combination of their revenue streams.

Choice D is incorrect. Although initial margins are collected by the CCP from its members as part of its risk management process, these funds serve as collateral and aren't considered direct sources of revenue for the CCP. The main sources of income would be fees charged on clearing trades and interest earned on margin deposits.

Things to Remember

- Central Counterparties (CCPs) act as intermediaries in the derivatives market, reducing counterparty risk.
- CCPs charge fees on clearing trades as a significant source of income.
- Interest earned on margins posted by members is another revenue source for CCPs.
- CCPs do not earn from trading spreads as they do not participate in trading activities.

- Initial margins collected by CCPs serve as collateral and are not direct sources of revenue.
-

Q.864 Tara Denis is the spokesperson for a central counterparty in one of the largest operating futures markets in Japan. During a Q&A session, one of the members of the public commented that the presence of CCPs in exchanges and OTC markets is creating a moral hazard. Which of the following is the most appropriate reference of the moral hazard pointed out by the member of the public?

- A. It is the moral hazard related to the standardization of products by the CCP. As the CCP standardizes all the products, the market participants use more and more alternative products that do not capture the true motive of the hedge.
- B. It is the moral hazard related to the effect of disincentivizing counterparty risk management practices by CCP members. Since the CCP acts as the counterparty to the transaction, the party or institution invests little resources in monitoring others parties' credit quality.
- C. It is the moral hazard related to the reduction in counterparty risk. Since the CCP assumes all the counterparty risk, it becomes cheaper for parties to enter into the contracts, which is unnecessary for them.
- D. It is a moral hazard related to the creation of liquidity. As the CCP creates and maintains liquidity in financial markets through margining, more and more participants enter the market with the intention of speculating.

The correct answer is **B**.

The moral hazard referred to by the member of the public is associated with the disincentivization of counterparty risk management practices by CCP members. A CCP, by its very nature, acts as the counterparty to every transaction. This means that the parties involved in the transaction have less incentive to invest resources in monitoring the credit quality of other parties. This is because the CCP assumes most of the counterparty risk. This can lead to a moral hazard as parties may engage in riskier behavior, knowing that the CCP will bear the brunt of any potential losses. This can potentially lead to systemic risk if many parties engage in such behavior and the CCP is unable to cover the losses.

Choice A is incorrect. The standardization of products by the CCP does not necessarily lead to moral hazard. While it's true that standardization can lead to market participants using alternative products, this doesn't inherently create a moral hazard situation. Moral hazard refers to the risk that one party has an incentive to take on excessive risk because they believe another party will bear the cost of that risk. In this case, there's no indication that CCP's product standardization leads parties to take on excessive risks believing that others will bear those costs.

Choice C is incorrect. While it's true that CCPs reduce counterparty risk, which could potentially make it cheaper for parties to enter into contracts, this doesn't necessarily constitute a moral hazard. The reduction in counterparty risk is one of the main functions and benefits of a CCP and does not inherently incentivize reckless behavior or excessive risk-taking.

Choice D is incorrect. The creation and maintenance of liquidity in financial markets by the CCP through margining does not directly foster a moral hazard situation either. Although more participants might enter the market with speculative intentions due to increased liquidity, this isn't directly linked with them taking on undue risks believing someone else will bear those costs.

Things to Remember

- Central Counterparties (CCPs) play a crucial role in mitigating counterparty risk in financial markets by becoming the buyer to every seller and the seller to every buyer.
 - Moral hazard refers to the risk that one party may take on excessive risks because it believes that another party will bear the costs of those risks.
 - Systemic risk is the risk of a failure in the financial system where a disturbance in one part of the system can spread and cause widespread financial instability.
 - Margining is the process of requiring participants in financial markets to deposit an initial amount of money to cover potential future losses.
 - Standardization of products by CCPs can help increase market efficiency and liquidity by making it easier for participants to trade and hedge risks.
-

Q.868 Edward Trott is a renowned anchor at a local business news channel. During a panel discussion in an evening news bulletin, he mentioned that central counterparties (CCPs) themselves are vulnerable to risks. One of the major risks faced by CCPs arises if the CCP does not receive reasonable economic bids for the contracts defaulted by clearing members. It then has to impose the significant losses of that member on another clearing member via loss allocation methods, which may result in financial distress and potential further defaults. Which of the following risks is the anchor referring to?

- A. Distress of other clearing members.
- B. Failed auction.
- C. Resignation of clearing members.
- D. Reputational risk.

The correct answer is **B**.

The risk that Edward Trott is referring to is known as the 'failed auction' risk. This risk arises when a central counterparty (CCP) does not receive reasonable economic bids for the contracts that have been defaulted by clearing members. In such a situation, the CCP is forced to impose the significant losses of the defaulting member on another clearing member through loss allocation methods. This can lead to financial distress and potentially trigger further defaults. The term 'failed auction' is used to describe this scenario because the CCP's attempt to auction off the defaulted contracts has failed to attract reasonable bids. This risk is a significant concern for CCPs as it can potentially destabilize the entire clearing system and lead to a chain reaction of defaults.

Choice A is incorrect. While the distress of other clearing members can be a consequence of the risk Mr. Trott is referring to, it is not the specific risk itself. The question asks for the specific risk that leads to this situation, which is a failed auction.

Choice C is incorrect. Resignation of clearing members could be an outcome or reaction to this risk but it's not the direct risk being referred to in this context. The specific risk that Mr. Trott refers to directly causes financial distress and potential defaults, which may lead some members to resign but it's not the primary issue at hand.

Choice D is incorrect. Reputational Risk might occur due to poor management or operational failures within CCPs but it does not directly relate with failure in receiving reasonable economic bids for contracts defaulted by clearing members leading them into financial distress and more defaults.

Things to Remember

- Central Counterparties (CCPs) play a crucial role in financial markets by acting as intermediaries in transactions, providing clearing and settlement services to reduce counterparty risk.
- CCPs use risk management tools such as margin requirements, default funds, and loss allocation methods to mitigate risks associated with clearing activities.
- Failed auction risk can have systemic implications as it can lead to a chain reaction of defaults and financial distress among clearing members.
- Loss allocation methods used by CCPs include mutualizing losses among clearing members, porting positions to solvent members, and auctioning off defaulted contracts.
- Reputational risk is a significant concern for financial institutions and CCPs, as it can impact their credibility and relationships with stakeholders.

Q.869 Theta Bank and Iota Bank have been engaging in bilateral credit default swap (CDS) trades. Following recent financial regulations aimed at reducing systemic risk, these trades are now required to be cleared through a recognized central counterparty (CCP). This regulatory shift involves a specific process where the CCP becomes the legal counterparty to both banks. Which of the following correctly describes this process?

- A. Netting
- B. Novation
- C. Margining
- D. Multilateral netting

The correct answer is **B**.

Novation refers to the replacement of an existing contract with a new one, with the consent of all parties involved. In this case, the existing bilateral CDS agreements between Theta Bank and Iota Bank are replaced with new contracts where the CCP becomes the legal counterparty to both banks. This reduces counterparty risk for each bank because they are no longer solely reliant on the other's creditworthiness.

A is incorrect. Netting is incorrect because it refers to the process of consolidating multiple financial obligations between two parties into a single net amount. Although netting reduces exposure and simplifies settlements, it does not involve replacing the counterparty with a CCP.

C is incorrect. Margining is incorrect as it pertains to the process of securing trades with collateral to manage potential credit risk. Margining is a risk management tool used within the context of cleared and non-cleared derivatives but does not describe the substitution of the contract parties.

D is incorrect. Multilateral netting is incorrect because this term describes the process of reducing obligations across multiple parties in a clearinghouse to a single net payment per party. While it is a function related to central clearing, it does not specifically denote the legal replacement of original contract parties with a CCP, which is the essence of novation.

Things to Remember

- **Central Counterparty (CCP):** A CCP is an entity that helps facilitate trading in various financial markets by becoming the legal counterparty to both sides of a trade. This helps reduce counterparty risk and increase market stability.
- **Counterparty Risk:** The risk that one party in a financial transaction will default or fail to meet its obligations, leading to financial losses for the other party.
- **Clearing:** The process of reconciling and settling trades to ensure smooth and efficient

transactions in financial markets.

- **Systemic Risk:** The risk of a widespread financial collapse or crisis that can impact the entire financial system and economy.
 - **Derivatives:** Financial instruments whose value is derived from an underlying asset, index, or rate. CDS is a type of derivative used for hedging credit risk.
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Q.871 Unlike in exchanges, the central counterparties of over-the-counter markets have to deal with complex transactions and projects. These CCPs are exposed to the risk of inconsistency in their margining functions. This is because the margin requirements of OTC products cannot be derived from market sources directly, but they require complex models to carry out the mark-to-market activities. This risk is most likely associated with:

- A. Distress risk
- B. Operational risk
- C. Legal risk
- D. Model risk

The correct answer is **D**.

Model risk is the risk that arises from the use of financial models in business activities. In the context of central counterparties (CCPs) in over-the-counter (OTC) markets, model risk is particularly relevant due to the complexity of OTC products. Unlike exchange-traded products, the initial margin and variation margin requirements of OTC products cannot be derived directly from market sources. Instead, they require the use of complex models to carry out mark-to-market activities. If these models are not robust or standardized, the CCPs may face difficulties in deriving the margin requirements in a timely and accurate manner. This could lead to inconsistencies in their margining functions, which could potentially result in significant financial losses. Therefore, managing model risk is a critical aspect of the operations of CCPs in OTC markets.

Choice A is incorrect. Distress risk refers to the risk that a company will default on its obligations due to financial distress. While CCPs do face this risk, it is not directly related to the complexity of OTC products and their margining functions.

Choice B is incorrect. Operational risk refers to the potential for loss resulting from inadequate or failed internal processes, people, and systems or from external events. Although operational risks may arise in handling complex OTC products, they are not specifically associated with inconsistencies in margining functions due to product complexity.

Choice C is incorrect. Legal risk pertains to the potential for loss due to legal or regulatory

action which can affect a company's operations or its financial position. This type of risk does not directly relate with the use of intricate models for mark-to-market activities in CCPs.

Things to Remember

- Central Counterparties (CCPs) play a crucial role in mitigating counterparty credit risk in OTC markets by becoming the buyer to every seller and the seller to every buyer.
 - Model risk can also arise from the assumptions and limitations of the models used, as well as from errors in implementation or interpretation.
 - CCPs typically use sophisticated risk management models to calculate margin requirements, stress testing, and other risk measures to ensure the stability of the financial system.
 - OTC derivatives are customizable contracts traded directly between two parties, making them more complex than standardized exchange-traded products.
 - Margin requirements are crucial for CCPs to cover potential losses in case of a counterparty default, and accurate calculation of these requirements is essential for the stability of the financial system.
-

Q.873 Which of the following model risks or model problems arise in the model-based initial margin estimation?

- A. In the model-based initial margin estimation, the defaulting party is unidentified until he defaults on its obligations in the contract.
- B. In the model-based initial margin estimation, the loss incurred due to the default of a member is allocated to the other clearing members.
- C. In the model-based initial margin estimation, the initial margins are estimated at a fixed dollar amount margin requirement set by the central counterparty (CCP).
- D. In the model-based initial margin estimation, the initial margins increase in proportion to the size of the position without considering that the risk of a large and concentrated position is adequately covered.

The correct answer is **D**.

In the model-based initial margin estimation, the initial margins increase in proportion to the size of the position without considering that the risk of a large and concentrated position is adequately covered. This is a significant risk as it can lead to underestimation of the risk associated with large and concentrated positions. The initial margin is meant to cover potential future exposure in the event of a default. However, if the margin increases merely in proportion to the size of the position, it may not adequately cover the risk of a large and concentrated position. This is because the risk of such a position is not linearly proportional to its size. For instance, a large and concentrated position may be subject to market illiquidity, making it difficult to liquidate without incurring significant losses. Therefore, the initial margin for such a position should be higher to cover this additional risk. If this is not taken into account in the model-based initial margin estimation, it can lead to significant losses for the central counterparty (CCP) in the event of a default.

Choice A is incorrect. The identification of the defaulting party is not a risk or problem that arises in the model-based initial margin estimation. This process focuses on calculating the initial margins for a financial contract, not identifying who might default on their obligations.

Choice B is incorrect. The allocation of loss due to a member's default to other clearing members does not pertain to the model-based initial margin estimation process. This issue relates more to the risk management practices of a central counterparty (CCP) rather than an inherent problem with estimating initial margins using models.

Choice C is incorrect. In model-based initial margin estimation, margins are not estimated at a fixed dollar amount set by CCPs but are calculated based on various factors such as market volatility and potential future exposure of the contract.

Things to Remember

- Model risk refers to the risk of financial loss resulting from errors in the development, implementation, or use of models and modeling techniques.
- Initial margin is the collateral that a party to a financial contract must deposit to cover potential future exposure in the event of a default.
- Central Counterparties (CCPs) play a crucial role in managing counterparty credit risk in financial markets by interposing themselves between the two parties to a trade, becoming the buyer to every seller and the seller to every buyer.
- Risk management practices in clearinghouses involve setting appropriate initial margin requirements to ensure the financial stability of the clearinghouse and protect it from losses due to defaults.
- Market illiquidity can pose significant challenges in liquidating large and concentrated positions, leading to potential losses if not adequately accounted for in risk management practices.

Q.874 A risk manager at a financial institution is reviewing the risks associated with clearing through a central counterparty (CCP). Which of the following statements about CCPs and their exposure to liquidity risk is correct?

- A. CCPs are not exposed to liquidity risk as they require their clearing members to post initial and variation margins.
- B. The default of a clearing member can lead to a loss of liquidity for the CCP.
- C. CCPs can only be exposed to liquidity risk if they have not properly calculated the margin requirements for their clearing members
- D. Liquidity risk for CCPs is solely related to the operational risk of the CCP.

The correct answer is **B**.

The default of a clearing member can indeed lead to a loss of liquidity for the CCP. When a clearing member defaults, the CCP is obligated to fulfill the defaulted member's obligations to the other clearing members. To do this, the CCP typically uses its own resources, such as its capital, guaranty funds, and default funds, to make the required payments. However, in the event of a large or unexpected default, these resources may not be sufficient to cover the losses. This insufficiency can lead to a loss of liquidity for the CCP, as it may struggle to meet its financial obligations. This scenario illustrates the exposure of CCPs to liquidity risk, despite the various risk mitigation measures they may have in place.

Choice A is incorrect. While it's true that CCPs require their clearing members to post initial and variation margins, this does not completely eliminate their exposure to liquidity risk. For instance, if a clearing member defaults and the value of the posted collateral falls significantly, the CCP could face a liquidity shortfall.

Choice C is incorrect. Even with properly calculated margin requirements, CCPs can still be exposed to liquidity risk due to various factors such as market volatility or default of multiple members which may lead to significant losses exceeding the available margins.

Choice D is incorrect. Liquidity risk for CCPs is not solely related to operational risk. It also arises from other sources such as credit risk (default by a member), market risk (changes in value of collateral), and even systemic risks (failure of financial institutions or disruption in financial markets).

Things to Remember

- Central Counterparties (CCPs) play a crucial role in mitigating counterparty credit risk in financial markets by interposing themselves between the two parties to a trade, becoming the buyer to every seller and the seller to every buyer.

- CCPs require clearing members to post initial margins (to cover potential future exposure) and variation margins (to cover current exposure) to mitigate risks.
 - CCPs also maintain guaranty funds and default funds to further protect against losses from clearing member defaults.
 - Liquidity risk for CCPs can arise from various sources, including credit risk, market risk, and operational risk, in addition to the risk of clearing member defaults.
 - CCPs are subject to regulatory oversight and are required to meet certain capital and liquidity requirements to ensure their stability and resilience.
-

Q.875 When trading standard transactions with an end user, a dealer is more likely to insist that the transactions are cleared through a CCP even when the credit quality of the end user is low. This is termed as:

- A. Moral hazard
- B. Adverse selection
- C. A tear up
- D. A pro-cyclical

The correct answer is **B**.

Adverse selection is a term commonly used in economics, insurance, and risk management that describes a situation where an individual's demand for insurance (either the propensity to buy insurance or the quantity purchased) is positively correlated with the individual's risk of loss (e.g., higher risks buy more insurance), and the insurer is unable to allow for this correlation in the price of insurance. This may be because of private information known only to the individual (information asymmetry), or because of regulations or social norms. In the context of this question, the dealer insisting on clearing the transactions through a Central Counterparty (CCP) despite the end user's low credit quality is a classic example of adverse selection. The dealer is aware of the higher risk associated with the end user due to their low credit quality, and therefore insists on using a CCP to mitigate this risk. This is a strategic move by the dealer to protect themselves from potential losses that could arise from the end user's inability to fulfill their financial obligations.

Choice A is incorrect. Moral hazard refers to the risk that one party behaves differently from how they would if they were fully exposed to the risk. In this case, the dealer insisting on clearing transactions through a CCP does not represent a change in behavior due to lack of exposure to risk.

Choice C is incorrect. A tear up refers to the cancellation of a contract or agreement, which is not what's happening in this scenario. The dealer and end user are still engaging in transactions; it's just that these transactions are being cleared through a CCP.

Choice D is incorrect. A pro-cyclical refers to any economic quantity that is positively correlated with the overall state of the economy i.e., it increases when economy booms and decreases during recessions. This term doesn't apply here as we're discussing credit quality and transaction clearance, not economic cycles.

Things to Remember

- **Central Counterparty (CCP):** A CCP is an entity that helps facilitate trading in various financial markets by acting as an intermediary between buyers and sellers. It helps reduce counterparty risk by becoming the buyer to every seller and the seller to every buyer.
 - **Credit Quality:** Refers to the assessment of the creditworthiness of a borrower or issuer of financial securities. It helps determine the likelihood of default on a financial obligation.
 - **Information Asymmetry:** This occurs when one party in a transaction has more or better information than the other party. It can lead to adverse selection and moral hazard issues.
 - **Risk Management:** The process of identifying, assessing, and prioritizing risks followed by coordinated and economical application of resources to minimize, monitor, and control the probability and impact of unfortunate events.
-

Q.876 The centralization feature of the central counterparty (CCP) increases the efficiency but at the same time also increases the operational risks for its members due to the concentration of the whole risk on the CCP. Which of the following situations can give rise to operational risk?

- A. Default by a clearing member.
- B. Increase in margin requirements.
- C. Failed auction.
- D. Infrastructure breakdown.

The correct answer is **D**.

The operational risk in a CCP can significantly increase due to the breakdown or failure of its infrastructure. This is because the CCP's infrastructure is the backbone of its operations, facilitating the clearing and settlement of transactions. If this infrastructure fails, it can disrupt the CCP's operations, leading to delays or failures in transaction processing. This can, in turn, lead to financial losses for the CCP's members, who rely on the CCP for the efficient clearing and settlement of their transactions. Therefore, an infrastructure breakdown is a situation that can give rise to operational risk in a CCP.

Choice A is incorrect. While a default by a clearing member can indeed pose significant risks, it is primarily a credit risk rather than an operational risk. Operational risks are associated with failures in systems, processes, or controls, including those caused by human error or system failures.

Choice B is incorrect. An increase in margin requirements can create liquidity pressures for members but does not directly escalate operational risk. Margin requirements are tools used to manage credit and market risks and do not pertain to the functioning of systems or processes.

Choice C is incorrect. A failed auction could potentially lead to losses for CCP members but this scenario represents more of a market risk than an operational one. The failure of an auction does not necessarily imply that there has been a failure in the CCP's systems or procedures.

Things to Remember

- Central Counterparty (CCP) is an entity that interposes itself between counterparties to contracts traded in various markets, becoming the buyer to every seller and the seller to every buyer.
- Operational risk is the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events.
- Infrastructure breakdown can include failures in technology systems, communication networks, or physical facilities that are essential for the operation of the CCP.
- Default by a clearing member is primarily a credit risk, where the member fails to meet its financial obligations, leading to potential losses for the CCP and its members.
- Margin requirements are collateral that clearing members must deposit to cover potential losses, helping to mitigate credit and market risks in the CCP.
- Failed auction can result in market risk, where the auction process for resolving trades or positions does not function as intended, potentially impacting the CCP and its members.

Q.877 Which of the following is the most accurate explanation of the sovereign risk faced by the central counterparty (CCP)?

- A. The sovereign risk faced by the CCP is referred to as the intervention of sovereign governments in operation and activities of the CCP.
- B. The sovereign risk faced by the CCP is due to the failure of members who have held sovereign bonds as margin, which may have declined in value due to sovereign failure.
- C. The sovereign risk faced by the CCP is referred to as the pressure and the undue influence that can arise when one of the members of the CCP is the agency of a sovereign fund/government.
- D. The sovereign risk faced by the CCP is related to changes in sovereign credit ratings impacting the valuation of government securities held as collateral.

The correct answer is **B**.

Sovereign risk, in the context of a Central Counterparty (CCP), is primarily associated with the potential failure of its members who have held sovereign bonds as margin. These bonds may decline in value due to a sovereign default or failure. This situation can significantly impact the financial stability of the CCP and its ability to manage counterparty risk. Members of a CCP often use repo rates as a risk management tool. During the Eurozone crises, it was observed that sovereign risk is strongly correlated with repo rates. This means that an increase in sovereign risk can lead to an increase in repo rates, which can further exacerbate the financial strain on CCP members holding sovereign bonds as margin. Therefore, managing sovereign risk is a critical aspect of the overall risk management strategy of a CCP.

Choice A is incorrect. While it's true that government intervention can impact the operations of a CCP, this is not what is typically referred to as 'sovereign risk'. Sovereign risk usually refers to the risk of default on sovereign bonds, which can impact the value of these bonds when they are held as margin by members of the CCP.

Choice C is incorrect. The term 'sovereign risk' does not refer to undue influence or pressure from a member who may be an agency of a sovereign fund/government. This could potentially be a conflict-of-interest issue, but it doesn't directly relate to sovereign risk in the context of a CCP.

Choice D is incorrect. This statement is partially correct but more specific to credit rating changes. The more encompassing risk includes the broader potential for sovereign default and the impact on the CCP's collateral.

Things to Remember

- Sovereign risk is the risk associated with the potential default of a government on its financial obligations.
- Central Counterparties (CCPs) are financial institutions that facilitate trading in various markets by becoming the buyer to every seller and the seller to every buyer.

- Members of a CCP often use government securities, such as sovereign bonds, as collateral for their transactions.
 - Sovereign risk can impact the financial stability of CCPs, especially when members use sovereign bonds as margin.
 - Managing sovereign risk is crucial for CCPs to ensure their financial stability and ability to manage counterparty risk effectively.
-

Q.878 The central counterparty (CCP) acquires banks services for the receipt and transfer of funds to and from its clearing members due to changes in variation margins and initial margins. These banks or large financial institutions can face technical failures or human failures that could stop them from processing the CCP's instructions to make cash payments and receive or deliver securities to its members. This could create liquidity problems on the end of the CCP and the clearing members that still have to fulfill obligations. This risk is most likely associated with:

- A. Sovereign risk
- B. Custodian risk
- C. Model risk
- D. Investment risk

The correct answer is **B**.

The risk described in the scenario is known as Custodian risk. Custodian risk, also known as custody risk, is associated with the inability of a custodian, in this case, the banks or large financial institutions, to perform its operations. These institutions act as the custodians of the CCPs, handling the receipt and transfer of funds due to changes in variation margins and initial margins. However, they can face technical or human failures that could prevent them from processing the CCP's instructions to make cash payments and receive or deliver securities to its members. This could create liquidity problems for both the CCP and its clearing members who still have obligations to fulfill. Moreover, it can also increase the systematic risk in the market. Despite the extensive regulatory controls that govern the custodian services of these banks, the risk of failure still exists, making custodian risk a significant concern in financial markets.

Choice A is incorrect. Sovereign risk refers to the risk of a country defaulting on its financial obligations or the change in a country's political or economic conditions that could adversely affect investments made in that country. This scenario does not involve any sovereign entity, hence sovereign risk is not applicable here.

Choice C is incorrect. Model risk pertains to the potential for different modeling assumptions and methodologies to produce different outputs. It arises due to errors in data, assumptions, or

model specifications which can lead to inaccurate predictions or estimates. In this scenario, there are no models being used for predictions or estimations; rather it's about operational failures at banks and financial institutions.

Choice D is incorrect. Investment risk refers to the possibility of losing invested capital due to various factors such as market volatility, inflation rates, interest rate changes etc., which can negatively impact investment returns. The situation described here does not involve any investment decisions but rather operational issues related with fund transfers and securities delivery.

Things to Remember

- **Central Counterparty (CCP):** A CCP is an entity that helps facilitate trading in financial markets by acting as an intermediary between buyers and sellers. It helps reduce counterparty risk by becoming the buyer to every seller and the seller to every buyer.
- **Variation Margins and Initial Margins:** Variation margins are additional funds required to be deposited by clearing members to cover losses in their positions, while initial margins are funds deposited upfront to cover potential future losses.
- **Liquidity Risk:** Liquidity risk is the risk of not being able to meet financial obligations when they come due without incurring unacceptable losses. It is a significant concern for financial institutions and markets.
- **Systematic Risk:** Systematic risk, also known as market risk, is the risk inherent to the entire market or an entire market segment. It cannot be diversified away and affects all securities in the market.
- **Regulatory Controls:** Regulatory controls are rules and regulations imposed by regulatory authorities to ensure the stability and integrity of financial markets and institutions.

Q.1138 The default of a clearing member could create further problems, including:

- A. A sudden influx of investment into safe-haven assets like gold.
- B. An immediate rise in global interest rates.
- C. An accelerated pace of financial regulation and policy changes worldwide
- D. Reduction in overall market liquidity due to increased counterparty risk.

The correct answer is **D**.

The default of a clearing member would likely shake confidence in the clearinghouse's ability to manage counterparty risk, leading to greater perceived risk in the market. This increased counterparty risk may lead other market participants to hold back from trading, leading to a decrease in market liquidity. This is the most immediate and direct consequence of the clearing member's default.

A is incorrect. While a default might increase market uncertainty and potentially lead some investors to shift towards safe-haven assets, this is not the most direct or immediate consequence of a clearing member's default.

B is incorrect. While a major default might potentially affect interest rates, it would not be an immediate effect, and the impact would depend on many other factors such as central bank policies and overall market conditions.

C is incorrect. Increased regulation isn't an immediate or direct effect of the default. Regulatory changes usually occur over longer time frames as they require a comprehensive understanding and assessment of the situation, drafting of new regulations, public commentary, and then eventual adoption and enforcement of those regulations.

Things to Remember

- Central Counterparty Clearing Houses (CCPs) play a crucial role in financial markets by managing and mitigating counterparty risk. They act as a buyer to every seller and a seller to every buyer, thus maintaining market stability.
- Counterparty risk is the risk that a party in a transaction will default on its contractual obligations. When a CCP member defaults, it shakes confidence in the CCP's ability to manage this risk, which can impact market liquidity.
- Market liquidity refers to the ability to buy or sell assets without causing a significant movement in the price. A decrease in liquidity could make it harder for market participants to execute their trades efficiently.
- The impact of a clearing member's default extends beyond the defaulting party and can cause ripple effects through the financial system due to the interconnectedness of financial institutions.

Q.1140 A large hedge fund, Alpha Strategies, operates heavily in derivative markets for risk management and speculative purposes. Due to the varying nature of their trades, Alpha

Strategies uses both bilateral markets and centrally cleared markets. The Risk Management team at Alpha Strategies is reviewing the fund's risk exposure in light of recent market volatility. The team is particularly focused on comparing the effect of netting in both types of markets. One of the new members of the team, Oliver, is assigned to evaluate the benefits and differences of netting in both types of markets. Based on his understanding of bilateral and centrally cleared markets, Oliver should conclude that:

- A. Both bilateral and centrally cleared markets offer equal levels of netting benefits due to standardized netting procedures.
- B. Netting in bilateral markets provides higher netting benefits compared to centrally cleared markets due to the ability to custom-design contracts.
- C. Centrally cleared markets offer superior netting benefits compared to bilateral markets due to multilateral netting.
- D. Neither market offers any significant netting benefits as netting is primarily influenced by the regulatory environment rather than the market structure.

The correct answer is C.

Centrally cleared markets, due to their structure and the involvement of a central counterparty (CCP), offer superior netting benefits compared to bilateral markets. This is primarily due to multilateral netting where a CCP aggregates and offsets all the receivables and payables from various transactions of a participant. This can significantly reduce the participant's counterparty risk and potentially lower required capital reserves.

A is incorrect. While both markets utilize netting, the level of benefits is not equal due to the different market structures. The involvement of a CCP in centrally cleared markets enables more efficient netting procedures.

B is incorrect. While it is true that bilateral contracts can be custom-designed, this does not necessarily lead to superior netting benefits. Instead, it can lead to increased counterparty risk due to the lack of a central counterparty and increased complexity.

D is incorrect. While the regulatory environment can influence netting procedures, it is not the primary determinant. The structure of the market - bilateral or centrally cleared - plays a crucial role in the extent of netting benefits.

Things to Remember

- Netting is the process of offsetting the value of multiple positions or payments due to be exchanged between two or more parties.
- Bilateral markets involve direct transactions between two parties without the involvement of a central counterparty.

- Centrally cleared markets involve a central counterparty (CCP) that interposes itself between the two parties to a trade, becoming the buyer to every seller and the seller to every buyer.
 - Multilateral netting in centrally cleared markets allows for the aggregation and offsetting of positions across multiple participants, leading to more efficient risk management.
 - Counterparty risk is the risk that one party in a financial transaction will default or fail to meet its obligations, leading to potential losses for the other party.
-

Q.1141 Central Counterparties (CCPs) play a crucial role in the financial markets by reducing counterparty risk and enhancing market liquidity. However, they are not immune to risks themselves. One of the significant risks they face is liquidity risk. Which of the following is the main source of liquidity risk in CCPs?

- A. High transaction costs.
- B. Varying margin payments.
- C. Overinvestment in long-term assets.
- D. High operating costs.

The correct answer is **B**.

The main source of liquidity risk in Central Counterparties (CCPs) is the varying margin payments. Margin payments are a type of financial guarantee required by CCPs from their members to cover potential future exposure in the period between the last margin collection and the liquidation of positions following a member's default. These payments vary as the positions of the members change. If a CCP does not have sufficient funds to meet these varying margin payments, it may fail to fulfill its obligations to the surviving members promptly. This situation can lead to liquidity risk. Therefore, CCPs must manage their liquidity risk effectively by ensuring they have enough liquid assets to meet their obligations.

Choice A is incorrect. While high transaction costs can impact the profitability of CCPs, they do not directly contribute to liquidity risk. Liquidity risk arises from the inability to meet short-term financial obligations, which is not necessarily related to transaction costs.

Choice C is incorrect. Overinvestment in long-term assets can lead to asset-liability mismatches and potential solvency issues for CCPs. However, it does not directly result in liquidity risk unless these assets cannot be easily liquidated or converted into cash without a significant loss of value.

Choice D is incorrect. High operating costs can strain the financial resources of CCPs but they are more related to operational efficiency rather than liquidity risk per se. These costs do not affect the ability of CCPs to meet their short-term financial obligations unless they lead to a depletion of liquid assets.

Things to Remember

- Central Counterparties (CCPs) are financial institutions that take on counterparty credit risk between parties in a transaction and aim to reduce the risk each party faces.
 - Liquidity Risk refers to the risk that a firm will not be able to meet its short-term financial demands due to an inability to convert assets into cash without incurring significant loss.
 - Varying margin payments can lead to liquidity risks as they can cause fluctuations in the amount of cash or collateral required, potentially leading CCPs unable to meet their obligations.
 - High transaction costs and operating costs do not directly contribute towards liquidity risk. They might affect profitability but do not necessarily impact short-term payment obligations.
 - Overinvestment in long-term assets could potentially lead towards liquidity risks if these assets cannot be quickly liquidated during times of need. However, this is more related with asset-liability management rather than day-to-day operations of CCPs.
-

Q.5358 A risk manager at XYZ bank is concerned with the advantages and disadvantages of using central clearing for OTC derivatives trades. One advantage of central clearing is that it can reduce systemic risk in the financial markets. Which of the following **correctly** describes how central clearing accomplishes this?

- A. By increasing the number of parties involved in a trade, thus spreading risk across multiple entities.
- B. By increasing the transparency of OTC derivative trades thus reducing the likelihood of fraud or manipulation.
- C. By providing a centralized mechanism for monitoring and managing counterparty credit risk.

D. By reducing the overall volume of OTC derivative trades, thus lowering the potential for market disruption in the event of a default.

The correct answer is C.

Central clearing provides a centralized mechanism for monitoring and managing counterparty credit risk, which in turn reduces the potential for systemic risk in the financial markets. By pooling risk and requiring collateral to be posted, central clearing houses can help to ensure that losses are contained and do not spread throughout the financial system. This is achieved by acting as the counterparty to all trades, thereby reducing the risk that a default by one party will have a domino effect on others. In addition, central clearing houses also have robust risk management systems in place to further mitigate counterparty credit risk.

Choice A is incorrect. Central clearing does not necessarily increase the number of parties involved in a trade. Instead, it introduces a central counterparty (CCP) that stands between the two original trading parties, thereby reducing direct exposure to each other's credit risk.

Choice B is incorrect. While central clearing can enhance transparency by providing a centralized mechanism for monitoring and managing counterparty credit risk, this does not directly contribute to the reduction of systemic risk. Transparency may help prevent fraud or manipulation but it doesn't inherently reduce systemic risk.

Choice D is incorrect. Central clearing does not aim to reduce the overall volume of OTC derivative trades. Rather, its primary function is to manage and mitigate counterparty credit risk which in turn reduces systemic risk within financial markets.

Things to Remember

- **Central Counterparty (CCP):** A central counterparty is an entity that interposes itself between the two counterparties in a trade, becoming the buyer to every seller and the seller to every buyer. This helps in reducing counterparty credit risk.
- **Collateral Management:** Central clearing houses require participants to post collateral, which acts as a buffer against potential losses in case of default.
- **Risk Management Systems:** Central clearing houses have robust risk management systems in place to monitor and manage risks associated with OTC derivative trades.
- **Systemic Risk:** Systemic risk refers to the risk of a widespread financial market disruption or collapse that can have a cascading effect on the entire financial system.
- **Counterparty Credit Risk:** Counterparty credit risk is the risk that one party in a trade may default on its obligations, leading to financial losses for the other party.

Q.5359 A risk manager at a multinational corporation is analyzing the currency exposure of the company's European operations. The company has receivables in euros, payables in US dollars, and a subsidiary in Germany that has both euro-denominated assets and liabilities. Which of the following is the most appropriate hedging strategy to manage this currency risk?

- A. Buying a currency option that gives the right to sell euros and buy US dollars at a predetermined exchange rate.
- B. Buying a currency option that gives the right to buy euros and sell US dollars at a predetermined exchange rate.
- C. Entering into a forward contract to sell euros and buy US dollars at a predetermined exchange rate.
- D. Entering into a forward contract to buy euros and sell US dollars at a predetermined exchange rate.

The correct answer is **A**.

The company's primary concern is the potential depreciation of the euro against the US dollar, given its receivables in euros and payables in US dollars. To hedge against this risk, the risk manager should consider purchasing a currency option that provides the right (but not the obligation) to sell euros and buy US dollars at a predetermined exchange rate. This strategy offers protection against unfavorable currency movements while still allowing the company to benefit from any favorable currency movements. Currency options provide a level of flexibility that is particularly useful in volatile currency markets. If the euro appreciates against the US dollar, the company can choose not to exercise the option and instead convert its euros at the more favorable current market rate. Conversely, if the euro depreciates against the US dollar, the company can exercise the option and convert its euros at the predetermined rate, thereby avoiding a larger loss.

Choice B is incorrect. Buying a currency option that gives the right to buy euros and sell US dollars at a predetermined exchange rate would not be suitable in this scenario. This strategy would be more appropriate if the company had payables in euros and receivables in US dollars, which is opposite to the given situation.

Choice C is incorrect. Entering into a forward contract to sell euros and buy US dollars at a predetermined exchange rate might seem like an appropriate strategy as it provides certainty about future exchange rates. However, it does not provide flexibility if the actual future spot rate turns out to be more favorable than the contracted forward rate.

Choice D is incorrect. Entering into a forward contract to buy euros and sell US dollars at a predetermined exchange rate would not be suitable for managing currency risk in this scenario as it implies that company has liabilities denominated in Euros while actually company has liabilities denominated in USD.

Things to Remember

- Currency exposure refers to the risk that a company's financial performance or position

will be impacted by fluctuations in exchange rates.

- Receivables are amounts owed to a company by its customers, typically in the form of invoices for goods or services provided.
 - Payables are amounts owed by a company to its suppliers or creditors, typically for goods or services received.
 - Currency options give the holder the right, but not the obligation, to buy or sell a specific amount of currency at a predetermined exchange rate within a specified period.
 - Forward contracts are agreements to buy or sell a specified amount of currency at a predetermined exchange rate on a future date.
 - Hedging is a risk management strategy used to offset potential losses from adverse price movements in financial instruments or currencies.
-

Q.5360 A CRO at a large bank is reviewing the bank's risk management practices for OTC derivatives. Which of the following is a regulatory initiative for OTC derivatives?

- A. All OTC derivatives must be traded bilaterally
- B. All OTC derivatives must be reported to a central trade repository
- C. OTC derivatives can only be cleared through individual banks
- D. OTC derivatives can only be traded over-the-counter

The correct answer is **B**.

One of the key regulatory initiatives for the OTC derivatives market is the requirement for all trades to be reported to a central trade repository. This initiative was put in place to provide regulators with comprehensive information on the risks being undertaken by participants in the OTC market. The central trade repository serves as a centralized database that collects and maintains records of all OTC derivative transactions. This allows for increased transparency and aids in the identification and mitigation of systemic risk. The central trade repository also facilitates the monitoring of market abuse and manipulation, thereby promoting market integrity. Furthermore, it enhances the efficiency of risk management practices by providing market participants with a consolidated view of their positions and exposures.

Choice A is incorrect. Not all OTC derivatives must be traded bilaterally. While bilateral trading is a common feature of OTC derivatives, it's not a regulatory requirement for all such instruments. Some OTC derivatives can also be cleared through central counterparties (CCPs).

Choice C is incorrect. It's not accurate to say that OTC derivatives can only be cleared through individual banks. In fact, the post-financial crisis regulatory reforms have encouraged the clearing of standardized OTC derivatives through CCPs to reduce counterparty risk.

Choice D is incorrect. Although the term "over-the-counter" suggests that these instruments are traded outside of formal exchanges, it doesn't mean they can only be traded over-the-counter. Some types of OTC derivatives may also be traded on organized exchanges.

Things to Remember

- Central Counterparties (CCPs) play a crucial role in the clearing of OTC derivatives by acting as an intermediary between the two parties to a trade, thereby reducing counterparty risk.
- Regulatory initiatives such as the Dodd-Frank Act in the United States and EMIR (European Market Infrastructure Regulation) in the European Union have introduced requirements for the central clearing of certain standardized OTC derivatives.
- OTC derivatives are typically customized contracts that are negotiated directly between two parties, allowing for flexibility in terms and structures compared to standardized exchange-traded derivatives.
- Counterparty credit risk is a significant concern in OTC derivatives trading, as the failure of one party to fulfill its obligations can lead to substantial losses for the other party.
- Margin requirements are often imposed on OTC derivatives transactions to mitigate counterparty credit risk and ensure that parties have sufficient collateral to cover potential losses.

Q.5361 A risk manager at a CCP is assessing the risks faced by the CCP. Which of the following is a risk faced by CCPs that arises from the use of subjective assumptions in determining the initial margin requirement and default fund contributions of traders?

A. Default Risk

B. Non-Default Events

C. Model Risk

D. Liquidity Risk

The correct answer is **C**.

Model Risk is a type of risk that CCPs face, which arises from the use of models to calculate initial margin requirements and default fund contributions. These models may not accurately capture the risks posed by clearing members and their portfolios. This inaccuracy can stem from the use of subjective assumptions in the models or from the models being based on historical data that may not be representative of current market conditions. Therefore, the risk that arises from the use of subjective assumptions in determining the initial margin requirement and default fund contributions of traders is referred to as Model Risk.

Choice A is incorrect. Default risk refers to the risk that a party will not meet its obligations under a financial contract. While this can be influenced by the initial margin requirement and default fund contributions, it is not directly related to the application of subjective assumptions in determining these amounts.

Choice B is incorrect. Non-default events refer to situations where a trader's position deteriorates but does not reach the point of default. These events are typically driven by market conditions and trader behavior, rather than subjective assumptions made when setting margin requirements or default fund contributions.

Choice D is incorrect. Liquidity risk pertains to the possibility that a CCP may not have sufficient funds available to meet its obligations as they come due. While liquidity needs can be influenced by margin requirements and default fund contributions, this risk arises from cash flow mismatches or market disruptions, rather than from subjective assumptions used in setting these amounts.

Things to Remember

- Central Counterparties (CCPs) play a crucial role in financial markets by acting as intermediaries in transactions, thereby reducing counterparty credit risk.
- Initial margin requirements are set by CCPs to cover potential future exposure to losses in the event of a counterparty default.
- Default fund contributions are made by clearing members to further protect the CCP in case of a default.
- Model Risk can arise from various sources, including model limitations, data quality issues, and the use of subjective assumptions.

- Model Risk can lead to inaccurate risk assessments, potentially exposing CCPs to unexpected losses.
-

Q.5362 Which of the following statements regarding margin requirements in central clearing and bilateral clearing is **correct**?

- A. In centrally cleared markets, margin requirements are typically lower compared to bilateral markets.
- B. In bilateral markets, margin requirements are typically lower compared to centrally cleared markets.
- C. Margin requirements are typically the same for both centrally cleared and bilateral markets.
- D. Margin requirements are not used in either centrally cleared or bilateral markets.

The correct answer is **B**.

Bilateral markets generally have lower margin requirements than centrally cleared markets. This is primarily because bilateral markets involve only two parties, and their creditworthiness is assessed by each other. This direct assessment allows for a more tailored approach to risk management, which can result in lower margin requirements. On the other hand, in centrally cleared markets, the credit risk is borne by all members. Therefore, margin requirements need to be higher to cover the potential losses from a defaulting member. This is a crucial aspect of risk management in centrally cleared markets, as the default of one member can have significant implications for all other members. Therefore, higher margin requirements are necessary to mitigate this risk and ensure the stability of the market.

Choice A is incorrect. While it might seem logical to assume that centrally cleared markets, with their added layer of security and risk management, would have lower margin requirements, this is not the case. In fact, centrally cleared markets often have higher margin requirements due to the increased risk associated with a larger number of participants and more complex transactions.

Choice C is incorrect. Margin requirements are not typically the same for both centrally cleared and bilateral markets. The level of risk involved in each type of market varies significantly, which leads to different margin requirements. Bilateral markets generally involve less participants and simpler transactions compared to centrally cleared markets, leading to lower margin requirements.

Choice D is incorrect. Margin requirements are used in both centrally cleared and bilateral markets as a key tool for managing risk and ensuring smooth trading activities. They act as a form of collateral that protects against potential losses from trading activities.

Things to Remember

- Margin requirements are a key tool for managing risk in financial markets.
 - In centrally cleared markets, the credit risk is borne by all members, leading to higher margin requirements.
 - Bilateral markets involve only two parties, allowing for a more tailored approach to risk management and lower margin requirements.
 - Margin requirements act as a form of collateral to protect against potential losses from trading activities.
 - The default of one member in a centrally cleared market can have significant implications for all other members, necessitating higher margin requirements.
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Reading 33: Futures Markets

Q.605 Jack Lee, a commodities investor at Singapore Investment Bank, instructs his team of traders to sell a September copper futures contract of 25,000 pounds in the COMEX (Commodities exchange, a sub-division of the NYMEX). Given these instructions, a bank's team of floor traders at the COMEX physically met the seller of the September Copper futures contract and determined the price of \$0.05 (5 cents) per pound. Looking at the nature of the transaction, one can say that the contract is being traded on:

- A. An over-the-counter market.
- B. An open outcry exchange.
- C. An electronic exchange.
- D. Private transaction.

The correct answer is **B**.

An open outcry exchange is a method of communication between professionals on a stock exchange or futures exchange typically on a trading floor. It involves shouting and the use of hand signals to transfer information primarily about buy and sell orders. The part of the trading floor where this takes place is called a pit. In this case, the team of floor traders physically met the seller of the September copper futures contract to determine the price of the contract, which is a characteristic of an open outcry exchange. Futures contracts are standardized agreements that typically trade on an exchange. One party agrees to buy a given quantity of securities or a commodity, and take delivery on a certain date. Physical meeting for price determination is a key feature of open outcry exchanges.

Choice A is incorrect. An over-the-counter (OTC) market is a decentralized market where trading of securities, commodities, or derivatives takes place directly between two parties without the supervision of an exchange. In this case, the transaction was executed on COMEX which is a centralized exchange and not an OTC market.

Choice C is incorrect. An electronic exchange operates through an electronic trading platform where buyers and sellers transact electronically without physical interaction. However, in this scenario, the bank's team of floor traders physically met with the seller to agree on a price which indicates that it was not traded on an electronic exchange.

Choice D is incorrect. The trade occurred on a public exchange (COMEX), not a private transaction, which would be a non-public and individualized agreement outside of an exchange.

Things to Remember

- Open outcry exchange involves shouting and hand signals on a trading floor to transfer information about buy and sell orders.
- Futures contracts are standardized agreements that typically trade on an exchange.

- Physical meeting for price determination is a key feature of open outcry exchanges.
 - Over-the-counter (OTC) markets are decentralized markets where trading occurs directly between two parties without exchange supervision.
 - Electronic exchanges operate through electronic trading platforms without physical interaction between buyers and sellers.
 - Private transactions are non-public and individualized agreements outside of an exchange.
-

Q.606 Matias, FRM, has recently joined the London office of Venture Financials as a junior derivatives trader. Currently, Matias has an open position of gold futures contracts on the London Exchange. According to the contract, Matias is obligated to sell 5,000 troy ounces of gold with delivery in April. Determine the appropriate position of Matias's trade.

- A. Hedged futures position.
- B. Short put option position.
- C. Short futures position.
- D. Short forwards position.

The correct answer is **C**.

Matias's position is a short futures position. In a futures contract, the investor agrees to sell the underlying asset at a predetermined price on a specific future date. In this case, Matias has agreed to sell 5,000 troy ounces of gold in April, which is a classic example of a short futures position. This position allows Matias to profit if the price of gold falls before the delivery date in April. It's important to note that futures contracts are standardized and traded on exchanges, which provides liquidity and reduces counterparty risk. However, it also exposes the investor to the risk of adverse price movements.

Choice A is incorrect. A hedged futures position involves taking a long and short position in two different but related markets to reduce risk. Matias only has a single open position on gold futures contracts, which does not constitute a hedged futures position.

Choice B is incorrect. A short put option position would mean that Matias has sold the right to another party to sell him gold at a predetermined price before the contract expires. However, as per the question, Matias is required to sell gold in future which aligns with the definition of a short futures contract rather than a put option.

Choice D is incorrect. While both forwards and futures contracts involve an agreement to buy

or sell an asset at a future date for an agreed price today, they are not identical. The key difference lies in their settlement process and standardization; forward contracts are private agreements between two parties and are not standardized, whereas future contracts are exchange-traded and thus highly standardized. In this case, since Matias's contract is traded on the London Exchange it can't be considered as forward contract.

Things to Remember

- Short Futures Position: Involves agreeing to sell an underlying asset at a predetermined price on a specific future date.
 - Futures Contracts: Standardized agreements traded on exchanges, providing liquidity and reducing counterparty risk.
 - Hedged Futures Position: Involves taking long and short positions in related markets to reduce risk.
 - Short Put Option Position: Selling the right to another party to sell the asset at a predetermined price before the contract expires.
 - Forwards Contracts: Private agreements between two parties, not standardized like futures contracts.
-

Q.607 Jack Manchester is a summer intern at one of the well-known investment banks in Canary Wharf. During lunch with his supervisor, Manchester was given a document pertaining to the futures contract's specifications. The excerpt from the document reads "... the tin must be a minimum of 99.85% purity conforming to BS EN 610:1996. All the tin deliverable against the London Metal Exchange (LME) contracts must be LME approved." This futures contract's specification is most likely related to:

- A. Position limits.
- B. Delivery arrangements.
- C. Contract size.
- D. Quality of the underlying.

The correct answer is **D**.

The excerpt from the document provided to Jack Manchester refers to the quality of the underlying asset, in this case, tin. Futures contracts, which are traded on exchanges, have

standardized specifications. These specifications include the quality of the underlying assets, delivery time and place, contract size, price, and quotation, among other things. In this particular excerpt, the reference to the tin's purity and the requirement for it to be approved by the London Metal Exchange (LME) clearly indicates that the specification is about the quality of the underlying asset. Therefore, choice D is the correct answer.

Choice A is incorrect. Position limits refer to the maximum number of speculative futures contracts one can hold. The information provided in the document does not pertain to position limits.

Choice B is incorrect. Delivery arrangements would typically include details about when and where delivery could take place, which are not mentioned in the document.

Choice C is incorrect. Contract size refers to the amount of the underlying asset that must be delivered upon contract expiration. The document does not provide any information regarding this aspect.

Things to Remember

- Futures contracts are standardized agreements to buy or sell a specific asset at a predetermined price at a specified time in the future.
- Underlying assets in futures contracts can vary from commodities like oil, gold, and wheat to financial instruments like stock indices and currencies.
- Quality of the underlying asset is a crucial specification in futures contracts to ensure uniformity and standardization in trading.
- Delivery arrangements in futures contracts specify when and where the delivery of the underlying asset will take place if the contract is held until expiration.
- Contract size refers to the quantity of the underlying asset that is traded in one futures contract.
- Position limits are restrictions on the maximum number of futures contracts that a trader can hold at any given time.
- Futures contracts are traded on exchanges like the Chicago Mercantile Exchange (CME) or the London Metal Exchange (LME).

Q.608 Nora Schneider is an experienced derivatives trader at a German commodities investing firm. Recently, she was given additional responsibilities to look after the trader's training

department. While training the newly employed derivatives traders, she instructed the trader to clearly read the terms and specifications of a futures contract. Which of the following feature is NOT specified in the contract specification details of a futures contract?

- A. Position limit.
- B. Delivery month.
- C. Price limits.
- D. Counterparty.

The correct answer is **D**.

The counterparty is not specified in the contract specification details of a futures contract. Futures contracts are standardized agreements that are traded on exchanges. The transactions are managed by clearinghouses, which serve as intermediaries between the buyer and the seller. As a result, the investors of futures contracts typically do not know the identity of the counterparty to the transaction. This is a key feature of futures contracts that distinguishes them from other types of financial contracts, such as forward contracts, where the parties are directly involved with each other and are aware of each other's identities. The absence of counterparty information in the contract specifications of a futures contract is a reflection of the role of the clearinghouse and the standardized nature of these contracts.

Choice A is incorrect. Position limit is typically included in the contract specification details of a futures contract. It refers to the maximum number of speculative futures contracts one can hold as determined by the Commodity Futures Trading Commission (CFTC).

Choice B is incorrect. Delivery month, which specifies when the commodity will be delivered, is also a standard feature in futures contracts.

Choice C is incorrect. Price limits are usually specified in futures contracts to prevent extreme price volatility within a single trading day.

Things to Remember

- Clearinghouses in futures contracts act as intermediaries between the buyer and the seller.
- Futures contracts are standardized agreements traded on exchanges.
- Counterparty information is not disclosed in futures contracts due to the role of clearinghouses.
- Position limits are set by regulatory bodies like the CFTC to control speculative positions in futures contracts.

- Delivery month specifies the month in which the commodity will be delivered in a futures contract.
 - Price limits are included in futures contracts to prevent extreme price volatility within a single trading day.
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Q.609 The Karachi Mercantile Exchange (KME) has set the daily price limit of rice futures contracts to Rs.4. The closing price of the rice futures contract on Monday was Rs.140 per 100 KG. If the evening newspaper on Wednesday reads that "Rice futures contracts closed limit down at Rs.138 per 100 KG," then which of the following is the most likely closing price of the rice futures contract on the preceding day?

- A. Rs. 136 per 100/KG
- B. Rs. 140 per 100/KG
- C. Rs. 142 per 100/KG
- D. Rs. 138 per 100/KG

The correct answer is **C**.

Price limits are the maximum price movement/limit set by exchanges. Suppose the intraday increase in the price of the futures contract is equal to the predefined price limit. In that case, the contract is said to be limit up, whereas if the intraday decrease in the price of the futures contract is equal to the predefined price limit, the contract is said to be limit down. Since the newspaper reads that the price of the contract closed limit down (price decreased by Rs. 4) at Rs. 138, the preceding day (i.e., Tuesday) price is:

Limit down close to the current day = Closing price of the preceding day - Price limit

The closing price of the preceding day = Rs. 138 + Rs. 4 = Rs. 142

Things to Remember

- Price limits are set by exchanges to control the volatility of futures contracts.
- Limit up occurs when the price of a futures contract increases by the predefined limit during a trading session.
- Limit down occurs when the price of a futures contract decreases by the predefined limit during a trading session.
- Price limits help prevent extreme price movements in a short period, protecting investors from potential losses.
- Traders need to be aware of price limits and their implications when trading futures contracts.

Q.610 Elif Makarov, a derivatives trader at one of the largest commodities trading firms in Moscow, is looking at a possible arbitrage trade in the copper futures contract. If the copper futures contract price is \$47.6 and the spot price of copper is \$48.9, then determine the appropriate strategy Makarov may take to earn the arbitrage profit.

- A. Take a short position in the copper futures contract and buy copper at the spot price.
- B. Take a long position in the copper futures contract and sell copper at the spot price.
- C. Wait for copper futures contracts to converge to the spot price and then take a short position in futures contracts.
- D. Wait for copper futures contracts to converge to the spot price and then take a long position in futures contracts.

The correct answer is **B**.

The optimal strategy for Makarov to secure an arbitrage profit in this scenario is to take a long position in the copper futures contract and sell copper at the spot price. Arbitrage is a trading strategy that involves buying and selling assets simultaneously in different markets to take advantage of price discrepancies. In this case, the futures price of copper (\$47.6) is lower than the spot price (\$48.9). This price discrepancy presents an opportunity for arbitrage. By taking a long position in the copper futures contract, Makarov is essentially agreeing to buy copper at a future date at the current futures price. At the same time, he can sell copper at the spot price, which is higher than the futures price. The difference between the spot price and the futures price represents the arbitrage profit. This strategy is known as 'cash-and-carry' arbitrage and is commonly used in commodities markets. It's important to note that this strategy assumes that the markets are efficient and that there are no transaction costs or restrictions on borrowing.

Choice A is incorrect. Taking a short position in the copper futures contract and buying copper at the spot price would not result in an arbitrage profit. This is because the futures price of copper is lower than its spot price, so selling futures (going short) and buying at the spot price would result in a loss rather than a profit.

Choice C is incorrect. Waiting for copper futures contracts to converge to the spot price and then taking a short position in futures contracts would not guarantee an arbitrage profit either. The convergence of future prices to spot prices does not necessarily mean that future prices will rise above current levels, which is necessary for this strategy to be profitable.

Choice D is incorrect. Similarly, waiting for copper futures contracts to converge with the spot price before taking a long position also does not ensure an arbitrage profit. This strategy assumes that future prices will fall below current levels after convergence, which may or may not happen.

Things to Remember

- Arbitrage is a trading strategy that involves exploiting price discrepancies in different markets to make a risk-free profit.

- 'Cash-and-carry' arbitrage involves buying an asset in the spot market and simultaneously selling it in the futures market to profit from price differentials.
 - Efficient markets hypothesis (EMH) states that asset prices reflect all available information and that it is impossible to consistently outperform the market.
 - Transaction costs, borrowing costs, and other frictions can impact the profitability of arbitrage strategies.
 - Convergence of futures prices to spot prices does not guarantee arbitrage opportunities as the direction of price movements after convergence is uncertain.
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Q.619 Alisha Gomez, head of the trading department, is interviewing with one of the potential candidates for a position as a junior trader in the derivatives units. Gomez asked the candidate to identify which of the following prices is used for calculating daily gains, losses, and margin requirements for the parties involved in the trading of futures contracts. Which of the following is the appropriate answer to Alisha Gomez's question?

- A. Opening price.
- B. High price.
- C. Closing price.
- D. Settlement price.

The correct answer is **D**.

The settlement price is the correct answer. In futures contract trading, the settlement price is used to calculate daily gains, losses, and margin requirements. It is not the closing price of the contract. Instead, it is the average price at which the contract is traded during the last period or before the end of a day's trading period. The exchange itself sets the settlement price. This is done to prevent traders from manipulating futures prices. The settlement price is a crucial component in futures trading as it determines the value of the contract at the end of each trading day. It is used to mark the positions to market, which means adjusting the value of a futures contract at the end of each trading day to reflect the profit or loss incurred on that day. This process ensures that losses and gains are recorded and accounted for daily, providing a transparent and fair trading environment.

Choice A is incorrect. The opening price is the price at which a security first trades upon the opening of an exchange on a trading day; however, it is not used in the calculation of daily gains, losses, and margin requirements in futures contract trading.

Choice B is incorrect. The high price refers to the highest traded price of a security during a trading day. While this information can be useful for other purposes, it does not play a role in calculating daily gains, losses or margin requirements for futures contracts.

Choice C is incorrect. The closing price refers to the final price at which a security trades during a regular trading session on any given day. Although important for various analyses and calculations, it's not used specifically for determining daily gains, losses or margin requirements in futures contract trading.

Things to Remember

- **Marking to market:** The process of adjusting the value of a futures contract at the end of each trading day to reflect the profit or loss incurred on that day.
 - **Manipulation prevention:** The exchange sets the settlement price to prevent traders from manipulating futures prices.
 - **Transparency:** Daily gains, losses, and margin requirements are calculated using the settlement price to provide a transparent and fair trading environment.
 - **Margin requirements:** The amount of funds that must be deposited by both parties involved in a futures contract to cover potential losses.
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Q.621 Vikram Pandit, a derivatives investor from Mumbai, instructs his broker to terminate his short position in 10 futures contracts of live cattle at Chicago's futures exchange. The broker proposed to him four alternatives for terminating the contract. Which of the following is the most appropriate method?

- A. Purchase cattle from Mumbai and physically deliver the cattle to the long party.
- B. Take a new short position of the same size in a live cattle futures contract with a different delivery date.
- C. Find a trader with a long position in live cattle and settle up between yourselves, off the floor of the exchange.
- D. Take a long position of the same size in a live cattle futures contract with the same delivery date.

The correct answer is **D**.

Taking a long position of the same size in a live cattle futures contract with the same delivery date is the most appropriate method to terminate a short position in futures contracts. This is

because futures contracts are standardized agreements that are traded on an exchange. When an investor takes a short position in a futures contract, they are agreeing to sell the underlying asset at a specific price on a specific date in the future. If the investor later decides to terminate this position, they can do so by taking an exactly opposite position (a long position) in the same futures contract with the same maturity date. This effectively cancels out the original short position, allowing the investor to exit the contract without having to deliver the physical goods. This method is commonly used in futures trading and is considered the most efficient and cost-effective way to terminate a futures contract.

Choice A is incorrect. Physically delivering the cattle from Mumbai to the long party in Chicago would not terminate the futures contract. The futures contract is a legal agreement to buy or sell a particular commodity at a predetermined price at a specified time in the future. Physical delivery of cattle does not nullify this agreement.

Choice B is incorrect. Taking a new short position of the same size in a live cattle futures contract with a different delivery date would not terminate the existing short position but rather create an additional one. This action would increase, not decrease, Vikram's exposure to live cattle prices.

Choice C is incorrect. Settling up between traders off the floor of exchange might be possible, but it's generally against exchange rules and could lead to penalties for both parties involved. Moreover, it doesn't guarantee that Vikram's original short position will be properly closed out on his broker's books.

Things to Remember

- **Futures Contracts:** Standardized agreements traded on an exchange to buy or sell an underlying asset at a specific price on a specific date in the future.
- **Short Position:** When an investor agrees to sell an asset in the future at a specific price.
- **Long Position:** When an investor agrees to buy an asset in the future at a specific price.
- **Terminating Futures Contracts:** Taking an opposite position (long for short, short for long) in the same futures contract with the same delivery date to cancel out the original position.
- **Physical Delivery:** Involves the actual transfer of the underlying asset from the seller to the buyer as specified in the futures contract.
- **Exchange Rules:** Rules and regulations set by the exchange to ensure fair and orderly trading of futures contracts.

Q.5363 A trader wants to sell a call option on a stock in a volatile market but wants to ensure that the option is sold above a certain price point. Which of the following order types should the trader use to guarantee execution at or above that price point?

- A. Market order
- B. Stop-limit order
- C. Discretionary order
- D. Fill-or-kill order

The correct answer is **B**.

A stop-limit order is an order to sell (or buy) at a specified price or better after the price has reached a specified stop price. This type of order combines both a stop order and a limit order, meaning that the order will be executed at a specified price or better but only after the stop price has been reached. In this case, the trader can set the stop price at the desired price point and the limit price at the minimum price they are willing to accept for the option. This ensures that the option is sold at or above the desired price point.

A is incorrect. A market order executes immediately at the best available price.

C is incorrect. A discretionary order, also known as a market-not-held order, refers to an order in which the broker has the option to delay execution with the expectation of obtaining a more favorable price.

D is incorrect. A fill-or-kill order requires immediate execution of the entire order or none at all, which may not be suitable in a volatile market.

Things to Remember

- Market orders execute immediately at the best available price.
 - Stop-limit orders combine a stop order and a limit order, ensuring execution at a specified price or better after the stop price has been reached.
 - Discretionary orders allow the broker to delay execution in the hope of obtaining a more favorable price.
 - Fill-or-kill orders require immediate execution of the entire order or none at all.
-

Reading 34: Using Futures for Hedging

Q.625 A Canadian importer has ordered \$1,000,000 US worth of oil drilling equipment to be delivered in six months. The current spot exchange rate is 1.3 CAD per 1.00 USD. However, the importer fears that the Canadian dollar will depreciate to 1.35 CAD per 1.00 USD in the next 6 months. As a result, the importer enters a forward contract to purchase \$1,000,000 at a forward rate of 1.32 CAD per 1.00 USD. If the Canadian dollar depreciates to \$1.35 CAD per 1.00 USD as predicted, what is the savings to the importer from his dealings in the forward market?

- A. \$350,000 CAD
- B. \$30,000 CAD
- C. \$50,000 CAD
- D. \$20,000 CAD

The correct answer is **B**.

The forward market will help the importer to lock in the exchange rate of 1.32 CAD per 1.00 USD. Thus, they will spend $1.32 * 1,000,000 = \$1,320,000$ CAD in the transaction.

Without the forward market, the importer would transact at 1.35 CAD per 1.00 USD, spending a total of $1.35 * 1,000,000 = \$1,350,000$ CAD

$$\text{Total savings} = 1,350,000 \text{ CAD} - 1,320,000 \text{ CAD} = 30,000 \text{ CAD}$$

Q.627 Colin Thomson, the risk manager of a tire manufacturing company, suggests that the company should focus its resources on its core business activities rather than investing resources in hedging the risks faced by the company. He further added that the shareholders have as much information as the management of the company. Therefore, shareholders can easily hedge the risks. Lastly, he argued that the shareholders hedge the company's stocks in much smaller quantities. Hence, it is cheaper for the shareholders to hedge the risk as compared to the company. Which of the following options is correct?

- A. Thomson's argument related to the availability of the company's information to the shareholders is incorrect. However, the argument related to the smaller costs incurred by shareholders for hedging risks is correct.
- B. Thomson's argument related to the availability of the company's information to the shareholders is correct. However, the argument related to the smaller costs incurred by shareholders for hedging risks is incorrect.
- C. Both arguments are correct.

D. None of Thomson's arguments are correct.

The correct answer is **D**.

None of Thomson's arguments are correct. The first argument made by Colin Thomson is that shareholders have as much information as the company's management. This is generally not the case. Shareholders may not have access to all the information that the management has, especially the internal information about the company's operations, strategies, and risk exposure. Therefore, they may not be able to hedge the risks as effectively as the company's management. The second argument made by Thomson is that it is cheaper for shareholders to hedge the risk as compared to the company. This is also incorrect. While it's true that shareholders can hedge the company's stocks in smaller quantities, the transaction costs and commissions for these smaller transactions are usually higher. On the other hand, the company, which hedges its risk in many transactions, pays much smaller per dollar transaction costs and commissions. Therefore, it is usually more cost-effective for the company to hedge its risks than for the shareholders.

Choice A is incorrect. Thomson's argument that shareholders have access to the same information as the company's management is not valid. In reality, there are often information asymmetries between a company's management and its shareholders due to insider knowledge and other factors. Furthermore, his argument that it would be more cost-effective for shareholders to hedge risks in smaller quantities is also flawed. The costs of hedging can be substantial for individual investors due to transaction fees and other expenses, which may outweigh any potential benefits.

Choice B is incorrect. As explained above, both of Thomson's arguments are flawed; hence this choice which suggests one of them being correct does not hold true.

Choice C is incorrect. As explained above, neither of Thomson's arguments are correct; hence this choice suggesting both being correct does not hold true.

Things to Remember

- **Information Asymmetry:** Shareholders may not have access to all the information that the management has, leading to information asymmetry in the market.
- **Transaction Costs:** Transaction costs and commissions for smaller transactions can be higher for individual shareholders compared to the company, which can affect the cost-effectiveness of hedging.
- **Insider Knowledge:** Management of a company may have insider knowledge about the company's operations, strategies, and risk exposure, giving them an advantage in making informed decisions.
- **Risk Management:** Companies often use hedging strategies to manage risks effectively

and protect their financial positions from adverse market movements.

Q.629 A risk analyst at a mid-sized alternative investment firm is responsible for hedging the company's exposure to oil prices. The analyst has taken two positions:

1. A long position in the spot oil market
2. short position in oil futures

Suppose the basis (spot price minus futures price) strengthens unexpectedly due to a significant increase in the spot price of oil, while the futures price remains relatively stable. Which of the following statements is true?

- A. The short futures position of the firm will improve.
- B. The long spot position of the firm will improve.
- C. The short futures position of the firm will worsen.
- D. Both the firm's positions will improve equally.

The correct answer is **B**.

Let's first understand the situation:

- The firm has a **long position** in the spot oil market (buying oil).
- The firm has a **short position** in oil futures (agreeing to sell oil at a future date).
- **Basis strengthening:** The **basis** (spot price minus futures price) strengthens unexpectedly due to a significant increase in the spot price of oil, while the futures price remains relatively stable.

Impact on positions:

a) Long spot position:

- As the spot price of oil has increased significantly, the value of the firm's long spot position will improve.
- The firm can now sell the oil it owns at a higher price in the spot market.

b) Short futures position:

- Since the futures price remains relatively stable, there's minimal immediate impact on this position.

- The firm's obligation to sell at the agreed-upon futures price is neither significantly better nor worse.

Evaluating the incorrect options:

- **A. Incorrect:** The short futures position doesn't improve as futures prices are stable.
 - **C. Incorrect:** The short futures position doesn't worsen as futures prices are stable.
 - **D. Incorrect:** While the long position improves, both positions don't improve equally.
-

Q.633 Futures contracts on jet fuel have maturity months in March, May, July, September, and December. An airline is hedging a purchase of 1 million barrels of fuel to be made on June 15 of this year. Which futures contract should it use?

- A. The July contract.
- B. The March contract.
- C. The December contract.
- D. The May contract.

The correct answer is **A**.

The July contract is the most appropriate choice for the airline company. In the context of futures contracts, most positions are closed out prior to the delivery period specified in the contract. Therefore, a practical guideline for hedgers is to choose the futures contract with the earliest possible maturity month following the maturity of the desired hedge. This approach helps to avoid the volatility that often occurs during the expiration month of the contract. In this scenario, the airline company plans to purchase the jet fuel in June. Therefore, the July contract, which is the earliest maturity month following June, is the most suitable choice for the company's hedging strategy.

Choice B is incorrect. The March contract would not be an appropriate choice for the airline company as it matures before the scheduled purchase date of June 15. This would expose the company to price risk between the maturity of the futures contract and the actual purchase date.

Choice C is incorrect. The December contract matures much later than the scheduled purchase date, which means that it could potentially expose the company to unnecessary price

risk in case jet fuel prices decrease after June.

Choice D is incorrect. Although May contract matures closer to June 15, it still exposes the airline company to a month's worth of price risk between its maturity and actual purchase date. Therefore, July contract (option A) which matures after June 15 provides a better hedge against potential increases in jet fuel prices.

Things to Remember

- Futures contracts are standardized agreements to buy or sell a specified asset at a predetermined price at a specified time in the future.
 - Hedging is a risk management strategy used to offset potential losses in one investment by taking an opposite position in a related security, such as using futures contracts to hedge against price fluctuations.
 - Maturity months in futures contracts represent the months in which the contract expires and delivery of the underlying asset can take place.
 - The concept of basis risk is important in hedging with futures contracts, as it refers to the risk that the price of the hedging instrument (futures contract) may not move in perfect correlation with the price of the asset being hedged.
 - Roll yield is the return generated by rolling a futures contract position from one maturity month to another, and it can impact the overall performance of a hedging strategy.
-

Q.634 A German electronic appliances manufacturer expects to receive 30 million Turkish Liras at the end of March. The Lira futures contracts on the Eurex Exchange are available for the delivery months of March, June, September, and December. The size of one contract is 10 million Turkish Liras. The company shorts three June contracts on February 1 with the futures price of 0.6500 cents per Lira. If the futures prices and spot price on the closing date are 0.6250 and 0.6150, respectively, then calculate the effective price received in Euros for 30 million Liras.

- A. The effective price is Euro 10,500.
- B. The effective price is Euro 192,000.
- C. The effective price is Euro 187,500.
- D. The effective price is Euro 184,500.

The correct answer is **B**.

Since the company had a short position in futures contracts and the price of the futures contracts has decreased over the period of the hedge, the company has gained on its exposure in the futures contracts. The effective price obtained in cents per Lira is the final spot price plus the gain on the futures:

$$0.6150 + (0.6500 - 0.6250) = 0.6400$$

The total amount received by the German manufacturer for the 30 million Liras is $30,000,000 * 0.6400$ cents = 19,200,000 cents or 192,000 Euros.

Q.635 Emanuel is a junior trader working in the derivatives and hedging unit of a brokerage firm. Emanuel's superior instructed him to take a hedged position for one of its clients who wants to hedge its exposure in 10 million tons of plastic. Since the underlying asset is plastic and difficult to find futures contracts with the underlying asset of plastic, the trader is advised to take a position in rubber futures contracts. The contract size of rubber is 45 tons. If the standard deviation of the spot prices of plastic is 0.019, the standard deviation of the futures prices of rubber is 0.032, and the correlation coefficient between the two is 0.87, then determine what should be the optimal hedge ratio.

- A. 0.52
- B. 0.59
- C. 1.46
- D. 1.68

The correct answer is **A**.

The optimal hedge ratio is calculated as:

$$HR = \rho * \left(\frac{\sigma_S}{\sigma_F} \right)$$

Where

σ_S is the standard deviation of the prices of the hedge asset;

σ_F is the standard deviation of the prices of futures asset; and

ρ is the correlation coefficient between σ_S and σ_F .

$$HR = 0.87 * \left(\frac{0.019}{0.032} \right) = 0.516.$$

Q.638 A portfolio manager has constructed a portfolio that perfectly mirrors the NASDAQ-100 index. The portfolio manager is worried about the changes in the portfolio's value, so he decides to hedge the portfolio using futures contracts on the mini NASDAQ-100 index. If the portfolio's value is \$16,165,000, the index futures price is 5,056 with each contract on \$20 times the index, then estimate the number of contracts required to hedge the portfolio.

- A. 138
- B. 142
- C. 160
- D. 101120

The correct answer is **C**.

The number of stock contracts required to hedge the portfolio is calculated as:

$$\text{Number of stock contracts} = \text{Beta of the portfolio} * \left(\frac{\text{Value of the portfolio}}{\text{Futures price} * \text{Contract multiplier}} \right)$$

Since the portfolio perfectly mirrors the index, the beta of the given portfolio is considered 1.

$$\text{The number of stock contracts required} = 1 * \left(\frac{16,165,000}{101,120} \right) = 159.86 \text{ or } 160 \text{ contracts.}$$

Q.639 Julia Lange, an investment manager, has constructed a portfolio that somewhat mirrors the S&P 500 index. The investment manager intends to hedge the portfolio by taking a short position in S&P 500 futures. The current worth of the portfolio is \$672,000,000, and the S&P 500 index futures price is 2,906, with each contract on \$250 times the index. If the portfolio's beta is 0.78, then estimate the number of contracts Lange should short to hedge her portfolio.

- A. 1,455 S&P 500 futures contracts
- B. 925 S&P 500 futures contracts
- C. 876 S&P 500 futures contracts
- D. 722 S&P 500 futures contracts

The correct answer is **D**.

The number of contracts required to hedge the portfolio is calculated as:

$$\text{Number of stock contracts} = \text{Beta of the portfolio} * \left(\frac{\text{Value of the portfolio}}{\text{Futures price} * \text{Contract multiplier}} \right)$$

Since the portfolio doesn't perfectly mirror the S&P 500 index the beta of the portfolio of 0.78 will be considered in the calculation.

$$\text{The number of contracts required} = 0.78 * \left(\frac{672,000,000}{726,500} \right) = 722 \text{ contracts.}$$

Q.640 Julia Lange, an investment manager, has constructed a portfolio with a beta of 0.78 that somewhat mirrors the S&P 500 index. The investment manager hedged the portfolio 1 month ago by taking a short position in the S&P 500 futures. The portfolio had a value of \$672,000,000, and the S&P 500 index futures price at the time of the purchase was 2,906, with each contract on 250 times the index. If the S&P 500 futures contract price fell to 2,715 this month, then estimate the number of additional contracts Lange should buy/short to hedge her portfolio, assuming that the portfolio value does not change.

- A. The manager must short an additional 50 S&P 500 futures contract to hedge the portfolio.
- B. The manager must buy 50 S&P 500 futures contracts to hedge the portfolio.
- C. The manager must short an additional 2,715 S&P 500 futures contracts to hedge the portfolio.
- D. The manager must short an additional 772 S&P 500 futures contracts to hedge the portfolio.

The correct answer is **A**.

The number of stock contracts required to hedge the portfolio is calculated as:

$$\text{Number of contracts} = \text{Beta of the portfolio} * \left(\frac{\text{Value of the portfolio}}{\text{Futures price} * \text{Contract multiplier}} \right)$$

Since the portfolio doesn't perfectly mirror the S&P 500 index, the beta of the portfolio of 0.78 will be considered in the calculation.

The initial number of contracts that the manager short at the time of purchasing the S&P 500 futures contract was:

$$\text{The number of contracts required} = 0.78 * \left[\frac{672,000,000}{(2,906 * 250)} \right] = 722 \text{ contracts.}$$

A month later, when the futures price fell from 2,906, to 2,715, the new number of contracts required to hedge the portfolio is now:

$$= 0.78 * \left[\frac{672,000,000}{(250 * 2,715)} \right] = 772 \text{ contracts}$$

Therefore, the manager must short an additional 50 futures contracts on the S&P 500 index.

Q.642 An investor owns a portfolio of some of the S&P 500 stocks that worth \$50 million. The systematic risk of the portfolio to the S&P 500 index is 0.96. The investor wants to remove the systematic risk from his portfolio completely, so he decides to reduce the portfolio's beta to zero. If the value of the S&P 500 index futures contracts is 1,111 and each index point costs \$250, how many contracts should he use to reduce the systematic risk?

- A. The investor must buy 173 index futures contracts
- B. The investor must short 173 index futures contracts
- C. The investor must buy 180 index futures contracts
- D. The investor must short 180 index futures contracts

The correct answer is **B**.

The investor must short 173 S&P 500 futures contracts to reduce the beta of the portfolio from 0.96 to 0.

$$\begin{aligned}\text{Number of contracts} &= (\text{Target beta} - \text{Portfolio beta}) * \left(\frac{\text{Value of the portfolio}}{\text{Futures price} * \text{Contract multiplier}} \right) \\ &= (0 - 0.96) * \left(\frac{50,000,000}{277,750} \right) = -173 \text{ contracts}\end{aligned}$$

The negative sign implies that the investor must short 173 S&P 500 index contracts to reduce the beta of the portfolio to 0.

Q.643 Adam Ryman was taking an aptitude test to join the graduate trainee program of a German investment bank. One of the questions in the exam asked to identify in which of the following processes an investor closes out the existing position as the maturity of the futures contract approaches and replaces it with another futures contract with a later delivery date or maturity. Which of the following is the correct answer to the question?

- A. Cross-over hedging.
- B. Rolling a hedge forward.
- C. Basis risk of hedging.
- D. Reducing the beta of the portfolio.

The correct answer is **B**.

The process described in the question is known as 'Rolling a hedge forward'. This is a common

practice in futures trading, where an investor closes out their existing position as the futures contract approaches its maturity date, and replaces it with another futures contract that has a later delivery date or maturity. This strategy allows the investor to maintain their position in the market, while avoiding the delivery of the underlying asset. It is particularly useful in situations where the investor's primary interest is in the price movements of the asset, rather than in owning the asset itself. The new contract that replaces the old one is typically identical in all respects except for its delivery date. This process of 'rolling forward' can be repeated indefinitely, allowing the investor to maintain a continuous presence in the futures market.

Choice A is incorrect. Cross-over hedging refers to the practice of hedging a position in one asset by taking a position in another asset. This is not the same as closing out a futures contract and replacing it with another one, which is described as rolling a hedge forward.

Choice C is incorrect. Basis risk of hedging refers to the risk that the value of a futures contract will not move in line with that of the underlying asset, leading to ineffective hedging. This concept does not involve replacing an expiring futures contract with another one.

Choice D is incorrect. Reducing the beta of a portfolio involves adjusting its sensitivity to market movements, typically through diversification or using derivatives like options and futures for hedging purposes. However, this does not specifically refer to closing out an existing futures position and opening another one with later maturity.

Things to Remember

- 'Rolling a hedge forward' is a common strategy used by investors in futures trading to maintain their position in the market as the maturity of the current futures contract approaches.
- This strategy involves closing out the existing futures position and replacing it with a new futures contract with a later delivery date.
- Investors use 'rolling a hedge forward' to avoid the delivery of the underlying asset and continue to participate in the price movements of the asset.
- The new futures contract used to replace the old one is typically identical in all aspects except for the delivery date.
- 'Rolling a hedge forward' can be repeated indefinitely, allowing investors to maintain a continuous presence in the futures market.

Q.644 A French carmaker expects to purchase 50,000 tons of copper at the end of April. The copper futures contracts on the Eurex Exchange are available for the delivery months of March, June, September, and December, and the size of one contract is for one ton of copper. The company took a long position in June contracts on March 1st at a futures price of 2.450 Euros per ton. If the futures price and spot price on the closing date are 2.42 and 2.30, respectively, then calculate the net cost in Euros and the gain or loss on the futures contract.

- A. The net cost is €116,500, and the loss on the contract is €6,000.
- B. The net cost is €121,000, and the loss on the contract is €6,000.
- C. The net cost is €116,500, and the loss on the contract is €1,500.
- D. The net cost is €121,000, and the loss on the contract is €7,500.

The correct answer is **C**.

Since the company has a long position in futures contracts and the futures contract price has decreased over the period of the hedge, the company has incurred a loss on its exposure in the futures contracts.

Note the effective price (Net cost of asset when a long hedge is used) is given by:

$$\begin{aligned}\text{Net cost of asset when long hedge is used} &= F_0 + (S_t - F_t) \\ &= F_0 + b_t\end{aligned}$$

Where

F_0 : Futures price at the time the hedge is initiated,

F_t : Futures price at the time the hedge is closed,

S_t : Spot price of the asset being hedged at the time the hedge is closed, and

$b_t = S_t - F_t$ = Basis at time t

So in this case we have: $F_0 = 2.45$, $F_t = 2.42$ and $S_t = 2.30$. Thus:

$$\text{Net cost of asset when long hedge is used} = 50,000 [2.45 + (2.30 - 2.42)] = 116,500$$

The loss (gain) for long futures is given by:

$$\text{loss} = F_t - F_0 = 2.42 - 2.45 = -0.03 \text{ per ton} \Rightarrow \text{Loss in this case} = -0.03 \times 50,000 = -1,500$$

Reading 35: Foreign Exchange Markets

Q.880 Iron Cement Co. is a Pakistani company. In 2015, the company obtained a €200 million loan facility from The Bavaria Bank (headquartered in Frankfurt, Germany). The proceeds were used to build a new cement plant in Pakistan. Due to Pakistan's fast-paced growth in the past few years, the Pakistani rupee has appreciated against most of the actively traded currencies, including the euro. Which of the following impacts is accurate, if the company pays back the loan today?

- A. The liability of the Pakistani firm has increased due to the currency appreciation.
- B. The liability of the Pakistani firm has decreased due to the currency appreciation.
- C. The liability of the Pakistani firm remains unaffected due to the currency appreciation.
- D. The liability of the Pakistani firm has increased since the loan is denominated in Euros.

The correct answer is **B**.

The liability of the Pakistani firm has decreased due to the currency appreciation. When a company borrows in a foreign currency, the amount it owes is fixed in that currency. However, the amount it will cost in the company's home currency to repay the loan can change with exchange rate movements. If the home currency appreciates against the foreign currency, it will take fewer units of the home currency to buy the foreign currency needed to repay the loan. Therefore, the real value of the liability in terms of the home currency decreases. In this case, the Pakistani rupee has appreciated against the euro, meaning it takes fewer rupees to buy each euro. Therefore, it will cost Iron Cement Co. fewer rupees to buy the euros needed to repay the €200 million loan, effectively reducing the company's liability.

Choice A is incorrect. The liability of the Pakistani firm would not increase due to the currency appreciation. In fact, it's quite the opposite. When a country's currency appreciates against another, it means that its purchasing power has increased relative to the other currency. Therefore, in this case, as the Pakistani rupee has appreciated against the euro, Iron Cement Co. would need fewer rupees to buy euros for repaying their loan.

Choice C is incorrect. The liability of Iron Cement Co does not remain unaffected due to currency appreciation. Currency exchange rates have a direct impact on liabilities denominated in foreign currencies because they determine how much of your home currency you need to repay those liabilities.

Choice D is incorrect. Although it's true that Iron Cement Co.'s loan is denominated in Euros, this doesn't necessarily mean that their liability has increased due to this fact alone. As explained above, since Pakistani Rupee has appreciated against Euro over time, it will require less amount of Rupees now than before for repayment which effectively reduces their liability rather than increasing it.

Things to Remember

- Foreign Exchange Risk: Companies that borrow in foreign currencies are exposed to

foreign exchange risk, where fluctuations in exchange rates can impact the real value of their liabilities.

- **Currency Appreciation:** When a currency appreciates, it means that it has strengthened in value relative to another currency, leading to potential benefits for entities with liabilities denominated in the appreciating currency.
 - **Currency Depreciation:** On the other hand, currency depreciation refers to a decrease in the value of a currency relative to another, which can increase the real value of liabilities denominated in the depreciating currency.
 - **Hedging:** Companies can use financial instruments such as forward contracts or options to hedge against foreign exchange risk and mitigate the impact of currency fluctuations on their liabilities.
-

Q.882 In recent years, India and China have become Asia's giants when it comes to information technology. India is a market leader in software development, while China is leading in the sector of IT. The two countries also have a bilateral trade agreement. India exports software to China and China sells hardware of the IT sector to Indian firms. If the Indian rupee has appreciated against the Chinese yuan, then determine which of the following effects of appreciation is correct.

- A. Indian goods will be cheaper for Chinese importers, while Chinese goods will become more expensive to Indian buyers.
- B. Indian goods will be more expensive for Chinese importers, while Chinese goods will be cheaper for Indian buyers.
- C. Indian goods will be cheaper for Chinese importers, and Chinese goods will also be cheaper for Indian buyers.
- D. Indian goods will be more expensive for Chinese importers, and Chinese goods will also be more expensive for Indian buyers.

The correct answer is **B**.

The appreciation of the Indian rupee against the Chinese yuan implies that the value of the Indian currency has increased relative to the Chinese currency. This means that for the same amount of yuan, a Chinese importer can now buy fewer Indian rupees. Consequently, the cost of Indian goods, which are priced in rupees, will effectively increase for Chinese importers. This is

because they now need more yuan to buy the same amount of rupees, and hence the same quantity of goods. On the other hand, for Indian buyers, Chinese goods will become cheaper. This is because the Indian rupee, having appreciated, now has more purchasing power. Therefore, for the same amount of rupees, an Indian buyer can now buy more yuan, and hence more Chinese goods. This dynamic is a fundamental principle of international trade and foreign exchange.

Choice A is incorrect. This choice suggests that Indian goods will be cheaper for Chinese importers, while Chinese goods will become more expensive to Indian buyers. However, this is not the case when the Indian rupee appreciates against the Chinese yuan. In fact, it's the opposite: Indian goods become more expensive for Chinese importers and Chinese goods become cheaper for Indian buyers.

Choice C is incorrect. This option implies that both Indian and Chinese goods will be cheaper for their respective importers due to currency appreciation of rupee against yuan. But this isn't accurate as an appreciation of a country's currency makes its exports more expensive and imports cheaper.

Choice D is incorrect. As explained above in Choice B explanation, when a country's currency appreciates, its exports become more expensive while its imports get cheaper. Therefore, it's not possible that both Indian and Chinese goods would be more expensive due to rupee appreciation against yuan.

Things to Remember

- **Exchange Rate:** The exchange rate between two currencies determines the value of one currency in terms of the other. It plays a crucial role in international trade and affects the competitiveness of exports and imports.
- **Appreciation vs. Depreciation:** Appreciation refers to an increase in the value of a currency relative to another, making imports cheaper and exports more expensive. Depreciation, on the other hand, is a decrease in the value of a currency, making exports cheaper and imports more expensive.
- **Trade Balance:** The trade balance of a country is the difference between its exports and imports. Exchange rate movements can impact the trade balance by influencing the cost of goods and services in international markets.
- **Competitiveness:** Currency appreciation can affect the competitiveness of a country's exports in global markets. A stronger currency may make exports more expensive and less competitive, while a weaker currency can boost export competitiveness.

Q.886 Jasmine Forst is a risk manager at Lifelong Insurance Company. The company has a

number of outstanding exposures in various foreign currencies. Today, she is analyzing the company's current outstanding exposures in foreign currencies to derive the possible effects of exchange rates on these exposures. Which of the following is true regarding Lifelong Insurance Company?

- A. If the company has a net short position in a specific foreign currency, then the company's risk increases if the value of the foreign currency depreciates against the dollar.
- B. If the company has a net short position in a specific foreign currency, then the company's risk increases if the value of the foreign currency appreciates against the dollar.
- C. If the company has a net long position in a specific foreign currency, then the company's risk increases if the value of the foreign currency appreciates against the dollar.
- D. If the company has a net long position in a specific foreign currency, then the company's risk increases if the value of the domestic currency depreciates against the dollar.

The correct answer is **B**.

When a company has a net short position in a specific foreign currency, it means that the company has more liabilities than assets in that currency. In this scenario, if the value of the foreign currency appreciates against the dollar, the company's risk increases. This is because the company would need more dollars to purchase the foreign currency to meet its liabilities. For example, if the company has a net short position of -€100, and the exchange rate is \$1.2/€, the company would need \$120 to purchase 100 euros to meet its liabilities. However, if the euro appreciates to \$1.5/€, the company would now need \$150 to purchase the same amount of euros. This increase in cost represents an increase in risk for the company.

Choice A is incorrect. If the company has a net short position in a specific foreign currency, it means that the company owes more of that currency than it owns. Therefore, if the value of this foreign currency depreciates against the dollar, Lifelong Insurance Company would actually benefit because it would need fewer dollars to settle its obligations in that foreign currency.

Choice C is incorrect. If Lifelong Insurance Company has a net long position in a specific foreign currency, it means that the company owns more of that currency than it owes. In this case, if the value of this foreign currency appreciates against the dollar, Lifelong Insurance Company's risk does not increase; instead, its potential profit increases because each unit of this foreign currency can now be exchanged for more dollars.

Choice D is incorrect. If Lifelong Insurance Company has a net long position in a specific foreign currency and if the value of domestic (U.S.) dollar depreciates against this particular foreign currency then again company's risk does not increase but rather its potential profit increases as each unit of this particular foreign can now be exchanged for more dollars.

Things to Remember

- Net short position in a specific foreign currency means the company owes more of that currency than it owns.
 - Net long position in a specific foreign currency means the company owns more of that currency than it owes.
 - Exchange rate movements can impact a company's risk exposure in foreign currencies.
 - Appreciation of a foreign currency against the dollar can increase a company's risk if it has a net short position in that currency.
 - Depreciation of a foreign currency against the dollar can benefit a company with a net short position in that currency.
 - Appreciation of a foreign currency against the dollar can increase potential profit for a company with a net long position in that currency.
 - Depreciation of the domestic currency against a foreign currency can increase potential profit for a company with a net long position in that currency.
-

Q.887 Sandy Lee is a junior economist at a mid-sized asset management company based in Boston. The company is considering making an investment in foreign bonds denominated in Swiss francs. For this type of investment, it is vital to estimate the impact of the exchange rate on the investment value. Which of the following is accurate?

- A. If the supply for Swiss francs increases, the value of the investment of the company will increase.
- B. If the demand for Swiss francs increases, the value of the investment of the company will increase.
- C. If the supply of the US dollar decreases, the value of the investment of the company will increase.
- D. If the demand for Swiss francs increases, the value of the investment of the company will decrease.

The correct answer is **B**.

The value of an investment in foreign bonds denominated in Swiss francs is directly influenced

by the exchange rate between the Swiss franc and the US dollar. When the demand for Swiss francs increases, it leads to an appreciation of the Swiss franc against the US dollar. This appreciation means that each Swiss franc is worth more in terms of US dollars. Therefore, the value of the investment, when converted back to US dollars, increases. This is because the same amount of Swiss francs now translates into a larger amount of US dollars. Hence, an increase in the demand for Swiss francs leads to an increase in the value of the company's investment.

Choice A is incorrect. An increase in the supply of Swiss francs would typically lead to a depreciation of the Swiss franc relative to other currencies, assuming demand remains constant. This depreciation would decrease the value of an investment denominated in Swiss francs when converted back into US dollars.

Choice C is incorrect. A decrease in the supply of US dollars could potentially lead to an appreciation of the dollar relative to other currencies, assuming demand remains constant. This appreciation would decrease the value of an investment denominated in Swiss francs when converted back into US dollars.

Choice D is incorrect. An increase in demand for Swiss francs would typically lead to an appreciation of the Swiss franc relative to other currencies, assuming supply remains constant. This appreciation would increase, not decrease, the value of an investment denominated in Swiss francs when converted back into US dollars.

Things to Remember

- Exchange rate risk: Investing in foreign bonds exposes investors to exchange rate risk, which can impact the value of the investment when converted back to the investor's home currency.
- Factors influencing exchange rates: Exchange rates are influenced by various factors such as interest rates, inflation, political stability, and market speculation.
- Appreciation vs. Depreciation: An appreciation in a currency's value means it has strengthened relative to another currency, while depreciation means it has weakened.
- Impact of supply and demand: Changes in the supply and demand for a currency can affect its value in the foreign exchange market.
- Hedging against exchange rate risk: Investors can use financial instruments like forward contracts or options to hedge against potential losses due to exchange rate fluctuations.

Q.893 An analyst is analyzing the exchange rate of the Turkish lira in terms of U.S. dollars. The current exchange rate is TRY 3.6 per USD, and the real interest rate in both countries is 2%.

Suppose that the prices of Turkish goods increased by 7%, and the prices of U.S goods increased by only 5.5%, then determine which of the following statements is true.

- A. The Turkish lira has depreciated in value against the USD.
- B. The Turkish lira has appreciated in value against the USD.
- C. The USD has depreciated in value against the Turkish.
- D. There has been no impact on the exchange rates as the real interest rate is identical in both countries.

The correct answer is **A**.

The Turkish lira has depreciated in value against the USD. This is because the inflation rate in Turkey is higher than that in the U.S. When inflation is higher in one country compared to another, the value of the currency in the country with the higher inflation rate tends to depreciate. This is because the goods in the country with higher inflation become more expensive relative to goods in the other country. As a result, demand for the more expensive goods decreases, which in turn decreases demand for the currency of that country. In this case, Turkish goods have become more expensive relative to U.S goods due to the higher inflation rate in Turkey. This has led to a decrease in demand for Turkish goods, and consequently, a decrease in demand for the Turkish lira. This decrease in demand for the Turkish lira has caused it to depreciate in value against the USD.

Choice B is incorrect. The Turkish lira has not appreciated in value against the USD. The increase in the price of goods in Turkey compared to the U.S indicates inflation, which typically leads to a depreciation of currency, not an appreciation.

Choice C is incorrect. Since the Turkish lira has depreciated against the USD, then the USD has appreciated against the Turkish lira.

Choice D is incorrect. While it's true that identical real interest rates can influence exchange rates, they do not negate the impact of inflation differentials on exchange rates. In this case, higher inflation in Turkey compared to the U.S would lead to a depreciation of Turkish lira against USD.

Things to Remember

- Real Interest Rate Parity (RIRP): RIRP states that the difference in real interest rates between two countries should equal the expected change in their exchange rates. This concept helps explain how interest rate differentials can impact exchange rates.
- Purchasing Power Parity (PPP): PPP theory suggests that exchange rates between two countries should equalize the prices of a basket of goods and services in each country. This theory can help understand how inflation differentials can affect exchange rates.

- Interest Rate Differentials: Differences in nominal and real interest rates between countries can impact capital flows and exchange rates. Higher interest rates in a country can attract foreign investment, leading to an appreciation of its currency.
 - Impact of Inflation: Higher inflation rates in a country can erode the value of its currency as the purchasing power of the currency decreases. This can lead to a depreciation of the currency relative to others with lower inflation rates.
-

Q.894 Which of the following theories suggests that the difference between the spot and the forward rates is due to the difference in interest rates?

- A. Purchasing power parity theory.
- B. Interest rate parity theory.
- C. Fischer theory.
- D. None of the above.

The correct answer is **B**.

The Interest Rate Parity (IRP) theory is a key concept in the foreign exchange markets, providing a framework for the relationship between interest rates and the exchange rate. According to the IRP theory, the difference between the spot and forward exchange rates for a currency pair should be equal to the difference between the interest rates of the two countries. This is because any discrepancy between these values would provide an arbitrage opportunity, where traders could borrow in the currency with the lower interest rate, convert to the other currency at the spot rate, lend in the other currency, and then convert back at the forward rate, making a risk-free profit. Therefore, in an efficient market, these arbitrage opportunities should not exist, leading to the interest rate parity condition. This theory is fundamental to the pricing of foreign exchange derivatives, and is widely used in international finance.

Choice A is incorrect. The Purchasing Power Parity theory is not related to the disparity between spot rates and forward rates due to differences in interest rates. Instead, it deals with the concept that the exchange rate between two countries should be equivalent to the ratio of their price levels.

Choice C is incorrect. The Fischer theory, also known as Fisher effect, primarily focuses on nominal interest rates, inflation and real interest rates. It does not explain the relationship between spot and forward rates based on differences in interest rates.

Choice D is incorrect. As explained above, there exists a specific theory - Interest Rate Parity Theory - which explains this relationship between spot and forward exchange rates based on differences in interest rate among countries.

Things to Remember

- Purchasing Power Parity (PPP) theory relates to the equilibrium exchange rate between two currencies based on their relative purchasing power.
 - Fisher Effect focuses on the relationship between nominal interest rates, inflation rates, and real interest rates.
 - Interest Rate Parity (IRP) theory is crucial in understanding the relationship between interest rates and exchange rates in the foreign exchange market.
 - IRP theory helps in explaining the equilibrium relationship between spot and forward exchange rates based on interest rate differentials.
 - Arbitrage opportunities arise when there are discrepancies between spot and forward rates, leading to market adjustments to eliminate such opportunities.
-

Q.895 A foreign currency analyst based in Dubai is forecasting the forward exchange rate USD/AED. Currently, the USD/AED spot exchange rate is 3.33. Suppose that the interest rate in the United Arab Emirates is 2% and the interest rate in the U.S. is 4%, determine the 1-year forward exchange rate of USD/AED.

- A. 3.455
- B. 3.4
- C. 3.395
- D. 3.266

The correct answer is **D**.

According to the interest rate parity theorem, the forward exchange rate must be derived with the following equation:

$$\begin{aligned} F &= \text{Spot Exchange Rate} \times \left(\frac{1 + R_{\text{AED}}}{1 + R_{\text{USD}}} \right)^T \\ &= 3.33 \times \left(\frac{1.02}{1.04} \right)^1 \\ &= 3.26596 \end{aligned}$$

Q.3573 Approximate the real interest rate if the nominal interest rate is 11% and inflation is 3.5%.

A. 14.5%

B. 13.5%

C. 3.5%

D. 7.25%

The correct answer is **D**.

To approximate the real interest rate, you can use the Fisher equation, which relates the nominal interest rate (i) to the real interest rate (r) and the inflation rate (π) as follows:

$$1 + r = \frac{1 + i}{1 + \pi}$$

Now, plug these values into the Fisher equation to calculate the real interest rate (r):

$$\begin{aligned} 1 + r &= \frac{1 + 0.11}{1 + 0.035} \\ \Rightarrow 1 + r &= \frac{1.11}{1.035} \\ \Rightarrow r &= 1.07246 - 1 \approx 0.07246 \end{aligned}$$

Q.3575 Due to the upcoming elections, the exchange rate USD/CAD has risen from 1.17 to 1.31. Which of the following statement is correct?

- A. The Canadian dollar has appreciated by 11.97%
- B. The US dollar has depreciated by 11.97%
- C. The US dollar has appreciated by 10.69%.
- D. The Canadian dollar has depreciated by 10.69%

The correct answer is **D**.

The Canadian dollar has depreciated by 10.69%. The depreciation or appreciation of a currency is calculated based on the percentage change in its exchange rate. In this case, the exchange rate of USD/CAD has increased from 1.17 to 1.31. This means that now you need more Canadian dollars to buy one US dollar, indicating that the value of the Canadian dollar has decreased relative to the US dollar. The percentage change in the CAD quote is calculated as follows:

$$\frac{\frac{1}{1.3100}}{\frac{1}{1.1700}} - 1 = -10.69\%$$

This negative value signifies a depreciation of the Canadian dollar by 10.69%.

Choice A is incorrect. The Canadian dollar has not appreciated by 11.97%. In fact, the increase in the exchange rate from 1.17 to 1.31 indicates that it now takes more Canadian dollars to buy one US dollar, which means the Canadian dollar has depreciated, not appreciated.

Choice B is incorrect. The US dollar has not depreciated by 11.97%. On the contrary, as explained above, an increase in the exchange rate signifies that it now takes more of the foreign currency (in this case CAD) to buy one unit of USD which implies that USD has actually appreciated.

Choice C is incorrect. The US dollar has not appreciated by 10.69%. While it's true that an increase in exchange rate indicates appreciation of USD against CAD, but this specific percentage change calculation does not match with given numbers and hence is inaccurate.

Q.3579 If the exchange rate quote for the euro (EUR/USD) changes from 1.3500 to 1.2600, then in approximate terms:

- A. The euro depreciated by 6.7%, and the dollar appreciated by 7.1%
- B. The dollar appreciated by 6.7%, and the euro depreciated by 7.1%
- C. The euro appreciated by 10.1%, and the dollar depreciated by 5%

D. The euro depreciated by 5%, and the dollar appreciated by 10.1%

The correct answer is **A**.

If the exchange rate quote for the euro (EUR/USD) changes from 1.3500 to 1.2600, then the approximate percentage change in the value of the euro and the US dollar can be calculated as follows:

Percentage Change in the US dollar:

- Initial exchange rate (EUR/USD): 1.3500
- New exchange rate (EUR/USD): 1.2600
- Percentage Change: $(\frac{1.2600-1.3500}{1.3500}) \times 100 \approx -6.67\%$

So, the euro has depreciated by approximately 6.67% against the dollar.

Percentage Change in the euro:

- Initial exchange rate (USD/EUR): $\frac{1}{1.3500} \approx 0.7407$
- New exchange rate (USD/EUR): $\frac{1}{1.2600} \approx 0.7937$
- Percentage Change: $(\frac{0.7937-0.7407}{0.7407}) \times 100 \approx 7.15\%$

So, the dollar has appreciated by approximately 7.15% against the US dollar.

Things to Remember

- Exchange rate quotes show the value of one currency in terms of another currency.
- A depreciation in a currency means that it has lost value relative to another currency, while an appreciation means it has gained value.
- Percentage change in exchange rates can be calculated using the formula:
$$(\frac{\text{New Value} - \text{Old Value}}{\text{Old Value}}) \times 100$$
- Understanding exchange rate movements is crucial for international trade, investment decisions, and risk management.
- Exchange rate fluctuations can impact the profitability of multinational companies and the value of investments denominated in foreign currencies.

Q.3581 If the exchange rate quote for the Mexican peso (USD/MXN) changes from 11.9500 to 12.4000, then in approximate terms:

- A. The peso depreciated by 3.8%, and the dollar appreciated by 3.6%
- B. The dollar depreciated by 3.8%, and the peso appreciated by 3.6%
- C. The peso appreciated by 3.8%, and the dollar depreciated by 3.6%
- D. The dollar appreciated by 3.8%, and the peso depreciated by 3.6%

The correct answer is **D**.

In the initial quote, you could buy 11.9500 pesos per US dollar.

After the change, the US dollar would buy 12.4000 pesos. The peso fell in value against the dollar.

To determine whether the peso appreciated or depreciated and whether the dollar appreciated or depreciated, we can calculate the percentage change.

Initial exchange rate (USD/MXN): 11.9500

New exchange rate (USD/MXN): 12.4000

To calculate the percentage change:

$$\begin{aligned}\text{Percentage Change} &= \left(\frac{\text{New Rate} - \text{Initial Rate}}{\text{Initial Rate}} \right) \times 100 \\ &= \left(\frac{12.4000 - 11.9500}{11.9500} \right) \times 100 = \left(\frac{0.4500}{11.9500} \right) \times 100 \approx 3.76\%\end{aligned}$$

So, the dollar has appreciated by 3.8%.

Since the exchange rate went from 11.9500 to 12.4000, you now need more pesos to get one US dollar. This means the peso has weakened, and the dollar has strengthened.

Now,

$$\frac{\frac{1}{12.4000} - \frac{1}{11.9500}}{\frac{1}{11.9500}} = -0.03629$$

Therefore, the peso has depreciated by 3.6%

Q.3814 The quote between currency X and Y is 1.3000, where currency Y is the base currency. What is the number of units of currency X required for the exchange of 50 units of currency Y?

- A. 65
- B. 70
- C. 85
- D. 64

The correct answer is **A**.

The base currency (in this case, Y) is always equal to one unit, and the quoted currency (in this case, X) is what that one base unit is equivalent to in the other currency.

So, the number of units of currency X in this case is:

$$1.3000 \times 50 = 65 \text{ units}$$

Q.3816 A UK-based company funds its Mexican investment by borrowing in euros (EUR) and buying the Mexican peso (MXN), and after some time, the company exchanges the money back to euros. What kind of transaction is this?

- A. Outright transaction
- B. FX swap
- C. Currency futures
- D. Forex forward

The correct answer is **B**.

An FX swap is a financial instrument that involves the exchange of an agreed amount of a certain currency for another currency, and then the reversal of this transaction at a later date. In this scenario, the UK-based company is essentially engaging in an FX swap. They are initially exchanging euros for Mexican pesos, and then at a later date, they are reversing this transaction by exchanging the Mexican pesos back into euros. This is the fundamental principle of an FX swap. The company is using the FX swap as a funding mechanism for their Mexican investment, which allows them to mitigate any potential foreign exchange risk associated with their investment.

Choice A is incorrect. An outright transaction involves the exchange of two currencies at a rate agreed on the date of the contract for value or delivery at some time in the future. This scenario

does not describe an outright transaction as there is no agreement on a future exchange rate.

Choice C is incorrect. Currency futures are standardized contracts to buy or sell a particular currency at a future date and price. In this scenario, there's no mention of any standardized contract to buy or sell currencies at a specific future date and price.

Choice D is incorrect. A forex forward involves an agreement to exchange specified amounts of two different currencies at a specific future date and exchange rate. The company in this scenario did not agree on any specific future date or exchange rate for converting euros into Mexican pesos and vice versa, hence it does not represent a forex forward transaction.

Things to Remember

- FX swaps are commonly used by companies to hedge against foreign exchange risk when conducting international business transactions.
 - FX swaps involve two transactions: an initial exchange of currencies and a subsequent reverse exchange at a later date.
 - Companies can use FX swaps to fund investments in foreign currencies without being exposed to fluctuations in exchange rates.
 - FX swaps are a popular tool in the foreign exchange market for managing currency exposure and liquidity needs.
-

Q.3817 Firms in the foreign exchange market are exposed to risks. What is the difference between translation and transaction risks?

- A. Transaction risks are prone to transactions that are aligned to domestic currency while translation risk arises due to transaction aligned to foreign currency.
- B. Transaction risks occur due to cash inflows and outflows in a foreign currency while the translation risk occurs due to exposure to FX gains and losses when the assets and liabilities dominated in a foreign currency are exchanged into domestic to generate financial statements.
- C. Translation risks arise due to cash inflows and outflows in a foreign currency while the transaction risk occurs due to exposure to FX gains and losses when the assets and liabilities dominated in a foreign currency are exchanged into domestic to generate financial statements.
- D. Transaction risks and translation risks both occur due to cash inflows and outflows in foreign currency transactions.

The correct answer is **B**.

Transaction risks and translation risks are two distinct types of risks that firms operating in the foreign exchange market often encounter. Transaction risks are associated with the cash inflows and outflows in a foreign currency. This type of risk arises when a firm engages in financial transactions that involve a foreign currency. The risk stems from the potential fluctuation in the foreign exchange rate between the time the transaction is initiated and when it is settled. This can lead to potential losses if the foreign exchange rate moves unfavorably. On the other hand, translation risk occurs due to exposure to foreign exchange gains and losses when the assets and liabilities dominated in a foreign currency are exchanged into the domestic currency to generate financial statements. This type of risk arises when a firm's financial statements, which include assets and liabilities denominated in foreign currencies, need to be converted back into the firm's domestic currency. If the foreign exchange rate has moved unfavorably, this can lead to potential losses on the firm's financial statements.

Choice A is incorrect. Transaction risks are not prone to transactions that are aligned to domestic currency. Instead, they occur due to cash inflows and outflows in a foreign currency. On the other hand, translation risk does not arise due to transactions aligned to foreign currency but occurs due to exposure to FX gains and losses when the assets and liabilities dominated in a foreign currency are exchanged into domestic currency for generating financial statements.

Choice C is incorrect. This choice incorrectly swaps the definitions of transaction risk and translation risk. Translation risks do not arise from cash inflows and outflows in a foreign currency; this is characteristic of transaction risks. Similarly, transaction risks do not occur due to exposure to FX gains or losses when assets and liabilities dominated in a foreign currency are exchanged into domestic ones; this describes translation risk.

Choice D is incorrect. Transaction risk is related to cash flow from transactions, while translation risk is related to the conversion of financial statements.

Things to Remember

- Transaction risk is also known as economic risk and refers to the risk of loss from exchange rate movements affecting the value of future cash flows.
- Translation risk is also known as accounting risk and refers to the risk of fluctuations in the value of foreign currency-denominated assets and liabilities when they are translated into the reporting currency for financial statements.
- Firms can use hedging techniques such as forward contracts, options, and swaps to manage both transaction and translation risks in the foreign exchange market.
- Transaction exposure is the sensitivity of a firm's cash flows and value to exchange rate movements, while translation exposure is the impact of exchange rate changes on the

reported financial statements of a firm.

- Managing foreign exchange risks is crucial for multinational corporations to protect their profitability and financial stability.
-

Q.3818 Assume the USD/CAD spot quote is bid 1.2800 and ask 1.2950. The six-month forward points quote is bid 40.60 and ask 56.50 The forward bid-ask spread is closest to:

- A. 0.00565
- B. 0.30065
- C. 0.00406
- D. 0.01659

The correct answer is **D**.

The spot bid-ask spread is 0.015 ($= 1.2950 - 1.2800$)

The six-month forward bid quote is $1.2800 + 0.00406 = 1.28406$, and the six-month forward ask quote is $1.2950 + 0.00565 = 1.30065$

Thus, the forward bid-ask spread is $1.30065 - 1.28406 = 0.01659$

Things to Remember

- Forward points are basis points that are added or subtracted to the spot rate to determine the forward rate for delivery on a specific valuation date.
 - The points can either be positive or negative, in conjunction with lower or higher interest rates.
 - A forward point is equivalent to 1/10,000 of a spot rate.
 - When points are added to the spot rate, this is called a forward premium; when points are subtracted from the spot rate, they constitute a forward discount. In this case, we have forward premiums.
-

Q.3819 In a particular year, the inflation rate in China is higher than in the US. What is likely to happen to the US-China exchange rate according to Purchasing power parity?

- A. The US currency will weaken with respect to the Chinese yuan
- B. The Chinese currency (yuan) will decline relative to the US dollar
- C. There will be no change in the US-China exchange rate
- D. The exchange rate will become more volatile

The correct answer is **B**.

According to the Purchasing Power Parity (PPP) theory, when the inflation rate in one country is higher than another, the currency of the country with the higher inflation rate is expected to depreciate relative to the other country's currency. This is because a higher inflation rate reduces the purchasing power of a country's currency. In this scenario, since the inflation rate in China is higher than in the US, the Chinese yuan is expected to decline relative to the US dollar. This depreciation of the yuan is necessary to maintain the equality of the price of a basket of goods in both countries, as suggested by the PPP theory. Therefore, the Chinese currency (yuan) will decline relative to the US dollar.

Choice A is incorrect. According to the PPP theory, if inflation in China is higher than in the United States, it means that goods and services are becoming more expensive in China relative to the US. This would lead to a depreciation of the Chinese yuan against the US dollar, not a weakening of the US currency.

Choice C is incorrect. The PPP theory suggests that changes in price levels (inflation) will affect exchange rates. Therefore, if there's a difference in inflation rates between two countries, it's unlikely that their exchange rate will remain unchanged.

Choice D is incorrect. In the context of PPP. While higher inflation can lead to changes in the exchange rate, PPP specifically predicts the direction of the change rather than volatility.

Things to Remember

- Purchasing Power Parity (PPP) theory states that exchange rates between two countries should equalize the price of a basket of goods and services in both countries.
- PPP theory suggests that if the inflation rate in one country is higher than another, the currency of the country with the higher inflation rate is expected to depreciate relative to the other country's currency.
- Exchange rates are influenced by various factors including interest rates, inflation, political stability, and economic performance.
- PPP is a long-term theory and may not hold in the short run due to factors like speculation, government intervention, and market psychology.

Q.3820 The spot rate for the EUR/USD is 1.3261. The interest rate in USD is 2% and in EUR is 3%. What is the 6-month EUR/USD forward rate?

- A. 1.3411
- B. 1.3326
- C. 1.3196
- D. 0.5321

The correct answer is **C**.

The forward rate for EUR/USD is given by:

$$\begin{aligned} F &= S \left(\frac{1 + i_{\text{USD}}}{1 + i_{\text{EUR}}} \right)^T \\ &= 1.3261 \times \left(\frac{1.02}{1.03} \right)^{0.5} = 1.3196 \end{aligned}$$

Q.3821 The nominal interest rate in the country is 3% and the inflation rate is 5%. What is the value of real interest rate?

- A. 10%
- B. 5%
- C. 2%
- D. -2%

The correct answer is **D**.

Recall that the real interest rate is approximated as:

$$r_{\text{real}} \approx r_{\text{nominal}} - r_{\text{inflation}}$$

So, in this case:

$$r_{\text{real}} \approx 3 - 5 = -2\%$$

Q.3822 The nominal interest rate in the country is 3% and the inflation rate is 5%. Which of the following statements is true about this country?

- A. When an investor earns at 3%, the investor's purchasing power decreases by 2% each year
- B. When an investor earns at 3%, the investor's purchasing power increases by 2% each year
- C. The total return by the investor in this country is equivalent to 8%
- D. An investor in this country should always expect a loss of 5% per year.

The correct answer is **A**.

When an investor earns at 3%, the investor's purchasing power decreases by 2% each year. This is because the real interest rate, which is the nominal interest rate adjusted for inflation, is negative in this case. The real interest rate can be approximated by subtracting the inflation rate from the nominal interest rate. In this scenario, the real interest rate is approximately $3\% - 5\% = -2\%$. This means that the purchasing power of the investor's money is decreasing by 2% each year, even though they are earning a nominal interest rate of 3%. This is because the inflation rate is higher than the nominal interest rate, which erodes the purchasing power of money over time. Therefore, even though the investor is earning interest, the real value of their money is decreasing due to the effects of inflation.

Choice B is incorrect. The investor's purchasing power does not increase by 2% each year. In fact, it decreases due to the inflation rate being higher than the nominal interest rate. The real interest rate (nominal interest rate - inflation) is negative (-2%), which means that the value of money is decreasing over time.

Choice C is incorrect. The total return by the investor in this country is not equivalent to 8%. This would be true if we were adding up the nominal interest and inflation rates, but this isn't how returns work in an economy with inflation. Instead, we need to consider how much purchasing power has changed, which involves subtracting the inflation rate from the nominal interest rate.

Choice D is incorrect. An investor in this country should not always expect a loss of 5% per year. While it's true that high inflation can erode purchasing power, it doesn't mean that investors will necessarily lose money at a constant percentage each year. In this case, because their investment return (3%) doesn't keep pace with inflation (5%), they're effectively losing 2% of their purchasing power annually—not 5%.

Things to Remember

- Real interest rate is the nominal interest rate adjusted for inflation.
- Real interest rate can be approximated by subtracting the inflation rate from the nominal interest rate.

- Inflation erodes the purchasing power of money over time.
 - High inflation can lead to a decrease in the real value of money.
 - Investors should consider the real interest rate when evaluating returns.
-

Q.3823 In a particular year, the interest rates for the EUR increase while that of the USD remain unchanged.

What will happen to the forward exchange rate EUR/USD?

- A. It will decrease.
- B. It will increase.
- C. It will increase and then decrease.
- D. It will not change.

The correct answer is **A**.

When the EUR interest rate increases while the USD interest rate remains constant, the forward rate, EUR/USD is expected to decrease. This relationship is explained by the interest rate parity theory, which suggests that the difference between the forward exchange rate and the spot exchange rate should be equal to the interest rate differential between the two currencies.

In this scenario, as the EUR interest rate increases while the USD interest rate remains unchanged, the interest rate differential between the two currencies widens. To ensure interest rate parity, the forward rate for EUR/USD must decrease.

Mathematically, this concept can be illustrated using the covered interest rate parity formula:

$$F = S \times \frac{1 + r_{USD}}{1 + r_{EUR}}$$

As the interest rate of the EUR increases, the denominator in the formula will increase, leading to a decrease in the forward rate.

This relationship is crucial for investors and businesses involved in currency trading and hedging, as it helps them understand how changes in interest rates can impact the forward exchange rates of different currency pairs.

Q.3824 The interest rate on the Nigerian Naira (NGN) is 2%, and that of the South African rand (ZAR) is 7%. Given that the spot rate NGN/ZAR is 1.3500, what is the 6-month forward exchange rate quoted as points?

- A. 135
- B. 327
- C. 365
- D. 478

The correct answer is **B**.

For an exchange rate XXXYYY, we have,

$$\begin{aligned}
 F &= S \frac{(1 + R_{YYY})^T}{(1 + R_{XXX})^T} \\
 &= S \frac{(1 + R_{ZAR})^T}{(1 + R_{NGN})^T} \\
 &= 1.3500 \frac{(1.07)^{\frac{1}{2}}}{(1.02)^{\frac{1}{2}}} \\
 &= 1.38269
 \end{aligned}$$

So, the points are given by:

$$10000(1.38269 - 1.3500) = 326.92 \approx 327$$

Q.3826 Consider the following information

Currency	Libor (annualized)	Currency Combinations	Spot Rate
CAD	0.62%	CAD/GBP	0.60
GBP	6.96%	AUD/GBP	0.0074
AUD	0.75%	AUD/CAD	0.95

If the covered interest rate parity holds, what is the forward rate of AUD/GBP currency for one year?

- A. 0.007
- B. 0.0079
- C. 0.0063
- D. 0.0054

The correct answer is **B**.

Since we have assumed that the covered interest rate parity holds, then the forward rate parity holds. That is, a one-year spot rate should be equal to the one-year forward rate. That is,

$$F_{\frac{\text{AUD}}{\text{GBP}}} = S_{\frac{\text{AUD}}{\text{GBP}}} \left(\frac{1 + i_{\text{GBP}}}{1 + i_{\text{AUD}}} \right) = 0.0074 \left(\frac{1.0696}{1.0075} \right) = 0.007856 \approx 0.0079$$

Currency	Libor (annualized)	Currency Combinations	Spot Rate
CAD	0.62%	CAD/GBP	2.4812
GBP	6.96%	AUD/GBP	0.0074
AUD	0.75%	AUD/CAD	3.8618

Q.3827 In a given market, the nominal rate of interest is 3% and the estimated inflation rate is 4%. Which of the following statements is correct regarding the market?

- A. When an investor earns a return of 3%, the purchasing power of the investor is reduced by 1%.
- B. When an investor earns a return of 3%, the purchasing power of the investor is reduced by 5%.

C. When an investor earns a return of 3%, the purchasing power of the investor is increased by 1%.

D. When an investor earns a return of 3%, the purchasing power of the investor is reduced by 7%.

The correct answer is **A**.

The real rate of interest is calculated by subtracting the inflation rate from the nominal interest rate. In this case, the nominal interest rate is 3% and the inflation rate is 4%. Therefore, the real rate of interest is $3\% - 4\% = -1\%$. This means that the purchasing power of the investor is reduced by 1%. When an investor earns a return of 3%, it is not enough to keep up with the inflation rate of 4%. As a result, the investor's purchasing power decreases by 1%.

Choice B is incorrect. The purchasing power of the investor would not be reduced by 5%. This would only be the case if the inflation rate was significantly higher than it is, or if the return on investment was significantly lower.

Choice C is incorrect. The purchasing power of the investor does not increase in this scenario. In fact, it decreases because the return on investment (3%) is less than the inflation rate (4%).

Choice D is incorrect. The purchasing power of an investor would not decrease by 7%. This percentage decrease in purchasing power would require a much higher inflation rate or a much lower return on investment than what has been given in this scenario.

Things to Remember

- **Real Rate of Interest:** The real rate of interest is the nominal interest rate minus the inflation rate. It represents the true return on an investment after adjusting for inflation.
- **Purchasing Power:** Purchasing power refers to the amount of goods and services that can be purchased with a unit of currency. Inflation erodes the purchasing power of money over time.
- **Impact of Inflation:** Inflation reduces the purchasing power of money. If the nominal rate of return on an investment is lower than the inflation rate, the investor's purchasing power decreases.
- **Relationship between Nominal Rate, Real Rate, and Inflation:** The nominal rate of interest is the sum of the real rate of interest and the inflation rate. Understanding this relationship is crucial for evaluating investment returns.

Q.3829 A forex trader noticed the EUR/USD spot rate was 1.2960 and expected to be 1.2863 after one year. What is the euro (EUR) expected appreciation/depreciation against the US dollar over the next year?

- A. -0.748%
- B. 0.748%
- C. 0.651%
- D. -0.651%

The correct answer is **A**.

We know that we are dealing with EUR/USD quotes. So, we calculate as:

$$\frac{1.2863}{1.2960} - 1 = -0.00748 = -0.748\%$$

This was expected because clearly, there was a decrease in EUR/USD, indicating that EUR is depreciating.

Note. The Euro is actually the base currency in line with the XXX/YYY format favored by FRM examiners, where X is the base currency and Y the quote currency. Clearly, the euro has depreciated against the dollar; after one year, it can only get you 1.2863USD from a previous high of 1.296USD. Negative appreciation = depreciation

Q.3830 What does a 4% appreciation in the CNY/ZAR exchange rate imply?

- A. It represents a 4 percent appreciation in the South African Rand (ZAR) as compared to the Chinese Yuan.
- B. It represents a 4 percent appreciation in the Chinese Yuan (CNY) as compared to the South African Rand.
- C. It represents a 4 percent depreciation in both the Chinese Yuan (CNY) and the South African Rand.
- D. It represents a 4 percent depreciation in the South African Rand as compared to the Chinese Yuan (CNY)

The correct answer is **B**.

An appreciation in the exchange rate of two currencies implies that the base currency (the first

currency in the pair) has increased in value relative to the quote currency (the second currency in the pair). In this case, the CNY/ZAR exchange rate refers to the amount of South African Rand (ZAR) one can get for one Chinese Yuan (CNY). Therefore, a 4% appreciation in this exchange rate means that the Chinese Yuan has increased in value by 4% against the South African Rand. This implies that you would now need more South African Rands to purchase one Chinese Yuan than you did before the appreciation. This is a fundamental concept in foreign exchange markets, where exchange rates are constantly fluctuating due to various economic factors such as interest rates, inflation rates, political stability, economic performance, etc.

Choice A is incorrect. An appreciation in the CNY/ZAR exchange rate signifies that the Chinese Yuan (CNY) has increased in value relative to the South African Rand (ZAR), not the other way around. Therefore, it does not represent a 4 percent appreciation in the South African Rand as compared to the Chinese Yuan.

Choice C is incorrect. The statement that there is a 4% depreciation in both currencies contradicts itself because if one currency depreciates, it means that it loses value against another currency which therefore appreciates. In this case, an appreciation of 4% in CNY/ZAR means that Chinese Yuan has appreciated or gained value against South African Rand and not both currencies depreciating at once.

Choice D is incorrect. This choice incorrectly interprets an appreciation of CNY/ZAR as a depreciation of ZAR relative to CNY. In reality, an increase or appreciation in the exchange rate indicates that one unit of CNY can now purchase more units of ZAR than before, meaning that CNY has appreciated or gained value against ZAR and not vice versa.

Things to Remember

- Exchange rates are quoted in pairs, where the base currency is the first currency and the quote currency is the second currency.
 - An appreciation in the exchange rate means the base currency has increased in value relative to the quote currency.
 - Factors influencing exchange rates include interest rates, inflation rates, political stability, economic performance, and market speculation.
 - Exchange rate movements can impact international trade, investment flows, and overall economic conditions.
-

Q.3834 In India, the price index was 117 last year, and the price level index this year is 125. If the real interest rate in India is 7.5%, what is the nominal interest rate?

- A. 15.40%
- B. 14.34%
- C. 8.00%
- D. 11.5%

The correct answer is **B**.

Recall that

$$r_{\text{real}} \approx r_{\text{nominal}} - r_{\text{inflation}}$$
$$\Rightarrow r_{\text{nominal}} = r_{\text{real}} + r_{\text{inflation}}$$

According to the information given in the question:

$$\text{Expected inflation} = \frac{125}{117} - 1 = 0.06838 = 6.838\%$$

So,

$$r_{\text{nominal}} = 7.5\% + 6.838\% = 14.338\%$$

Q.3835 A multinational uses options for hedging against FX risk on its monthly transactions. Which of the following does the multinational need to do to minimize hedging costs?

- A. Purchasing options on each currency in the market.
- B. Purchasing options on a basket of currencies rather than individual currency
- C. Buy an option on a single exposure that applies in one time period (like one month)
- D. Buy an option on a single exposure that applies several months

The correct answer is **B**.

Purchasing options on a basket of currencies rather than an individual currency is the most

effective strategy for a multinational corporation to minimize hedging costs. This is because the corporation is exposed to several currencies each month, and by buying options on a collection (basket) of currencies, it can distribute the risk, thereby minimizing it. This strategy allows the corporation to hedge against the risk of multiple currencies simultaneously, which is more cost-effective than hedging against each currency individually. Furthermore, this approach provides a more diversified risk profile, which can help to further reduce potential losses.

Choice A is incorrect. Purchasing options on each currency in the market would not be a cost-effective strategy for the corporation. This approach would require significant resources and may expose the corporation to unnecessary risks associated with individual currency fluctuations.

Choice C is incorrect. Buying an option on a single exposure that applies in one time period (like one month) may not provide adequate coverage for the corporation's foreign exchange transactions, especially if these transactions occur over multiple time periods or involve multiple currencies.

Choice D is incorrect. While buying an option on a single exposure that applies several months might provide some level of protection, it does not necessarily minimize costs associated with hedging as it does not take into account potential changes in currency values over time.

Things to Remember

- FX risk hedging involves using financial instruments like options to protect against adverse movements in exchange rates.
- Options give the holder the right, but not the obligation, to buy or sell a currency at a specified exchange rate within a set period.
- By hedging with a basket of currencies, a multinational can diversify its risk exposure and potentially reduce costs compared to hedging each currency individually.
- Timing and duration of options contracts are important considerations in FX risk management.
- Cost-effective hedging strategies aim to balance protection against risk with minimizing expenses related to hedging instruments.

Q.4453 Consider a hypothetical world of two countries only. A fund manager borrows funds from a country with interest rate X and invests in another country with interest rate Y , where $X < Y$. He intends to generate profit from the interest differential between the two countries. The major risk in this strategy is:

- A. None – it's a riskless strategy.
- B. The interest differential may increase.
- C. The exchange rate of the currencies may change.
- D. The interest differential may become negative.

The correct answer is **C**.

The fund manager borrows in one currency, converts the currency into another currency, and invests the converted amount in the country with a higher interest rate. Thus, the profit generated in the transaction can be reduced/affected in case the exchange rate changes.

A is incorrect. This option is incorrect because the strategy involves borrowing in one currency and investing in another currency. The exchange rate between these two currencies can fluctuate, which introduces risk. Even if the interest rate differential is positive ($X < Y$), an adverse movement in the exchange rate can erode or even eliminate the profit, making it a risky strategy.

B is incorrect. This option is incorrect because an increase in the interest differential (where Y becomes even greater than X) would actually benefit the fund manager, increasing the profit from the strategy. The risk lies in factors that would negatively impact the profit, not those that would enhance it.

D is incorrect. This option is incorrect because while it is theoretically possible for the interest differential to become negative, it is not the primary risk in this carry trade strategy. The primary risk is the exchange rate movement, which can fluctuate more unpredictably and have an immediate and significant impact on the profit generated from the strategy. The interest rate differential changing to a negative value is less likely and is usually slower to occur compared to exchange rate changes.

Things to Remember

- Carry Trade: Strategy where an investor borrows funds in a currency with a low interest rate and invests in a currency with a higher interest rate to profit from the interest rate differential.
- Interest Rate Parity: Theory that suggests that the difference in interest rates between two countries should be equal to the difference between the forward exchange rate and the spot exchange rate.
- Exchange Rate Risk: The risk that an investment's value will decrease due to changes in the value of the currency exchange rates.
- Forward Exchange Rate: The exchange rate at which a foreign exchange contract is

agreed for future delivery.

Q.4893 If the interest rates of currency AAA increases and that of currency BBB remains constant, what will happen to forward rates if the exchange rate is quoted as AAA/BBB?

- A. Forward rates increase.
- B. Forward rates decrease.
- C. Forward rates remain constant.
- D. Forward rates increase then decrease.

The correct answer is **B**.

When the interest rate of currency AAA increases while the interest rate of currency BBB remains constant, the forward rate for the currency pair, AAA/BBB, is expected to decrease. This is based on the interest rate parity theory, which states that the difference between the forward exchange rate and the spot exchange rate should be equal to the interest rate differential between the two currencies.

In this case, since the interest rate of currency AAA has increased, and the interest rate of currency BBB remains the same, the interest rate differential between the two currencies widens. To maintain interest rate parity, the forward rate for AAA/BBB must decrease.

Mathematically, this can be illustrated using the covered interest rate parity formula:

$$\text{Forward Rate} = \text{Spot Rate} \times \frac{1 + \text{Interest Rate BBB}}{1 + \text{Interest Rate AAA}}$$

As the interest rate of currency AAA increases, the denominator in the formula will increase, leading to a decrease in the forward rate.

Q.4895 Which of the following is the correct difference between translation and transaction risk?

- A. As compared to transaction risk, translation risk does not affect the cash flows of a company.
- B. As compared to translation risk, transaction risk does not affect the cash flows of a company.
- C. Translation risk is hedged using outright forward transactions and swaps.

D. Transaction risk is hedged using the forward contracts.

The correct answer is **A**.

Translation risk and transaction risk are two types of foreign exchange risks that multinational corporations often face. Translation risk arises when a company's financial statements, which are in a foreign currency, are converted back into the parent company's currency. This type of risk does not directly affect the cash flows of a company, but it can impact the reported earnings and equity. On the other hand, transaction risk is associated with future cash flows that might change due to changes in exchange rates. This type of risk directly affects the cash flows of a company.

Choice B is incorrect. Transaction risk does affect the cash flows of a company. It arises from the effect of unexpected currency fluctuations on a company's future cash transactions, and can have significant impacts on the profitability and overall financial position of a company.

Choice C is incorrect. While it's true that translation risk can be hedged using various financial instruments, outright forward transactions and swaps are not exclusively used for this purpose. Other hedging strategies such as money market hedges or futures contracts may also be employed depending on the specific circumstances and needs of the corporation.

Choice D is incorrect. Although forward contracts can indeed be used to hedge transaction risk, this statement is too narrow in scope as it implies that only forward contracts are used for this purpose which isn't accurate. A variety of other derivative instruments like options or futures could also be utilized to hedge against transaction risk.

Things to Remember

- Translation risk arises from the conversion of financial statements in a foreign currency back into the parent company's currency.
- Translation risk impacts reported earnings and equity but does not directly affect cash flows.
- Transaction risk is associated with future cash flows that may change due to exchange rate fluctuations.
- Transaction risk directly affects the cash flows of a company and can impact profitability and financial position.
- Translation risk can be hedged using various financial instruments such as outright forward transactions, swaps, money market hedges, or futures contracts.
- Transaction risk can be hedged using not only forward contracts but also options, futures, or other derivative instruments.

Reading 36: Pricing Financial Forwards and Futures

Q.665 An investment manager at Galaxy Asset Management instructs his broker to short sell 2,000 shares of Solar Computer Corp. in April. The broker borrowed the shares from another client and shorted the 2,000 shares of Solar at €456.8 per share. The manager then asked the broker to close the short position in mid-September when the price per share got to €455.8. If the shares paid a dividend of €1.85 per share in July, then calculate the net payoff of the investment manager after closing out the position. (For this question, assume there are no fees, commissions, or margins.)

- A. The investment manager will receive a net cash inflow of €913,600
- B. The investment manager will pay a net cash outflow of €911,600
- C. The investment manager will pay a net cash outflow of €1,700.
- D. The investment manager will pay a net cash outflow of €3,700

The correct answer is **C**.

The following are the cash flows of the investor during the months:

In April, the manager received a cash inflow of €913,600 as a result of shorting the 2,000 shares of Solar Computers Corp:

$$2,000 \text{ Short shares} * €456.8 \text{ per share} = €913,600$$

In July, the shares earned a dividend of €1.85 per share, which was transferred to the lender of the shares:

$$2000 \text{ shares} * €1.85 = €3,700$$

In September, the shares were purchased at a price of €455.8 to close the short position. Thus, the cash outflow for September is €911,600.

And the net cash flow of the investor is:

$$€913,600 - €3,700 - €911,600 = -€1,700.$$

Things to Remember

- **Short Selling:** Short selling is a trading strategy where an investor borrows shares and immediately sells them, hoping to buy them back later at a lower price.
- **Dividends:** Dividends are payments made by a corporation to its shareholders, usually in the form of cash or additional shares.
- **Net Payoff:** The net payoff is the final result of a financial transaction after considering

all cash inflows and outflows.

- **Short Position:** A short position is created when an investor sells a security that they do not own with the intention of buying it back at a lower price in the future.
 - **Cash Flow:** Cash flow is the movement of money into or out of a business, project, or financial product.
-

Q.667 Which of the following situations correctly depicts a short squeeze scenario?

- A. The prices of shares of a specific firm are continuously decreasing causing more and more investors to short sell the shares of that firm.
- B. The prices of shares of a specific firm are decreasing rapidly, and the supply of shares is greater than its demand.
- C. The prices of shares of a specific firm are increasing rapidly, forcing short sellers to closeout their positions.
- D. The process of borrowing the shares of a specific firm from a client and selling them at the current rate with the expectation of purchasing the same shares at lower prices in the future.

The correct answer is **C**.

A short squeeze occurs when the prices of shares of a specific firm are increasing rapidly, forcing short sellers to closeout their positions. Short selling is a trading strategy where investors sell shares they do not own, with the expectation that the price will fall and they can buy the shares back at a lower price, thereby making a profit. However, if the price of the shares increases instead of falling, the short sellers are forced to buy back the shares at a higher price to close their positions. This buying pressure further increases the price of the shares. This situation, where short sellers are squeezed out of their positions due to rapidly increasing prices, is known as a short squeeze. It is a risky situation for short sellers as they stand to incur significant losses if they are unable to close their positions before the prices rise too high.

Choice A is incorrect. While it is true that a decrease in share prices may encourage more investors to short sell, this does not represent a short squeeze. A short squeeze occurs when the price of an asset increases rapidly, forcing those who have short sold the asset to buy it back at higher prices to cover their positions.

Choice B is incorrect. This scenario describes a situation where there's an oversupply of shares in the market leading to a decrease in price. However, this does not constitute a short squeeze which specifically refers to rapid increase in share prices causing distress for those who have taken short positions.

Choice D is incorrect. This choice simply describes the process of short selling but does not depict a situation of 'short squeeze'. In fact, if the expectation (of purchasing same shares at lower prices) mentioned here fails and instead share prices rise sharply, then that would lead to a 'short squeeze'.

Things to Remember

- Short selling is a trading strategy where investors sell borrowed shares with the expectation of buying them back at a lower price in the future.
 - A short squeeze occurs when the price of the asset being shorted increases rapidly, forcing short sellers to buy back the shares at higher prices to cover their positions.
 - Short squeezes can lead to significant losses for short sellers if they are unable to close their positions quickly as prices rise.
 - Short squeezes can be triggered by positive news, strong buying pressure, or a combination of factors that drive up the price of the asset.
-

Q.669 Kevin Rodriguez is a candidate for the position of a junior trader at a mid-sized investment bank in Mexico. The bank's hiring process is rigid, consisting of 1 written exam and 2 interviews. Rodriguez has cleared the written exam and is currently being interviewed by the recruitment committee. The committee asked Kevin to describe the situation where an investor can make a risk-free profit on a forward contract. Kevin presented the following two scenarios:

- I. If the forward price of the stock is greater than the current price, the investor can profit by purchasing shares at the current price and shorting shares at the forward price.
- II. If the current price of the stock is greater than the forward price, the investor can profit by purchasing shares at the current price and shorting shares at the forward price.

Assuming that the forward price being considered differs from the forward price implied by the spot price and the current interest rate, which of the above-mentioned scenarios will generate profit?

- A. Scenario I will generate a profit, and scenario II will generate a loss
- B. Scenario I will generate a loss, and scenario II will generate a profit
- C. Scenario I will generate a profit, and scenario II will also generate a profit
- D. Scenario I will generate a loss, and scenario II will also generate a loss

The correct answer is **A**.

Scenario I will generate a profit, and scenario II will generate a loss. This is because in Scenario I, if the forward price of the stock is higher than the current price, the investor can borrow funds at the risk-free rate to buy shares at the current price and short forward contracts to sell the asset at the higher forward price. This strategy allows the investor to earn a risk-free return. The investor is essentially locking in a higher selling price for the shares in the future, which, if the shares are bought at a lower price now, will result in a profit. This is a classic example of arbitrage, where the investor takes advantage of price discrepancies in different markets to make a risk-free profit.

On the other hand, Scenario II will generate a loss. If the current price of a stock is higher than the forward price, the investor would need to short the shares now, invest the proceeds at the risk-free rate, and use the proceeds to take a long position in the forward contract. However, this strategy would not result in a profit because the investor would be buying high (current price) and selling low (forward price), which is the opposite of the profit-making strategy of buying low and selling high. Therefore, Scenario II will generate a loss, making Choice A the correct answer.

Choice B is incorrect. Scenario I will not generate a loss. If the forward price of the stock is greater than the current price, an investor can profit by buying shares at the current price and shorting shares at the forward price. This arbitrage opportunity exists because of mispricing in the market.

Choice C is incorrect. Both scenarios will not generate a profit. In Scenario II, if an investor buys shares at a higher current price and shorts them at a lower forward price, they would incur a loss rather than making a profit.

Choice D is incorrect. It's not true that both scenarios will generate losses. As explained above, in Scenario I where the forward price of stock is greater than its current spot rate, there exists an arbitrage opportunity for risk-free profits by buying low (at spot rate) and selling high (at future date with higher forward rate).

Things to Remember

- **Forward Contracts:** These are agreements between two parties to buy or sell an asset at a specified future time for a price agreed upon today. They are used to hedge against price fluctuations and to speculate on future price movements.
- **Arbitrage:** This is the practice of taking advantage of a price difference between two or more markets. It involves buying low in one market and selling high in another to make a risk-free profit.
- **Risk-Free Rate:** This is the theoretical rate of return of an investment with zero risk. It is often used as the benchmark for pricing financial instruments and determining arbitrage opportunities.
- **Spot Price:** The current market price of an asset for immediate delivery or settlement.

- Interest Rate Parity: This concept states that the difference in interest rates between two countries should be equal to the difference between the forward exchange rate and the spot exchange rate.
-

Q.670 Priyanka Singh is a derivative investment manager at Hind Investments based in Mumbai. Priyanka is analyzing the shares of Cosmetic World Company that are currently trading at \$76.2 per share. Cosmetic World has the largest market shares in the cosmetics market of Asia, and the company has been profitable for over a decade. The 3-month forward contract on the stock is being offered for the price of \$86.8. Priyanka Singh wants to trade 5,000 shares of Cosmetic World with the intention of closing the position in 3 months. If the risk-free interest rate is 5% compounded annually, determine the arbitrage profit.

- A. \$77,158
- B. \$68,485
- C. \$50,798
- D. \$48,324

The correct answer is **D**.

Since the forward price of the contract over Cosmetic World is higher than its current price, the investor can profit by borrowing the funds at the current rate to purchase shares at current prices and short forward contracts to sell the shares at higher forward prices.

If the investor borrows \$76.2 dollar at the risk-free rate to purchase shares now, the loan amount, in 3 months, will grow to:

$$\$76.20 \times (1.05)^{3/12} \times 5,000 \text{ shares} = \$385,675.72$$

Since the investor will use the proceedings to short a forward contract (sell the shares at the forward price of \$86.8), the cash flow from the short sell is:

$$\$86.80 \times 5,000 \text{ shares} = \$434,000$$

$$\text{The arbitrage net cash flow is } \$434,000 - \$385,675.72 = \$48,324.28$$

Things to Remember

- Arbitrage: Arbitrage is the practice of taking advantage of a price difference between two or more markets to generate a profit.
- Forward Contracts: Forward contracts are agreements between two parties to buy or sell an asset at a specified future time for a price agreed upon today.

- **Risk-free Interest Rate:** The risk-free interest rate is the theoretical rate of return of an investment with no risk of financial loss.
 - **Market Efficiency:** Market efficiency is a concept in financial economics that states that asset prices fully reflect all available information.
 - **Short Selling:** Short selling is the practice of selling borrowed assets in the hope of buying them back later at a lower price, thereby making a profit.
-

Q.672 George Brown, a fixed-income investment analyst, is determining the price of a 6-month forward contract on a unique asset. The risk-free rate of interest is 12% per year, compounded semi-annually, whereas the dividend yield on the asset is 7% p.a, with semi-annual compounding. If the asset price is \$95, then what is the price of the forward contract?

- A. \$100.8
- B. \$97.29
- C. \$93.96
- D. \$96.14

The correct answer is **B**.

The price of a forward contract when the underlying pays a dividend is given by:

$$F = S \left(\frac{1 + r}{1 + q} \right)^T$$

where:

S= current asset price;

r = risk-free rate;

q =dividend yield paid by the underlying asset expressed on a per annum basis; and

T = time to maturity (in years) of the forward contract

The price of the forward contract is:

$$\$95 \left(\frac{1.06}{1.035} \right) = \$97.29$$

Q.676 Ellen Harper, a portfolio manager at Deutsch Investments Group (DIG), is considering investing in the 6-months futures contract on the German DAX-30 index. The DAX-30 is currently valued at 12,240 with a dividend yield of 1.7% per year. If the risk-free rate in Germany is 3.2%, then the price of the futures contract should be:

- A. 12,241
- B. 12,330
- C. 13,080
- D. 12,578

The correct answer is **B**.

The price of the futures contract on the DAX-30 index is calculated as:

$$\text{Price of futures contract} = \text{Current index} \times \left(\frac{1 + f}{1 + q} \right)^T$$

Where f is the risk-free rate and q is the annual dividend yield.

$$\text{Price of futures contract} = 12,240 \left(\frac{1.032}{1.017} \right)^{0.5} = 12,329.935$$

Q.677 Amy Damian is a portfolio manager at a local pension fund. She has recently received great appreciation from the upper management of the fund because of her arbitrage profit of \$1.6 million on index futures. She earned arbitrage profit during the period where the prices of futures contracts on the S&P 500 were trading lower than the current prices of the index. Which of the following trading strategies must she have used?

- A. Purchasing the stocks whose movement closely mirrors the S&P 500 index and short-selling S&P 500 futures
- B. Short-selling the stocks whose movement closely mirrors the S&P 500 index and taking a long position in S&P 500 futures
- C. Purchasing the stocks whose movement closely mirrors the S&P 500 index and taking a long position in S&P 500 futures

D. Short-selling the stocks whose movement closely mirrors the S&P 500 index and taking a short position in S&P 500 futures

The correct answer is **B**.

Amy Damian would have short-sold the stocks whose movement closely mirrors the S&P 500 index and taken a long position in S&P 500 futures. This strategy is known as 'index arbitrage' and is used when the futures contracts are trading at a price lower than the current price of the index. The idea behind this strategy is to exploit the price difference between the index and its futures contract. By short-selling the stocks of the index, Amy is betting that the prices of these stocks will fall. At the same time, by taking a long position in the futures contract, she is betting that the price of the futures contract will rise. When the futures contract expires, the price of the futures contract and the index should converge, allowing Amy to profit from the price difference. It's important to note that it's not necessary to short-sell all the stocks in the index. Instead, an investor can short-sell a representative sample of stocks whose price movements closely mirror those of the index. This strategy is often used by institutional investors and hedge funds to exploit pricing inefficiencies in the market.

Choice A is incorrect. Purchasing the stocks whose movement closely mirrors the S&P 500 index and short-selling S&P 500 futures would not yield a profit in this scenario. This is because when futures contracts on the S&P 500 are trading at a lower price than the current index prices, it indicates that market participants expect future prices to be lower than current prices. Therefore, buying stocks and short-selling futures would result in losses if future prices indeed fall.

Choice C is incorrect. Purchasing the stocks whose movement closely mirrors the S&P 500 index and taking a long position in S&P 500 futures would also not yield a profit under these circumstances. This strategy involves betting on an increase in both stock and future prices, which contradicts with market expectations of lower future prices.

Choice D is incorrect. Short-selling the stocks whose movement closely mirrors the S&P 500 index and taking a short position in S&P 500 futures implies betting on falling stock and future prices simultaneously. However, given that futures are already trading at a discount to spot (current) price, further decrease may not be substantial enough to generate significant profits from this strategy.

Things to Remember

- Index arbitrage is a trading strategy used to exploit price differences between an index and its futures contract.
- Short-selling involves selling an asset that the seller does not own, with the intention of buying it back later at a lower price.
- Long position refers to buying an asset with the expectation that its price will rise in the future.

- Short position refers to selling an asset with the expectation that its price will fall in the future.
 - Arbitrage opportunities arise when there are price discrepancies between related assets or securities.
-

Q.680 When considering short-selling as an investment strategy, it's crucial to understand the risks and mechanisms involved. In short-selling, an investor

- A. sells an asset that he does not own with the intent of buying it back in the future at a lower price.
- B. buys an asset with the intent of selling it in the future at a higher price for profit.
- C. sells an asset that he does not own with the intent of buying it back in the future at a higher price.
- D. owns the asset but sells it with the intent of buying it back at a higher price.

The correct answer is **A**.

Short-selling is a trading strategy where an investor sells an asset that they do not own with the intention of buying it back later at a lower price. The investor borrows the asset (usually from a broker) and sells it on the market. Later, when the price of the asset has fallen, the investor buys it back to return to the lender. The profit from this transaction is the difference between the selling price and the buying price. This strategy is used when the investor believes that the price of the asset will decrease in the future. It's a risky strategy because if the price of the asset increases instead of decreasing, the investor will have to buy it back at a higher price, resulting in a loss.

Choice B is incorrect. This option describes a long position, not short-selling. In a long position, an investor buys an asset with the expectation that its price will rise in the future, allowing them to sell it for a profit. This is fundamentally different from short-selling where the investor sells an asset they do not own with the intent of buying it back at a lower price.

Choice C is incorrect. This choice incorrectly describes short-selling as selling an asset one does not own with the intention of buying it back at a higher price. The goal of short-selling is to profit from a decrease in the price of an asset, not an increase.

Choice D is incorrect. This option refers to selling assets that one already owns with the intention of buying them back at a higher price later on which contradicts the concept of short selling where you sell assets you don't own and buy them back when their prices fall.

Things to Remember

- Short-selling involves selling an asset that the investor does not own, with the intention of buying it back at a lower price in the future.
 - Short-selling is a strategy used when an investor believes that the price of the asset will decrease in the future.
 - Short-selling can be risky because if the price of the asset increases instead of decreases, the investor will incur a loss.
 - Short-selling requires borrowing the asset from a broker to sell it on the market.
 - Short-selling is subject to margin requirements and potential margin calls if the price of the asset rises significantly.
-

Q.3530 Long positions in futures contracts are more desirable to forward contracts when the correlation between futures prices and interest rates is:

- A. Zero
- B. Positive
- C. Negative
- D. Known in advance using the spot curve

The correct answer is **B**.

When the correlation between interest rates and futures prices is positive, futures contracts are more desirable to holders of long positions than forward contracts. This is because rising prices will lead to futures profits that are reinvested in periods of rising interest rates and falling prices will lead to losses that occur during periods of falling interest rates. Therefore, it is far better to receive cash flows in the interim than the expiration under such conditions. This is due to the fact that futures contracts are marked to market daily, meaning that gains and losses are realized and can be reinvested daily. This allows the holder of a long futures contract to potentially earn interest on their gains, which can be particularly beneficial in a rising interest rate environment.

Choice A is incorrect. A zero correlation between futures prices and interest rates would not make a long position in futures contracts more desirable than forward contracts. This is because the value of a futures contract is not influenced by changes in interest rates when there's no correlation.

Choice C is incorrect. Negative correlation between futures prices and interest rates would

make a long position in forward contracts more desirable than futures contracts, not the other way around. This is because as interest rates decrease (increase), the price of future contracts also increases (decreases).

Choice D is incorrect. Knowing the correlation in advance using the spot curve does not necessarily make a long position in futures more desirable than forwards. The desirability depends on whether this known correlation is positive or negative, rather than just knowing it.

Things to Remember

- Forward contracts are private agreements between two parties to buy or sell an asset at a specified future date at a price agreed upon today.
 - Futures contracts are standardized agreements to buy or sell an asset at a specified price on a specified future date.
 - Marking to market is the process of adjusting the value of the contract to reflect the current market price.
 - Positive correlation between futures prices and interest rates can make long positions in futures contracts more desirable due to the potential for reinvestment of gains in a rising interest rate environment.
 - Negative correlation between futures prices and interest rates can make long positions in forward contracts more desirable.
-

Q.4673 An investor considers investing in a forward contract to buy an asset currently valued at USD 500 for USD 800 in 2 years. Given that the current interest rate is 5% with annual compounding, what is the current value of the forward contract?

- A. 346.49
- B. -225.62
- C. 235.62
- D. -220.45

The correct answer is **B**.

A forward contract is a linear derivative whose value is given by:

$$S - PV(K)$$

S is the current asset price, and PV (K) denotes the present value of the asset's future price.

So in this, the value is given by:

$$500 - 800(1.05)^{-2} = -225.62$$

Q.4897 Suppose that Paul enters into a 3-year forward contract on a stock that pays no dividends and that the current stock price is USD 35 and the annually-compounded risk-free rate is 5%. What is the forward price of this forward contract?

- A. \$40.52
- B. \$36.75
- C. \$30.23
- D. \$40.67

The correct answer is **A**.

This is a no-income case, and the forward price is given by:

$$F = S(1 + R)^T$$

Where,

S = Spot Price

F = Forward Price

R = Risk-free interest rate per year compounded annually

T = Time to maturity,

Thus,

$$\begin{aligned} F &= 35(1.05)^3 \\ &= 40.5168 \approx \$40.52 \end{aligned}$$

Q.4898 Suppose that John enters into a 3-year forward contract on a bond. The spot price of the bond is USD 80. The bond is expected to provide a coupon of USD 5 at the end of the 1st year and the 2nd year. The annually compounded risk-free rate for all maturities is 4% per year. What is the 3-year forward price?

- A. \$89.99
- B. \$79.38
- C. \$84.79
- D. \$ 83.90

The correct answer is **B**.

We first calculate the present value of income:

$$I = 5(1.04)^{-1} + 5(1.04)^{-2} = 9.43047 \approx \$ 9.43$$

Now, since this is a known-income case, we use the formula:

$$F = (S - I) (1 + R)^T$$

Where:

S = Spot Price

F = Forward Price

R = Risk-free interest rate per year compounded annually

T = Time to maturity

I = Income

Thus,

$$\begin{aligned} F &= (80 - 9.43) (1.04)^3 \\ &= \$79.38 \end{aligned}$$

Q.4899 What is the 2-year futures price on the stock index, whose value is USD 2,000, and that the annually compounded risk-free rate is 5% per annum, and the annual dividend yield on the index is 2%?

- A. \$2,119.38
- B. \$2,289.80
- C. \$2,121.80
- D. \$1,887.34

The correct answer is **A**.

The futures price is given by the formula:

$$F = S \left(\frac{1 + R}{1 + Q} \right)^T$$

Where:

S = Spot Price

F = Forward Price

R = Risk-free interest rate per year compounded annually

Q = Yield

T = Time to maturity

Thus, the two-year futures price is:

$$F = 2,000 \left(\frac{1.05}{1.02} \right)^2 = \text{USD } 2119.38$$

Q.4900 Paul enters into a two-year forward contract on a stock that pays no dividends and that the current stock price is USD 33 and the annually-compounded risk-free rate is 5% per year. Suppose that one year after the forward contract was initiated, the spot price is USD 35, and the risk-free rate has changed to 6% per annum. What is the value of this forward contract?

- A. 2.22
- B. 0.68
- C. 1.9
- D. 2.33

The correct answer is **B**.

The forward price, initially when the contract is initiated, is given by:

$$K = S(1 + R)^T = 33(1.05)^2 = \$36.38$$

After one-year, the forward price is given by:

$$F = S(1 + R)^T = 35(1.06)^1 = \$37.10$$

The value of the forward contract is

$$\frac{37.10 - 36.38}{1.06^1} = 0.6792 \approx 0.68$$

Q.4901 What would you expect when there is a negative correlation between return on assets and interest rates?

- A. The forward price is greater than the futures price.
- B. The forward price is less than the futures price.
- C. The forward price is equal to the futures price.
- D. The relationship between the forward price and the futures price cannot be determined.

The correct answer is **A**.

The forward price is greater than the futures price. This is because when there is a negative correlation between return on assets and interest rates, it implies that when the price of an asset rises, funds are typically invested at a lower rate. Conversely, if the asset's price falls, funds may be invested at a higher rate. As a result, a long futures contract is less appealing than a long forward contract, as the futures price tends to be lower than the forward price. This is due to the daily settlement of futures contracts, which can lead to a lower futures price compared to the forward price in a negatively correlated environment. Therefore, in a situation where there is a negative correlation between return on assets and interest rates, one would expect the forward price to be greater than the futures price.

Choice B is incorrect. The forward price is not less than the futures price in this scenario. This is because the futures price takes into account daily settlement, which can lead to a higher overall cost due to interest rate fluctuations. In a negative correlation scenario where asset returns increase and interest rates decrease, the futures price would typically be higher than the forward price.

Choice C is incorrect. The forward price and futures prices are not equal in this case. While both are derivative contracts that allow for buying or selling an asset at a future date, they differ in how they account for interest rates and dividends. In particular, futures prices incorporate daily settlement which can cause them to diverge from forward prices when there's volatility in interest rates.

Choice D is incorrect. As explained above, one of the options (A) accurately describes how negative correlation between return on assets and interest rates affects the relationship between forward and future prices.

Things to Remember

- Forward contracts are customizable agreements between two parties to buy or sell an asset at a specified price on a future date.
- Futures contracts are standardized agreements traded on exchanges, with daily settlement and margin requirements.
- Correlation measures the relationship between two variables, with negative correlation indicating an inverse relationship.
- Return on assets (ROA) is a financial ratio that measures a company's profitability by showing how efficiently it generates profits from its assets.
- Interest rates impact the cost of borrowing and investing, affecting asset prices and returns.

Reading 37: Commodity Forwards and Futures

Q.805 Henry Luis is a commodity trader at a mutual fund that focuses on derivatives and commodities investments. Luis was instructed to pay special attention to the storage costs and cost of carry while valuing the commodities futures contract. Which of the following commodities is likely to have the smallest storage costs?

- A. Crude oil
- B. Corn
- C. Livestock
- D. Gold

The correct answer is **D**.

Gold is a precious metal that is highly valued for its properties and uses. It is a dense, soft, shiny, malleable, and ductile metal. Gold is also a good conductor of electricity and is resistant to corrosion and tarnish. These properties make gold an ideal commodity for storage as it does not require special conditions or maintenance. Furthermore, gold is highly liquid and readily available, which means that investors can easily purchase gold for delivery at the maturity of the contract. This reduces the need for storage and consequently, the storage costs. In comparison to other commodities like crude oil, corn, and livestock, gold has the smallest storage costs. Crude oil requires special storage facilities to prevent leakage and environmental damage. Corn is a perishable commodity and requires conditions that prevent spoilage. Livestock requires feed, care, and suitable living conditions. All these factors contribute to their higher storage costs.

Choice A is incorrect. Crude oil incurs significant storage costs due to the need for specialized storage facilities that meet safety and environmental regulations. Additionally, the cost of insuring these facilities can also be substantial.

Choice B is incorrect. Corn, being a perishable commodity, requires controlled environments for storage to prevent spoilage and infestation. This results in high storage costs.

Choice C is incorrect. Livestock also incurs high storage costs as it requires feeding, care and appropriate living conditions which are expensive to maintain over time.

Things to Remember

- Storage costs are an important factor to consider when valuing commodities futures contracts as they can significantly impact the overall profitability of the investment.
- Cost of carry refers to the costs associated with holding a physical commodity or financial instrument over a period of time, including storage costs, insurance, financing, and other expenses.

- Gold is often considered a safe haven asset and a hedge against inflation, which can also influence its storage costs and demand.
 - Commodities like crude oil, corn, and livestock have specific storage requirements due to their nature, which can result in higher storage costs compared to other commodities like gold.
 - Investors and traders need to carefully assess storage costs along with other factors like market demand, supply dynamics, and geopolitical events when trading commodities futures contracts.
-

Q.806 Commodities futures markets consist of hundreds of different commodities with different properties and attributes. Some commodities do not consider the storage costs separately because, in those commodities, the forward price of the commodity compensates the commodity owner for the cost of storage. Such commodity markets are referred to as:

- A. Discount markets.
- B. Free markets.
- C. Carry markets.
- D. Forwards markets.

The correct answer is **C**.

The term 'carry' in financial markets refers to the cost of holding a financial instrument or commodity. In the context of commodities, a commodity is said to be in 'carry' when it is being stored rather than being traded. This concept is akin to the financial cost of carry in financial markets. The forward price of a commodity in a carry market compensates the owner for the cost of storage. This becomes more apparent in the case of commodities, as the process of producing and distributing them often involves storing them. Therefore, the term 'carry markets' accurately describes commodity markets where the forward price of the commodity compensates the owner for the storage cost.

Choice A is incorrect. Discount markets refer to markets where securities are bought and sold at prices lower than their face value, which is not related to the concept of storage costs in commodity futures markets.

Choice B is incorrect. Free markets are economic systems where prices for goods and services are self-regulated by buyers and sellers negotiating in an open market. This term does not specifically refer to commodities that include storage costs in their forward price.

Choice D is incorrect. Forwards markets involve contracts that agree on a set price for a future transaction, but this term does not imply whether or not storage costs are included in the forward price of commodities.

Things to Remember

- Storage costs play a crucial role in commodity markets, especially for physical commodities that require storage over time.
 - Contango and backwardation are important concepts in commodity futures markets, where the forward price of a commodity is higher (contango) or lower (backwardation) than the spot price.
 - Hedging strategies using commodity futures contracts help market participants manage price risk associated with fluctuations in commodity prices.
 - Speculators in commodity markets take on risk in the hope of making a profit from price movements, adding liquidity to the market.
 - Factors such as supply and demand dynamics, geopolitical events, weather conditions, and economic indicators can influence commodity prices.
-

Q.808 John Sinclair is a high net worth individual investor and the owner of a chain of independent fossil fuel power plants in Saint Petersburg, Russia. During a web conference with his investment advisor from Canada's largest investment bank, the advisor advised Sinclair to invest in futures contracts on Crude and Brent oil. He mentioned that apart from the monetary gains, the investor might also receive nonmonetary benefits from the physical possession of these commodities. Which of the following benefits is the investment advisor referring to?

- A. Risk-free rate
- B. Lease yield
- C. Convenience yield
- D. Roll-over yield

The correct answer is **C**.

The nonmonetary benefit that the investment advisor is referring to is known as the convenience yield. This is a term used in the commodities market to describe the benefits that an investor

derives from physically holding a commodity. These benefits are nonmonetary in nature and can include factors such as ensuring supply during shortages, avoiding costs associated with storage and transportation, and providing a hedge against price volatility. In the context of the question, since John Sinclair owns a business that is directly related to the underlying commodities of Crude and Brent oil, he stands to gain from the convenience yield. For instance, in the event of a short-term oil supply shortage, he could use the oil from his futures contracts to continue producing power, thereby avoiding potential losses or disruptions to his business.

Choice A is incorrect. The risk-free rate is not a nonmonetary benefit. It refers to the theoretical rate of return of an investment with no risk of financial loss, which is not applicable in this context as investing in futures contracts on Crude and Brent oil involves significant risks.

Choice B is incorrect. Lease yield refers to the return obtained from leasing a commodity, which does not apply here as John Sinclair's investment advisor has suggested investing in futures contracts, not leasing commodities.

Choice D is incorrect. Roll-over yield pertains to the gains or losses experienced when an investor moves his position from a front month contract to another contract set for a future date. This concept does not relate to any nonmonetary benefits that could be derived from physically possessing these commodities.

Things to Remember

- **Convenience Yield:** The convenience yield is the benefit or premium associated with holding a physical commodity rather than a futures contract. It represents the nonmonetary advantages of holding the physical asset, such as operational flexibility, supply security, and hedging against price spikes.
- **Risk-free Rate:** The risk-free rate is the theoretical rate of return of an investment with zero risk of financial loss. It is often used as a benchmark for evaluating the performance of other investments and plays a crucial role in pricing financial instruments.
- **Lease Yield:** Lease yield refers to the return obtained from leasing a commodity to another party. It is a source of income for commodity owners who choose to lease out their assets rather than physically hold them.
- **Roll-over Yield:** Roll-over yield is the gain or loss experienced when an investor shifts their position from one futures contract to another with a later expiration date. It reflects the impact of rolling over positions in futures markets.

Q.809 Busra Turkmen is a business newscaster and an economic analyst at one of the leading business and finance-focused news channel in Germany. While writing the evening business report, she noticed that the prices in gold forwards contracts are upward sloping, which means the forward prices of longer maturity gold contracts are higher than the prices of shorter maturity gold contracts. Which of the following terms can she use to define the given trend in gold forward prices?

- A. Convenience yield.
- B. Backwardation.
- C. Upwardation.
- D. Contango.

The correct answer is **D**.

In the context of commodity markets, the term 'Contango' is used to describe a situation where the forward price of a commodity is higher than the spot price. This is typically observed when the costs associated with storing the commodity (such as warehousing and insurance costs) are significant. In such a scenario, the forward price needs to be higher than the spot price to compensate for these costs. This results in an upward sloping forward curve, where the forward prices of contracts with longer maturity are higher than those with shorter maturity. This is exactly the trend that Busra Turkmen observed in the gold forwards contracts. Therefore, the term 'Contango' accurately describes this trend.

Choice A is incorrect. Convenience yield refers to the benefits or advantages that a company gains by holding a physical commodity, instead of the contract for the commodity. It does not describe the trend in forward prices of commodities.

Choice B is incorrect. Backwardation is a situation where the spot price is higher than the forward price. This scenario contradicts Busra's observation where forward prices are increasing with maturity, hence it cannot be correct.

Choice C is incorrect. There's no term as 'Upwardation' in commodity markets, making this option invalid.

Things to Remember

- Contango is a term used in commodity markets to describe a situation where the forward price of a commodity is higher than the spot price.
- Backwardation is the opposite of contango, where the spot price is higher than the forward price.
- Convenience yield refers to the benefits or advantages gained by holding a physical commodity instead of a contract for the commodity.

- Forward curves can provide important insights into market expectations and supply-demand dynamics for commodities.
 - Storage costs, interest rates, and market expectations play a significant role in determining the shape of forward curves for commodities.
-

Q.810 In the United States, forward and futures contracts are available for trading on various commodities. These commodities are classified in the categories of extractable or renewable and primary or secondary. Which of the following commodities can be classified as renewable as well as a primary commodity?

- A. Oil
- B. Copper
- C. Livestock
- D. Gasoline

The correct answer is **C**.

Livestock is a renewable and primary commodity. Renewable commodities are those that can be replenished over time through natural processes. Livestock, being a product of agricultural activities, falls under this category. They are bred and raised, and their population can be replenished over time, making them renewable. Primary commodities, on the other hand, are commodities that are traded in their raw, unprocessed form. Livestock, being traded in their live form without any processing, are considered primary commodities. Therefore, livestock is both a renewable and a primary commodity.

Choice A is incorrect. Oil is an extractable commodity, not renewable. It is formed from the remains of ancient marine plants and animals, a process that takes millions of years. Therefore, it cannot be replenished in a short period of time.

Choice B is incorrect. Copper is also an extractable commodity as it comes from the earth's crust and cannot be renewed once extracted.

Choice D is incorrect. Gasoline, although derived from oil which makes it primary in terms of processing stage, it's not renewable because its source (oil) isn't renewable.

Things to Remember

- Commodities can be classified based on various criteria such as extractable or renewable, primary or secondary, hard or soft, etc.

- Renewable commodities are those that can be replenished over time through natural processes.
 - Primary commodities are traded in their raw, unprocessed form.
 - Extractable commodities are those that are obtained through extraction processes like mining or drilling.
 - Secondary commodities are those that are derived from primary commodities through processing.
-

Q.813 A commodities trader at an investment bank has analyzed the forward prices of gold contracts and realized that there might be an arbitrage profit present in gold futures contracts. The spot price for one ounce of gold is \$1,205 and the 6-month futures contract is quoted as \$1,253 per ounce. If the risk-free rate is 6% (compounded continuously), then the arbitrage profit for trading one gold futures contract is:

- A. \$7.81
- B. \$23.61
- C. \$11.31
- D. \$10.96

The correct answer is **C**.

The futures price of the 6-month gold contract is greater than the spot price. Therefore, an arbitrage opportunity exists.

An investor should take the following steps at time 0:

1. Borrow \$1,205 for 6 months at the risk-free rate
2. Buy gold at the spot price of \$1,205
3. Take a short exposure in the 6-month gold futures contract for the futures price of \$1,253

At the expiration of the futures contract:

1. Sell gold at the futures price of \$1,253
2. Pay the borrowed money with interest of $\$1,205e^{0.06 \times 0.5} = \$1,241.69$
3. Earn the cash-and-carry arbitrage profit of $(\$1,253 - \$1,241.69) = \$11.31$

Things to Remember

- Arbitrage: Arbitrage is the practice of simultaneously buying and selling an asset to

profit from a difference in price. In this case, the commodities trader is exploiting an arbitrage opportunity between the spot price and the futures price of gold.

- **Spot Price:** The spot price is the current market price at which an asset can be bought or sold for immediate delivery. In this case, the spot price of one ounce of gold is \$1,205.
 - **Forward Price:** The forward price is the price at which a contract obligates the buyer to purchase an asset at a future date. In this case, the 6-month futures contract for gold is quoted at \$1,253 per ounce.
 - **Risk-Free Rate:** The risk-free rate is the theoretical rate of return of an investment with zero risk of financial loss. In this scenario, the risk-free rate is 6% compounded continuously.
 - **Cash-and-Carry Arbitrage:** Cash-and-carry arbitrage involves simultaneously buying an asset in the spot market and selling it in the futures market to profit from price differentials. This strategy exploits pricing inefficiencies in the market.
-

Q.814 Branden Berger is an active trader at Eclipse Funds. He has been closely monitoring the spot prices and forward prices of corn bushel for a long time. He has noticed that the spot price and the 1-year forward price of a corn bushel contract are identical at \$6.90 per corn bushel. If the risk-free rate is 8%, then which of the following strategy will earn him arbitrage profit?

- A. Borrow the amount equal to the spot price of corn for 1 year at the risk-free rate of 8%, buy a corn bushel at the spot price, and take a short position in a 1-year corn forward contract. At the expiration of the contract, the investor will sell the corn bushel at the futures price and pay off the borrowed money with interest.
- B. Borrow the amount equal to the spot price of the corn bushel, lend the money for one year at the risk-free of 8%, and take a long position in a corn forward contract. After one year, the investor will receive the lent money with interest, receive the corn bushel at the expiration of the contract, and deliver the corn bushel.
- C. Short sell corn at the spot price of \$6.90 per bushel, lend the money for one year at the risk-free of 8% and take a long position in a corn forward contract. After one year, the investor will receive the lent money with interest, receive the corn bushel at the expiration of the contract, and deliver the corn bushel.
- D. Since the spot price of the corn bushel is equal to the 1-year forward price of the corn

forward contract, an arbitrage profit is not possible.

The correct answer is **C**.

The strategy outlined in choice C allows for an arbitrage profit. The trader begins by short selling corn at the spot price of \$6.90 per bushel. Short selling involves borrowing an asset and selling it with the expectation that the price will decrease, allowing the trader to buy it back at a lower price and return it to the lender, keeping the difference as profit. In this case, the trader then lends the proceeds from the short sale for one year at the risk-free rate of 8%. Simultaneously, the trader takes a long position in a corn forward contract for the price of \$6.90. After one year, the trader will receive the lent money with interest, which would amount to \$7.45. The trader will also receive the corn bushel at the expiration of the contract for \$6.90, and deliver the corn to the lender. The difference between the amount received from lending and the cost of the corn bushel is the arbitrage profit.

Choice A is incorrect. This strategy would not generate an arbitrage profit because the spot and forward prices are identical. Borrowing money to buy corn at the spot price and then selling it at the same price in the future will only result in paying interest on the borrowed money, without any profit.

Choice B is incorrect. This strategy involves borrowing money to lend it out again, which does not make sense as it would just incur unnecessary transaction costs without generating any profits. Furthermore, taking a long position in a forward contract when the spot and forward prices are identical will not yield any arbitrage opportunities.

Choice D is incorrect. While it's true that when spot and forward prices are identical there usually isn't an arbitrage opportunity, this statement ignores other factors such as risk-free rate of return that can create such opportunities under certain conditions.

Things to Remember

- **Arbitrage:** Arbitrage is the practice of taking advantage of a price difference between two or more markets to generate profit with little to no risk.
- **Spot Price:** The spot price is the current market price of a security, commodity, or currency for immediate delivery and payment.
- **Forward Price:** The forward price is the predetermined delivery price for an asset in a forward contract set at the time of the contract.
- **Risk-Free Rate:** The risk-free rate is the theoretical rate of return of an investment with zero risk of financial loss, typically represented by government bonds.
- **Short Selling:** Short selling is the practice of selling borrowed assets with the expectation that the price will decrease, allowing the seller to buy back the assets at a

lower price and return them to the lender, profiting from the difference.

- **Long Position:** A long position is the buying of an asset with the expectation that its value will increase over time, allowing the investor to sell it at a higher price in the future.
 - **Transaction Costs:** Transaction costs are the expenses incurred when buying or selling assets, including brokerage fees, taxes, and other charges.
-

Q.815 Branden Berger is an active derivative trader at the Eclipse Funds. He has been closely monitoring the spot prices and forward prices of corn bushel for a long time. He has noticed that the spot price and 1-year forward price of a corn bushel contract are identical at \$6.90 per corn bushel. If the risk-free rate is 8%, then the arbitrage profit is equal to:

- A. \$0
- B. \$0.53
- C. \$0.55
- D. \$1.06

The correct answer is **C**.

The investor should take the following steps at time 0:

1. Short sell corn at the spot price of \$6.90 per bushel
2. Lend the \$6.90 for 1 year at the risk-free rate
3. Take a long position in a 1-year corn forward contract

At the expiration of the futures contract:

1. Receive the lent money with interest for $\$6.90(1.08) = \7.45
2. Purchase the corn bushel at the forward price of \$6.90 and deliver the corn bushel
3. Earn the reverse cash-and-carry arbitrage profit of $(\$7.45 - \$6.9) = \$0.55$

Note. Selling short comes with a fee, but we ignore that for exam purposes.

Things to Remember

- **Arbitrage:** Arbitrage is the practice of taking advantage of a price difference between two or more markets. Traders engage in arbitrage to make a profit by simultaneously buying and selling an asset in different markets to exploit price differences.

- **Spot Price:** The spot price is the current market price of a security, commodity, or currency that is settled immediately. It is the price at which an asset can be bought or sold for immediate delivery.
 - **Forward Price:** The forward price is the predetermined delivery price for an asset in a forward contract. It is agreed upon at the time of the contract and is used to eliminate the risk of price fluctuations.
 - **Risk-Free Rate:** The risk-free rate is the theoretical rate of return of an investment with zero risk of financial loss. It is often used as the benchmark for pricing various financial instruments.
 - **Short Selling:** Short selling is the practice of selling borrowed assets with the expectation that the price will decline in the future. Traders profit by buying back the assets at a lower price to return to the lender.
 - **Cash-and-Carry Arbitrage:** Cash-and-carry arbitrage is a trading strategy that involves buying an asset in the spot market and simultaneously selling the same asset in the futures market to profit from price differentials.
-

Q.816 An investor is analyzing a 6-month oil forward contract that is quoted as \$54 per barrel. The spot price of oil is \$55 per barrel, and the risk-free rate is 10%. In order to earn risk-free profits, the investor short sells oil at the spot price of \$55, lends the money for 6 months at risk-free of 10%, and takes a long position in an oil forward contract for the price of \$54 per barrel. After 6 months, the investor receives the lent money with interest equaling \$57.68, purchases the oil at the forward price of \$54, and delivers the oil to earn an arbitrage profit of \$3.68 per barrel. Which of the following strategies has he most likely implemented?

- A. Cash-and-carry arbitrage strategy.
- B. Arbitrage-free strategy.
- C. Reverse cash-and-carry strategy.
- D. Binominal arbitrage strategy.

The correct answer is **C**.

The investor has implemented a reverse cash-and-carry arbitrage strategy. This strategy involves three steps at the initial time: short selling the commodity, lending the proceeds from the short sale at market interest rates for the duration of the forward or futures contract, and taking a long position in the futures contract at the market price. At the expiration of the contract, the investor collects the lent proceeds with interest, takes delivery of the commodity at the futures price, and delivers the commodity to fulfill the short sale commitment. This strategy is typically used when the futures price is lower than the spot price, adjusted for the cost of carry, which includes interest on the funds used to finance the holding of the commodity and any storage costs. In this case, the investor was able to earn an arbitrage profit of \$3.68 per barrel by implementing this strategy.

Choice A is incorrect. A cash-and-carry arbitrage strategy involves buying an asset at the spot price, storing it, and selling a forward contract on the asset. In this case, the investor is short selling oil at the spot price and taking a long position in the oil forward contract, which is not consistent with a cash-and-carry arbitrage strategy.

Choice B is incorrect. An arbitrage-free strategy implies that there are no opportunities to make risk-free profits. However, in this scenario, the investor earns an arbitrage profit of \$3.68 per barrel by short selling oil at the spot price and taking a long position in the oil forward contract.

Choice D is incorrect. A binomial arbitrage strategy involves creating a riskless portfolio by combining positions in options and their underlying assets to exploit mispriced options for profit. This scenario does not involve any options; hence it cannot be described as a binomial arbitrage strategy.

Things to Remember

- Cash-and-carry arbitrage strategy involves buying an asset at the spot price, storing it, and selling a forward contract on the asset.
- Arbitrage-free strategy implies that there are no opportunities to make risk-free profits.
- Binomial arbitrage strategy involves creating a riskless portfolio by combining positions in options and their underlying assets to exploit mispriced options for profit.
- Reverse cash-and-carry arbitrage strategy involves short selling the commodity, lending the proceeds from the short sale at market interest rates for the duration of the forward or futures contract, and taking a long position in the futures contract at the market price.
- Cost of carry includes interest on the funds used to finance the holding of the commodity and any storage costs.

Q.818 An investor is interested in taking a long position in a 12-month cotton forward contract. Estimate the 12-month forward price for cotton that has a spot price of \$37 per pound and an annual lease rate of 5% if the risk-free rate for the commodity is equivalent to 7.5% with annual compounding.

- A. \$39.88
- B. \$38.89
- C. \$37.88
- D. \$36.08

The correct answer is **C**.

The lease rate is used by the lender of the commodity to calculate the lease payment for lending the commodity to borrow. The lease rate must be incorporated into the equation to calculate the forward price of the commodity.

The forward price of cotton is calculated using the following equation:

$$F = S \left(\frac{1 + R}{1 + L} \right)^T$$

Where:

F = forward price

S = spot price

R = risk-free rate

L = lease rate

Thus,

$$F = 37 \left(\frac{1.075}{1.05} \right)^1 = \$37.88 \text{ per pound}$$

Q.819 Ahmet Abdullah is a research analyst at Klosky Investment Company. He is interested in analyzing the forward price curve trend of silver prices. Due to a lack of trading, he is unable to get the forward prices for silver. However, he found out that there is an established lending market for silver and the silver lease rate is 7.9%. If the risk-free rate is 8.2%, then which of the following is true?

- A. The market for silver is said to be in backwardation

- B. The market for silver is said to be in contango
- C. The market for silver is said to be in upwardation
- D. It cannot be ascertained without forward prices

The correct answer is **B**.

The market for silver is said to be in contango. In the context of commodity markets, contango is a situation where the forward price of a commodity is higher than the spot price. This is typically indicative of an upward sloping forward curve. The relationship between the forward and spot prices of a commodity can be expressed using the formula: $F = Se^{(r - l)T}$, where F is the forward price, S is the spot price, r is the risk-free rate, l is the lease rate, and T is the time to maturity. In this scenario, given that the risk-free rate (8.2%) is higher than the lease rate (7.9%), the forward prices will be greater than the spot prices, suggesting that the market for silver is in contango. This is consistent with the general principle that in a contango market, investors are willing to pay more for a commodity in the future than the actual expected price of the commodity. This could be due to various factors such as storage costs, convenience yield, and expectations of future price movements.

Choice A is incorrect. Backwardation refers to a market condition where the futures prices are lower than the spot prices. In this case, without knowing the forward prices, we cannot determine if the market for silver is in backwardation.

Choice C is incorrect. Upwardation is not a recognized term in finance or commodity markets and thus does not describe any market condition for silver.

Choice D is incorrect. Even though we do not have direct information about forward prices, given that there's a well-established lending market for silver with a lease rate of 7.9% and risk-free rate of 8.2%, it can be inferred that the cost of carry (risk-free rate - lease rate) is positive which indicates that the market for silver could be in contango.

Things to Remember

- Contango: Situation where the forward price of a commodity is higher than the spot price, indicating an upward sloping forward curve.
- Backwardation: Market condition where futures prices are lower than spot prices.
- Cost of Carry: Calculated as the the risk-free rate minus the lease rate, used to determine market conditions like contango or backwardation.
- Storage Costs: Costs associated with holding a physical commodity until delivery.
- Convenience Yield: Benefit or advantage of holding a physical commodity rather than a futures contract.

- Expectations of Future Price Movements: Factors influencing investors' willingness to pay more for a commodity in the future.
-

Q.821 An analyst is analyzing the effect of storage costs and convenience yields on the forward prices of livestock. The storage cost of the livestock is 1.3%, and the convenience yield is 2.9%. Evaluate the final impact on the forward price of livestock if the risk-free rate is 1.4%.

- A. The forward price of livestock will be greater than the spot price of livestock.
- B. The forward price of livestock will be lower than the spot price of livestock.
- C. The forward price of livestock will be equal to the spot price of livestock.
- D. The forward price of livestock cannot be ascertained.

The correct answer is **B**.

The forward price of livestock will be lower than the spot price of livestock. This is because the convenience yield is greater than the storage cost and risk-free rate. This relation can be demonstrated with this equation.

$$F = S * e^{(\text{Risk-free rate} + \text{Storage cost} - \text{Convenience yield})}$$

Since the convenience yield is greater than the sum of storage cost and the risk-free rate, the net effect will be negative, and the forward price will be lower than the spot price.

Q.822 Garry Johnson has recently joined the derivatives unit of Brilliance Investment Bank as a research analyst. He has been assigned to focus his research on the energy sector. Johnson is analyzing spot prices and forward prices of Crude and Brent oil contracts in the futures markets, and he notices a trend in forward prices. The forward prices of oil contracts are in a downward sloping curve, which means the forward prices with larger maturities are lower than the forward prices of oil futures with shorter maturities. This trend in forward prices is referred to as:

- A. Diminishing curve
- B. Backwardation
- C. Upwardation
- D. Contango

The correct answer is **B**.

The term used to describe the observed trend in forward prices is 'backwardation'. Backwardation is a situation in the futures market where the forward prices of commodities with distant maturities are lower than the forward prices of the same commodity with shorter maturities. This phenomenon is typically observed when the market expects the spot price of the commodity to decrease over time. In such a scenario, market participants are willing to sell the commodity at a lower price in the future, resulting in a downward sloping forward price curve. This is a common occurrence in the energy sector, particularly in the oil markets, where supply and demand dynamics can significantly influence price trends. Backwardation can provide valuable insights into market expectations about future price movements and can be a useful tool for traders and investors in formulating their trading and investment strategies.

Choice A is incorrect. There is no term such as "Diminishing curve" in the context of futures markets. It seems to be a made-up term and does not describe the observed trend in forward prices.

Choice C is incorrect. The term "Upwardation" does not exist in financial terminology related to futures markets. This choice appears to be a play on words combining 'upward' and 'backwardation', but it's not a recognized concept or phenomenon.

Choice D is incorrect. Contango refers to an upward sloping forward curve, where future prices are higher than spot prices, which is opposite of what Johnson observed in his analysis.

Things to Remember

- Backwardation is often associated with expectations of a decrease in the spot price of the underlying commodity.
- Contango is the opposite of backwardation, where future prices are higher than spot prices.
- Market participants use backwardation and contango signals to make informed decisions about trading and investing in futures markets.
- Factors such as supply and demand dynamics, geopolitical events, and economic indicators can influence the shape of the forward price curve in futures markets.
- Understanding the term structure of futures prices is crucial for risk management and hedging strategies in the energy sector and other commodity markets.

Q.823 An investment manager is analyzing the forward curve of a specific commodity, which will help him identify if the forward price of the commodity will be higher or lower than the spot price. Suppose that the manager has figured that the lease rate of the specific commodity is 6.5% and the risk-free rate is 6%, then determine which of the following option is true.

- A. The market of the commodity is in contango.
- B. The market of the commodity is in backwardation.
- C. The market of the commodity is in upwardation.
- D. It could not be ascertained without established forward quotes.

The correct answer is **B**.

Backwardation is a term used in the commodity market to describe a situation where the forward prices of a commodity are lower than the spot prices. This is typically indicated by a downward sloping forward curve. The relationship between the forward and spot prices of a commodity can be expressed using the formula:

$$F = Se^{(\text{extRisk-free rate} - \text{extLease rate})T}$$

In this scenario, the risk-free rate (6%) is lower than the lease rate (6.5%). According to the formula, this would result in forward prices that are lower than the spot prices, indicating a market in backwardation. Therefore, the investment manager's analysis suggests that the commodity market is in a state of backwardation.

Choice A is incorrect. The market of the commodity is not in contango. Contango occurs when the forward price of a commodity is higher than the spot price, which typically happens when the lease rate is less than the risk-free rate. In this case, however, we know that the lease rate (6.5%) exceeds the risk-free rate (6%), suggesting that backwardation rather than contango should be expected.

Choice C is incorrect. There's no such term as "upwardation" in financial markets or commodity trading terminology, making this choice invalid.

Choice D is incorrect. It can be ascertained without established forward quotes because we have enough information to determine whether it's in contango or backwardation based on given lease and risk-free rates.

Q.3527 The spot price of oil is USD 95 per barrel, and the six-month futures price is USD 100 per barrel. The cost of storing oil for six months has a present value of USD 10 per barrel, and the risk-free rate is 5% per year.

Determine the convenience yield, Y.

- A. 7.6%
- B. 5.0%
- C. 4.5%
- D. 15.8%

The correct answer is **D**.

$$F = (S + U) \left[\frac{1 + R}{1 + Y} \right]^T$$

where:

F = futures price

S = spot price

U = present value of the storage cost

R = risk-free rate

Y = convenience yield

T = time to maturity of futures contract

We can rewrite the above equation So that:

$$\begin{aligned} Y &= \left[\frac{S + U}{F} \right]^{\frac{1}{T}} (1 + R) - 1 \\ &= \left[\frac{105}{100} \right]^{\frac{1}{2}} (1.05) - 1 \\ &= 0.157625 \end{aligned}$$

That is, the convenience yield is 15.8%

Reading 38: Options Markets

Q.724 Jessie Leeson has spent her last semester studying abroad at the University of Vienna. She took advanced finance and derivatives courses during her semester abroad where she studied pricing, valuing, hedging, and trading strategies involving derivatives. During a meeting with her colleagues, one of them asked her to outline the similarities and differences between options, futures, and forwards. She made the following statements:

- I. The cost of entering into an option or a futures contract is zero, but forward contracts have a cost.
- II. The holder of an option has a right but not an obligation to exercise the option, while the holder of a futures/forward contract has an obligation to honor the contract.

Which of the following is correct?

- A. Only statement I is correct.
- B. Only statement II is correct.
- C. Both statements are correct.
- D. None of the statements is correct.

The correct answer is **B**.

Only statement II is correct. An option contract gives the holder the right, but not the obligation, to buy or sell the underlying asset at a specified price within a certain period of time or on a specific date. This means that the holder can choose to exercise the option if it is beneficial to them, or let it expire worthless if it is not. On the other hand, futures and forward contracts obligate the holder to buy or sell the underlying asset at a specified price on a specific future date. This means that the holder is legally bound to fulfill the terms of the contract, regardless of whether it is beneficial to them or not. Therefore, Jessie's statement II accurately describes the fundamental difference between options and futures/forward contracts.

Choice A is incorrect. Statement I is not correct. While it's true that the cost of entering into a futures contract is typically zero, this isn't the case for options or forward contracts. Options require an upfront premium to be paid by the buyer to the seller, and while forward contracts don't usually require an upfront payment, they do have a cost in terms of opportunity cost or potential credit risk.

Choice C is incorrect. As explained above, statement I is not correct which makes this choice incorrect as well.

Choice D is incorrect. Statement II made by Jessie Leeson is accurate - holders of options have a right but not an obligation to exercise their option, whereas holders of futures/forward contracts are obligated to honor their contract. Therefore, this choice stating that none of the statements are correct would be inaccurate.

Things to Remember

- Options, futures, and forwards are all types of derivative contracts that derive their value from an underlying asset.
 - Options give the holder the right, but not the obligation, to buy or sell the underlying asset at a specified price within a certain period of time.
 - Futures and forward contracts obligate the holder to buy or sell the underlying asset at a specified price on a specific future date.
 - Options typically require an upfront premium payment, while futures contracts may not have an upfront cost but still involve potential costs.
 - Forward contracts may have a cost in terms of opportunity cost or potential credit risk.
-

Q.725 Adam Smith is a former computer engineer who has been actively trading stocks and derivatives after his early retirement from a 35-year engineering career. Smith holds 5,000 stocks of Banana Computers. He recently entered into a transaction where he has the right to buy 1,000 stocks from another investor if the value of the stock increases beyond a predefined price. Which of the following accurately defines the transaction?

- A. Smith is long put options.
- B. Smith is short on equity swaps.
- C. Smith is long on equity forwards.
- D. Smith is long call options.

The correct answer is **D**.

Adam Smith is long call options. A long call option position is a strategy that an investor uses when they believe the price of the underlying asset or security will rise. In this case, the underlying asset is the stock of Banana Computers. The investor, Adam Smith, has bought the right, but not the obligation, to buy a specified amount of these stocks (1,000 in this case) at a predetermined price from another investor. This right is valid until the expiration date of the option. If the price of the stocks increases beyond the predetermined price (also known as the strike price), Adam can exercise his option to buy the stocks at the lower strike price and then sell them at the current market price, making a profit. The profit would be the difference between the market price and the strike price, minus the premium paid for the option. If the price of the stocks does not increase beyond the strike price, Adam can choose not to exercise his option. In this case, his loss would be limited to the premium he paid for the option. This strategy is known as being 'long' on call options because the investor benefits from a long-term

rise in the price of the underlying asset.

Choice A is incorrect. A long put option gives the holder the right, but not the obligation, to sell a specified amount of an underlying security at a specified price within a specified time frame. This is different from Adam's transaction where he has the right to purchase additional stocks.

Choice B is incorrect. Being short on equity swaps means that one party agrees to pay another party the return on an equity (like dividends and capital gains or losses), in exchange for receiving a fixed or floating rate of interest over a certain period. This does not match with Adam's transaction which involves having the right to buy more stocks if their value surpasses a certain price.

Choice C is incorrect. Long on equity forwards implies that one has agreed to buy an asset at a future date for a price agreed upon today. However, in Adam's case, he has only obtained the right (not obligation) to purchase additional stocks if their value exceeds some predetermined level.

Things to Remember

- Call options: A call option gives the holder the right, but not the obligation, to buy an underlying asset at a specified price within a specified time frame.
 - Long position: Being "long" on an asset means that an investor benefits from an increase in the price of that asset.
 - Strike price: The predetermined price at which the option holder can buy (in the case of a call option) or sell (in the case of a put option) the underlying asset.
 - Premium: The price paid by the option holder to the option writer for the right to buy or sell the underlying asset.
 - Expiration date: The date at which the option contract expires and the holder must decide whether to exercise the option or not.
-

Q.727 Emmy Annie, a finance student at the University of Kennesaw, regularly invests her extra income in stocks and derivatives. She owns stocks of ABC Inc., a cleaning company, which are currently trading at \$45. She believes the stock will trade below \$45 if new regulations on cleaning companies are introduced next month. She is interested in entering an option position that gives her the right to sell her stocks at \$45. If the price of the stock goes below \$45, suggest the most appropriate option position for Annie.

- A. A long call option with the strike price of \$45.
- B. A short call option with the strike price of \$45.
- C. A Long put option with the strike price of \$45.
- D. A short put option with the strike price of \$45.

The correct answer is **C**.

A long put option with the strike price of \$45 is the most suitable option for Annie. A put option gives the holder the right, but not the obligation, to sell a specified amount of an underlying security at a specified price within a specified time. This is known as the strike price. In this case, the strike price is \$45. If Annie anticipates that the price of the ABC Inc. stock will fall below \$45 due to the introduction of new regulations, she can protect her investment by buying a put option. If the stock price does fall, she can exercise her option and sell her shares at the strike price of \$45, thereby avoiding a loss. This strategy is known as a protective put. It is used when an investor wants to protect an existing or planned purchase of a stock from falling below the current price. The cost of this protection is the price of the put option, also known as the premium. If the stock price does not fall as anticipated, Annie will only lose the premium paid for the put option, but her shares will still be worth \$45 each.

Choice A is incorrect. A long call option gives the holder the right to buy an asset at a specified price within a specific time period. In this case, Annie anticipates a fall in the stock price and hence, buying a call option would not be beneficial as it would only be profitable if the stock price increases.

Choice B is incorrect. A short call option involves selling the right to buy an asset at a certain price within a specific time period. This strategy could potentially expose Annie to unlimited losses if the stock price were to increase significantly, which is not her expectation.

Choice D is incorrect. A short put option involves selling the right to sell an asset at a certain price within a specific time period. If Annie sells (writes) put options on ABC Inc., she will have an obligation to buy shares of ABC Inc., if they fall below \$45 per share, which contradicts her anticipation of falling prices.

Things to Remember

- Options are financial derivatives that give the holder the right, but not the obligation, to buy or sell an underlying asset at a specified price within a specified time period.
- A put option gives the holder the right to sell the underlying asset at a specified price, while a call option gives the holder the right to buy the underlying asset at a specified price.
- The strike price of an option is the price at which the underlying asset can be bought or sold when the option is exercised.
- A long position in an option involves buying the option, while a short position involves selling the option.
- A protective put strategy involves buying a put option to protect against a potential

decrease in the price of the underlying asset.

- The premium is the price paid for an option, which represents the cost of buying or selling the option.
-

Q.728 Franky M. purchased an American put option from Lee V. on the stocks of Fast Cars Co. to sell 2,000 shares of stock at a price of \$3.30 per share. The put option has a strike price of \$31.70. If the stock price at the expiration of the option is \$30, then which of the following statements is true?

- A. Franky lost \$6,600 on the short position while Lee gained \$6,600 on the long position.
- B. Franky lost \$6,600 on the long position while Lee gained \$6,600 on the short position.
- C. Franky lost \$3,200 on the long position while Lee gained \$3,200 on the short position.
- D. Franky lost \$3,200 on the short position while Lee gained \$3,200 on the long position.

The correct answer is **C**.

Franky M., as the buyer of the put option, holds the long position. This means he has the right, but not the obligation, to sell the underlying asset (in this case, the stocks of Fast Cars Co.) at the strike price. The strike price is \$31.70, and the stock price at the expiration of the option is \$30. This means that Franky can sell the stocks for \$1.70 more per share than their current market price. Therefore, his gain from exercising the option is $(\$31.70 - \$30) \times 2,000 = \$3,400$. However, Franky had to pay a premium to buy the option, which cost him \$3.30 per share, or $\$3.30 \times 2,000 = \$6,600$ in total. Therefore, his net loss from this transaction is $\$6,600 - \$3,400 = \$3,200$. On the other hand, Lee V., as the seller of the put option, holds the short position. He is obligated to buy the stocks if Franky decides to exercise the option. Lee's gain from selling the option is the premium he received, which is \$6,600. However, he loses on the underlying asset because he has to buy the stocks for more than their current market price. His loss from the underlying asset is the same as Franky's gain, which is \$3,400. Therefore, his net gain from this transaction is $\$6,600 - \$3,400 = \$3,200$.

Choice A is incorrect. Franky did not lose \$6,600 on the short position. In fact, Franky was in a long position as he had the right to sell the shares at a predetermined price. The amount lost also does not match with the given scenario.

Choice B is incorrect. While it's true that Franky was in a long position and Lee was in a short position, they did not gain or lose \$6,600 respectively. The difference between the strike price and stock price multiplied by number of shares gives us \$3,400 $(\$1.70 \times 2000)$, which is what Franky gained and Lee lost.

Choice D is incorrect. This choice incorrectly states that Franky was in a short position when he actually held an American put option which puts him in a long position with rights to sell at strike price.

Things to Remember

- Long position: In the context of options, a long position refers to the holder of the option who has the right (but not the obligation) to buy or sell the underlying asset at a specified price within a specified period.
 - Short position: In the context of options, a short position refers to the writer (seller) of the option who is obligated to buy or sell the underlying asset if the option holder decides to exercise their right.
 - American put option: An American put option gives the holder the right to sell the underlying asset at the strike price before or at the expiration date.
 - Strike price: The strike price is the predetermined price at which the holder of an option can buy or sell the underlying asset.
 - Premium: The price paid by the option buyer to the option seller for the right to buy or sell the underlying asset at a specified price.
-

Q.730 Nina Singh has recently started investing in options after watching some options investing tutorial videos, so her knowledge of options is limited. She has analyzed that she can take one of the four positions in the options market. She can take either a long/buyer position or a short/seller position in a call option. Similarly, she can also take either a long or a short position in put options. Which of the abovementioned positions has unlimited potentials for loss?

- A. A long position in a call option.
- B. A short position in a call option.
- C. A long position in a put option.
- D. A short position in a put option.

The correct answer is **B**.

A short position in a call option has unlimited loss potential. This is because the seller of a call option, or the person in a short position, is obligated to sell the underlying asset at the strike price if the buyer decides to exercise the option. If the price of the underlying asset increases significantly, the seller will have to purchase the asset at the higher market price to fulfill their obligation. Since there is no upper limit to how high the price of an asset can rise, the potential

loss for the seller of a call option is theoretically unlimited. This is a significant risk that sellers of call options must be aware of when entering into such contracts. It is also why many brokers require traders to have a certain level of experience and financial resources before they can sell call options.

Choice A is incorrect. A long position in a call option does not carry the potential for unlimited losses. The maximum loss that can be incurred by an investor who buys a call option (long position) is the premium paid to purchase the option, which occurs if the option expires worthless.

Choice C is incorrect. A long position in a put option also does not carry the potential for unlimited losses. Similar to a long call, an investor who buys a put option (long position) can only lose up to the premium paid for purchasing the options contract.

Choice D is incorrect. A short position in a put option carries substantial risk but it's not unlimited. The maximum loss occurs when the price of underlying asset falls to zero and it equals to strike price minus premium received.

Things to Remember

- Options are financial derivatives that give the holder the right, but not the obligation, to buy or sell an underlying asset at a specified price before or on a specified date.
- Call options give the holder the right to buy the underlying asset, while put options give the holder the right to sell the underlying asset.
- Long positions in options involve buying the option, while short positions involve selling the option.
- Options trading involves understanding and managing risks, including the potential for unlimited losses in certain positions.
- Option premiums are the prices paid for options contracts, and they represent the maximum potential loss for buyers of options.

Q.731 Franklin Cole, an investment manager at Small Lounge Investments Co., has conducted a fundamental research on the shares of Red Hat Corp and instructed his assistant to sell put options on the shares with a strike of \$30. The assistant receives a premium of \$0.50. If the price of the stock increases from \$30 to \$33, determine the position's payoff and profit.

A. payoff = \$0; profit = \$0.50

B. payoff = \$0.5; profit = \$0.50

C. payoff = \$0; profit = \$0

D. payoff = \$3; profit = \$0.50

The correct answer is **A**.

$$\begin{aligned}\text{Payoff to the long position} &= P_T = \max(0, X - S_T) \\ &= \max(0, \$30 - \$33) = 0\end{aligned}$$

$$\begin{aligned}\text{Payoff the short position, payoff} &= -P_T = -0 = 0 \\ \text{Profit to the short position} &= P_0 - P_T\end{aligned}$$

where P_0 is the option premium

$$\text{Profit} = \$0.50 - 0 = \$0.50$$

Detailed Response

In option contracts, there are always two parties:

(I) the buyer, who takes the long position, and

(II) the seller (writer) who takes the short position

It follows that the seller (short position) of a put option is the trader that promises to buy the underlying stocks at the expiry of the contract. The buyer (long position) is the party that has a right but not the obligation to sell the stocks at expiry. The buyer is also called the holder.

Holder.

At expiration, the holder will only benefit if the prevailing market price is less than the exercise/strike price. The payoff is equal to $P_t = X - S_T$, i.e., strike price minus the market price. If the stock stays at X or above, the payoff will be zero.

The holder's profit is always given as the option payoff minus the premium paid at time 0.

Seller.

At expiration, the seller will only “benefit” if the prevailing market price is greater than the exercise/strike price. The payoff is equal to the negative value of the holder's payoff.

Also, the seller's profit will be the negative value of the holder's profit.

In short, options are a zero-sum game. What's lost by one party is gained by the other.

Franklin and his assistant sell the option; they hold a short position.

So they receive a premium of 0.50, and to their luck, the stock price actually rises, meaning that the buyer cannot exercise their right to sell.

Thus, the payoff to the buyer is zero, and the payoff to Franklin is “-0”

The holder's profit: option payoff less the premium paid, i.e., $0 - 0.50 = -0.50$.

The seller's profit is $-(-0.50) = 0.5$

Again, the zero-sum game is at play here.

Q.732 Management at Digi Computational Investments has analyzed that the finance and banking sector of the U.S. is currently in turmoil. The sector has not properly recovered from the last financial crisis, and the new variables underlying the financial sector have already started tumbling. Taking this into consideration, Digi Computational Investments took a long position in a European call option on the Nasdaq-100 Index (NDX) which is composed of 108 non-financial companies at a price of \$20 per index option. The strike price of the index option is 3,355, and the option expires in March 2020. If the current index price is 3,457, then estimate the total gain or loss for the buyer of the call option.

- A. \$ 102
- B. \$ 98
- C. \$ 78
- D. \$ 82

The correct answer is **D**.

Since the current index price is higher than the strike price, the buyer of the call option will exercise the index option. If the option is exercised, the gain to the buyer of the index option is:

Net gain on the call option = Gain on the option - Premium on the option

Net gain on the call option = $3,457 - 3,355 - 20 = \$ 82$

Things to Remember

- **European Call Option:** A type of financial derivative that gives the holder the right, but not the obligation, to buy the underlying asset at a specified price on the option's expiration date.
 - **Strike Price:** The price at which the option holder can buy or sell the underlying asset when the option is exercised.
 - **Expiration Date:** The date on which the option contract expires and becomes void.
 - **Long Position:** A trading position where an investor buys an asset with the expectation that its value will increase.
 - **Nasdaq-100 Index (NDX):** An index that includes 100 of the largest non-financial companies listed on the Nasdaq stock exchange.
-

Q.733 A treasury manager at a large manufacturing firm believes that the price of the shares of Bright Star Hospitals Group (HBHG) will increase by at least 30% in value in the coming 2 to 3 years. The manager, therefore, is interested in taking a long position in an option that allows him to purchase the stock anytime it increases in value above some determined strike price, and the option should have an expiry of at least 38 months. Which of the following options is suitable for the manager?

- A. Equity call option
- B. Index call option
- C. Long-term equity anticipation securities
- D. Covered call option

The correct answer is C.

Long-term equity anticipation securities (LEAPS) are the most suitable option for the manager. LEAPS are a type of options contract with an expiration date that can extend up to 39 months. They allow investors to speculate on the long-term performance of a company's stock, which aligns with the manager's investment strategy. The manager believes that the price of HBHG's shares will increase significantly in the next 2 to 3 years, and LEAPS would provide him with the opportunity to benefit from this potential price increase. Furthermore, LEAPS would allow the manager to purchase the stock anytime its value exceeds a predetermined strike price, which is exactly what the manager is looking for. Therefore, LEAPS are the most appropriate option for the manager given his investment objectives and market outlook.

Choice A is incorrect. An equity call option would allow the manager to buy the stock at a predetermined price, but these options typically have shorter expiry periods (usually less than a year). Therefore, an equity call option would not meet the manager's preference for a minimum expiry period of 38 months.

Choice B is incorrect. An index call option gives the holder the right to buy an index at a specified price before expiration. However, this type of option wouldn't allow the manager to capitalize on his specific prediction about Bright Star Hospitals Group's stock performance as it is based on an entire index rather than individual stocks.

Choice D is incorrect. A covered call strategy involves owning or buying shares of stock and selling an equivalent number of call options on that same stock. This strategy does not align with the treasury manager's objective because it limits potential upside if HBHG's share price increases significantly, which contradicts his optimistic outlook for HBHG.

Things to Remember

- Options are financial derivatives that give the holder the right, but not the obligation, to buy or sell an underlying asset at a specified price on or before a specified date.
- Call options give the holder the right to buy the underlying asset, while put options

give the holder the right to sell the underlying asset.

- Expiration date is the date on which the option contract expires and becomes void.
- Strike price is the price at which the option holder can buy or sell the underlying asset.
- LEAPS (Long-term Equity Anticipation Securities) are long-term options contracts with expiration dates that can extend up to 39 months.
- Covered call option strategy involves owning the underlying asset (e.g., stock) and selling call options on that asset.

Q.734 The following are the strike prices, current prices, and expiration dates for options on equities of various companies.

	Call option on Sun Inc.	Call option on Moon Corp.	Put option on Pluto Co.
Strike price	\$109	\$113	\$87
Current price	\$111	\$109	\$89

If you have long positions in all of these options, then which of the following options is true?

- A. The call on Sun Inc. is in the money, the call on Moon Corp. is in the money, and the put on Pluto Co. is out of the money.
- B. The call on Sun Inc. is in the money, the call on Moon Corp. is out of the money, and the put on Pluto Co. is in the money.
- C. The call on Sun Inc. is in the money, the call on Moon Corp. is out of the money, and the put on Pluto Co. is out of the money.
- D. The call on Sun Inc. is out of the money, the call on Moon Corp. is in the money, and the put on Pluto Co. is out of the money.

The correct answer is **C**.

The call option on Sun Inc. is in the money, the call on Moon Corp. is out of the money, and the put on Pluto Co. is out of the money. This is because a call option is considered 'in the money' when the current price of the underlying asset is higher than the strike price of the option. In this case, the current price of Sun Inc. (\$111) is higher than the strike price (\$109), hence the call option is 'in the money'. On the other hand, a call option is considered 'out of the money' when the current price of the underlying asset is lower than the strike price of the option. In this

case, the current price of Moon Corp. (\$109) is lower than the strike price (\$113), hence the call option is 'out of the money'. Similarly, a put option is considered 'out of the money' when the current price of the underlying asset is higher than the strike price of the option. In this case, the current price of Pluto Co. (\$89) is higher than the strike price (\$87), hence the put option is 'out of the money'.

Choice A is incorrect. The call option on Moon Corp. is not in the money. A call option is considered to be in the money when the current price of the underlying asset is higher than the strike price of the option, which does not hold true for Moon Corp.

Choice B is incorrect. While it correctly states that the call on Sun Inc. is in-the-money and that Moon Corp.'s call option is out-of-the-money, it incorrectly states that Pluto Co.'s put option is in-the-money. A put option would be considered as being 'in-the-money' if its strike price was higher than current market price, which isn't true for Pluto Co.

Choice D is incorrect. This choice incorrectly states that Sun Inc.'s call option and Pluto Co.'s put options are out-of-the-money while stating correctly about Moon Corp.'s position being 'in-the-money'. For a call to be 'out-of-the-money', its strike price should be more than current market value which isn't true for Sun Inc., and similarly for a put to be 'out-of-the-money', its strike price should be less than current market value which again doesn't hold true for Pluto Co.

Q.735 Xiamen Lee has a long position in an American call option on oil futures contract with a strike price of \$40 per contract expiring in September. The current price of the oil futures contract has increased to \$46, but the investor believes that the price of the contract can further increase. Since it is an American option, the investor can exercise the contract anytime until its expiration in September. Which of the following is the last day on which Xiamen can trade his call option?

- A. The third Friday of August is the last trading day of the call option.
- B. The third Monday of September is the last trading day of the call option.
- C. The third Monday of August is the last trading day of the call option.
- D. The third Friday of September is the last trading day of the call option.

The correct answer is **D**.

The last trading day for American options, such as the one held by Xiamen Lee, is typically the third Friday of the month in which the contract is set to expire. This is a standard practice in the United States and most other exchanges around the world. Therefore, since Xiamen's contract is set to expire in September, the last day he can trade his call option is the third Friday of September. It's important to note that while the last trading day is the third Friday of the month, the actual expiration date of the contract is the Saturday following this Friday. However, since trading does not occur on Saturdays, the practical deadline for trading the option is the preceding Friday.

Choice A is incorrect. The third Friday of August cannot be the last trading day for the call option as American options can be exercised at any point until their expiration, which in this case is September. Therefore, it would not make sense for the last trading day to be in August.

Choice B is incorrect. The third Monday of September cannot be the last trading day of the call option because typically, options expire on the third Friday of their expiration month, not Monday.

Choice C is incorrect. Similar to choice A, it would not make sense for the last trading day to be in August when Xiamen's contract expires in September.

Things to Remember

- American options can be exercised at any point until their expiration date.
 - The last trading day for American options is typically the third Friday of the month in which the contract expires.
 - The expiration date of the contract is the Saturday following the last trading day, but trading does not occur on Saturdays.
 - Options typically expire on the third Friday of their expiration month.
-

Q.738 An investor is considering an option on the stock of a specific company, which has the strike price of \$29 per share and the option expiry date of March. The option is constructed in a way that if the final per share price of the stock reaches \$71 at the expiration, the option will give a payoff of \$100 to the buyer.

Which of the following best describes this type of option?

- A. European call option
- B. American call option
- C. Binary call option
- D. LEAPS call option

The correct answer is **C**.

A binary call option is the correct answer. This type of option is structured in a way that the payoff is a fixed amount of money if either (I) a specified price is reached/exceeded at the time of expiration or (II) the option is simply in-the-money at the time of expiration. In this case, the investor will receive a payoff of \$100 if the stock price reaches \$71 at the time of expiration, which is a characteristic of a binary call option. Binary options are also known as all-or-nothing

options because they either pay the full amount or nothing at all. This type of option is often used when an investor believes that the price of an underlying asset will reach a certain level in the future, but is unsure about the sustainability of the price increase.

Choice A is incorrect. A European call option allows the holder to buy the underlying asset at a specified price within a specific time period, but only on the expiration date. However, in this scenario, the payout is not dependent on the difference between the stock price and strike price at expiration but rather it's fixed at \$100 if stock price hits \$71 which aligns with characteristics of a binary call option.

Choice B is incorrect. An American call option also allows its holder to buy an underlying asset at a specified price before or on its expiration date. But similar to European options, their payoff depends on difference between stock and strike prices which doesn't match with given scenario.

Choice D is incorrect. LEAPS (Long-term Equity Anticipation Securities) are long-dated options that have an expiry period longer than one year. The type of payoff structure described in this question does not relate to whether an option has long or short term maturity but rather it's about how payoff is calculated at expiry which makes LEAPS irrelevant here.

Things to Remember

- European call options allow the holder to buy the underlying asset at a specified price within a specific time period, but only on the expiration date.
 - American call options allow the holder to buy the underlying asset at a specified price before or on its expiration date.
 - Binary options are also known as all-or-nothing options because they either pay the full amount or nothing at all based on specific conditions being met.
 - LEAPS (Long-term Equity Anticipation Securities) are long-dated options that have an expiry period longer than one year.
-

Q.741 A portfolio manager at Sea Breeze bank owns 10,000 shares of PNG Corp. The manager does not expect an appreciable price increase in the next six months. To diminish his volatility, the manager has written an option over the shares of PNG Corp. that expires in 6 months. Which of the following defines the manager's position in the option?

- A. Covered put option
- B. Covered call option
- C. Protective put option
- D. Protective call option

The correct answer is **B**.

The portfolio manager's position is a covered call option. A covered call option is a financial strategy in which an investor sells, or 'writes', call options on an asset they already own or hold in their portfolio. This strategy is used when the investor anticipates that the price of the underlying asset will remain relatively stable over the life of the options contract. The investor earns income from the premium received from selling the call option, which can help to offset potential losses if the price of the underlying asset falls. However, the investor also limits their potential gains if the price of the underlying asset rises, as they are obligated to sell the asset at the strike price of the call option. In this case, the portfolio manager owns the underlying shares of PNG Corp and has written a call option on these shares, indicating a covered call position.

Choice A is incorrect. A covered put option involves selling a put option on an asset while also shorting the same asset. In this case, the manager owns shares of PNG Corp and has written an option on them, which does not align with the definition of a covered put.

Choice C is incorrect. A protective put strategy involves buying a put option to hedge against potential losses in an owned stock. The manager in this scenario has written (sold) an option, not bought one, so it cannot be a protective put position.

Choice D is incorrect. A protective call strategy would involve buying a call option to protect against potential increases in the price of a shorted stock. However, the manager owns shares and has written an option on them; he hasn't bought any options or shorted any stocks.

Things to Remember

- A covered call option involves selling call options on an asset that the investor already owns in their portfolio.
- A covered put option involves selling put options on an asset while also shorting the same asset.
- A protective put option involves buying a put option to hedge against potential losses in an owned stock.
- A protective call option involves buying a call option to protect against potential increases in the price of a shorted stock.
- Options can be used to manage risk, generate income, or speculate on price movements.

Q.742 Which of the following is responsible for ensuring that the writer of the option honors the option or fulfills the obligations determined under the terms of the option?

- A. Exchange
- B. Market maker
- C. Options clearing corporation
- D. Buyer of the option

The correct answer is **C**.

The Options Clearing Corporation (OCC) is the entity that ensures the writer of an option fulfills the obligations determined under the terms of the option. The OCC acts as a guarantor, ensuring that the obligations of the contracts they clear are fulfilled. It operates as a clearinghouse for futures markets, providing a system of guarantees that reduces the risk of default. The OCC works with its members, who are required to maintain a specific amount of capital and a special fund that can be used in case of default. This structure ensures the integrity of the options market by guaranteeing the performance of every option contract.

Choice A is incorrect. While the exchange provides a platform for trading options, it does not ensure that the obligations under the terms of an option are fulfilled by the writer of the option. This responsibility lies with another entity.

Choice B is incorrect. The market maker's role is to provide liquidity in the market by buying and selling securities, including options, but they do not guarantee that obligations under an option contract will be fulfilled by its writer.

Choice D is incorrect. The buyer of an option has no obligation to ensure that terms of an option are fulfilled by its writer. Their only obligation is to pay for and accept delivery if they choose to exercise their right under the contract.

Things to Remember

- The Options Clearing Corporation (OCC) functions as a central clearinghouse for option transactions, guaranteeing that the obligations of options contracts are met if a party defaults.
- OCC's role includes managing risk through a rigorous system of margins and guarantees, ensuring market stability and integrity in options trading.
- The clearing process by OCC is critical in maintaining trust and efficiency in the options market, as it removes the counterparty risk that one party will fail to perform as obligated.

Q.743 Harry Wilson owns a call option on the shares of Blue Company and a put option on the shares of Green Company. Both companies belong to the booming communication sector of Holland. The sector has been growing at a rate of 7% for the last 5 years. Blue Company recently announced a cash dividend of \$1.20 on its stock while Green Company announced a 4-for-3 stock split. Which of the following impacts on the options is accurate?

- A. Only the call option on Blue Company will be adjusted for the cash dividend.
- B. Only the put option on Green Company will be adjusted for the stock split.
- C. Options on both of these companies will be adjusted for the cash dividend and the stock split.
- D. None of the companies' options will be adjusted.

The correct answer is **B**.

Only the put option on Green Company will be adjusted for the stock split. This is because options are typically adjusted for events that change the number of shares in circulation, such as stock splits or stock dividends. In the case of Green Company, the 4-for-3 stock split will increase the number of shares, which will necessitate an adjustment of the put option held by Harry Wilson. This adjustment is necessary to maintain the economic equivalence of the option contract before and after the stock split. It ensures that the holder of the option is not disadvantaged by the increase in the number of shares. The adjustment usually involves changing the strike price and the number of shares that each contract represents.

Choice A is incorrect. While it's true that dividends can impact the price of an option, standard exchange-traded options are not adjusted for regular cash dividends. This is because the declaration of a dividend is usually factored into the option's price prior to its announcement.

Choice C is incorrect. As explained above, standard exchange-traded options are not adjusted for regular cash dividends. Therefore, the call option on Blue Company will not be adjusted for the cash dividend. Similarly, while stock splits do result in adjustments to options contracts, they only affect the specific company's options where split has happened and not all companies' options in general.

Choice D is incorrect. This statement contradicts with how stock splits affect options contracts. In case of a 4-for-3 stock split announced by Green Company, there will indeed be an adjustment to Harry Wilson's put option on Green Company shares.

Things to Remember

- Stock splits, like the 4-for-3 split announced by Green Company, require an adjustment to option contracts to reflect the increased number of shares while maintaining the economic equivalence of the option terms.
- Adjustments to options due to stock splits commonly involve modifying the strike price

and the quantity of shares each option contract involves, ensuring that the value of the option remains proportional post-split.

- Regular cash dividends, such as the one announced by Blue Company, do not typically lead to adjustments in standard exchange-traded options, as their impact is often anticipated and priced into the options before the dividend's announcement.
-

Q.3561 Jason Briggs purchased a 3-month call option by paying \$0.08. The exercise price of the option is \$1.32 while the underlying is priced at \$1.35.

Is the option currently in-the-money and at what price will break-even occur?

- A. In-the-money: No; Break-even price: \$1.27
- B. In-the-money: Yes; Break-even price: \$1.40
- C. In-the-money: Yes; Break-even price: \$1.35
- D. In-the-money: No; Break-even price: \$1.40

The correct answer is **B**.

The call option is in-the-money as the underlying price is greater than the exercise price (\$1.35 vs. \$1.32, respectively).

Break-even price = $X + C_0 = \$1.32 + \$0.08 = \$1.40$

Things to Remember

- Options are considered in-the-money when the underlying asset's price is higher than the option's strike price for call options.
- Break-even price for a call option is the sum of the exercise price and the premium paid for the option.
- Break-even analysis is important for options traders to understand at what price the option position will start making a profit.
- Options provide the right, but not the obligation, to buy (in the case of a call option) or sell (in the case of a put option) an underlying asset at a specified price within a specified time frame.

Q.3563 When trading options, understanding the specific obligations associated with different types of options is key. Which of the following best describes the obligation of the writer of a put option?

- A. The obligation to buy the underlying security at the option's strike price if the option is exercised
- B. The obligation to sell the underlying security at the option's strike price if the option is exercised
- C. The right, but not the obligation, to buy the underlying security at the option's strike price if the option is exercised
- D. The right, but not the obligation, to sell the underlying security at the option's strike price if the option is exercised

The correct answer is **A**.

The writer of a put option has the obligation to buy the underlying security at the option's strike price if the option is exercised. This is because when an investor writes a put option, they are essentially selling someone else the right to sell a security at a predetermined price (the strike price). If the buyer of the put option decides to exercise their option, the writer is obligated to buy the security at the agreed-upon strike price, regardless of the current market price. This could potentially lead to a loss for the option writer if the market price of the security is lower than the strike price at the time of exercise. However, the writer does receive a premium for writing the option, which can offset some or all of this potential loss.

Choice B is incorrect. The obligation to sell the underlying security at the option's strike price if the option is exercised is not correct. This describes a call option, not a put option. In a put option, the writer has an obligation to buy, not sell.

Choice C is incorrect. The right, but not the obligation, to buy the underlying security at the option's strike price if the option is exercised does not describe an obligation of any kind and hence it cannot be correct. This choice describes an aspect of options for buyers rather than sellers or writers.

Choice D is incorrect. The right, but not the obligation, to sell the underlying security at the option's strike price if the option is exercised also does not describe an obligation and hence it cannot be correct either. Similar to Choice C this choice also describes aspects of options for buyers rather than sellers or writers.

Things to Remember

- Options are financial derivatives that give the holder the right, but not the obligation, to buy or sell an underlying asset at a specified price within a specific time period.

- A put option gives the holder the right to sell the underlying asset at a specified price within a specific time period.
 - The writer of an option is the one who sells the option and takes on the obligation associated with it.
 - Option writers receive a premium for taking on the obligation, which can help offset potential losses.
 - Understanding the rights and obligations of options is crucial for effective risk management and trading strategies.
-

Reading 39: Properties of Options

Q.744 Which of the variables given below is likely to have the smallest impact on the prices of plain vanilla stock options?

- A. The strike price of the option
- B. Expected dividend
- C. Risk-free rate
- D. The creditworthiness of the counterparty

The correct answer is **D**.

The creditworthiness of the counterparty is the least likely to affect the prices of plain vanilla stock options. This is because these options are typically standardized and traded on exchanges, where the Options Clearing Corporation (OCC) guarantees that the writer of the option will not default on their obligations. Therefore, the creditworthiness of the counterparty becomes less relevant in this context, as the risk of default is mitigated by the OCC's guarantee.

It's important to note that while the creditworthiness of the counterparty can be a significant factor in over-the-counter (OTC) options, where the parties deal directly with each other without an intermediary, it's not a major factor in exchange-traded options. This is one of the key differences between OTC options and exchange-traded options, and it's crucial for financial risk managers to understand this distinction when assessing the risks associated with different types of options.

Choice A is incorrect. The strike price of the option has a significant impact on the price of plain vanilla stock options. The higher the strike price, the lower the value of a call option (and vice versa for put options). This is because the strike price represents the price at which an investor can buy (call) or sell (put) an asset in future.

Choice B is incorrect. Expected dividends also have a substantial influence on option prices. If dividends are expected to be high, this will reduce the expected future stock price and hence reduce call prices and increase put prices.

Choice C is incorrect. The risk-free rate also affects option pricing significantly as it represents time value of money in financial markets. Higher risk-free rates increase call prices and decrease put prices as they increase cost of carry.

Things to Remember

- Plain vanilla options are standardized exchange-traded options with no special or unusual features.
- The Options Clearing Corporation (OCC) acts as a guarantor for exchange-traded options, reducing counterparty credit risk.

- Over-the-counter (OTC) options are customized contracts traded directly between two parties, where counterparty credit risk is a significant factor.
 - The strike price of an option is the price at which the underlying asset can be bought or sold, affecting the option's value.
 - Expected dividends impact option prices by affecting the expected future stock price.
 - The risk-free rate influences option pricing by factoring in the time value of money in financial markets.
-

Q.745 Raj Kumar is an individual options investor. He recently started investing in equity options because of his informal experience in investing in equities. He opened two options positions on the stock of Red Horse Auto Inc. Kumar purchased a call option on the company's stock with a specific strike price and later also purchased a put option on the same stock with the same strike price. If the current stock price decreases, then which of the following shows the accurate effect on option prices?

- A. The price of the call option will increase while the price of the put option will decrease.
- B. The price of the call option will decrease while the price of the put option will increase.
- C. The price of the put option will increase while there will be no impact on the price of the call option.
- D. The price of the call option will increase while there will be no impact on the price of the put option.

The correct answer is **B**.

The price of a call option is likely to decrease while the price of a put option is likely to increase when the current stock price decreases. This is because the value of a call option is derived from the potential for the stock price to increase above the strike price. If the stock price decreases, the likelihood of the stock price rising above the strike price also decreases, thereby reducing the value of the call option. On the other hand, the value of a put option increases when the stock price decreases. This is because a put option gives the holder the right to sell the stock at the strike price. If the stock price decreases, the holder of the put option can still sell the stock at the higher strike price, making the put option more valuable.

Choice A is incorrect. The price of a call option will not increase when the underlying stock price decreases. This is because the value of a call option increases as the underlying stock price

increases, giving the holder the right to buy at a lower strike price and sell at a higher market price. Therefore, if the stock price decreases, it reduces this potential benefit and thus reduces the value of the call option.

Choice C is incorrect. While it's true that put options increase in value when stock prices decrease (as they give holders an opportunity to sell at a higher strike price), there would indeed be an impact on Raj's call option as well. As explained above, if Red Horse Auto Inc.'s current stock prices decrease, this would reduce Raj's potential benefit from his call option and thus its value would also decrease.

Choice D is incorrect. Similar to Choice A, this choice incorrectly suggests that a decrease in Red Horse Auto Inc.'s current stock prices would lead to an increase in Raj's call option value - which contradicts basic options pricing theory. Furthermore, while it correctly states that there will be no impact on his put option (which actually increases in such scenarios), it fails to capture how both types of options are affected by changes in underlying asset prices.

Things to Remember

- **Call Option:** A call option gives the holder the right, but not the obligation, to buy the underlying asset at a specified price within a specific time period.
 - **Put Option:** A put option gives the holder the right, but not the obligation, to sell the underlying asset at a specified price within a specific time period.
 - **Strike Price:** The price at which the option holder can buy (for a call option) or sell (for a put option) the underlying asset.
 - **Current Stock Price:** The market price of the underlying stock at the current moment.
 - **Options Pricing:** Options prices are influenced by factors such as the underlying asset price, volatility, time to expiration, and interest rates.
-

Q.746 The prices of options on the stocks or equities are affected by a number of variables. Some variables directly impact the options while others have an impact on the underlying stock, which in turn also affects the prices of stock options. If the strike price of an option is increased, then which of the following depicts the accurate impact of this increase?

- A. The increase in strike price will decrease the price of European put and American put options but increase the price of European call and American call options.
- B. The increase in strike price will increase the price of European put and American put options but decrease the price of European call and American call options.
- C. The increase in strike price will increase the price of European put and American call

options but decrease the price of European call and American put options.

D. The increase in strike price will decrease the price of European put and American call options but increase the price of European call and American put options.

The correct answer is **B**.

The increase in the strike price of an option will have a similar impact on American and European call options, and an opposite impact on American and European put options. An increase in strike price will decrease the likelihood of call options to be in the money, and thus, decrease the price of American and European call options. Conversely, an increase in strike price will increase the likelihood of put options to be in the money, and thus, it will increase the price of American and European put options. This is because the strike price is the price at which the holder of an option can buy (in case of a call option) or sell (in case of a put option) the underlying security when the option is exercised. For a call option, a higher strike price means that the option holder has to pay more to buy the underlying security, which makes the option less attractive and decreases its price. For a put option, a higher strike price means that the option holder can sell the underlying security at a higher price, which makes the option more attractive and increases its price.

Choice A is incorrect. An increase in the strike price does not decrease the price of European and American put options, but rather increases them. This is because a higher strike price makes these options more valuable as they provide the right to sell at a higher price. Similarly, it does not increase the prices of European and American call options; instead, it decreases them since a higher strike price makes these options less attractive as they provide the right to buy at a higher cost.

Choice C is incorrect. While an increase in strike price indeed increases the value of American call options due to their early exercise feature, it does not decrease the value of European call and American put options. Instead, it decreases the value of European call option (as explained above) and increases that of an American put option (due to its early exercise feature).

Choice D is incorrect. An increase in strike price does not decrease both European put and American call option prices nor does it increase both European call and American put option prices. As explained earlier, an increased strike price will lead to increased values for both types of put options while decreasing values for both types of call options.

Things to Remember

- Strike price is the price at which the option holder can buy (in case of a call option) or sell (in case of a put option) the underlying security when the option is exercised.
- An increase in strike price decreases the likelihood of call options to be in the money and increases the likelihood of put options to be in the money.
- American options can be exercised at any time before expiration, while European options can only be exercised at expiration.

- Options pricing is influenced by factors such as underlying asset price, volatility, time to expiration, interest rates, and dividends.
 - Put options give the holder the right to sell the underlying asset at the strike price, while call options give the holder the right to buy the underlying asset at the strike price.
-

Q.747 Jack Anderson, a portfolio manager at Vito Investment Company, manages an \$800 million mutual fund that invest in a large variety of financial instruments. A significant portion of the portfolio is invested in call and put options on the S&P 500 index (SPX), NASDAQ-100 Index (NDX), and Dow Jones Industrial Average (DJX). However, due to upcoming elections in the U.S., the volatility of these indices has increased. Which of the following best describes the impact on index options?

- A. The increase in volatility will increase the prices of call index options but decrease the prices of put index options.
- B. The increase in volatility will decrease the prices of call index options but increase the prices of put index options.
- C. The increase in volatility will decrease the prices of both call and put index options.
- D. The increase in volatility will increase the prices of both call and put index options.

The correct answer is **D**.

The increase in volatility will increase the prices of both call and put index options. The reason for this is that volatility is a key factor in the pricing of options. When volatility increases, the potential range of movement for the underlying asset (in this case, the indices) also increases. This means that there is a greater chance that the option will end up in the money, which makes the option more valuable. Therefore, both call and put options will increase in price. The downside risk for both calls and puts is limited to the premium paid, but the upside potential is unlimited. This makes the upside risk more rewarding while limiting the downside risk. Therefore, when volatility increases, the prices of both call and put options tend to increase.

Choice A is incorrect. The increase in volatility does not have a differential impact on call and put options. Both types of options become more valuable as the potential range of underlying index values increases, which is reflected in higher option prices.

Choice B is incorrect. This choice incorrectly suggests that increased volatility will decrease the prices of call index options while increasing the prices of put index options. In reality, an increase in volatility makes both call and put options more valuable due to the greater potential for price swings in the underlying indices.

Choice C is incorrect. Contrary to this choice, increased volatility does not decrease but rather increases the prices of both call and put index options. Higher volatility implies a wider range of possible outcomes for the underlying indices, thus making both types of options more valuable.

Things to Remember

- Volatility is a key factor in the pricing of options
 - Higher volatility leads to a greater potential range of movement for the underlying asset
 - Both call and put options become more valuable as volatility increases
 - The upside potential for options is unlimited while the downside risk is limited to the premium paid
 - Increased volatility widens the range of possible outcomes for the underlying indices, making both call and put options more valuable
-

Q.748 Kelly Jackson is a junior research analyst at an Asian Investment Fund. The fund has a large exposure to US stocks and options. Jackson is given the task of analyzing the impact on the prices of call and put options if the underlying stock pays a cash dividend. Jackson came up with the following scenarios that show the impact of an increase in cash dividends on call and put options. Which of the four scenarios is consistent with the principles of option pricing?

- A. An increase in the expected cash dividend will increase the price of put and call stock options.
- B. An increase in the expected cash dividend will decrease the price of put and call stock options.
- C. An increase in the expected cash dividend will decrease the price of put options but increase the price of call stock options.
- D. An increase in the expected cash dividend will increase the price of put options but decrease the price of call stock options.

The correct answer is **D**.

An increase in the expected cash dividend will increase the price of put options but decrease the price of call stock options. This is because the stock price typically drops by the amount of the dividend on the ex-dividend date. Higher expected cash dividends imply lower call premiums because the options are now more likely to be in the money, and higher put premiums because

the options are now more likely to be out of the money. The stock price reflects the value of a company. When a company pays a cash dividend, it depletes its reserves, which reduces the value of the firm. Consequently, the stock price also declines to reflect this reduction in value. Additionally, the share price can be viewed as the present value of all future cash flows. Once a dividend has been paid out, that's one less cash flow in the present value equation, which also contributes to the decrease in stock price.

Choice A is incorrect. An increase in the expected cash dividend does not increase the price of both put and call stock options. This is because when dividends increase, it reduces the stock price which in turn decreases the value of a call option while increasing the value of a put option.

Choice B is incorrect. An increase in expected cash dividends does not decrease the price of both put and call stock options. As explained above, an increased dividend decreases the stock's price which increases a put option's value but decreases a call option's value.

Choice C is incorrect. While it correctly states that an increase in expected cash dividends will decrease the price of put options, it incorrectly suggests that this would also lead to an increase in the price of call options. In reality, as mentioned earlier, higher dividends lower stock prices which subsequently reduce a call option's value.

Things to Remember

- Dividends impact option pricing by affecting the stock price
 - Higher expected cash dividends lead to lower call premiums and higher put premiums
 - Stock price drops by the amount of the dividend on the ex-dividend date
 - Stock price reflects the value of the company and decreases when dividends are paid
 - Share price is the present value of all future cash flows
-

Q.749 Mehmet Orkan, a junior investment analyst at an Istanbul-based investment company, is analyzing various call options on U.S. stocks. He has obtained the following call options quotes on some blue-chip companies in the U.S consumer goods sector. Which of the following options has the highest value?

Name	Nature of the option	Expiration date
A	European call option	3 months
B	European call option	9 months
C	American call option	3 months
D	American call option	9 months

A. Option A

- B. Option B
- C. Option C
- D. Option D

The correct answer is **D**.

The value of an option is influenced by several factors, including the time to expiration and the nature of the option. As the time to expiration increases, the value of the option generally increases or remains the same. This is because the longer the time to expiration, the greater the chance that the underlying asset's price will move in a favorable direction, thereby increasing the option's value. Therefore, the 9-month options (both American and European) will have a higher value than the 3-month options. Between the 9-month American and European options, the American option will have a higher value. This is because an American option gives the holder the right to exercise the option at any time up to the expiration date, providing more flexibility and potential for profit. In contrast, a European option can only be exercised at the expiration date. This lack of flexibility can limit the potential for profit, thereby reducing the value of the option. Therefore, the 9-month American call option (Choice D) will have the highest value. **Choice A is incorrect.** Although Option A may have certain attractive features, it does not hold the highest value. This could be due to a variety of factors such as its strike price, time to expiration, or the volatility of the underlying asset. **Choice B is incorrect.** Despite any potential benefits that Option B might offer, it does not hold the highest value among these options. The value of an option is determined by several factors including its intrinsic value and time value which are influenced by elements like the price of the underlying asset, strike price, time until expiration and volatility. **Choice C is incorrect.** While Option C might seem appealing for various reasons, it does not possess the highest value in comparison to other options listed here. Factors such as its moneyness status (whether it's in-the-money or out-of-the-money), remaining time until expiration and implied volatility play a crucial role in determining an option's worth.

Q.751 Upper and lower pricing bounds for American and European call and put options are important in order to prevent investors from earning arbitrage profits. If option prices are above the upper boundary or below the lower boundary, an investor can earn arbitrage profit by opening a position in an options contract and in the underlying stock simultaneously. Which of the following statements is consistent with the upper boundary limit of the American put option?

- A. The price of an American put option should be equal to or higher than the current price of the underlying stock.
- B. The price of an American put option should be equal to or higher than the strike price of the option on the underlying stock.
- C. The price of the American put option should be equal to or lower than the current price of the underlying stock.

D. The price of the American put option should be equal to or lower than the strike price of the option on the underlying stock.

The correct answer is **D**.

The price of an American put option should be equal to or lower than the strike price of the option on the underlying stock. This is because the value of a put option is derived from the right it provides to its holder to sell the underlying asset at the strike price. If the price of the put option were to exceed the strike price, it would mean that the holder is paying more for the option than they would receive from exercising it. This would not be rational, as the holder could instead sell the asset in the market at a higher price. Therefore, the price of a put option should never exceed the strike price of the option. If it does, an arbitrage opportunity would arise. An investor could sell the overpriced put option, invest the proceeds at the risk-free rate, and make a risk-free profit. This is why the price of an American put option should be equal to or lower than the strike price of the option on the underlying stock.

Choice A is incorrect. The price of an American put option should not be equal to or higher than the current price of the underlying stock. This would imply that the holder could sell the stock for its full market value and still retain a valuable put option, which contradicts no-arbitrage principles.

Choice B is incorrect. The price of an American put option should not be equal to or higher than the strike price of the option on the underlying stock. If this were true, it would mean that exercising the put option immediately would yield a profit greater than or equal to zero regardless of what happens in future, which again contradicts no-arbitrage principles.

Choice C is incorrect. While it's true that an American put option's value will generally decrease as its underlying asset's price increases (all else being equal), saying that its price "should" be lower than its underlying asset's current market value implies a stronger relationship between these two prices than actually exists in practice.

Things to Remember

- Upper and lower pricing bounds for options are important to prevent arbitrage opportunities in the market.
- American options can be exercised at any time before expiration, while European options can only be exercised at expiration.
- Put options give the holder the right to sell the underlying asset at a specified price (strike price).
- Arbitrage opportunities arise when mispricing occurs in the options market, allowing investors to make risk-free profits.

- Risk-free rate is an important factor in determining arbitrage opportunities in the options market.
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Q.752 Adam Gilbert is a risk manager that works with Global Trade Brokerage Firm in New York City. Global Trade Brokerage is a member of the options exchange which provides brokerage services to option traders and also works as the market maker in the exchange. Gilbert's job responsibility is to derive upper and lower boundaries for options so the prices are arbitrage-free. Which of the following is an accurate estimation of the lower band for European call option prices on a non-dividend-paying stock if the current stock price is \$92 and the strike price of the option on that stock is \$89? Suppose the option is expiring in 6 months, and the risk-free rate is 8% with annual compounding.

- A. \$3
- B. \$3.97
- C. \$6.49
- D. \$6.36

The correct answer is **D**.

The lower bound of the European call option on a non-dividend paying stock is equal to:

$$S - K(1 + r)^{-t} = 92 - 89 * 1.08^{-0.5} = 6.3597$$

Where S = current price; K = strike price; r = risk-free rate; and t = time to expiration.

Therefore, an arbitrage opportunity exists if the value of the European call option is below \$6.3597

Q.753 Vijay Singh works as an investment manager at Global Investment Company in New York. Global also provides brokerage services to its clients. Therefore, it is a usual task at Global to derive upper and lower boundaries for options so the prices are arbitrage-free. Which of the following given options is the accurate estimation of the lower price boundary for European put options on a non-dividend paying stock that expires in 3 months, if the current stock price is \$31, the strike price is \$33, and the risk-free rate is 10% with annual compounding?

- A. \$2.37
- B. \$1.22
- C. \$2.80
- D. \$1.19

The correct answer is **B**.

The lower bound of the European put option on a non-dividend paying stock is equal to:

$$p \geq K(1 + r)^{-t} - S = 33(1.10)^{-0.25} - 31 = 1.22298$$

Where p = put option price; S = current stock price; K = strike price; r = risk-free rate; and t = time to expiration.

An arbitrage opportunity exists if the value of the European put option is below \$1.22

Q.755 Johanna Smith is a treasury manager at Easy Bank. She manages the treasury affairs and also the investment advisory activities of the bank. She invests in treasury and money market instruments to manage short-term cash, but for long-term cash management, she uses other instruments. Currently, she has invested in a zero-coupon bond with the face value of X , and at the same time, she has also taken a long exposure in call options on the stocks of a specific firm with the strike price of X . Suppose that at the time of expiration of the call options, the final price of the stock is A , which is below X . If the bond matures on the options expiration date, then estimate the payoff of the combination of the bonds and call options.

- A. The payoff is A
- B. The payoff is X
- C. The payoff is $A+X$
- D. The payoff is zero

The correct answer is **B**.

The net payoff of the combination of a zero-coupon bond and a call option is X (the face value of the bond). This is further described below:

The combination of a zero-coupon bond which pays X amount at maturity and a call option with the strike price of X is called a fiduciary call.

The payoff of the zero-coupon bond at the maturity is X.

The payoff of the call option, if the option is in the money, is $S - X$ (where S is the current price and X is the strike price). If the option is out of the money, the payoff of the option is 0.

Therefore, the payoff of a fiduciary call with an in the money option = $(S - X) + X = S$

And the payoff of a fiduciary call with an out of the money option = $(0) + X = X$

As mentioned in the question the current price of the stock is A which is below X, so the option expired out of the money, and the payoff of the fiduciary call is X.

Things to Remember

- Zero-coupon bond: A bond that does not pay periodic interest but is issued at a discount to its face value, which is paid at maturity.
 - Call option: A financial contract that gives the holder the right, but not the obligation, to buy an asset at a specified price within a specific time period.
 - Strike price: The price at which the holder of an option can buy (in the case of a call option) or sell (in the case of a put option) the underlying asset.
 - In the money option: An option that would lead to a positive payoff if it were exercised immediately.
 - Out of the money option: An option that would lead to a negative payoff if it were exercised immediately.
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Q.756 An investor has constructed a fiduciary call which consists of a Millers Corp. zero-coupon bond with a face value of \$95 and the bond is expected to mature in March 2017, and a call option on Millers Corp.'s common stock with the strike price of \$95 and the option expiring in March 2017. Suppose that in March 2017, the current price of the stock is \$90, then what is the total cash inflow that the investor will receive at maturity?

A. \$90

B. \$95

C. \$100

D. \$185

The correct answer is **B**.

The net payoff of the combination of a zero-coupon bond with a face value of \$95 and a call option with a strike price of \$95 is \$95 (the face value of the bond). This is further described below:

The combination of a zero-coupon bond which pays X amount at maturity, and a call option with a strike price of X is called a fiduciary call.

The zero-coupon bond will mature at its face value, which is \$95. Since it's a zero-coupon bond, it doesn't pay periodic interest but is instead sold at a discount and pays its face value at maturity. In this case, the investor will receive \$95 at maturity.

The current price of the stock is \$90, which is lower than the strike price. Since it wouldn't make sense to exercise the option to buy the stock at \$95 when it's available in the market for \$90, the option will expire worthless. Therefore, there is no cash inflow from this component of the fiduciary call. In other words, the option is out of the money, and its payoff is $(90 - 95) = 0$.

Considering both components, the total cash inflow at maturity for the investor will be solely from the zero-coupon bond, which is \$95. The call option does not contribute to the cash inflow since it expires worthless.

Things to Remember

- A fiduciary call is a combination of a zero-coupon bond and a call option with the same strike price.
- Zero-coupon bonds do not pay periodic interest but are sold at a discount and pay their face value at maturity.
- The payoff of a call option is the difference between the stock price at maturity and the strike price, if positive.
- If the stock price is below the strike price at maturity, the call option expires worthless.
- Investors use fiduciary calls to lock in a future purchase price for a stock while earning a fixed return from the bond component.

Q.757 The put-call parity is an important relationship in options pricing. The put-call parity

relationship is established on the payoff of the combination of two portfolios, a fiduciary call, and a protective put. The fiduciary call is composed of a risk-free discount bond and a call option, while the protective put consists of a put option and a stock. Which of the following principle must hold true in the put-call parity?

- A. The face value of the discount bond must be below the strike price of call and put options.
- B. The face value of the discount bond must be above the strike price of call and put options.
- C. The face value of the discount bond must be equal to the final price of call and put options.
- D. The face value of the discount bond must be equal to the strike price of call and put options.

The correct answer is **D**.

The put-call parity is a principle in options pricing that states the price of a call option implies a certain fair price for the corresponding put option and vice versa. The put-call parity relationship is established based on the payoff of two portfolios: a fiduciary call and a protective put. The fiduciary call consists of a risk-free discount bond and a call option, while the protective put is made up of a put option and the underlying stock. In this context, the face value of the discount bond at maturity must be equal to the strike price of the call and put options at expiration. This is because the strike price is the price at which the holder of the option can buy (in case of a call option) or sell (in case of a put option) the underlying security when the option is exercised. Hence, for the put-call parity to hold, the face value of the discount bond (which is the amount that will be returned to the bondholder at maturity) must be equal to the strike price of the call and put options.

Choice A is incorrect. The face value of the discount bond being below the strike price of call and put options would not uphold the principle of put-call parity. This is because it would imply that the risk-free return from holding a fiduciary call (bond + call option) would be less than that from holding a protective put (put option + stock), which contradicts the concept of arbitrage-free pricing in financial markets.

Choice B is incorrect. Similarly, if the face value of the discount bond were above the strike price, it would mean that there's an opportunity for arbitrage as one could earn more by investing in a fiduciary call than in a protective put, which again contradicts with no-arbitrage principle.

Choice C is incorrect. The face value of discount bond being equal to final price of options does not make sense as final prices are uncertain and depend on various factors like underlying asset's price at maturity, volatility etc., whereas face value of bond is known upfront and does not change over time.

Things to Remember

- Put-call parity is a fundamental concept in options pricing that helps ensure no arbitrage opportunities exist in the market.
 - The relationship between call and put options, along with the underlying asset and risk-free rate, is crucial for understanding put-call parity.
 - Put-call parity can be used to calculate the theoretical value of options and identify mispricings in the market.
 - Understanding the components of the fiduciary call and protective put portfolios is essential for grasping the concept of put-call parity.
 - Arbitrage opportunities arise when put-call parity is violated, leading to market corrections to restore equilibrium.
-

Q.758 Vijay Mehta is a portfolio manager at First American Investments. He manages a portfolio that invests in a wide variety of financial instruments. Currently, a large portion of his portfolio consists of stocks and options. Recently, he purchased the stock of Jack Ville Inc. and at the same time, he also took a long exposure in a put option on the stocks of Jack Ville Inc. with the strike price of X . Suppose that, at the time of expiration, the final price of the stock, S , is below the strike price, X , then estimate the payoff of the combination of the stock and the put option.

- A. The payoff of the combination of the stock and the put option is S .
- B. The payoff of the combination of the stock and the put option is X .
- C. The payoff of the combination of the stock and the put option is $S+X$.
- D. The payoff of the combination of the stock and the put option is zero.

The correct answer is **B**.

By investing in the stock of Jack Ville Inc. and taking a long exposure in a put option on Jack Ville Inc. stock, the portfolio manager has constructed a protective put.

A protective put consists of a stock and a put option on the same stock with a strike price of X .

The payoff of the put option if the option is in the money is $X-S$ (where S is the current price and X is the strike price).

If the option is out of the money, the payoff of the option is 0.

The payoff of the stock will be equal to the current price or the final price of stock i.e. S

Therefore, the payoff of the protective put with in-the-money put option $= (X-S) + S = X$

And the payoff of a fiduciary call with an out of the money option $= (0) + S = S$

As mentioned in the question, the current price of the stock, S , is below X , so the option is in the money, and the payoff of the protective put is X .

Things to Remember

- A put option gives the holder the right, but not the obligation, to sell a specified amount of an underlying asset at a specified price within a specified time.
 - In a protective put strategy, an investor buys a put option to protect against a drop in the stock price of the underlying asset they own.
 - The payoff of a put option is calculated as the difference between the strike price and the final price of the underlying asset at expiration.
 - If the final price of the underlying asset is below the strike price, the put option is in the money and the payoff is positive.
 - The payoff of a combination of a stock and a put option is the sum of the payoffs of each individual component.
-

Q.760 The put-call parity relation suggests that the portfolios with identical payoffs must sell for the same price in order to prevent arbitrage profit or riskless gains. The put-call parity is, therefore, constructed of the fiduciary call and protective put options. Which of the following equation is inconsistent with the put-call parity equation?

A. $S = c - p + X(1 + r)^{-t}$

B. $p = c - S + X(1 + r)^{-t}$

C. $X(1 + r)^{-t} = S + c - p$

D. $c = S + p - X(1 + r)^{-t}$

The correct answer is **C**.

The equation $X(1 + r)^{-t} = S + c - p$ is inconsistent with the put-call parity equation. The put-call parity equation is given by $c + X(1 + r)^{-t} = S + p$, where 'c' is the call price, 'p' is the put price,

$X(1+r)^{-t}$ is the present value of a zero-coupon bond, and 'S' is the current price of the stock. Rearranging this equation, we get $c = S + p - X(1+r)^{-t}$, which is the price of a call option. The equation in choice C rearranges the terms in a way that does not represent any of the individual securities in the put-call parity relationship, making it inconsistent with the put-call parity equation.

Choice A is incorrect. This equation correctly represents the put-call parity principle. It states that the price of the underlying asset (S) is equal to the price of a call option (c) minus the price of a put option (p), plus the present value of strike price ($X(1+r)^{-t}$).

Choice B is incorrect. This equation also correctly represents the put-call parity principle, but from a different perspective. It states that the price of a put option (p) equals to call option's price (c) minus underlying asset's current market value(S), plus present value of strike price($X(1+r)^{-t}$).

Choice D is incorrect. This equation accurately reflects put-call parity as well, stating that call option's cost(c) equals to current market value(S) plus cost of protective put(p), minus present discounted value of exercise or strike Price($X(1+r)^{-t}$).

Things to Remember

- Put-call parity is a fundamental concept in options pricing that establishes a relationship between the prices of European call and put options with the same strike price and expiration date.
- The put-call parity equation helps in understanding the relationship between the prices of call and put options, the underlying asset price, the strike price, and the risk-free rate.
- Arbitrage opportunities arise when the put-call parity relationship is violated, allowing traders to make riskless profits by exploiting the price discrepancies.
- The put-call parity equation is essential for pricing options and understanding the dynamics of the options market.
- Understanding put-call parity can help investors and traders in making informed decisions regarding options trading strategies and risk management.

Q.761 Jacob Clarke is an investment manager at one of the largest investment banks in Canada. Clarke has a wide variety of investment options to invest in. However, he is interested in constructing the payoff of a synthetically created long position in call options. Which of the following positions should he take to create the payoff of a synthetic call option?

- A. Long a stock, short a put option, and short a zero-coupon bond.
- B. Long a call option, short a put option, and short a zero-coupon bond.
- C. Short a stock, long a put option, and long a zero-coupon bond.
- D. Long a stock, long a put option, and short a zero-coupon bond.

The correct answer is **D**.

A synthetic call option can be created by taking a long position in a stock, a long position in a put option, and a short position in a zero-coupon bond. This combination of positions replicates the payoff of a long position in a call option. The underlying principle behind this strategy is the put-call parity, which is a fundamental concept in options pricing. The put-call parity states that the price of a call option (c) plus the present value of the strike price (X) discounted at the risk-free rate (r) for the time to maturity (T) is equal to the price of the stock (S) plus the price of a put option (p). Mathematically, this is represented as $c + X(1+r)^{-T} = S + p$. Rearranging this equation to solve for the price of a call option, we get $c = S + p - X(1+r)^{-T}$. This equation shows that a call option can be synthetically created by taking a long position in a stock, a long position in a put option, and a short position in a zero-coupon bond. This strategy allows an investor to replicate the payoff of a call option without actually owning the option.

Choice A is incorrect. While a long position in a stock and shorting a put option can replicate the payoff of a call option, shorting a zero-coupon bond would not contribute to this synthetic strategy. Shorting the bond would create an obligation to pay at maturity, which is not characteristic of a call option's payoff.

Choice B is incorrect. This choice involves both long and short positions in options (call and put respectively), which does not replicate the payoff of just one call option. Additionally, shorting a zero-coupon bond again introduces an obligation that does not align with the desired payoff.

Choice C is incorrect. Shorting a stock and going long on both put options and zero-coupon bonds do not mimic the characteristics of being long on call options. In fact, this combination could potentially lead to unlimited losses if the stock price increases significantly - something that doesn't happen when holding only call options.

Things to Remember

- Put-call parity is a fundamental concept in options pricing that relates the prices of call and put options with the price of the underlying asset.
- A synthetic call option can be created by combining a long stock position, a long put option, and a short position in a zero-coupon bond.
- Shorting a zero-coupon bond introduces an obligation to pay at maturity, which is not characteristic of a call option's payoff.

- Combining long and short positions in options (call and put) does not replicate the payoff of just one call option.
 - Shorting a stock and going long on both put options and zero-coupon bonds do not mimic the characteristics of being long on call options.
-

Q.3413 A stock is currently trading at \$60 per share. A European call option having an exercise price of \$71 and one year to maturity is currently trading at \$10. If the risk-free rate is 7%, per annum, then what is the put option price?

- A. \$3.80
- B. \$16.36
- C. \$16.20
- D. \$5.06

The correct answer is **B**.

Using the put call-parity equation:

$$\begin{aligned}
 S_0 + P &= C + X(1 + r)^{-T} \\
 P &= C + X(1 + r)^{-T} - S_0 \\
 &= 10 + 71 \times (1.07)^{-1} - 60 = 16.35514 \approx 16.36
 \end{aligned}$$

Q.3514 Rabi Koch took a long position in a March put option with the strike price of \$65. What is the outcome of the position if the spot price is \$78 at expiration?

- A. \$11 positive payoff
- B. \$13 negative payoff
- C. \$13 positive payoff
- D. \$0 payoff

The correct answer is **D**.

Since the spot price of the put option is higher than the strike price the option is out of the money. The payoff to the option buyer is:

$$P_T = \max(0, X - S_T) = \max(0, 65 - 78) = 0$$

A note on puts

A short put refers to the opening of an options trade by selling or writing a put option. The trader who buys the put option is long that option (holds the long position), and the trader who wrote that option is short (holds the short position).

For the long position (buyer), the option is in the money (ITM) if and only if the prevailing spot price at expiry is less than the strike price. In such circumstances, the buyer would be able to "cut" their loss by selling the underlying at the strike price which would be considerably higher than the prevailing market price. Buyers of puts are bearish, i.e, they expect the underlying to lose value over time.

Things to Remember

- Options are financial derivatives that give the holder the right, but not the obligation, to buy or sell an underlying asset at a specified price within a specific time period.
 - A put option gives the holder the right to sell the underlying asset at a specified price (strike price) within a specific time period.
 - The payoff for a put option at expiration is calculated as the maximum of 0 or (strike price - spot price).
 - If the spot price is higher than the strike price at expiration, the put option is out of the money and the payoff is 0.
 - Long position in options refers to buying options, while short position refers to selling options.
 - Options can be used for speculation, hedging, or generating income through option premiums.
-

Q.3564 Which of the following conditions will create the biggest discrepancy in price between a long-term European put option and an otherwise identical short-term put option?

- A. The volatility in the market is low
- B. Interest rates are lower than they have ever been in the past
- C. Interest rates are higher than they have ever been in the past

D. Both A and B

The correct answer is **B**.

When interest rates are lower than they have ever been in the past, the price discrepancy between a long-term European put option and an otherwise identical short-term put option will be the greatest. European options can only be exercised at their expiration date. Therefore, a longer time to expiration implies that the option holder will have to wait longer to receive money from the sale of the underlying asset. This waiting period represents a lost interest opportunity. When interest rates are low, this lost interest is reduced, making the long-term option more valuable compared to the short-term option. Therefore, in a low-interest-rate environment, the price discrepancy between the long-term and short-term options will be the greatest.

Choice A is incorrect. While it's true that lower volatility would decrease the price of both options, it wouldn't necessarily increase the price discrepancy between them. The impact of volatility on option pricing is more complex and depends on other factors such as the strike price and time to maturity.

Choice C is incorrect. Higher interest rates could potentially increase the price discrepancy between long-term and short-term options, but not necessarily to a greater extent than extremely low interest rates. The relationship between interest rates and option prices isn't linear, so we can't say definitively that higher rates will lead to a larger price discrepancy.

Choice D is incorrect. As explained above, neither low market volatility nor high interest rates would necessarily lead to a greater price discrepancy between long-term and short-term options than extremely low interest rates would.

Things to Remember

- Volatility in the market affects option prices -higher volatility generally leads to higher option prices
- Interest rates have an impact on option prices -lower interest rates tend to increase the value of long-term options
- Time to expiration is a key factor in option pricing -longer time to expiration generally increases the value of an option
- European options can only be exercised at expiration, unlike American options which can be exercised at any time before expiration

Q.3567 When evaluating call options, it's important to recognize the factors that influence their value. Which of the following conditions will increase the value of a call option?

- A. A decrease in volatility
- B. An increase in risk-free rate
- C. A decrease in stock price
- D. An increase in the dividend rate

The correct answer is **B**.

An increase in the risk-free rate will increase the value of a call option. The risk-free rate is the theoretical rate of return of an investment with zero risk, typically associated with high-quality government bonds. In the context of options pricing, an increase in the risk-free rate increases the present value of the expected payoff from the option, thereby increasing the value of the option. This is because the holder of the option can invest the money that would otherwise have been spent on purchasing the underlying asset at the risk-free rate until the option's expiration date. The higher the risk-free rate, the higher the potential return from this alternative investment, making the option more valuable.

Choice A is incorrect. A decrease in volatility would not increase the value of a call option. In fact, it's the opposite: higher volatility generally increases the value of options because it implies a greater range of potential outcomes for the underlying asset's price, thus increasing the likelihood that the option will be in-the-money at expiration.

Choice C is incorrect. A decrease in stock price would not lead to an increase in the value of a call option. The value of a call option increases as the price of underlying asset (in this case, stock) increases because it gives holder right to buy at lower specified price and sell at current market price which is higher.

Choice D is incorrect. An increase in dividend rate would actually decrease, not increase, the value of a call option. This is because when dividends are paid out on stocks, they reduce company's share prices by roughly equivalent amount which reduces potential upside for call options holders.

Things to Remember

- Volatility: Higher volatility generally increases the value of options because it implies a greater range of potential outcomes for the underlying asset's price.
- Stock Price: The value of a call option increases as the price of the underlying asset (stock) increases.
- Dividend Rate: An increase in dividend rate would decrease the value of a call option as dividends reduce the potential upside for call option holders.
- Risk-Free Rate: An increase in the risk-free rate will increase the value of a call option

as it increases the present value of the expected payoff from the option.

Q.3568 Leslie Hower is a junior trader at a derivatives dealer firm. During her first week at the firm, Hower attempts to synthetically sell a risk-free bond using call and put options. She purchases call and put options with the same exercise price and time to maturity. She simultaneously buys the underlying.

With respect to her attempts in creating a synthetic short position in a risk-free bond, Hower is accurate regarding her decision to:

- A. Purchase call options
- B. Purchase put options
- C. Buy the underlying short
- D. Sell put options

The correct answer is **B**.

Creating a synthetic short position in a bond (or any asset) typically involves using derivatives to mimic the payoff of a short sale of the asset. To synthetically short a bond, the position should provide the payoff that mirrors the loss in value of the bond if the bond's price were to rise (since when you short-sell something, you profit if the price goes down and lose if the price goes up).

Here's what Leslie Hower would need to do to create a synthetic short position in a risk-free bond using options:

- **Sell Call Options.** By selling call options, Hower obligates herself to sell the bond at the exercise price if the options are exercised. If the bond's price increases, the calls will be exercised, and she will "lose" money, analogous to a short position's loss when the price of the underlying asset increases.
- **Buy Put Options.** Buying put options gives Hower the right, but not the obligation, to sell the bond at the exercise price. If the bond's price falls, she can exercise the puts and sell at the higher exercise price, profiting from the decline, just as a short seller would.
- **Do not buy the underlying.** If she buys the underlying, she is going long on the bond, which contradicts the intention of creating a short position.

A is incorrect Purchasing call options would create a long position, not a short one.

C is incorrect Buying the underlying is a long position, not a short one.

D is incorrect Selling put options would be a bet on the bond price staying the same or increasing, not decreasing.

Things to Remember

- **Short Selling:** Short selling is the practice of selling assets that the seller does not own at the time of the sale. This is done with the expectation that the price of the asset will decrease, allowing the seller to buy back the assets at a lower price.
 - **Options:** Options are financial derivatives that give the buyer the right, but not the obligation, to buy or sell an underlying asset at a specified price before or on a specified date.
 - **Synthetic Positions:** Synthetic positions are created using a combination of financial instruments to replicate the payoff of another position. They are often used to hedge or speculate on the price movements of an underlying asset.
 - **Call Options:** Call options give the holder the right to buy an asset at a specified price within a specified time frame.
 - **Put Options:** Put options give the holder the right to sell an asset at a specified price within a specified time frame.
-

Q.3569 According to the put-call parity, a long position in a put option can be replicated by going:

- A. Short a call option, short the underlying, and long a risk-free bond
- B. Short a call option, long the underlying, and short a risk-free bond
- C. Long a call option, short the underlying, and short a risk-free bond
- D. Long a call option, short the underlying, and long a risk-free bond

The correct answer is **D**.

According to the put-call parity, a long position in a put option can be replicated by going long a call option, short the underlying, and long a risk-free bond. This is derived from the put-call

parity equation: $S_0 + p_0 = c_0 + X/(1+r)^T$. Rearranging the equation to solve for p_0 (the price of the put option), we get: $p_0 = c_0 + X/(1+r)^T - S_0$. This implies that the price of a put option is equal to the price of a call option plus the present value of the strike price minus the price of the underlying. Therefore, an investor can replicate a long put option position by going long a call option (buying a call option), short the underlying (selling the underlying), and long a risk-free bond (buying a risk-free bond).

Choice A is incorrect. Shorting a call option and the underlying while going long on a risk-free bond does not create a synthetic long position in a put option according to the put-call parity. This combination would rather create a synthetic short position in a put option.

Choice B is incorrect. Shorting a call option, going long on the underlying, and shorting a risk-free bond also does not align with the put-call parity for creating a synthetic long position in a put option. This combination would result in an undefined or non-standard options strategy.

Choice C is incorrect. Going long on both, the call option and shorting the underlying while also shorting on risk-free bond does not satisfy the equation of put-call parity for creating synthetic long position in put-option. This combination would lead to an undefined or non-standard options strategy.

Things to Remember

- Put-call parity is a fundamental concept in options pricing that shows the relationship between the price of a European call option and a European put option with the same strike price and expiration date.
 - Put-call parity helps in understanding the relationship between call and put options and can be used for arbitrage opportunities in the market.
 - Risk-free bond is often used in options trading to replicate certain positions or create synthetic positions.
 - Understanding the relationship between different options strategies and their payoff structures is crucial for successful options trading.
-

Q.3570 A three-month call option with an exercise price of \$55 is being sold for \$8. A three-month Treasury bond is being sold in the marketplace with the same face value as the option's exercise price. The underlying is currently worth \$60, and the risk-free rate is 4.30%. Assuming the put-call parity holds, a put option is being sold for:

- A. \$0.73
- B. \$2.42
- C. \$12.34
- D. \$8.48

The correct answer is **B**.

Using the put-call parity relationship:

$$\begin{aligned}c_0 + X(1 + r)^{-T} &= S_0 + p \\&= p_0 = \$8 + (\$55(1.043)^{-0.25}) - \$60 = \$2.4241\end{aligned}$$

Reading 40: Trading Strategies

Q.754 Johanna Smith is a treasury manager at Easy Bank. She manages the treasury affairs and also the investment advisory activities of the bank. She invests in treasury and money market instruments to manage short-term cash, but for long-term cash management, she uses other instruments. Currently, she has invested in zero-coupon bonds with the face value of \$100, and at the same time, she has also taken a long exposure in call options with the strike price of \$100. Which of the following accurately depicts Smith's strategy?

- A. Smith has constructed a put-call parity.
- B. Smith has constructed a covered call.
- C. Smith has constructed a fiduciary call.
- D. Smith has constructed a protective put.

The correct answer is **C**.

Smith has constructed a fiduciary call. A fiduciary call is a portfolio that consists of a zero-coupon bond that pays a certain amount (X) at maturity and a call option with a strike price of X . The payoff of the fiduciary call is X if the call option expires out of the money, and it is S ($X + S - X$) if the call option expires in the money (S = current stock price). This strategy is used to reduce and control losses using options rather than stocks. It allows the investor to control the same amount of stocks using call options as they would have if they had bought the shares directly. Instead of buying 100 shares of a stock using all the money, the investor buys 1 contract of its call options (1 contract = 100 shares) with just a small fraction of the money and invests the rest into an interest-bearing risk-free investment vehicle. This way, the investor increases their overall profits and reduces their maximum risk exposure to just the amount of money they used towards buying the call options.

Choice A is incorrect. Put-call parity is a principle that defines the relationship between the price of European put options and calls of the same class, that is, with the same underlying asset, strike price, and expiration date. In this scenario, Smith has not constructed a put-call parity as she has only taken a long position in call options and invested in zero-coupon bonds.

Choice B is incorrect. A covered call strategy involves owning or buying shares of stock and selling an equivalent number of call options on that stock. This strategy generates income from the premiums earned from selling call options. However, Smith's strategy does not involve owning or buying shares; instead she has invested in zero-coupon bonds.

Choice D is incorrect. A protective put strategy involves purchasing a put option to hedge against potential losses in an owned stock's value. In this case, Smith has taken a long position in call options rather than purchasing put options; hence her strategy does not constitute a protective put.

Things to Remember

- Zero-coupon bonds are fixed-income instruments that do not pay interest but are issued

at a discount to their face value and mature at par.

- Call options give the holder the right, but not the obligation, to buy an underlying asset at a specified price within a specific time period.
 - Fiduciary call strategy involves combining a zero-coupon bond with a call option to control the same amount of stock using options rather than directly buying shares.
 - Put-call parity is a fundamental principle in options pricing that relates the prices of European put and call options with the same strike price and expiration date.
 - Covered call strategy involves owning the underlying stock and selling call options against it to generate income from premiums.
 - Protective put strategy involves buying a put option to hedge against potential losses in the value of an owned stock.
-

Q.765 Mahesh Kumar has recently joined Singapore Standard Bank, the largest investment banks in South-East Asia. Kumar has analyzed an open position his bank has in the stock of a Singaporean carmaker. The current value of the stock is \$44, but he believes that the price of the stock will have trouble reaching above \$48 because of technical and fundamental factors. Kumar called one of the bank's traders and shared his analysis regarding the stock. The analyst informed the manager that he is going to lock the profit with a covered call strategy. How exactly is he going to apply the covered call?

- A. Since the bank already owns the stocks, the trader is going to buy out-of-the-money call options at the strike price of \$48.
- B. Since the bank already owns the stocks, the trader is going to sell in-the-money call options at the strike price of \$44.
- C. Since the bank already owns the stocks, the trader is going to buy at-the-money call options at the strike price of \$44.
- D. Since the bank already owns the stocks, the trader is going to sell out-of-the-money call options at the strike price of \$48.

The correct answer is **D**.

A covered call strategy involves owning the underlying asset and selling call options on that

same asset. The goal of this strategy is to generate additional income from the option premium, while also providing some protection against potential losses if the asset's price falls. In this case, the trader believes that the stock price will not exceed \$48. Therefore, he decides to sell out-of-the-money call options with a strike price of \$48. Out-of-the-money options are those with a strike price that is higher (for call options) or lower (for put options) than the current market price of the underlying asset. By selling these options, the trader can earn the premium, which provides additional income. If the stock price does not exceed \$48, the options will expire worthless, and the trader will keep the premium as profit. However, if the stock price does rise above \$48, the options will be exercised, and the trader will have to sell the stock at the strike price. But since the trader already owns the stock, this is not a problem. The profit from selling the stock and the premium received from selling the options will offset any potential losses. This strategy is a common way to generate income and hedge against potential losses in the stock market.

Choice A is incorrect. The trader would not buy out-of-the-money call options at the strike price of \$48. In a covered call strategy, the trader sells call options on an asset they already own to generate income from the option premium. Buying a call option would not serve this purpose.

Choice B is incorrect. Selling in-the-money call options at the strike price of \$44 is also not correct because it does not align with Kumar's analysis that suggests that the stock price may struggle to surpass \$48. Selling in-the-money calls would mean that the bank believes that stock prices will decrease, which contradicts Kumar's analysis.

Choice C is incorrect. Buying at-the-money call options at a strike price of \$44 does not represent a covered call strategy either as it involves buying rather than selling options and thus does not generate income from premiums.

Things to Remember

- A covered call strategy involves owning the underlying asset and selling call options on that same asset
- Out-of-the-money options have a strike price higher than the current market price of the underlying asset
- This strategy helps generate additional income from the option premium and provides some protection against potential losses
- If the stock price does not exceed the strike price, the options will expire worthless, and the trader keeps the premium as profit
- If the stock price rises above the strike price, the options will be exercised, and the trader will have to sell the stock at the strike price

Q.766 Investments banks create customized derivative products for risk-averse retail investors. These products have features of multiple instruments. The payoff on these customized products depends on underlying assets like stocks, indices, and other risky assets. However, the investments in these assets cannot decrease below the initial principal. Which of the following products have these features?

- A. Covered call
- B. Protective put
- C. Principal protected notes
- D. Straddle notes

The correct answer is **C**.

These are specialized investment products that banks create to cater to risk-averse retail investors. The payoffs or the return on these products depends on underlying assets like stock, indices, and other risky assets, but these products do not decrease in value beyond the principal investment. This means that the investor's initial investment is protected, regardless of the performance of the underlying assets. This feature makes principal protected notes particularly attractive to risk-averse investors who want to participate in potentially high-return markets without risking their initial investment.

Choice A is incorrect. A covered call is a financial market transaction in which the seller of call options owns the corresponding amount of the underlying instrument, such as shares of a stock or other securities. It does not guarantee that the investment value cannot fall below the initial principal amount.

Choice B is incorrect. A protective put is an investment strategy that involves adding a put option to a security that an investor already owns in order to protect against falling prices. However, it does not ensure that the investment value will not fall below the initial principal amount.

Choice D is incorrect. Straddle notes are complex financial products that involve buying both a call and put option on an underlying asset with identical strike prices and expiration dates, allowing investors to profit from large movements in either direction but do not guarantee protection of initial principal invested.

Things to Remember

- Principal protection notes are structured products that guarantee the return of the initial investment amount, regardless of the performance of the underlying assets.
- These products are popular among risk-averse investors who want exposure to potentially high-return markets without risking their principal investment.

- Covered calls involve selling call options on underlying securities that the investor already owns, providing some downside protection but not guaranteeing the initial principal.
 - Protective puts involve buying put options on securities to protect against downside risk, but they do not ensure the return of the initial principal.
 - Straddle notes involve buying both call and put options on an underlying asset, allowing investors to profit from large price movements but not protecting the initial principal.
-

Q.767 John Greenwood has recently joined A.K.K. Investment Company as a junior investment analyst. Greenwood has very little past experience in trading options. Therefore, he frequently has to refer to his superiors for trading strategies and terminologies. Recently Greenwood was instructed to apply a bull spread strategy on Blue Balloon Corp. stock options. Which of the following transactions correctly depicts the bull spread strategy?

- A. Taking a long position in a European put option with a specific strike price and simultaneously taking a short position in a European call option with a higher strike price
- B. Taking a long position in a European put option with a specific strike price and simultaneously taking a short position in a European put option with a lower strike price
- C. Taking a long position in a European call option with a specific strike price and simultaneously taking a short position in a European call option with a higher strike price
- D. Taking a long position in a European call option with a specific strike price and simultaneously taking a short position in a European call option with a lower strike price

The correct answer is **C**.

A bull spread strategy is a type of options strategy that is used when an investor expects a moderate rise in the price of the underlying asset. This strategy is constructed by buying a call option with a certain strike price, and selling another call option with a higher strike price. Both options must have the same underlying asset and expiration date. In the context of the question, Greenwood is expected to implement a bull spread strategy on Blue Balloon Corp. stock options. Therefore, he should take a long position in a European call option with a specific strike price and simultaneously take a short position in a European call option with a higher strike price. This strategy would allow Greenwood to profit from a moderate increase in the price of Blue Balloon Corp. shares, while limiting his potential losses. The maximum profit in a bull call spread is the difference between the strike prices, less the cost of initiating the trade. The maximum loss is limited to the initial cost of the trade.

Choice A is incorrect. This choice suggests taking a long position in a European put option and shorting a European call option with a higher strike price. This does not represent a bull spread strategy, but rather it represents an example of straddle or strangle strategies which are used when the investor expects high volatility in the market.

Choice B is incorrect. Taking a long position in a European put option and simultaneously taking a short position in another European put option with lower strike price represents bear spread strategy, not bull spread. In bear spreads, investors expect the price to fall and hence they buy put options at higher strike prices and sell at lower ones.

Choice D is incorrect. As explained above, this choice suggests buying call options at one strike price and selling call options at lower strike prices which contradicts the concept of bull spread strategy where we expect prices to rise so we buy call options at lower prices (long) and sell them (short) when their prices are expected to be higher.

Things to Remember

- A bull spread strategy is used when an investor expects a moderate rise in the price of the underlying asset.
- It involves buying a call option with a certain strike price and selling another call option with a higher strike price.
- Both options in a bull spread strategy must have the same underlying asset and expiration date.
- The maximum profit in a bull call spread is the difference between the strike prices, less the cost of initiating the trade.
- The maximum loss in a bull spread strategy is limited to the initial cost of the trade.
- Straddle and strangle strategies are used when the investor expects high volatility in the market, not for moderate price rises like in a bull spread strategy.
- Bear spread strategies involve buying put options at higher strike prices and selling at lower ones, as investors expect the price to fall.

Q.768 An investment manager at Skyline Bank frequently invests in stocks and derivatives. He is always testing different options strategies to maximize the value of the assets under management. Recently, he applied a bull spread strategy on Ocean Shipping Co. stock options. The manager applied a strategy by purchasing European call options on the stock of the firm with a strike price of \$89, and at the same time, he sold European call options on the same stocks with a strike price of \$92. Suppose that the final price of the stocks at expiration is \$97, then estimate the payoff of the strategy. Ignore the cost of the strategy.

- A. \$8
- B. \$5
- C. \$3
- D. -\$2

The correct answer is C.

In a bull spread strategy, an investor buys European call options with a specific strike price (\$89) and simultaneously sells European call options with a higher strike price (\$92).

If the current price (\$97) is higher than the strike price of the short call option (\$92), both call options will be exercised, and the payoff of the investor will be $X_2 - X_1$ or $\$92 - \$89 = \$3$.

Things to Remember

- Bull spread strategy involves buying a call option with a lower strike price and selling a call option with a higher strike price on the same underlying asset
- Payoff for a bull spread strategy is calculated as the difference between the strike prices of the call options involved
- European options can only be exercised at expiration, unlike American options which can be exercised at any time before expiration
- Options strategies involve combinations of buying and selling options to achieve specific risk-return profiles

Q.769 Saddam Ahmed is a junior portfolio manager at Westend Investments. His investing activities are focused on equities and options. Recently, he purchased a 6-month European call option on a specific stock for \$5 with a strike price of \$110. At the same time, he sold a 6-month European call option on the same stocks for \$3 with a strike price of \$115. Suppose that the final price of the stock at expiration is \$113, then estimate the profit/loss of the strategy.

- A. -\$2
- B. \$1
- C. \$3
- D. \$6

The correct answer is **B**.

The investor has applied a bull spread strategy. In a bull spread strategy, an investor buys a European call option with a specific strike price (\$110) and simultaneously sells a European call option with a higher strike price (\$115).

Since the investor paid \$5 to buy the call option and received \$3 for selling the other call option, the net cash outflow or the cost of the strategy is \$2.

Since the current price of the stock is \$113, which is higher than the strike price of the long call option but lower than the strike price of the short call option, the payoff of the investor is:

Profit/Loss = Current price - Strike price of the long call - Net cost of the strategy

Profit/Loss = \$113 - \$110 - \$2 = \$1

Additional explanation on how bull spread strategies work

In a bull-spread strategy, this is what happens:

(I) You buy a call option- giving you the right to buy the underlying at some point in the future at a specified strike price x. To do so, you pay a premium A

(II) You sell a call option - requiring you to sell the underlying at some point in the future at a specified strike price y, where $y > x$. Here, you receive a premium B

This is a bullish strategy meaning you expect the stock price to rise. If that happens, you will exercise the call option (I), i.e. buy low and sell the underlying at the prevailing market price, making a profit. However, your profit has a ceiling. If the market price soars above the strike price of the short position, you will not make any more money because the holder of the second option will most likely exercise the right to buy (and you will have no choice but to sell to them).

The maximum loss, on the other hand, is equal to B-A; the net cost of the two positions; the difference between what you receive for the short position (B) and what you pay for the long position (A).

Things to Remember

- European call option
- Strike price
- Bull spread strategy
- Net cost of the strategy
- Payoff calculation for options

Q.770 Nancy Smith is an independent individual investor. She has 5 years of experience trading equities, bonds, and options. Smith has recently learned about spread strategies in options that could be implemented to earn protected profits. She is particularly interested in implementing the bear spread strategy. Keeping in view Smith's intended spread strategy, determine how she can implement the bear spread strategy.

- A. She can implement the bear spread strategy by buying a European put option with a specific strike price and simultaneously selling a European put option with a higher strike price.
- B. She can implement the bear spread strategy by selling a European put option with a specific strike price and simultaneously buying a European put option with a higher strike price.
- C. She can implement the bear spread strategy by buying a European put option with a specific strike price and simultaneously buying a European call option with a lower strike price.
- D. She can implement the bear spread strategy by selling a European put option with a specific strike price and simultaneously buying a European put option with a lower strike price.

The correct answer is **B**.

In a bear spread strategy, an investor sells a European put option with a specific strike price and simultaneously buys a European put option with a higher strike price. This strategy is designed to profit from a decrease in the price of the underlying asset. The sold put option (with a lower strike price) will generate income, which helps to offset the cost of the purchased put option (with a higher strike price). If the price of the underlying asset falls below the strike price of the purchased put option, the investor can exercise this option, selling the asset at the higher strike price. However, if the price of the underlying asset is above the strike price of the sold put option at expiration, this option will be worthless, and the investor will only lose the net premium paid for the spread. Therefore, the maximum loss for the investor is limited to the net premium paid for the spread, while the maximum profit is the difference between the two strike prices minus the net premium paid. This strategy is called a bear spread because it profits when the price of the underlying asset falls, similar to how a bear market is characterized by falling prices.

Choice A is incorrect. This choice suggests buying a European put option with a specific strike price and selling another with a higher strike price. However, this would result in a bull spread strategy, not a bear spread. In the bear spread strategy, the investor expects the price of the underlying asset to fall and hence would sell an option at a lower strike price while buying another at a higher strike price.

Choice C is incorrect. This choice suggests implementing the bear spread by buying both put and call options which is not correct. The bear spread strategy involves only put options or only call options but not both simultaneously.

Choice D is incorrect. It describes a bull put spread, which is used when the investor's outlook is neutral to bullish, not bearish. This configuration seeks to profit from the decay of options

premiums provided the underlying stays above the higher put strike price, opposite to the bearish sentiment implied by seeking a bear spread.

Things to Remember

- Spread strategies in options involve taking positions in two or more options of the same type (either calls or puts) on the same underlying asset.
 - A bear spread strategy is used when an investor expects the price of the underlying asset to decrease.
 - In a bear spread strategy using put options, the investor sells a put option with a lower strike price and buys a put option with a higher strike price.
 - The maximum loss for the investor in a bear spread strategy is limited to the net premium paid for the spread, while the maximum profit is the difference between the two strike prices minus the net premium paid.
 - Options can be European or American style, with European options only exercisable at expiration, while American options can be exercised at any time before expiration.
-

Q.771 Ahmet Gogh believes the price of the stocks of Red Bus Co. has more downside potential than upside potential. Therefore, he has purchased a European put option on the stock of Red Bus with a strike price of \$42 and simultaneously sold a European put option with a strike price of \$38. At expiration, the final price of the stock is \$43. What is the payoff of the strategy?

- A. \$0
- B. \$1
- C. \$4
- D. \$5

The correct answer is **A**.

The payoff always ignores the cost of the option(s).

The payoff for each of these options at expiration can be calculated as follows:

For the bought put option with a strike price of \$42:

As the stock price at expiration (\$43) is higher than the strike price (\$42), this option is out of

the money, and Ahmet will not exercise it. Therefore, the payoff is \$0.

For the sold put option with a strike price of \$38:

Again, since the stock price at expiration (\$43) is higher than the strike price (\$38), this option is out of the money, and it will not be exercised by the option buyer. Therefore, the payoff for Ahmet is also \$0.

Given these results, the total payoff for the strategy is \$0 ($0 + 0$). This means Ahmet doesn't gain or lose anything from the options at expiration.

Things to Remember

- European put option: A type of financial derivative that gives the holder the right, but not the obligation, to sell a specific quantity of an underlying asset at a specified price within a specific time frame.
 - Strike price: The price at which the option holder can buy or sell the underlying asset when the option is exercised.
 - Out of the money option: An option that would lead to a negative cash flow if it were exercised immediately.
 - Payoff of an option: The amount of profit or loss that the option holder gains or loses at expiration.
 - Option buyer vs. option seller: The buyer of an option pays a premium to the seller for the right to exercise the option, while the seller receives the premium in exchange for taking on the obligation to potentially buy or sell the underlying asset.
-

Q.773 Phillip Harris is a senior arbitrageur investor at Dynamic Arbitrage Investment Co. He recently found out that if the value of a box spread is not equal to the present value of the payoff of the box spread, an investor could earn an arbitrage profit. He also found out that if the market value of a box spread is too high, it is profitable to sell the box spread. What positions should Harris take in call and put options to sell a box spread?

A. Harris must buy a European call option and buy a European put option with a specific strike price (X1), and simultaneously sell a European call option and sell a European put option with a higher strike price (X2).

B. Harris must sell a European call option and sell a European put option with a specific strike price (X1), and simultaneously buy a European call option and buy a European put option with a higher strike price (X2).

C. Harris must buy a European call option and sell a European put option with a specific strike price (X1), and simultaneously sell a European call option and buy a European put option with a higher strike price (X2).

D. Harris must sell a European call option and buy a European put option with a specific strike price (X1), and simultaneously buy a European call option and sell a European put option with a higher strike price (X2).

The correct answer is **D**.

To sell a box spread, an investor must sell a European call option and buy a European put option with a specific strike price (X1), and simultaneously buy a European call option and sell a European put option with a higher strike price (X2). This strategy is known as a box spread strategy. A box spread is a complex strategy that involves four options with the same expiration date but different strike prices. The goal of this strategy is to create a risk-free position that can potentially generate arbitrage profits. The box spread strategy is often used by sophisticated investors like Phillip Harris who have a deep understanding of options pricing and can identify mispriced options in the market. When the market value of a box spread is higher than the present value of its payoff, it indicates that the options are overpriced. By selling the box spread, the investor can lock in a guaranteed profit regardless of the future price movements of the underlying asset. This is because the payoff from the box spread at expiration will always be equal to the difference between the two strike prices, regardless of the price of the underlying asset. Therefore, if the investor can sell the box spread for more than this amount, they can earn an arbitrage profit.

Choice A is incorrect. This choice suggests buying both a call and a put option at strike price X1, and selling both a call and a put option at strike price X2. This strategy would actually create a long box spread position, not the short box spread position that Harris needs to profit from an inflated market value of the box spread.

Choice B is incorrect. Selling both a call and put option at strike price X1, while buying both at strike price X2 would also result in creating a long box spread position. Again, this is contrary to what Harris needs to do in order to exploit the arbitrage opportunity.

Choice C is incorrect. This choice suggests buying a call and selling a put at strike price X1, while selling another call and buying another put at higher strike price X2. However, this combination does not form any type of box spread strategy (neither long nor short), hence it cannot be used by Harris for his purpose.

Things to Remember

- Box spread strategy involves four options with the same expiration date but different strike prices
- The goal of a box spread strategy is to create a risk-free position that can potentially generate arbitrage profits

- Market value of a box spread being higher than the present value of its payoff indicates overpriced options
 - Arbitrage opportunities arise when the market value of a box spread is not equal to the present value of the payoff
 - Phillip Harris, as a senior arbitrageur investor, needs to understand options pricing and identify mispriced options in the market
-

Q.776 An investor is interested in a spread trading where he can sell a European call option and buy a European call. If the investor wishes for the expiration date of the long call to be greater than the short call, then which of the following is the strategy he is interested in?

- A. Bull spread
- B. Bear spread
- C. Butterfly spread
- D. Calendar spread

The correct answer is **D**.

A calendar spread, also known as a horizontal spread or a time spread, is a strategy that involves buying and selling two options of the same type (calls or puts), same underlying security, same strike price, but with different expiration dates. The investor sells an option with a near-term expiration date and buys an option with a longer-term expiration date. The strategy is named 'calendar spread' because it profits from the passage of time - the time decay of options. The near-term option that the investor sells will decay (lose value) faster than the long-term option that the investor buys. This differential decay rate is what allows the investor to potentially profit from a calendar spread.

In the context of the question, the investor is interested in selling a European call option and buying another European call option with a longer expiration date. This aligns perfectly with the definition and structure of a calendar spread, making choice D the correct answer.

Choice A is incorrect. A bull spread involves buying a call option with a certain strike price and selling another call option with a higher strike price, both having the same expiration date. This strategy is used when the investor expects a moderate rise in the price of the underlying asset.

Choice B is incorrect. A bear spread involves buying a put option and selling another put option with a lower strike price, both having the same expiration date. This strategy is used when an investor expects a moderate decline in the price of an underlying asset.

Choice C is incorrect. A butterfly spread involves positions in options with three different strike prices, all having the same expiration date. It can be created by purchasing one call (put) at the lowest (highest) strike, selling two calls (puts) at the middle strike and purchasing one final call (put) at the highest (lowest) strike.

Things to Remember

- Calendar spreads involve options with the same underlying security, same strike price, but different expiration dates
 - Calendar spreads profit from the time decay of options
 - Bull spreads involve buying a call option and selling another call option with a higher strike price
 - Bear spreads involve buying a put option and selling another put option with a lower strike price
 - Butterfly spreads involve options with three different strike prices and the same expiration date
-

Q.779 Suppose an individual investor has implemented a straddle trading strategy. The investor purchased a 6-month European call option with a strike price of \$58 on the stock of a specific firm for \$5, and simultaneously purchased a 6-month European put option on the stock of the same firm for \$4 with a strike price of \$58. If the price of the underlying stock after 6-month period is \$65, which of the following is closest to the profit of the straddle strategy?

- A. \$7
- B. \$0
- C. -\$2
- D. -\$9

The correct answer is C.

In a straddle strategy, the investor buys a European call option and put option with the same strike price and expiration. In the provided case, the investor purchased a 6-month European call option for \$5 with the strike price of \$58, and at the same time, purchased a 6-month put option for \$4 with a strike price of \$58.

The total cost of the straddle is the sum of the call premium and the put premium.

Cash outflow (cost) = \$5 + \$4 = \$9

Payoff on a put option = $(X - S, 0) = 0$

Payoff on a call option = $(S - X, 0) = (65 - 58) = \7

Profit of the straddle = \$7 - \$9 = -\$2

Things to Remember

- A straddle strategy involves buying both a call option and a put option with the same strike price and expiration date.
- The investor profits from a straddle strategy when there is a significant price movement in the underlying asset, regardless of the direction.
- The maximum loss for a straddle strategy is limited to the initial cost of purchasing both the call and put options.
- The breakeven points for a straddle strategy are the strike price plus the total cost of the options and the strike price minus the total cost of the options.
- Volatility plays a crucial role in the profitability of a straddle strategy, as higher volatility increases the likelihood of significant price movements.

Q.781 An investor has recently learned about spread trading strategies. To test one of the spread combinations, the investor purchased a 3-month European call option on stocks of Big Corp. with a strike price of \$101. At the same time, he also took a long position in two 3-month European put options on the stocks of Big Corp. with a strike price of \$101. Which of the following strategies is he most likely testing?

- A. Straddle strategy
- B. Butterfly strategy
- C. Strap strategy
- D. Strip strategy

The correct answer is **D**.

The investor is most likely testing a strip strategy. A strip strategy is a type of options trading strategy that involves purchasing a European call option and two European put options with the same strike prices and expiration dates. This strategy is typically used when an investor believes that the price of the underlying asset will experience significant volatility in the near future, and they believe that the price is more likely to fall than to rise. The two put options provide a higher level of protection against a price drop than the single call option provides potential for profit if the price rises. Therefore, the strip strategy is a bearish strategy that provides protection against downside risk while still allowing for profit if the price rises.

Choice A is incorrect. A straddle strategy involves buying a call and a put option on the same underlying asset with the same strike price and expiration date. In this case, the investor has bought two put options, not one, which does not align with a straddle strategy.

Choice B is incorrect. A butterfly spread strategy involves buying or selling multiple options of the same underlying asset with different strike prices but at the same expiration date. The investor in this scenario has purchased options with the same strike price, hence it cannot be a butterfly spread strategy.

Choice C is incorrect. A strap strategy involves buying two call options and one put option on the same underlying asset with identical strike prices and expiration dates. However, in this case, it's reversed - two put options are bought instead of two call options - so it doesn't match up to a strap strategy.

Things to Remember

- Straddle strategy involves buying a call and a put option on the same underlying asset with the same strike price and expiration date.
- Butterfly spread strategy involves buying or selling multiple options of the same underlying asset with different strike prices but at the same expiration date.

- Strap strategy involves buying two call options and one put option on the same underlying asset with identical strike prices and expiration dates.
 - Strip strategy involves purchasing a European call option and two European put options with the same strike prices and expiration dates.
 - Options trading strategies can be used to speculate on the direction of price movements, hedge against potential losses, or generate income.
-

Q.782 An investment manager has realized that there is a great potential for profits in the options market without tying up much capital. To test the potential of options trading, he implemented a spread strategy by purchasing two 6-month European call options on stocks of a specific firm with the strike price of X and, at the same time, buying a 6-month European put option on the stocks of the same firm with the same strike price.

Which strategy is he most likely using?

- A. Straddle strategy
- B. Strip strategy
- C. Strap strategy
- D. Strangle strategy

The correct answer is **C**.

A strap strategy is an options strategy that involves holding two calls and one put with the same strike price and expiration date. This strategy is used when the trader believes that the underlying stock will experience significant volatility in the near term. The two calls provide profit if the stock price increases, while the put limits the loss if the stock price decreases. In this case, the investment manager purchased two 6-month European call options and one 6-month European put option on the stocks of a specific firm with the same strike price. This aligns with the definition of a strap strategy, making choice C the correct answer.

Choice A is incorrect. A straddle strategy involves buying a call and put option with the same strike price and expiration date. This strategy is used when an investor believes there will be a large price movement but is unsure of the direction. In this case, the manager bought two call options and one put option, which does not align with a straddle strategy.

Choice B is incorrect. A strip strategy involves buying two put options and one call option with the same strike price and expiration date. This strategy is used when an investor believes that there will be a significant downward movement in the stock's price. However, in this scenario, the manager bought two call options and one put option which corresponds to a strap strategy rather than a strip.

Choice D is incorrect. A strangle strategy involves buying out-of-the-money call and put options with different strike prices but same expiration date. The aim of this approach is to profit from large movements in either direction of underlying asset's price while limiting potential losses if no such movement occurs. But here, all three options purchased by investment manager have same strike prices which doesn't match characteristics of strangle.

Things to Remember

- Options trading allows investors to leverage their capital and potentially earn higher returns compared to traditional stock trading.
 - European options can only be exercised at expiration, while American options can be exercised at any time before expiration.
 - Spread strategies involve taking positions in multiple options contracts to hedge risk or speculate on price movements.
 - Volatility plays a crucial role in options pricing, with higher volatility leading to higher option premiums.
 - Understanding different options strategies and their characteristics is essential for successful trading in the options market.
-

Q.783 An investment manager has realized that there is a great potential for profits in the options market without tying up much capital. To test the potential of options trading, he implemented one of the spread strategies by purchasing a 9-month European call option on the stocks of Petro Co. with a strike price of \$37, and at the same time, buying a 9-month European put option on the stocks of the same firm with a strike price of \$32.

Which of the following strategies is the investment manager most likely testing?

- A. Calendar spread strategy
- B. Straddle strategy
- C. Strip strategy
- D. Strangle strategy

The correct answer is **D**.

The investment manager is testing the Strangle strategy. In a Strangle strategy, an investor purchases a European call and a European put option on the stock of a specific firm. These

options have the same expiration date but different strike prices. This strategy is used when the investor believes that the stock price will experience significant movement but is unsure of the direction. The call option allows the investor to profit if the stock price rises above the strike price, while the put option provides profit if the stock price falls below the strike price. The maximum loss for the investor is the total premiums paid for the options, which occurs if the stock price at expiration is between the strike prices of the call and put options. The potential profit is unlimited on the upside and significant on the downside.

Choice A is incorrect. A calendar spread strategy involves buying and selling the same type of options (calls or puts) on the same underlying asset with the same strike price but different expiration dates. In this case, both options are for 9 months and have different strike prices, so it cannot be a calendar spread strategy.

Choice B is incorrect. A straddle strategy involves buying a call and put option on the same underlying asset with the same strike price and expiration date. Here, although both options are for 9 months, they have different strike prices (\$37 for call option and \$32 for put option), hence it does not represent a straddle strategy.

Choice C is incorrect. A strip strategy involves buying one at-the-money call option and two at-the-money put options on the same underlying asset with identical expiration dates. In this scenario, only one call option and one put option were purchased with differing strike prices which does not align with a strip strategy.

Things to Remember

- Options trading allows investors to leverage their capital and potentially earn high profits with limited risk.
- European options can only be exercised at expiration, while American options can be exercised at any time before expiration.
- Strike price is the price at which the option holder can buy (call option) or sell (put option) the underlying asset.
- Calendar spread strategy involves options with the same underlying asset, same strike price, but different expiration dates.
- Straddle strategy involves buying a call and put option on the same underlying asset with the same strike price and expiration date.
- Strip strategy involves buying one call option and two put options on the same underlying asset with identical expiration dates.
- Strangle strategy involves buying a call and put option on the same underlying asset

with the same expiration date but different strike prices.

Q.4623 A portfolio X consists of a five-year zero-coupon bond and a five-year call option on portfolio Y. The current price of portfolio Y is \$20,000, and the strike price of the option is also \$20,000. The interest rate is 7% per annum. To ensure no losses to a trader while still providing the trader with room for profits, the premium paid to secure the call option should cost less than:

- A. 20000
- B. 5740.28
- C. 15740.28
- D. 14259.72

The correct answer is **B**.

Principal Protected Notes act by reducing losses while still providing room for potential gains.

To hedge against losses, the trader should buy a zero-coupon bond that will yield, at maturity, the \$20,000 needed to exercise the option.

Therefore, the price of the bond should be equal to the present value of the strike price.

$$\begin{aligned}PV &= 20,000(1 + 0.07)^{-5} \\ &= 14,259.72\end{aligned}$$

To make the portfolio profitable, the premium paid to secure the call option should cost less than:

$$\$20,000 - \$14,259.72 = \$5,740.28$$

Q.4624 With respect to interest rates, when are Principal Protected Notes (PPNs) most profitable?

- A. When interest rates increase
- B. When interest rates decrease
- C. When interest rates remains constant
- D. When interest rates are volatile

The correct answer is **A**.

Principal Protected Notes (PPNs) are most profitable when interest rates increase. This is because the profitability of PPNs is tied to the price of the option, which is less than $K - PV(K)$, where K is the strike price needed to purchase the underlying asset in the portfolio. As interest rates increase, the present value (PV) of K decreases, leading to a larger difference between K and $PV(K)$. This larger difference translates into higher profits for PPNs. The following table illustrates this concept:

Interest rates	Time (years)	K	PV(K)	$K - PV(K)$
0%	3	100	100	0
5%	3	100	86.38	13.62
10%	3	100	75.13	24.87

As shown in the table, the difference between K and $PV(K)$ increases as the interest rate increases, leading to higher profits for PPNs.

Choice B is incorrect. When interest rates decrease, the profitability of PPNs does not necessarily increase. This is because lower interest rates mean that the cost of borrowing for companies decreases, which can lead to an increase in corporate bond prices. However, this does not directly translate into higher profits for PPNs as they guarantee only the return of the principal amount and not any potential gains from increased bond prices.

Choice C is incorrect. If interest rates remain constant, it doesn't necessarily yield higher profits for PPNs. The profitability of PPNs depends on how well the underlying investment performs and not solely on stable interest rates.

Choice D is incorrect. Volatile interest rates do not guarantee higher profits for Principal Protected Notes (PPNs). In fact, volatility in interest rates can introduce uncertainty and risk into the performance of these notes as it affects both their cost structure and their returns from underlying investments.

Q.4625 The strike price of a three-month call and a three-month put option with the same time to maturity is \$50. The cost of the call option is \$4, whereas the cost of the put option is \$6. Using a short straddle strategy of trading, by how much should the asset price move in order to incur a loss?

- A. 4
- B. 6
- C. 2
- D. 10

The correct answer is **D**.

The question is testing the use of the short straddle trading strategy, a trading strategy that involves selling a call and a put option with the same strike price and time to maturity.

Strike price = \$50

Benefit of setting up the short straddle = \$4 + \$6 = \$10

Upper price bound of the asset price at maturity = \$50 + \$10 = \$60

Lower price bound of the asset price at maturity = \$50 - \$10 = \$40

Therefore, for a loss to be incurred, the asset price at maturity should be either above \$60 or below \$40.

Things to Remember

- Short straddle strategy involves selling a call option and a put option with the same strike price and time to maturity
 - Maximum profit for a short straddle is the sum of the premiums received for selling the call and put options
 - Maximum loss for a short straddle is unlimited as the asset price can move significantly in either direction
 - Short straddle strategy is used when the trader expects low volatility in the underlying asset
-

Q.4626 Consider two call options with strike prices of \$30 and \$35 and two put options with

strike prices \$30 and \$35. How can a trader create a bear spread trading strategy using two options?

- A. Buy the put option with a strike price of \$30 and sell the put option with a strike price of \$35.
- B. Buy the put option with a strike price of \$35 and sell the put option with a strike price of \$30.
- C. Buy the put option with a strike price of \$30 and sell the call option with a strike price of \$35.
- D. Buy the put option with a strike price of \$35 and sell the call option with a strike price of \$30.

The correct answer is **B**.

A bear spread strategy can be created by buying a put option with a higher strike price and selling a put option with a lower strike price. In this case, the trader would buy the put option with a strike price of \$35 and sell the put option with a strike price of \$30. This strategy is designed to profit from a decline in the price of the underlying asset. If the price of the asset falls below \$30, the trader can exercise the \$35 put option, selling the asset for \$35. Meanwhile, the \$30 put option that the trader sold will either expire worthless (if the price of the asset is above \$30) or will be exercised by its owner, requiring the trader to buy the asset for \$30. In either case, the trader profits from the difference between the strike prices of the two options.

Choice A is incorrect. Buying a put option with a strike price of \$30 and selling a put option with a strike price of \$35 would not create a bear spread. This strategy would actually result in an unlimited potential loss if the underlying asset's price increases, which is contrary to the objective of a bear spread strategy that aims to profit from declining prices.

Choice C is incorrect. Buying the put option with a strike price of \$30 and selling the call option with a strike price of \$35 does not constitute as constructing a bear spread strategy. This combination could lead to significant losses if the underlying asset's price rises above \$35, as there would be an obligation to sell at this lower strike price while having only limited potential profit from exercising the bought put at \$30.

Choice D is incorrect. Buying the put option with a strike price of \$35 and selling the call option with a strike price of \$30 also does not form an effective bear spread strategy. In this case, if prices rise above 30\$, trader will have unlimited loss due to sold call while limited gain due to bought put at 35\$.

Things to Remember

- A bear spread strategy involves buying a put option with a higher strike price and selling a put option with a lower strike price to profit from a decline in the price of the underlying asset.

- Options trading strategies like bear spreads are used by traders to manage risk and potentially profit from market movements.
 - Understanding the relationship between strike prices and the direction of the market is crucial in selecting the appropriate options trading strategy.
 - Risk management is essential in options trading to protect against potential losses and maximize potential gains.
-

Q.4627 Consider two call options with strike prices of \$30 and \$35 and two put options with strike prices of \$30 and \$35. How can a trader create a bull spread trading strategy using two options?

- A. Buy the call option with a strike price of \$30 and sell the call option with a strike price of \$35.
- B. Buy the call option with a strike price of \$35 and sell the call option with a strike price of \$30.
- C. Buy the put option with a strike price of \$30 and sell the call option with a strike price of \$30.
- D. Buy the put option with a strike price of \$35 and sell the call option with a strike price of \$35.

The correct answer is **A**.

A bull spread strategy is typically constructed by buying a call option with a lower strike price and selling a call option with a higher strike price. This strategy is used when the trader expects a moderate increase in the price of the underlying asset. In this case, buying the call option with a strike price of \$30 and selling the call option with a strike price of \$35 creates a bull spread. The maximum profit in this strategy is the difference between the two strike prices minus the net premium paid. The maximum loss is limited to the net premium paid for the options. This strategy provides a trader with a way to profit from a moderate rise in the price of the underlying asset while limiting potential losses.

Choice B is incorrect. Buying a call option with a higher strike price and selling a call option with a lower strike price would create a bear spread, not a bull spread. This strategy would be profitable if the trader expects the price of the underlying asset to fall, not rise.

Choice C is incorrect. Buying a put option and selling a call option both with the same strike price does not create any kind of spread strategy. This combination of options is known as 'straddle' which is used when high volatility in either direction is expected in the market.

Choice D is incorrect. Similar to choice C, buying and selling options (whether calls or puts) at the same strike price does not result in any type of spread strategy but rather creates another form of straddle or strangle depending on whether they are at-the-money or out-of-the-money respectively.

Things to Remember

- A bull spread strategy involves buying a call option with a lower strike price and selling a call option with a higher strike price.
 - The maximum profit in a bull spread strategy is the difference between the two strike prices minus the net premium paid.
 - The maximum loss in a bull spread strategy is limited to the net premium paid for the options.
 - A bear spread strategy is the opposite of a bull spread, involving buying a call option with a higher strike price and selling a call option with a lower strike price.
 - A straddle strategy involves buying a put option and a call option with the same strike price, used when high volatility in either direction is expected in the market.
 - A strangle strategy involves buying a put option and a call option with different strike prices, used when moderate volatility is expected in the market.
-

Q.4873 An investor creates a bull put spread by purchasing a put option for a premium of \$25. The put option comes with a strike price of \$95 and expires in July 2022. At the same time, the investor sells a put option for a premium of \$50. The put option comes with a strike price of \$140 and expires in July 2022. The underlying asset is the same and is currently trading at \$145. Determine the maximum loss.

- A. 40
- B. 20
- C. 25
- D. 15

The correct answer is **B**.

$$\begin{aligned}\text{Max loss} &= \text{strike price of short put} - \text{strike price of long put} - \text{net premium received} \\ &= 140 - 95 - (50 - 25) = 20\end{aligned}$$

Note: A bull put spread is a strategy utilized by an investor when they believe the underlying stock will exhibit a moderate increase in price. It involves purchasing an out-of-the-money (OTM) put option and selling an in-the-money (ITM) put option with a higher strike price but with the same underlying asset and expiration date. A bull put spread works when the market is exhibiting an upward trend. The maximum gain is the net premium received.

Q.4912 Peter creates a bear spread using put options with strike prices of USD 35 and USD 40 with the same time to maturity. The options cost USD 3 and USD 5, respectively under what circumstances will Peter make a profit?

- A. If the asset price is less than 40.
- B. If the asset price is less than 38.
- C. If the asset price lies between 38 and 40.
- D. If the asset price is greater than 38.

The correct answer is **B**.

If the asset price is less than 38, Peter will make a profit. The cost of setting up the bear spread is the net cost, which is the difference between the cost of the two options, i.e., $\text{USD } 5 - \text{USD } 3 = \text{USD } 2$. When the asset price is less than 35, the payoff is $40 - 35 = \text{USD } 5$. After subtracting the cost of setting up the bear spread, we have a profit of $\text{USD } 5 - \text{USD } 2 = \text{USD } 3$. When the asset price lies between 35 and 40, the payoff is $40 - S$ (where S is the asset price). This payoff must be greater than 2 (the cost of setting up the bear spread) in order to make a profit. Therefore, when

the asset price is less than 38, we have a profit of $40 - S - 2$. Thus, Peter will make a profit when the asset price is less than 38.

Choice A is incorrect. Although the asset price being less than 40 would result in both options being in-the-money, it does not guarantee a profit for Peter. The total cost of the options is USD 8 (USD 3 + USD 5). Therefore, to realize a profit, the asset price must be sufficiently below USD 40 such that the combined payoff from both options exceeds this cost.

Choice C is incorrect. If the asset price lies between USD 38 and USD 40 at expiration, only one of Peter's put options (the one with a strike price of USD 40) will be in-the-money. The payoff from this option may not cover the total cost of both options (USD 8), resulting in a loss for Peter.

Choice D is incorrect. If the asset price is greater than USD 38 at expiration, neither of Peter's put options will be in-the-money and they will expire worthless. This means that he would lose his entire initial investment on purchasing these two put options.

Things to Remember

- Bear spread involves buying a put option with a higher strike price and selling a put option with a lower strike price.
 - The net cost of setting up a bear spread is the price difference between the two options.
 - Profit from a bear spread occurs when the asset price is below the higher strike price but above the lower strike price.
 - Maximum profit for a bear spread is achieved when the asset price is below the lower strike price at expiration.
 - Maximum loss for a bear spread is limited to the initial net cost of setting up the spread.
-

Q.4913 Hassan intends to create a bull spread on put options on an asset with strike prices of USD 20 and USD 25 and the same time to maturity. The options cost USD 2 and USD 2.5, respectively. How can Hassan create the intended bull spread?

- A. Buying the option with a lower strike price and simultaneously selling the put option with a higher strike price.
- B. Buying the option with a higher strike price and simultaneously selling the put option with a lower strike price.
- C. Buying the option with a lower strike price, and at the same time, buying the put

option with a higher strike price.

D. Selling the option with a lower strike price, and at the same time, selling the put option with a higher strike price.

The correct answer is **A**.

A bull spread is a type of options strategy that is used when the investor expects a moderate rise in the price of the underlying asset. This strategy is created by buying an option with a lower strike price and simultaneously selling an option with a higher strike price. In the context of this question, Hassan can create a bull spread by buying the put option with a strike price of USD 20 (which costs USD 2) and selling the put option with a strike price of USD 25 (which costs USD 2.5). The lower strike price put option (USD 20) is bought because Hassan expects the price of the underlying asset to increase. On the other hand, the higher strike price put option (USD 25) is sold to offset the cost of buying the lower strike price put option. This strategy allows Hassan to profit from a moderate increase in the price of the underlying asset while limiting his potential loss to the net premium paid for the options (i.e., the cost of the bought option minus the income from the sold option).

Choice B is incorrect. Buying the option with a higher strike price and simultaneously selling the put option with a lower strike price would not create a bull spread. This strategy would actually result in a bear spread, as it profits when the asset's price decreases.

Choice C is incorrect. Buying both options, regardless of their strike prices, does not create a bull spread either. This strategy is known as a straddle or strangle, depending on whether the strike prices are the same or different, respectively, which aims to profit from large movements in either direction and not specifically from an increase in an asset's price.

Choice D is incorrect. Selling both options does not create a bull spread; instead, it creates a neutral or undefined position, potentially exposing Hassan to unlimited risk if the market moves unfavorably..

Things to Remember

- A bull spread is a type of options strategy used when the investor expects a moderate rise in the price of the underlying asset.
- A bull spread involves buying an option with a lower strike price and simultaneously selling an option with a higher strike price.
- Options strategies like straddles and strangles are used for profiting from large movements in either direction, not specifically from an increase in an asset's price.
- Selling both options creates a neutral or undefined position, potentially exposing the investor to unlimited risk if the market moves unfavorably.

Q.4914 Suppose a strangle is created from a call option with a strike price of USD 35, which costs USD 1, and a put option with a strike price of USD 20, which costs USD 2. The two options have the same time to maturity. What is the profit as a function of the asset price, S_T at option maturity, when $S_T \leq 20$?

- A. 0
- B. $17 - S_T$
- C. -3
- D. $S_T - 38$

The correct answer is **B**.

When $S_T \leq K_1$, then the payoff from a long call with a strike price $K_2 = 35$, will be,

$$\max(0, S_T - 35) = 0$$

and the payoff from a long put option with a strike price $K_1 = 20$, will be,
 $\max(0, 20 - S_T) = 20 - S_T$

But the total cost of setting up the strategy is USD 3; thus, the profit is given by,

$$20 - S_T - 3 = 17 - S_T$$

Q.4915 A straddle is created from a call and a put with the same strike price of USD 45. If the options have the same time of expiration and that the cost of setting up the straddle is 5, what conditions will lead to a profit on the straddle?

- A. If $S_T > 50$ or if $S_T < 40$
- B. If $S_T < 40$
- C. $S_T < 10$
- D. $10 < S_T < 50$

The correct answer is **A**.

A straddle is created by buying the call and buying the put.

In a straddle strategy, a profit can be made if $S_T > K + C$ or $S_T < K - C$ where S_T is the asset price at time T, K is the strike price, and C is the cost of setting up the straddle.

In our case, a profit is made if, $S_T > 50$ or if $S_T < 40$

Note that a straddle involves two transactions on the same security, with positions that offset one another.

A long straddle is created by purchasing a call and a put with the same strike price and expiration. A short straddle is created by selling a call and a put with the same strike price and expiration.

Things to Remember

- Straddle strategy involves buying a call option and a put option with the same strike price and expiration date.
 - A long straddle is profitable if the underlying asset price at expiration is either significantly higher or lower than the strike price plus the cost of setting up the straddle.
 - A short straddle is profitable if the underlying asset price at expiration remains relatively stable and close to the strike price.
 - Implied volatility plays a crucial role in the profitability of a straddle strategy, as it affects the potential price movements of the underlying asset.
 - Straddles are used by investors and traders to profit from significant price movements in either direction, without having to predict the direction of the movement.
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Reading 41: Exotic Options

Q.784 Susanne Alexander is a junior investment analyst at JCB Investment Bank in Tokyo. She has no prior experience in trading equities and derivatives. She has been assigned to the trading unit of the bank where she is taking her initial training from the senior management of the trading unit. In one of the training sessions, she was asked to identify the option that most likely trades in the over-the-counter options market. Identify for Alexander the correct option.

- A. Plain vanilla options
- B. Covered calls
- C. European options
- D. Exotic options

The correct answer is **D**.

Exotic options are a type of derivative that are more complex than the commonly traded 'vanilla' options. They are called 'exotic' because they are more specialized and less commonly traded than the standard options. Exotic options are customized to meet the specific needs of the investor, which can include different payoff structures, cost structures, or expiration dates. Because of their complexity and customization, exotic options are typically traded in over-the-counter (OTC) markets rather than on standardized exchanges. OTC markets allow for greater flexibility and negotiation between the buyer and seller, which is necessary for the trading of these complex and customized options. Therefore, Susanne Alexander should identify exotic options as the type of option that is most likely to be traded in the OTC market.

Choice A is incorrect. Plain vanilla options are not the most likely to be traded in the over-the-counter options market. These types of options have standard features and are usually traded on organized exchanges, not over-the-counter.

Choice B is incorrect. Covered calls are a strategy that involves owning the underlying asset and selling call options on that same asset. This strategy is typically used by investors who own a stock and want to generate additional income from it, rather than being a type of option commonly traded in the over-the-counter market.

Choice C is incorrect. European options can only be exercised at expiration, which makes them less flexible than American options which can be exercised at any time before expiration. While European options do trade in over-the-counter markets, they are not the most likely type of option to be traded there.

Things to Remember

- Over-the-counter (OTC) markets allow for greater flexibility and negotiation between the buyer and seller
- Exotic options are more specialized and less commonly traded than standard options

- Exotic options are customized to meet specific investor needs
 - Plain vanilla options have standard features and are usually traded on organized exchanges
 - Covered calls involve owning the underlying asset and selling call options on that same asset
 - European options can only be exercised at expiration
-

Q.787 Ganesh Singh is a junior options trader at an Indian brokerage house. He was recently promoted from the equity division to the derivatives unit of the firm. One of her clients asked Singh to take a long position on his behalf in a Bermudan option. Since Singh is unfamiliar with Bermudan options, describe the unique feature of these types of options.

- A. A Bermudan option is a non-standard European option, which can be exercised any time until its expiration.
- B. A Bermudan option is a non-standard American option, which can be exercised any time until its expiration.
- C. A Bermudan option is a non-standard European option, which can only be exercised at certain dates until its expiration.
- D. A Bermudan option is a non-standard American option, which can only be exercised at certain dates until its expiration.

The correct answer is **D**.

A Bermudan option is a non-standard American option, which can only be exercised at certain dates until its expiration. This type of option is a hybrid of American and European options. Like American options, Bermudan options can be exercised before the expiration date. However, unlike American options, which can be exercised at any time before expiration, Bermudan options can only be exercised on specific dates stipulated in the contract. This feature makes Bermudan options more flexible than European options but less flexible than American options. The pricing of Bermudan options is typically more complex due to this unique feature. It's important to note that the premiums for Bermudan options are lower than those of American options, but they are more expensive than European options.

Choice A is incorrect. A Bermudan option is not a non-standard European option that can be exercised any time until its expiration. In fact, European options can only be exercised at the time of expiration, not anytime before it.

Choice B is incorrect. While it's true that American options can be exercised any time until

their expiration, this does not define a Bermudan option. The distinctive feature of Bermudan options is that they can only be exercised at certain dates until their expiration.

Choice C is incorrect. Although this choice correctly states that Bermudan options can only be exercised at certain dates until their expiration, it incorrectly classifies them as non-standard European options when they are actually closer to American style in terms of exercise rights.

Things to Remember

- European options can only be exercised at the expiration date.
 - American options can be exercised at any time until their expiration date.
 - Bermudan options are a hybrid of American and European options.
 - Bermudan options can only be exercised on specific dates stipulated in the contract.
 - The pricing of Bermudan options is more complex due to their unique exercise feature.
 - Premiums for Bermudan options are lower than American options but higher than European options.
-

Q.789 A gap option is a non-standard option that is created with a European call option. However, the European call option used in the construction of a gap option is different from the regular European call option. Which of the following is the accurate difference between a gap European call option and a regular call option?

- A. A gap option has two strike prices i.e. X_1 and X_2 (where $X_2 > X_1$). When the final stock price is greater than X_2 ($S > X_2$), the payoff of the gap call option is $S - X_1$.
- B. A gap option has two strike prices i.e. X_1 and X_2 (where $X_2 > X_1$). When the final stock price is greater than X_2 ($S > X_2$), the payoff of the gap call option is $S - X_2$.
- C. A gap option has two strike prices i.e. X_1 and X_2 (where $X_2 > X_1$). When the final stock price is greater than X_1 ($S > X_1$), the payoff of the gap call option is $S - X_1$.
- D. A gap option has two strike prices i.e. X_1 and X_2 (where $X_2 > X_1$). When the final stock price is greater than X_1 ($S > X_1$), the payoff of the gap call option is $S - X_2$.

The correct answer is **A**.

A gap option is a type of derivative that differs from a regular European call option in its structure and payoff mechanism. A regular European call option has a single strike price, X . If the final price of the underlying asset is greater than X , the payoff of the option is $S - X$, where S

is the final price of the underlying asset. However, a gap option has two strike prices: X_1 and X_2 , where X_2 is greater than X_1 . The payoff of a gap call option is triggered when the final price of the underlying asset is greater than X_2 . In this case, the payoff is $S - X_1$. This means that the holder of the gap option will receive the difference between the final price of the underlying asset and the lower strike price (X_1), provided that the final price of the asset is greater than the higher strike price (X_2). This structure allows the holder of the gap option to potentially realize a higher payoff than with a regular call option, depending on the final price of the underlying asset.

Choice B is incorrect. The payoff of a gap call option is not $S - X_2$ when the final stock price is greater than X_2 . In a gap option, the payoff is determined by subtracting the lower strike price (X_1) from the final stock price (S), not the higher strike price (X_2).

Choice C is incorrect. This statement incorrectly suggests that when the final stock price exceeds X_1 , then there will be a payoff for a gap call option. However, in reality, for a gap call option to have any value at expiration, it requires that the underlying asset's final price must exceed X_2 .

Choice D is incorrect. Similar to choice B, this statement incorrectly states that if $S > X_1$ then payoff would be $S - X_2$ which contradicts with how Gap options work as explained above.

Things to Remember

- Regular European call options have a single strike price, while gap options have two strike prices (X_1 and X_2).
- The payoff of a regular European call option is $S - X$, where S is the final price of the underlying asset and X is the strike price.
- For a gap call option, the payoff is triggered when the final price of the underlying asset is greater than X_2 , and the payoff is $S - X_1$.
- Gap options provide the holder with the potential for higher payoff compared to regular call options, depending on the final price of the underlying asset.
- Understanding the difference in payoff structures between regular call options and gap options is crucial for pricing and trading these derivatives.

Q.790 Hannah Bruce is a derivatives investment adviser at Dot Investments in New York. She provides advisory services to retail as well as institutional investors. One of her clients, a small size community insurance company, intended to invest in equities option that starts at some future date and expires at an expiration date further in the future. Which of the following options should Bruce recommend?

- A. A European gap option
- B. An employee option
- C. A futures option
- D. A forward start option

The correct answer is **D**.

A forward start option is the most suitable choice for the client's needs. This type of option is a non-standard option that allows the holder to exercise the option starting from a future date (T1) and ending at another future date (T2). This type of option is often used in employee stock options, where the employer commits to grant an at-the-money option at a future date. In this case, the small community insurance company can benefit from the flexibility of choosing when to exercise the option within the specified time frame. This can provide the company with the opportunity to maximize its returns based on market conditions during the option period.

Choice A is incorrect. A European gap option is a type of exotic option that has a strike price, but the payoff depends on the difference between the asset's price at expiration and a 'gap' price. This does not align with the company's need for an option that begins at a future date and expires at a later date.

Choice B is incorrect. An employee option, typically part of an employee stock ownership plan (ESOP), allows employees to buy shares of their employer's company at a predetermined price. This type of option does not meet the requirement specified by the insurance company.

Choice C is incorrect. A futures option gives its owner the right to buy or sell a futures contract on an underlying asset at a specific price before it expires, but it doesn't necessarily start in future as required by this scenario.

Things to Remember

- Options are financial derivatives that give the holder the right, but not the obligation, to buy or sell an underlying asset at a specified price on or before a specified date.
- European options can only be exercised at expiration, while American options can be exercised at any time before expiration.
- Exotic options, such as gap options, have non-standard features that differentiate them from traditional options like European or American options.
- Employee stock options are a common form of compensation used by companies to attract and retain talent.
- Futures options are options on futures contracts, which are agreements to buy or sell

an asset at a predetermined price on a specified date in the future.

- Forward start options allow the holder to start exercising the option at a future date, providing flexibility in timing for the option holder.
-

Q.791 Gabriela Clarke is a senior derivatives investment manager at one of the largest investment banks in London. She specializes in constructing complex exotic options for her clients. Currently, she is investing in a series of call options with a strategy in which she purchases an option with the strike price of K and expiry date of T1. She then invests in another option that starts at T1 and expires at T2. This option will have a strike price equal to the price of the underlying at T1. She invests in many such options with the same strategy, where one option starts as the last option expires. The series of such options is called a:

- A. Cliquet option
- B. Gap option
- C. Forward start option
- D. Futures option

The correct answer is **A**.

The series of options described in the question is known as a Cliquet option. A Cliquet option is an exotic option that consists of a series of consecutive forward start options. The first option in the series is active immediately, and the subsequent options become active as the previous one expires. Each option is struck at the money when it becomes active. This means that the strike price of each option is equal to the market price of the underlying asset at the time the option becomes active. This type of option allows investors to lock in gains periodically, providing protection against significant market downturns. However, it also limits the potential upside if the market performs exceptionally well. The Cliquet option is particularly useful in volatile markets, where the price of the underlying asset is expected to fluctuate significantly over the life of the option series.

Choice B is incorrect. A Gap option is a type of exotic option where the payoff depends on the difference between the asset price at the time of exercise and a reference price, which could be either the price at which it was bought or another specified price. This does not match with Gabriela's strategy where each new option starts as soon as previous one expires.

Choice C is incorrect. A Forward start option is an options contract that begins at a predetermined date in future. The strike price of this type of options contract is determined after its inception, typically based on prevailing market prices at that time. While Gabriela's strategy involves starting new options after previous ones expire, it doesn't involve pre-determining future dates for these options to start.

Choice D is incorrect. Futures Option gives its holder the right but not obligation to buy or sell a futures contract at a specific price before expiration, while Gabriela's strategy involves buying call options with different expiry dates and strike prices equivalent to underlying asset's price when previous option expires.

Things to Remember

- Cliquet options are also known as ratchet options.
 - These options are popular in structured products and are used for capital protection and periodic gains locking.
 - Investors can benefit from Cliquet options in volatile markets where there is expected fluctuation in the underlying asset's price.
 - Understanding the mechanics of exotic options like Cliquet options is crucial for derivatives investment managers like Gabriela Clarke.
-

Q.792 Jiao Bu is a Chinese retail investor who has recently moved to the United States. Bu mistakenly invested in an exotic option that has two strike prices and two exercise dates. On the first exercise date T1, Jiao is entitled to pay the first strike price of X1 and receive a call option, which will give her the right to purchase the underlying asset for the second strike price of X2 on the second exercise date T2. In which of the following exotic options has she mistakenly invested?

- A. Compound option
- B. Cliquet option
- C. Forward start option
- D. Barrier option

The correct answer is **A**.

A compound option is an option on an option that has two strike prices and two expiration dates. On the first exercise date T1, if the current price of the underlying asset is above X1, Bu is entitled to pay the first strike price of X1 and receive a call option, which will give her the right to purchase the underlying asset for the second strike price of X2 on the second exercise date T2. This is called a call on-call compound option. Compound options can also be call on put, put on call, and put on put options. Compound options are often used in markets where uncertainty is high, such as in the commodities and foreign exchange markets. They provide investors with additional flexibility to manage their risk exposure and can be tailored to meet specific

investment objectives.

Choice B is incorrect. A Cliquet option, also known as a ratchet or ladder option, is an exotic option where the payoff is determined by the difference in the asset price at various points in time. It does not involve two distinct strike prices and two separate exercise dates like in Jiao Bu's case.

Choice C is incorrect. A Forward start option is an option that starts at a future date with its strike price determined at that future date based on some reference point. This does not match with Jiao Bu's situation where there are two distinct strike prices and two separate exercise dates.

Choice D is incorrect. A Barrier option becomes active or inactive if the price of the underlying asset crosses a certain level (the barrier). This type of exotic options doesn't have features like multiple strike prices and exercise dates as described in Jiao Bu's investment scenario.

Things to Remember

- Compound options are options on options that have two strike prices and two expiration dates.
- Compound options can be call on call, call on put, put on call, or put on put options.
- Compound options provide investors with additional flexibility to manage their risk exposure.
- Cliquet options, also known as ratchet or ladder options, have payoffs determined by the difference in asset prices at various points in time.
- Forward start options start at a future date with the strike price determined at that future date based on a reference point.
- Barrier options become active or inactive based on whether the price of the underlying asset crosses a certain level (the barrier).

Q.794 Adam McGill is a hedge fund manager who is interested in purchasing an exotic option on the stock of Turkish Airlines stocks. Turkey is currently holding a referendum and the referendum results will either have a significantly positive or negative impact on the tourism industry as well as the Turkish Airlines stocks. Since Adam is not sure about the direction of the prices of the stocks, he intends to purchase an option that gives him the right to decide if the option is a call or a put at a specific date. Determine which of the following options is suitable for him.

- A. Call on call compound option
- B. Chooser option
- C. Ratchet option
- D. Forward start option

The correct answer is **B**.

Choice A is incorrect. A call on call compound option gives the holder the right to buy another call option at a specific future date. However, it does not provide the flexibility to choose between a call or put option, which is what Adam requires.

Choice C is incorrect. A ratchet option adjusts its strike price at specified periods based on the underlying asset's performance. While this type of exotic option provides some level of flexibility, it does not allow Adam to decide whether the option is a call or put at a specific future date.

Choice D is incorrect. A forward start option begins at a future date with an exercise price determined then. Although this type of exotic option offers some degree of uncertainty management, it still doesn't provide Adam with the ability to choose between a call and put position as required.

Things to Remember

- Exotic options are non-standard options with complex features, often tailored to specific investor needs.
- Chooser options allow the holder to decide at a specific future date whether the option will be a call or put.
- Risk management in exotic options involves understanding the potential outcomes and designing strategies to mitigate risks.
- Exotic options can be used for speculation, hedging, or arbitrage purposes.
- Understanding the payoff structure and characteristics of exotic options is crucial for effective decision-making.

Q.796 A type of barrier option in which a regular put option on the underlying asset comes into existence when the price of the underlying asset reaches a specific barrier, which is set equal to or above its initial level is called a:

- A. Down-and-in put option.
- B. Down-and-out put option.
- C. Up-and-in put option.
- D. Up-and-out put option.

The correct answer is **C**.

An up-and-in put option is a type of barrier option that comes into existence when the price of the underlying asset reaches a specific barrier, which is set equal to or above its initial level. This means that the put option is activated or 'knocked in' when the price of the underlying asset rises to the barrier level. The holder of this option gains the right to sell the asset at the strike price if the asset's price rises to the barrier level. This type of option is used by investors who believe that the price of the underlying asset will rise to the barrier level and then fall. The up-and-in put option provides a safety net, allowing the investor to limit potential losses if their prediction is correct.

Choice A is incorrect. A Down-and-in put option becomes active when the price of the underlying asset falls to a predetermined barrier level. This is not consistent with the scenario described in the question where the barrier is set at a level equal to or higher than the initial level of the asset.

Choice B is incorrect. A Down-and-out put option ceases to exist when the price of an underlying asset falls to a certain barrier level, which contradicts with our scenario where activation occurs when reaching a certain threshold.

Choice D is incorrect. An Up-and-out put option becomes inactive once it reaches or exceeds its barrier, which does not align with our case where activation happens upon reaching or exceeding a specific threshold.

Things to Remember

- **Barrier Options:** Barrier options are a type of exotic option that comes into existence or ceases to exist based on the price of the underlying asset reaching a certain barrier level.
- **Knock-in Option:** A knock-in option is an option that becomes active only if the price of the underlying asset reaches a certain barrier level.
- **Knock-out Option:** A knock-out option is an option that ceases to exist if the price of the underlying asset reaches a certain barrier level.
- **Barrier Level:** The barrier level is the specific price level set for the underlying asset, which determines the activation or deactivation of the barrier option.

- Exotic Options: Exotic options are non-standard options with complex features, such as barriers, that differentiate them from traditional options like vanilla options.
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Q.798 Hakim Ahmed is a junior derivatives trader who has recently started trading exotic options. A week ago, he purchased an exotic option that pays off nothing if the price of the underlying asset exceeds the strike price at a predetermined date and pays a fixed amount if the underlying asset price is below the strike price. Which of the following options has he purchased?

- A. Asset-or-nothing call binary option.
- B. Asset-or-nothing put binary option.
- C. Cash-or-nothing call binary option.
- D. Cash-or-nothing put binary option.

The correct answer is **D**.

A cash-or-nothing put is a type of binary option that pays a fixed amount if the price of the asset is below the strike price at the expiration date and nothing if the price is above the strike price at expiry.

Option A is incorrect: An asset-or-nothing call pays off the value of the underlying asset if the price of the asset is above the strike price at expiration, and nothing if the asset price is less than the strike price.

Option B is incorrect: An asset-or-nothing put binary option pays an amount equal to the asset price if the price of the underlying asset is below the strike price at the expiration date and nothing if the price is above the strike price at expiry.

Option C is incorrect: A cash-or-nothing call pays a fixed amount of cash if the price of the underlying asset is above the strike price at expiration, and pays nothing if the asset price is below the strike price.

Things to Remember

- Binary options are a type of option where the payoff is either a fixed amount or nothing at all.
- Asset-or-nothing options pay off the value of the underlying asset, while cash-or-nothing options pay a fixed amount of cash.
- Binary options are often used for hedging or speculation in financial markets.

- Exotic options have complex payoff structures compared to standard options like vanilla options.
 - Understanding the payoff structure of exotic options is crucial for traders to make informed decisions.
-

Q.799 In which of the following exotic options is the payoff the final current price of the asset minus the minimum or lowest asset price that the underlying asset has achieved during the life of the option?

- A. Floating lookback call option.
- B. Floating lookback put option.
- C. Fixed lookback call option.
- D. Fixed lookback put option.

The correct answer is **A**.

The payoff of a floating lookback call option is indeed the final current price of an asset minus the minimum or lowest asset price that the underlying asset has achieved during the life of the option. This is because a floating lookback call option allows the holder to 'look back' over the life of the option and select the lowest price as the strike price. This means that the holder can benefit from the maximum possible upward movement of the asset's price. This type of option provides a safety net for the holder, as they can choose the most favorable price during the option's life, maximizing their potential profit.

Choice B is incorrect. A floating lookback put option's payoff is calculated as the maximum asset price during the life of the option minus the final current price of the asset, which does not match with our given payoff structure.

Choice C is incorrect. A fixed lookback call option's payoff is calculated as the final current price of the asset minus a predetermined strike price or alternatively, it can be calculated as maximum value of underlying asset during life of option minus strike price. This also does not align with our given payoff structure.

Choice D is incorrect. A fixed lookback put option's payoff would be either zero or strike price minus minimum value of underlying asset during life of option, whichever is higher. This too doesn't correspond to our given exotic options' payoff structure.

Things to Remember

- Exotic options are non-standard options with complex features, often tailored to

specific needs or market conditions.

- Lookback options allow the holder to 'look back' over the life of the option and select the most favorable price as the strike price.
 - Floating lookback options have strike prices based on the extreme values of the underlying asset during the option's life.
 - Fixed lookback options have predetermined strike prices set at the beginning of the option's life.
 - Lookback options can provide flexibility and potential for higher profits, but they can also be more complex and difficult to price.
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Q.801 Katja Firos is an investment analyst at Frankfurt Securities, a brokerage and investment company. She was instructed by the head of the investments unit to hedge the portfolio of exotic options through static options replication. Which of the following steps should she take to implement the static replication method?

- A. The static replication method involves searching for a portfolio of the same exotic options which other market participants are using for hedging.
- B. The static replication method involves searching for a portfolio of plain vanilla options with opposite attributes that inversely replicates the exotic options and then taking a short position in this portfolio in order to hedge the exotic options.
- C. The static replication method involves searching for a portfolio of plain vanilla options with similar attributes that approximately replicates the exotic option and then taking a short position in this portfolio in order to hedge the exotic options.
- D. The static replication method involves searching for a portfolio of similar exotic options with similar attributes and then taking a long position in this portfolio in order to hedge the exotic options.

The correct answer is **C**.

The static replication method involves searching for a portfolio of plain vanilla options with similar attributes that approximately replicate the exotic option and then taking a short position in this portfolio in order to hedge the exotic options. This method is used to create a portfolio that mimics the payoff of the exotic option. By taking a short position in this replicating portfolio, the risk associated with the exotic options can be offset. This is because the gains from the short

position in the replicating portfolio will offset the losses from the exotic options and vice versa. Therefore, this method effectively hedges the risk associated with the exotic options.

A, B and D are incorrect following the above explanation.

Things to Remember

- Static replication is a hedging technique used to offset the risk associated with exotic options by creating a portfolio of plain vanilla options that replicates the payoff of the exotic option.
 - Plain vanilla options are standard options with simple features such as fixed expiration dates and strike prices.
 - Exotic options have complex features such as barriers, Asian options, and lookback options, making them harder to hedge directly.
 - By using static replication, investors can hedge the risk of exotic options without directly trading the exotic options themselves.
 - Static replication requires careful selection of plain vanilla options with similar attributes to the exotic option to create an effective hedge.
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Q.802 Exotic options are customized and designed to meet the requirements of investors, which is why these options trade on OTC markets. These options have features that allow them to change the expiration date and strike prices. Forward start options are also non-standard options that allow the investor to purchase an option that will start at a future date. Which of the following options is most similar to a forward start option?

- A. Warrants
- B. Butterfly spread options
- C. Gap options
- D. Employee options

The correct answer is **D**.

Employee options are the most similar to forward start options. Forward start options are a type of exotic option that starts at a future date, known as T1, and expires at another future date, T2.

This feature allows investors to benefit from future price movements. Employee options share a similar characteristic. In the case of employee options, the employer pledges to grant an at-the-money option at a future date. This means that the strike price of the option is the same as the market price of the underlying asset at the time of the grant. This similarity in structure and functionality makes employee options the most analogous to forward start options.

Choice A is incorrect. Warrants are a type of derivative that confers the right, but not the obligation, to buy or sell a security – most commonly an equity – at a certain price before expiration. The price at which the underlying security can be bought or sold is referred to as the exercise price or strike price. While they do offer flexibility in terms of timing, they do not have the specific feature of setting a future start date for the option like forward start options.

Choice B is incorrect. Butterfly spread options involve positions in options with three different strike prices and are typically used when anticipating minimal movement in the price of the underlying asset. They do not bear resemblance to forward start options as they don't allow for altering expiration dates and strike prices at future dates.

Choice C is incorrect. Gap options are contracts where payout depends on whether the underlying asset's price changes beyond a certain level (gap level). They differ from forward start options because their payoff structure depends on whether this gap level has been reached rather than having an option that commences at a future date.

Things to Remember

- Exotic options are highly customized options that have non-standard features, such as the ability to change expiration dates and strike prices.
- OTC markets (Over-the-Counter markets) are decentralized markets where exotic options are traded directly between two parties, allowing for more flexibility and customization.
- Forward start options are a type of exotic option that starts at a future date and expires at another future date, providing investors with the opportunity to benefit from future price movements.
- Employee options are a type of compensation given to employees in the form of stock options, which can be similar to forward start options in terms of starting at a future date.

Reading 42: Properties of Interest Rates

Q.645 Donald Gregg is a senior professor of economics at the University of Vikings. He has authored various books on the subject of macroeconomics, financial instruments, and derivatives. He is famous for conducting a bi-yearly informative seminar where he delivers his analysis on finance-related topics. In his last seminar, he said that the government also borrows funds from public institutions in exchange for their guarantee to return the funds with interest. These transactions are considered risk-free as governments are not likely to default. Which of the following rates do governments use to borrow funds denominated in their own currency?

- A. LIBOR.
- B. Fed funds rate.
- C. Repo rate.
- D. Treasury rate.

The correct answer is **D**.

The Treasury rate is the interest rate that the government pays to borrow money from investors. This borrowing is done through the issuance of treasury bills and treasury bonds. These are considered risk-free financial instruments as they are backed by the full faith and credit of the government. The government sells these securities to investors to raise funds. The Treasury rate is determined through auctions conducted by the Department of the Treasury. The rate is influenced by various factors including the overall demand for these securities, the economic outlook, and the monetary policy stance of the central bank. The Treasury rate serves as a benchmark for other interest rates in the economy as it is considered the risk-free rate of return.

Choice A is incorrect. LIBOR (London Interbank Offered Rate) is a benchmark interest rate at which major global banks lend to one another in the international interbank market for short-term loans. It does not represent the rate at which governments borrow funds.

Choice B is incorrect. The Fed funds rate, set by the Federal Reserve, is the interest rate at which depository institutions lend reserve balances to other depository institutions overnight on an uncollateralized basis. While it influences other interest rates and thus borrowing costs, it's not directly used as a borrowing rate by governments.

Choice C is incorrect. The Repo (Repurchase Agreement) rate refers to the discount rate at which a central bank repurchases government securities from commercial banks, aiming to regulate money supply in the economy. It's not directly related to government borrowing from public institutions.

Things to Remember

- Government borrowing through treasury securities is a key mechanism for raising funds to finance budget deficits.

- The Treasury rate serves as a benchmark for other interest rates in the economy.
 - Government securities are considered risk-free due to the backing of the government's credit.
 - Central banks play a crucial role in regulating money supply through various tools like repo rates.
 - Understanding different interest rates like LIBOR, Fed funds rate, and Repo rate is essential for comprehending the financial markets.
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Q.648 Xiaojun Lee is the treasury manager at the Atlanta Small Business Bank. She works in a team that supervises all the branches of the banks in Atlanta. Her core responsibility is to look after treasury transactions and to make sure the bank, at all time, meets its reserve requirements with the Federal Reserve. Today, Lee has analyzed that the bank will fall short \$200 million from its reserve requirements. In order to avoid penalties, the bank must borrow some funds from another bank. Which of the following rate must Lee use as the reference rate for borrowing \$200 million overnight?

- A. Treasury rate.
- B. Fed funds rate.
- C. Repo rate.
- D. It cannot be determined.

The correct answer is **B**.

The Fed funds rate is the rate that banks use for overnight borrowing or lending of surplus funds to meet the Federal Reserve's reserve requirements. This rate is crucial for banks to maintain their reserve requirements and avoid penalties. In this scenario, since the Atlanta Small Business Bank is short of its reserve requirements, it needs to borrow funds overnight. Therefore, the Fed funds rate is the appropriate reference rate for this transaction.

Choice A is incorrect. The Treasury rate is the yield on a U.S. government debt obligation (a U.S. Treasury note or bond) that is considered risk-free as it is backed by America's full faith and credit. It does not apply to interbank lending, which is what Lee needs for meeting the reserve requirements.

Choice C is incorrect. The Repo rate, or repurchase rate, refers to the interest rate at which commercial banks borrow short-term funds from a central bank by selling securities with an agreement to repurchase them at a later date. This does not apply in this scenario as Lee needs

to borrow from another bank, not from the central bank.

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Things to Remember

- **Reserve Requirements:** Regulations set by the Federal Reserve that require banks to hold a certain amount of funds in reserve to ensure liquidity and stability in the banking system.
 - **Interbank Lending:** The borrowing and lending of funds between banks to manage their reserve requirements and maintain liquidity.
 - **Overnight Borrowing:** Short-term borrowing or lending of funds that occurs overnight to meet immediate funding needs.
 - **Risk-Free Rate:** The rate of return on an investment with no risk of financial loss, often represented by the Treasury rate for U.S. government securities.
-

Q.649 Mohan Das is the treasury manager of a bank based in Frankfurt. He is responsible for looking at the bank's treasury operations and the compliance unit of the bank closely supervises his department. Today, Das is informed by the front office that the bank has to disburse a large fund to an institutional client which they believe will affect the bank's reserves with the central bank. The management suggested borrowing the funds from another bank to meet the central bank's reserve requirements, but he argues that the bank, instead, should sell its securities to another bank with the promise to purchase the securities back at a higher price. Which of the following interest rates is the manager most likely to use for the given transaction?

- A. LIBOR.
- B. Fed funds rate.
- C. Repo rate.
- D. Treasury rate.

The correct answer is **C**.

The Repo Rate, also known as the Repurchase Agreement rate, is the interest rate that Das is most likely referring to in this scenario. The Repo Rate is a borrowing rate used by financial institutions when they sell their securities for a certain price and agree to buy them back at a later date for a higher price. This type of transaction is known as a 'repurchase agreement' or

'repo'. In a repo transaction, the seller is effectively borrowing money and using the securities as collateral, while the buyer is lending money and has the security of the collateral. The difference between the sale price and the repurchase price represents the interest on the loan, which is the Repo Rate. This rate is used as a short-term monetary policy instrument by central banks to control the money supply in the economy. In the given scenario, Das is suggesting a repo transaction to raise the necessary funds without borrowing from another bank, which aligns with the definition and use of the Repo Rate.

Choice A is incorrect. LIBOR (London Interbank Offered Rate) is the interest rate at which banks offer to lend funds (in marketable size) to other banks in the London interbank market. It does not directly relate to repurchase agreements as suggested by Das.

Choice B is incorrect. The Fed funds rate refers to the interest rate at which depository institutions lend reserve balances to other depository institutions overnight on an uncollateralized basis. While it does involve interbank lending, it doesn't specifically pertain to transactions involving securities repurchase agreements.

Choice D is incorrect. The Treasury rate refers to the yield on government securities, specifically U.S Treasury bonds, notes and bills. It's not directly related with repurchase agreement transactions as proposed by Das.

Things to Remember

- Repurchase Agreement (Repo) transactions involve selling securities with an agreement to repurchase them at a later date at a higher price, effectively acting as a short-term collateralized loan.
- Repo Rate is the interest rate associated with repurchase agreements and is used by financial institutions for short-term borrowing and lending of funds.
- Central banks use the Repo Rate as a monetary policy tool to manage liquidity and control the money supply in the economy.
- LIBOR (London Interbank Offered Rate) is a benchmark interest rate used for interbank lending, not specifically for repurchase agreements.
- Fed funds rate is the interest rate at which depository institutions lend reserve balances to other depository institutions overnight, not directly related to repo transactions.
- Treasury rate refers to the yield on government securities like U.S. Treasury bonds, notes, and bills, and is not directly linked to repurchase agreements.

Q.650 Gamze Goc is an independent wealth advisor that focuses on providing investment and savings advice to professionals. She advised one of her clients to invest \$10,000 for 5 years into a government's national saving plan which pays a monthly interest of 8% per year. This rate is fixed regardless of the tenure of the investment. Since the client does not have an alternative option to invest his savings, he asked what interest rates he would earn if the rate was compounded continuously. Identify the most appropriate answer to the client's inquiry.

- A. The continuously compounded interest rate is 9.23%.
- B. The continuously compounded interest rate is 8.33%.
- C. The continuously compounded interest rate is 8.05%.
- D. The continuously compounded interest rate is 7.97%.

The correct answer is **D**.

The formula used to convert a rate that is compounded at a certain frequency into a continuously compounded rate is:

$$R_c = m * \ln \left(1 + \frac{R_m}{m} \right)$$

Where

m is the compounding frequency,

R_m is the compounded rate at m frequency

And R_c is the continuously compounded rate

$$R_c = 12 \times \ln \left(1 + \frac{0.08}{12} \right) = 0.07973$$

Q.651 Ahmed Hatti is an undergrad business and finance student at the University of Millennials. Along with his friend, he manages a small hypothetical fund from his dorm room. The fund consists of small investments from his colleagues and family. As a fund manager, he is also responsible for generating a quarterly income newsletter, which he has to email to all fund contributors.

Recently, Hatti decided to invest a small portion of his fund into an interest-bearing account that quotes an interest rate of 16% compounded continuously. In order to add the interest rate into the quarterly newsletter, he must convert the continuously compounded rate into a quarterly compounded rate. Which of the following is the most appropriate conversion of the rate?

- A. The quarterly compounded rate is 15.6%.

- B. The quarterly compounded rate is 16%.
- C. The quarterly compounded rate is 16.3%.
- D. The quarterly compounded rate is 17.1%.

The correct answer is **C**.

The formula used to convert a continuously compounded rate into a rate that is compounded at a certain frequency is:

$$R_m = m \left\{ e^{\frac{R_c}{m}} - 1 \right\}$$

Where

m is the compounding frequency,

R_m is the annual rate, compounded m times a year

And R_c is the continuously compounded rate

$$R_m = 4 \left(e^{\frac{0.16}{4}} - 1 \right) = 16.32\%$$

Things to Remember

- Continuous compounding is a method of calculating interest where interest is calculated and added to the principal continuously, rather than at specific intervals.
- The formula for continuously compounded interest is: $A = P \cdot e^{rt}$, where A is the amount of money accumulated after t years, including interest, P is the principal amount, r is the annual interest rate (in decimal form), and t is the time the money is invested for in years.
- Compounding frequency refers to how often the interest is calculated and added to the principal amount. Common compounding frequencies include annually, semi-annually, quarterly, and monthly.
- When converting a continuously compounded rate to a rate compounded at a specific frequency, the formula $R_m = m \left\{ e^{\frac{R_c}{m}} - 1 \right\}$ is used, where R_m is the annual rate compounded m times a year, R_c is the continuously compounded rate, and m is the compounding frequency.

Q.653 An investor has invested \$1,000 in a 7-year zero-coupon bond with continuous compounding. If the bond is quoted as 9% per year compounded continuously, then estimate the value of the investment at the end of 7 years.

- A. The future value of a zero-coupon is \$1,656.0.
- B. The future value of a zero-coupon is \$1,828.0.
- C. The future value of a zero-coupon is \$1,877.6.
- D. The future value of a zero-coupon is \$1,912.0.

The correct answer is **C**.

Since the zero-coupon bond is compounded continuously, the future value of this 7-years 9% per annum bond is:

$$\begin{aligned} \text{FV} &= \text{Face value} * e^{\text{rate} * \text{period}} \\ &= 1000 * e^{0.09 * 7} \\ &= 1,877.6 \end{aligned}$$

Q.655 The process by which traders can use the quotes of treasury bills and coupon-bearing treasury bonds to derive a zero-coupon yield curve or spot curve is referred to as:

- A. Interpolation
- B. Duration
- C. Bootstrapping
- D. Calibration

The correct answer is **C**.

Bootstrapping is a financial method used to derive a zero-coupon yield curve, also known as a spot rate curve, from the rates and quotes of zero-coupon and coupon-bearing Treasury bonds. The term 'bootstrapping' refers to the process of 'pulling oneself up by one's bootstraps', which is a metaphor for achieving a complex task starting from basics without external help. In the context of finance, bootstrapping is a method of constructing a yield curve. The yield curve, in turn, is a graphical representation of the interest rates on debt for a range of maturities. It shows the yield an investor is expected to earn if he lends money for a given period of time. The spot rate curve or zero-coupon yield curve represents the yields of hypothetical zero-coupon bonds. Since they are not directly observable in the market, they are derived from the yields of coupon-bearing bonds, and this process is known as bootstrapping.

Choice A is incorrect. Interpolation is a statistical method used to estimate values between two known values. While it can be used in the process of constructing a yield curve, it is not the specific process referred to in this question.

Choice B is incorrect. Duration refers to the sensitivity of a bond's price to changes in interest rates. It does not refer to the process of constructing a zero-coupon yield curve or spot curve using treasury bills and bonds that bear coupons.

Choice D is incorrect. Calibration involves adjusting model parameters within certain bounds until model outputs match observed data. Although calibration might be part of building financial models, it does not specifically refer to the construction of a zero-coupon yield curve or spot curve.

Things to Remember

- Zero-coupon bonds are bonds that do not pay periodic interest but are issued at a discount to their face value and redeemed at face value.
 - Spot rates are the yields on zero-coupon bonds maturing at different points in time.
 - The zero-coupon yield curve is used in various financial calculations, such as pricing fixed-income securities, valuing interest rate derivatives, and assessing the term structure of interest rates.
 - Bootstrapping is a common method used in fixed income markets to derive the zero-coupon yield curve.
 - The process of bootstrapping involves solving for the spot rates at different maturities by using the prices of coupon-bearing bonds and applying the law of one price.
-

Q.656 Every year, thousands of students in Turkey take the Certified Trader exam. The exam tests in detail the knowledge of students who are willing to join the banking sector. In last year's exam, a question asked the students to calculate a 1-year forward rate 2 years from now. The question also provided the following table of zero spot rates p.a:

Year	Zero rates (per year)
1	6%
2	6.5%
3	7.2%

Using the information provided in the table, which of the following is the accurate 1-year forward

rate 2 years from now?

- A. 6.5%
- B. 7.2%
- C. 8.6%
- D. 9.3%

The correct answer is **C**.

The 1-year forward rate 2 years from now is calculated as:

$$\text{Forward Rate} = \text{Spot Rate}_{\text{year 3}} + (\text{Spot Rate}_{\text{year 3}} - \text{Spot Rate}_{\text{year 2}}) * \left(\frac{\text{Time}_2}{(\text{Time}_3 - \text{Time}_2)} \right)$$

$$\text{Forward Rate} = 0.072 + (0.072 - 0.065) * \left(\frac{2}{(3 - 2)} \right) = 0.086 \text{ or } 8.6\%$$

Alternative Approach

Let's start off by assuming that there are two people, each with \$100

A chooses to invest his cash for 3 years straight at the 3-year spot rate.

B opts to invest his cash for 2 years at the 2-year spot rate and then reinvest his proceeds for a further year at the 1-year forward rate at that point.

In a fair market, both A and B should have the same amount of money at the end of year 3.

What that means, therefore is that the 3-year spot rate is equal to the 2-year spot rate multiplied by the 1-year forward rate two years from today.

In other words,

$$(1 + 3 - \text{year spot})^3 = (1 + 2 - \text{year spot})^2 (1 + 1 - \text{year forward})$$

We should set the forward rate such that B ends with the same amount of money as A.

$$1.072^3 = 1.065^2 \times 1 - \text{year forward rate}$$

$$1 - \text{year forward rate} = 0.086$$

Q.657 Beijing Shipping Corp. enters into a forward rate agreement with Geneva Bank to receive a 7% fixed rate on the principal of \$50 million based on a three-month rate beginning in six-month time. If the three-month rate in six-month time is 6.8%, then what is the cash inflow/outflow for the Beijing Shipping at the end of the sixth month?

- A. Cash outflow of \$25,000.
- B. Cash inflow of \$24,582.
- C. Cash outflow of \$24,500.
- D. Cash outflow of \$24,582.

The correct answer is **B**.

The FRA's payoff will take place in the ninth month. The net payoff will be the difference between the receipt of the fixed rate of 7% and the floating rate payment. As given in the question, if the floating rate is 6.8% in six months, so the payoff at the end of the ninth month is calculated as:

$$\begin{aligned}\text{Payoff} &= \text{Principal} \times (\text{Fixed rate} - \text{Floating rate}) \times \text{Time} \\ &= 50,000,000 \times (0.07 - 0.068) \times 0.25 \\ &= 25,000\end{aligned}$$

Although any interest should be due at the end of the FRA period (e.g., 9 months in this question), the common practice is for the FRA to be settled at the beginning of the FRA period or rather as soon as the floating rate becomes known. Therefore, we can find the payoff at the end of the sixth month by discounting any payoffs for three months, with the floating rate as the discount rate.

$$\text{Payoff} = \frac{\$25,000}{\left(1 + \frac{0.068}{4}\right)} = \$24,582$$

Since the payoff is positive, Beijing Corp. will receive a cash inflow of \$24,582 at the six-month point.

Q.660 Hina Bibi is a fixed-income analyst at Vio Investment Company. She is responsible for analyzing the risk and return of a company's portfolio of fixed income investments. She is analyzing the change in the price of a hypothetical 7-year bond with the face value of 100 and the price of 96.86. If the duration of the bond that she analyzing is 1.962, then which of the following options presents the accurate change in the price of the bond if the yield on the bond increases by 50 basis points?

- A. The price of the bond will increase by \$0.95
- B. The price of the bond will decrease by \$0.95
- C. The price of the bond will increase by \$0.98
- D. The price of the bond will decrease by \$0.98

The correct answer is **B**.

The change in the price of a bond given the change in the yield can be predicted by duration. In the given question, where the price of the bond is 96.86, and the duration of the bond is 1.962, a 50 basis point increase in the yield will decrease the price of the bond by:

Change in price = -Bond price * Duration * Change in yield

Change in price = $-96.86 * 1.962 * 0.005 = -0.950$

Therefore, the price of the bond after the 50 basis point increase in yield is $96.86 - 0.950 = 95.91$.

Things to Remember

- Duration measures the sensitivity of a bond's price to changes in interest rates.
- A bond's price and yield have an inverse relationship -when yields increase, bond prices decrease, and vice versa.
- Modified duration is a measure of a bond's sensitivity to changes in interest rates and is calculated as $\text{Duration} / (1 + \text{Yield})$.
- Convexity is a measure of the curvature of the price-yield relationship of a bond and helps refine the estimate of the bond's price change given a change in yield.
- For a bond with positive convexity, the actual price change will be slightly higher than what is predicted by duration alone when yields change.

Q.662 There are different measures available that are used to measure the change in the price of

the bond given the change in the yield curve. Which of the following measures is used for the purpose of estimating changes in bond prices if the changes in the yield curve are larger?

- A. Duration
- B. Convexity
- C. Modified duration
- D. Concavity

The correct answer is **B**.

Convexity is a measure of the curvature in the relationship between bond prices and bond yields that demonstrates how the duration of a bond changes as the interest rate changes. This measure is used to assess the sensitivity of the price of a bond to changes in interest rates. Convexity is particularly useful when there are larger changes in interest rates, as it provides a more accurate estimate of bond price changes than duration alone. Convexity takes into account the effects of changes in the yield curve on bond prices, which can be significant when the changes in the yield curve are large. Therefore, convexity is the most appropriate measure to use when estimating changes in bond prices due to larger changes in the yield curve.

Choice A is incorrect. Duration is a measure of the sensitivity of the price of a bond to changes in interest rates. However, it assumes that this relationship is linear and does not account for substantial changes in the yield curve.

Choice C is incorrect. Modified duration, like duration, measures the price sensitivity of a bond to interest rate changes but it also assumes a linear relationship between bond prices and yields. It fails to capture the effects of larger shifts in the yield curve.

Choice D is incorrect. Concavity isn't used as a metric in bond investing for predicting alterations in bond prices due to yield curve fluctuations. It's rather an attribute related to convexity which describes how the change in yield affects convexity itself.

Things to Remember

- Duration is a measure of the sensitivity of the price of a bond to changes in interest rates, assuming a linear relationship.
- Modified duration adjusts the basic duration measure to account for changes in interest rates.
- Convexity measures the curvature in the relationship between bond prices and bond yields, providing a more accurate estimate of bond price changes than duration alone.
- Convexity is particularly useful when there are larger changes in interest rates, as it

captures the effects of changes in the yield curve on bond prices.

- Concavity is a related attribute to convexity, describing how the change in yield affects convexity itself.
-

Q.663 Fredrick Hessen is a senior professor in the department of macroeconomics at Welth Business School. In the current semester, his course focuses on interest rates and the term structure of interest rates. One day, he made the following comment:

“There is no relationship between short-term, medium-term, and long-term interest rates. These interest rates are independently determined by the supply and demand in their specific bond market. For instance, the short-term interest rate is determined by the supply and demand of short-term bonds”.

Which of the following theories is associated with the professor’s comment?

- A. Expectation theory
- B. Market segmentation theory
- C. Liquidity preference theory
- D. None of the above

The correct answer is **B**.

The Market Segmentation Theory is the correct answer. This theory posits that the markets for different maturity bonds are completely segmented and that the interest rates for each segment are determined by the supply and demand within that segment. This means that the interest rates for short-term, medium-term, and long-term bonds are determined independently of each other. This theory aligns with Professor Hessen's statement that there is no relationship between short-term, medium-term, and long-term interest rates and that these rates are independently determined by the supply and demand in their specific bond markets. The Market Segmentation Theory also explains why certain investors, such as pension funds and insurance companies, prefer bonds of a certain maturity and are unlikely to switch from one maturity to another based on liquidity considerations.

Choice A is incorrect. The Expectation theory suggests that long-term interest rates are an average of expected future short-term rates, implying a direct relationship between short, medium and long term rates. This contradicts Professor Hessen's viewpoint which suggests these rates are independently determined.

Choice C is incorrect. The Liquidity preference theory posits that investors demand a premium for longer term bonds due to the increased risk associated with holding assets over a longer period of time. This implies an interrelationship between different term interest rates, which again contradicts Professor Hessen's viewpoint.

Choice D is incorrect. As explained above, there exists a theory in economics - the Market Segmentation Theory - that aligns with Professor Hessen's viewpoint suggesting that different term interest rates are independently determined by their respective bond markets' supply and demand dynamics.

Things to Remember

- Expectation Theory: Suggests that long-term interest rates are an average of expected future short-term rates, implying a direct relationship between short, medium, and long-term rates.
 - Liquidity Preference Theory: Posits that investors demand a premium for longer-term bonds due to the increased risk associated with holding assets over a longer period of time.
 - Term Structure of Interest Rates: Refers to the relationship between short-term, medium-term, and long-term interest rates.
 - Segmented Markets: The concept that different segments of the bond market are isolated from each other, leading to independent determination of interest rates within each segment.
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Q.664 Transactions worth billions of dollars depend on the shape of the zero rate curve. The shape of the zero curve has gained the attention of economists, mathematicians, and investors. Many theories exist that present their perspective about the shape of the zero curve. One of those theories suggests that investors are likely to invest their funds for a shorter period while borrowers are more willing to borrow the funds at long-term fixed rates. The theory also concludes that the forward rates are greater than the future spot rates, which justifies the empirical result that the yield curve tends to be upward sloping. Which of the following theories provides the above-mentioned conclusion?

- A. Expectation theory
- B. Market segmentation theory
- C. Liquidity preference theory
- D. None of the above

The correct answer is **C**.

The Liquidity Preference Theory is the correct answer. This theory suggests that investors prefer to invest their funds for shorter periods, while borrowers are more inclined to borrow at long-term fixed rates. The theory also asserts that forward rates exceed future spot rates, providing an empirical explanation for the upward slope of the yield curve. This theory is based on the premise that investors require a premium, or a higher return, for holding long-term securities because they are considered riskier. The risk arises from the uncertainty about future interest rates and inflation. Therefore, to compensate for this risk, long-term rates are generally higher than short-term rates, resulting in an upward-sloping yield curve.

Choice A is incorrect. The Expectation theory suggests that the yields on long-term bonds are an average of expected future short-term rates. It does not account for the preference of investors to invest their funds for shorter periods or borrowers' inclination to borrow at long-term fixed rates.

Choice B is incorrect. The Market segmentation theory posits that financial markets are segmented and each segment (short, intermediate, and long term) has its own supply and demand dynamics influencing interest rates. This theory does not explain the preference of investors or borrowers as described in the question.

Choice D is incorrect. As explained above, the description provided in the question matches with Liquidity preference theory which makes option D - 'None of the above' incorrect.

Things to Remember

- **Yield Curve:** A graphical representation of the relationship between the interest rates (or yields) and the time to maturity of debt securities.
 - **Forward Rates:** The interest rates implied by current interest rates and expected future interest rates.
 - **Spot Rates:** The interest rates applicable to a specific period in the future.
 - **Risk Premium:** The additional return that investors require for holding a risky asset compared to a risk-free asset.
 - **Interest Rate Risk:** The risk that arises from changes in interest rates impacting the value of fixed-income securities.
-

Q.3533 Stock IIK is currently selling for \$80. The 28 analysts offering 12-month price targets for IIK have a median target of \$91. Given that the stock reaches the median target of \$91 in 12 months, what is the continuously compounded return of this asset?

- A. 0.1375
- B. 0.1198
- C. 0.1319
- D. 0.1288

The correct answer is **D**.

$$\text{Future value} = \text{Present Value} \times e^{RT}$$

Where R is the continuously compounded rate of interest, and T is the time to maturity.

$$\begin{aligned} 91 &= 80e^{R \cdot 1} \\ \Rightarrow e^R &= \frac{91}{80} \\ \Rightarrow R &= \ln\left(\frac{91}{80}\right) \\ &= 0.12883 \end{aligned}$$

Alternatively,

$$\begin{aligned} \text{Holding period return} &= \frac{S_T}{S_0} - 1 \\ &= \frac{\$91}{\$80} - 1 \\ &= 1.1375 - 1 = 13.75\% \end{aligned}$$

$$\text{Continuous Return} = \ln(1 + i) = \ln(1 + 13.75\%) = 0.12883$$

Q.3534 An investor invested \$154,856 into a mutual fund 5 years ago. If the investment is now worth \$201,694, what is the compound annual growth rate of the investment?

- A. 6%
- B. 7.9%
- C. 5.4%
- D. 5.9%

The correct answer is **C**.

$$\begin{aligned}\text{Compound annual growth rate (CAGR)} &= (\text{Ending value}/\text{Beginning value})^{1/n} \\ &= (201,694/154,856)^{1/5} - 1 = 5.4\%\end{aligned}$$

Things to Remember

- Compound Annual Growth Rate (CAGR) is a useful measure of an investment's annual growth rate over time, taking into account the effect of compounding.
 - CAGR is often used to compare the returns on different investments over the same period of time.
 - It is important to note that CAGR assumes that the investment has been compounding steadily over the entire period.
 - CAGR is a geometric progression ratio that provides a constant rate of return over the time period being measured.
 - Investors use CAGR to determine the annual growth rate needed to reach a future value from an initial investment.
-

Q.3535 A bank advertises that it pays an annual interest of 10% with semi-annual compounding on its savings account. What is the effective annual rate?

A. 10.375%

B. 10.25%

C. 10.5%

D. 10.42%

The correct answer is **B**.

$$\begin{aligned}\text{EAR} &= \left(1 + \frac{\text{Annual rate}}{\text{Compounding frequency}}\right)^{\text{Compounding frequency}} - 1 \\ &= (1 + 10\%/2)^2 - 1 = 0.1025 = 10.25\%\end{aligned}$$

Q.3536 The price of a stock increases from \$24 to \$40 in two years. What is the continuously compounded annual return for the stock?

- A. 43.10%
- B. 28.00%
- C. 51.08%
- D. 25.54%

The correct answer is **D**.

The continuously compounded 2-year return = $\ln(40/24) = 51.08\%$

Annually compounded rate of return = $(51.08\% / 2) = 25.54\%$

Things to Remember

- Continuous compounding is a method of calculating interest where the interest is added to the principal balance continuously, rather than at specific intervals.
 - The continuously compounded rate of return can be calculated using the formula: $\ln(\text{Ending Value} / \text{Beginning Value})$.
 - Annual compounding is another method of calculating interest where the interest is added to the principal balance once a year.
 - To convert a continuously compounded rate to an annually compounded rate, you can divide the continuously compounded rate by the number of compounding periods in a year.
-

Q.3537 In order to have liquid cash at hand, a company always keeps \$200,000 in its bank account. The stated annual interest rate quoted by the bank is 8%. Assuming that compounding is done continuously and there have been no withdrawals and additions, what is the balance in the company's bank account after one year?

A. \$200,321

B. \$216,657

C. \$202,149

D. \$217,985

The correct answer is **B**.

Using the formula for the future value with continuous compounding (N is the number of years in the expression)

$$FV_N = \$200,000e^{0.08(1)} = \$216,657.41$$

Q.3538 Jose Calzon currently has \$5,040.11 in his bank account. If he plans to buy a car for \$5,500 next year, what is the annual interest rate (compounded monthly), that a bank must pay so that James receives a sum of \$5,500 next year?

- A. 0.76%
- B. 9.12%
- C. 0.73%
- D. 8.76%

The correct answer is **D**.

Interest rate can also be considered as the required rate of return. In the above case, James wants his \$5,040.11 to grow to \$5,500. The rate required to achieve this return can be calculated as under: Amount deposited today $\times$ (1 + Rate of interest) = Amount next year

$$\text{Rate of interest} = \frac{\text{Amount next year}}{\text{Amount deposited}} - 1$$

$$= \left(\frac{\$5,500}{\$5,040.11} \right) - 1 = 0.0912 \text{ or } 9.12\%$$

To turn the annual interest rate into a monthly rate, we use the relationship:

$$1 + i = \left(1 + \frac{i_m}{m} \right)^m$$

$$\Rightarrow 1.0912 = \left(1 + \frac{i_{12}}{12} \right)^{12}$$

$$\Rightarrow m_{12} = [1.0912^{\frac{1}{12}} - 1] \times 12 = 0.08759$$

Q.3539 An investor received \$100,000 after five years from a certificate of deposit which paid him an interest of 12% with monthly compounding. What is the sum deposited by the investor at the beginning of the 5 years?

- A. \$79,670
- B. \$55,045
- C. \$56,743
- D. \$68,856

The correct answer is **B**.

Given an initial investment of A that earns an annual rate R, compounded m times a year for a total of n years, then we can compute the future value, FV, as follows:

$$FV = A\left[1 + \frac{R}{m}\right]^{m \times n}$$
$$100,000 = A\left[1 + \frac{0.12}{12}\right]^{60}$$
$$A = \$55,045$$

Things to Remember

- **Compound Interest:** Compound interest is the interest calculated on the initial principal, which also includes all of the accumulated interest from previous periods on a deposit or loan.
- **Effective Annual Rate (EAR):** The effective annual rate is the rate of interest that is actually earned or paid on an investment, loan, or other financial product due to the result of compounding over a given time period.
- **Time Value of Money:** The concept that a dollar today is worth more than a dollar in the future due to its potential earning capacity is known as the time value of money.
- **Present Value:** The present value is the current worth of a future sum of money or stream of cash flows given a specified rate of return.

Q.3540 A 3-year bond with a face value of \$100 offers a 7% coupon rate with interest paid annually. Assuming the following sequence of spot rates, the price of the bond is *closest to*:

Time to Maturity	Spot Rate (%)
1	4
2	5
3	5.5

- A. 102.48
- B. 106.74
- C. 103.56
- D. 104.2

The correct answer is **D**.

The price of a bond is the present value of its future cash flows, which in this case are the coupon payments and the face value return at maturity. Given the coupon rate of 7% and the face value of 100, the annual coupon payment is 7 (100) = 7 dollars. The spot rate for each year is given in the table. To find the present value of the cash flows, we can use the formula:

$$PV = \frac{C}{(1+r)^1} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \frac{F}{(1+r)^3}$$

Where:

C = annual coupon payment

F = face value

r = spot rate Plugging in the values we get:

$$\text{price} = \frac{7}{(1+0.04)^1} + \frac{7}{(1+0.05)^2} + \frac{107}{(1+0.055)^3} = 104.20$$

Therefore, the price of the bond is closest to \$104.20\$

Q.3541 An analyst has gathered the following estimated series of spot rates for a developing country:

- 0.5-year: 2%
- 1-year: 3%
- 1.5-year: 3.55%

- 2-year: 4%
- 2.5-year: 4.5%
- 3-year: 5%
- 3.5-year: 5.45%

Given that the information is accurate, what is the price of a 3-year, 1,000 face value, 5% annual coupon paying bond?

- A. 1115.3
- B. 995.65
- C. 1001.8
- D. 998.51

The correct answer is **C**.

The general approach to bond valuation is to utilize a series of spot rates to reflect the timing of future cash flows.

$$(50/1.03) + (50/1.04^2) + (1,050/1.05^3) = 1,001.80$$

Things to Remember

- Bond valuation involves calculating the present value of all future cash flows generated by the bond.
- Spot rates are the yields-to-maturity on zero-coupon bonds maturing at different points in time.
- The spot rate curve is used to discount each cash flow of a bond to determine its present value.
- Coupon payments are typically fixed and paid semi-annually or annually, depending on the bond's terms.
- Yield to maturity (YTM) is the total return anticipated on a bond if it is held until it matures.

Q.3542 The 2-period spot rate, S_2 is 9%, and the 1-period spot rate, S_1 is 4%. Calculate the forward rate for one period, one period from now, $f_{1,1}$.

- A. 5%
- B. 4.8%
- C. 14.24%
- D. 8.73%

The correct answer is **C**.

$$(1 + S_2)^2 = (1 + S_1)(1 + f_{1,1})$$

$$f_{1,1} = [(1.09)^2/1.04] - 1 = 14.24\%$$

Alternatively, we can denote this as:

$$F = \frac{V_2}{V_1}$$

Where V_1 is the value to which one dollar grows by time T_1

V_2 is the value to which one dollar grows by time T_2

In this case,

$$F = \frac{V_2}{V_1} = \frac{1.09^2}{1.04} = 1.1424$$

So that $F = 1.1424 - 1 = 14.24\%$

Q.3543 If the current 1-year spot rate is 3%, the 1-year forward rate one year from today ($f_{1,1}$) is 4%, and the 1-year forward rate two years from today ($f_{2,1}$) is 5%, then the 3-year spot rate is *closest to*:

- A. 4%
- B. 12.5%
- C. 4.2%
- D. 8.42%

The correct answer is **A**.

There is a connection between spot rates and forward rates. The accumulation amount at time k of an investment of \$1 at time $t=0$ is given by $(1 + s_k)^k$. If we had agreed to invest this amount at time k for $n-k$ years, then the accumulation at time n would be: $(1 + s_k)^k \cdot (1 + f_{[k,n]})$

We are also aware that \$1 invested at time 0 for n years will be $(1 + s_n)^n$

$$\begin{aligned} &= (1 + s_k)^k \cdot (1 + f_{[k,n]}) = (1 + s_n)^n \\ &= (1 + f_{[k,n]})^{n-k} = \frac{(1 + s_n)^n}{(1 + s_k)^k} \end{aligned}$$

For a one-period rate, $f_k = f_{k,1}$. Also, define $f_0 = s_1$. Using the same arguments,

$$\Rightarrow (1 + s_n)^n = (1 + f_0)(1 + f_1)(1 + f_2) \dots (1 + f_{n-1})$$

Therefore,

$$\begin{aligned} (1 + s_3)^3 &= [(1.03)(1.04)(1.05)] \\ \Rightarrow s_3 &= [(1.03)(1.04)(1.05)]^{\frac{1}{3}} - 1 = 3.997\% \end{aligned}$$

Q.3545 A corporate bond has the following characteristics:

- Price: USD 106.50
- Coupon rate: 5%
- Duration: 7.5 years
- Convexity: 101

If the credit spreads narrow by 175 basis points, then what will be the price of the bond?

- A. USD 114.68
- B. USD 122.13
- C. USD 123.78
- D. USD 117.68

The correct answer is **B**.

The change in bond's price resulting from the change in the yield can be estimated using the following formula:

$$\Delta B = -DB\Delta y + \frac{1}{2}CB(\Delta y)^2$$

Where

ΔB = The change in the bond's price resulting from the change in the yield.

Δy = The change in the bond's yield.

B = the price of the bond.

C = the convexity of the bond.

D = the duration of the bond.

From the information given in the question,

$$\Delta B = -7.5 \times 106.50 \times (-0.0175) + \frac{1}{2} \times 101 \times 106.50 \times (-0.0175)^2 = 15.62522$$

The change in bond's price is positive so that the new price of the bond is:

$$15.62522 + 106.50 = 122.12522 \approx 122.13$$

Q.3548 You have been provided the following information on a bond:

Period	PV of cash flow (USD Mn)
1	3.7
2	4.9
3	22.3

If the yield to maturity is 6%, then what is the modified duration of the bond?

- A. 2.45 years
- B. 2.65 years
- C. 2.30 years
- D. 2.25 years

The correct answer is **A**.

Macaulay Duration = $(1 \times 3.7 / 30.9) + (2 \times 4.9 / 30.9) + (3 \times 22.3 / 30.9) = 2.60$ years

Modified Duration = Macaulay Duration / $(1 + \text{YTM}) = 2.60 / 1.06 = 2.45$ years

Q.3551 A 4-year, 5% annual-pay bond has a face value of \$1,000. The interest rate stands at 3% per annum.

Determine the approximate modified duration?

A. 0.97

B. 7.8

C. 3.62

D. 3.73

The correct answer is **C**.

$$\text{Modified duration} = \frac{\text{Macaulay duration}}{1 + \frac{y}{m}}$$

Where y is the yield and m is the compounding frequency per annum. We first calculate the Macaulay duration:

$$\text{Macaulay duration} = \frac{\sum_{t=1}^n \text{PV}(C_t)T}{\text{Market Price of Bond}}$$

Where $\text{PV}(C_t)$ is the present value of coupon payments at time t, and T is the time to maturity

$$\text{Price of the bond} = \frac{50}{1.03^1} + \frac{50}{1.03^2} + \frac{50}{1.03^3} + \frac{1050}{1.03^4} = 1074.34$$

Thus,

$$\text{Macaulay duration} = \frac{1}{1074.34} \times \frac{50}{1.03^1} + \frac{2}{1074.34} \times \frac{50}{1.03^2} + \frac{3}{1074.34} \times \frac{50}{1.03^3} + \frac{4}{1074.34} \times \frac{1050}{1.03^4} = 3.73412$$

And so the modified duration is given by:

$$\begin{aligned} \text{Modified duration} &= \frac{\text{Macaulay duration}}{1 + \frac{y}{m}} \\ &= \frac{3.73412}{1 + \frac{0.03}{1}} \\ &= 3.62535 \end{aligned}$$

Q.3552 A bond has a duration of 10.62 and a convexity of 91.46. For a 200 bps increase in yield, what is the bond's approximate percentage price change?

- A. -19.41%
- B. -24.90%
- C. -1.62%
- D. -4.51%

The correct answer is **A**.

The estimated price change = $-(\text{Duration}) \times (\text{Change in yield}) + (1/2) \times (\text{Convexity}) \times (\text{Change in yield})^2$
 $= -10.62 \times 0.02 + 0.5 \times 91.46 \times 0.02^2 = -19.41\%$

Things to Remember

- Duration measures the sensitivity of a bond's price to changes in interest rates.
 - Convexity is a measure of the curvature of the price-yield relationship of a bond and helps refine the estimate of price changes calculated using duration.
 - For small changes in interest rates, convexity helps improve the accuracy of the estimated percentage price change.
 - A positive convexity means that the bond's price increases at an increasing rate as yields decrease, and vice versa.
 - Convexity is especially important when interest rates change significantly or when the bond has a long maturity.
-

Q.3553 A 9% bond has a full price of \$905 and a YTM of 10%. Estimate the percentage change in the full price of the bond for a 30 basis point increase in YTM, assuming the bond's modified duration is 9.42, and its convexity is 68.33.

- A. -2.65%
- B. 2.83%
- C. -2.80%
- D. 2.86%

The correct answer is **C**.

Expected change in bond's price = $\Delta P/P = -D_{\text{mod}} * \Delta y + 0.5 * C * \Delta y^2$

Duration effect = $-D_{\text{mod}} * \Delta y = -9.42 * 0.003 = 0.02826 = -2.826\%$

Convexity effect = $0.5 * C * \Delta y^2 = 0.5 * 68.33 * 0.003^2 = 0.000307 = 0.0307\%$

Expected change in bond's price = $(-0.02826 + 0.000307) = -2.79530\%$

Things to Remember

- Modified duration measures the percentage change in the bond's price for a 1% change in yield to maturity.
 - Convexity is a measure of the curvature in the relationship between bond prices and bond yields.
 - Convexity helps to refine the estimate of the percentage change in bond price when there are large changes in interest rates.
 - When interest rates rise, bond prices fall and vice versa.
 - Duration and convexity are important risk measures used in fixed income portfolio management.
-

Q.3554 A bond valued at \$200,000 has a duration of 8 and a convexity of 20. Assuming that the bond's spread relative to the benchmark curve increases by 25 basis points due to a credit downgrade, then what is the approximate change in the bond's market value?

- A. \$3,988
- B. \$3,960
- C. \$3,970
- D. \$3,368

The correct answer is **A**.

Price Change = (-Duration * Yield change) + (0.5 * Convexity * Yield change²)

Price change = $(-8 * 0.0025) + (0.5 * 20 * 0.0025^2) = -1.99\%$

The bond's value will fall by approximately $1.990\% * 200,000 = \$3,988$.

Things to Remember

- Duration measures the sensitivity of a bond's price to changes in interest rates.
 - Convexity measures the curvature of the price-yield relationship of a bond.
 - An increase in spread relative to the benchmark curve indicates a higher credit risk for the bond.
 - Price change calculations using duration and convexity help estimate the impact of interest rate changes on bond values.
-

Q.4821 The expectations hypothesis (theory) states that:

- A. We can forecast future interest rates by looking at the term structure of interest rates since the return on a long-term bond is, in essence, the average return on short-term bonds over the same period.
- B. We can forecast future interest rates by looking at past returns on similar instruments.
- C. We can forecast future interest rates by looking at the behavior of the stock market.
- D. We can forecast future interest rates by looking at the term structure of interest rates since the return on a short-term bond is essentially the average return on long-term bonds over the same period.

The correct answer is **A**.

The Expectations Hypothesis, also known as the Pure Expectations Theory, posits that the return on a long-term bond is essentially the average return on short-term bonds over the same period. This theory asserts that expected future spot rates of interest are equal to the forward rates that can be calculated today. In other words, the forward rates are unbiased predictors for making expectations of future spot rates. Therefore, we can forecast future interest rates by looking at the term structure of interest rates. The term structure of interest rates, also known as the yield curve, is a graphical representation that shows the relationship between interest rates and the time to maturity of a debt for a given borrower in a given currency. The term structure reflects expectations about future interest rates, inflation rates, and real rates of return. Hence, by analyzing the term structure of interest rates, we can derive insights into future interest rate movements.

Choice B is incorrect. The Expectations Hypothesis does not suggest that future interest rates can be forecasted by looking at past returns on similar instruments. This theory posits that the expected future interest rates are reflected in the term structure of current interest rates, not past returns.

Choice C is incorrect. According to the Expectations Hypothesis, future interest rates are not predicted based on the behavior of the stock market. The theory focuses on bond markets and their term structures for predicting future interest rates.

Choice D is incorrect. The statement contradicts the essence of Expectations Hypothesis which suggests that long-term bond yields are an average of expected short-term yields and not vice versa.

Things to Remember

- The Expectations Hypothesis is based on the idea that the term structure of interest rates reflects market expectations about future interest rates.
- Forward rates are unbiased predictors of future spot rates according to the Expectations Hypothesis.
- The term structure of interest rates, also known as the yield curve, is a crucial tool for analyzing interest rate movements and expectations.
- Market participants use the yield curve to make decisions about borrowing, lending, and investing based on their expectations of future interest rates.

Q.4822 The annual yield of a one-year government bond is 10% and the expected yield on a one-year bond starting one year from now is 11.0%. The expected yield on a one-year bond starting two years from now is 11.5%. According to the expectations hypothesis, what would be the annual yield of a three-year government bond?

- A. 36.14%
- B. 12.04%
- C. 11.04%
- D. 10.83%

The correct answer is **D**.

According to the Expectations Theory, the forward rates are unbiased predictors for making expectations of future spot rates. We can therefore forecast future interest rates by looking at the term structure of interest rates since the return on a long-term bond is in essence the average return on short-term bonds over the same period.

$$\begin{aligned}
 (1 + r_3)^3 &= (1 + r_1)(1 + f_{1,1})(1 + f_{2,1}) \\
 (1 + r_3)^3 &= (1 + 0.1)(1 + 0.11)(1 + 0.115) \\
 1 + r_3 &= 1.1083 \\
 r_3 &= 10.83\%
 \end{aligned}$$

Note: We can as well simply find the average of the short-term rates:

$$\text{Three-year rate} = \frac{(10\% + 11\% + 11.5\%)}{3} = 10.83\%$$

Q.4825 Four U.S. T-bonds currently on the market have the following characteristics:

Price per \$100 par value	Coupon (paid semiannually)	Maturity (yrs.)	Semiannual period
\$101.50	5.0%	0.5	1
\$102.60	5.25%	1.0	2
\$103.15	5.75%	1.5	3
\$103.95	6.20%	2.0	4

Determine the one-year spot rate via the bootstrapping method, assuming semiannual compounding.

- A. 1.98%

- B. 2.61%
- C. 1.31%
- D. 2.25%

The correct answer is **B**.

Bootstrapping is the process of carving out spot rates from the market prices of a set of coupon-paying bonds. The spot rates are determined in a consequential manner: after obtaining the first spot rate, say, for six months, we use that to obtain the spot rate for one year. Then we can use the six-month and one-year spot rates to obtain the one and a half-year spot rate, and so on.

We know that the price of a bond is the discounted value of all of its future cashflows.

For the six month T-bond, therefore,

$$\begin{aligned}
 101.5 &= \frac{(100 + 5/2)}{(1 + \frac{Z_{0.5}}{2})^1} \\
 101.5[(1 + \frac{Z_{0.5}}{2})^1] &= \$102.5 \\
 (1 + \frac{Z_{0.5}}{2})^1 &= \frac{102.5}{101.5} \\
 (1 + \frac{Z_{0.5}}{2})^1 &= 1.0099 \\
 Z_{0.5} &= 0.0099 \times 2 = 1.98\%
 \end{aligned}$$

Thus, the six-month spot rate is 1.98% with semiannual compounding.

For the 1-year T-bond,

$$\begin{aligned}
102.6 &= \frac{5.25/2}{(1 + \frac{z_{0.5}}{2})^1} + \frac{\$100 + \frac{5.25}{2}}{(1 + \frac{z_1}{2})^2} \\
102.6 &= \frac{\frac{\$5.25}{2}}{(1 + \frac{0.0198}{2})^1} + \frac{\$100 + \frac{\$5.25}{2}}{(1 + \frac{z_1}{2})^2} \\
102.6 &= \$2.6 + \frac{\$102.63}{(1 + \frac{z_1}{2})^2} \\
100(1 + \frac{z_1}{2})^2 &= \$102.63 \\
(1 + \frac{z_1}{2})^2 &= 1.0263 \\
(1 + \frac{z_1}{2}) &= \sqrt{1.0263} = 1.0131 \\
\frac{z_1}{2} &= 0.0131 \\
z_1 &= 2.61\%
\end{aligned}$$

Q.4826 The **main** advantage of the secured overnight financing rate (SOFR) over LIBOR is that:

- A. It's based on the average borrowing rates across a larger collection of banks.
- B. It's easier to compute.
- C. It's based on actual observable transactions, not estimates.
- D. It incorporates a built-in hedge against default risk.

The correct answer is **C**.

The Secured Overnight Financing Rate (SOFR) is a benchmark interest rate for dollar-denominated derivatives and loans, which is based on actual observable transactions. Unlike LIBOR, which is based on estimates provided by a panel of banks, SOFR is derived from the Treasury repurchase market, which is a highly liquid and active market. This means that SOFR is based on a large volume of actual transactions, making it a more reliable and transparent benchmark. The use of actual transaction data reduces the risk of manipulation that was a concern with LIBOR, which was based on estimates rather than actual transactions. Therefore, the main advantage of SOFR over LIBOR is its basis on actual observable transactions, not estimates.

Choice A is incorrect. While it's true that SOFR is based on a larger collection of banks, this isn't the primary advantage of SOFR over LIBOR. The main advantage lies in its basis on actual observable transactions rather than estimates.

Choice B is incorrect. The ease of computation isn't the primary reason for the transition from LIBOR to SOFR. Both rates involve complex calculations and require sophisticated financial knowledge to understand and compute.

Choice D is incorrect. Although SOFR incorporates some risk measures, it does not provide a built-in hedge against default risk. Its main advantage over LIBOR lies in its transparency and reliability as it's based on actual observable transactions.

Things to Remember

- LIBOR (London Interbank Offered Rate) is a benchmark interest rate that was commonly used for short-term loans and derivatives. However, due to concerns about its reliability and susceptibility to manipulation, global regulators have been pushing for its replacement with alternative rates like SOFR.
 - SOFR (Secured Overnight Financing Rate) is based on the overnight Treasury repurchase market, where participants borrow and lend cash overnight using US Treasury securities as collateral. This market is considered more transparent and less prone to manipulation compared to the interbank lending market that LIBOR is based on.
 - Transitioning from LIBOR to alternative rates like SOFR involves significant operational and contractual challenges for financial institutions, as many existing contracts and financial products are tied to LIBOR. Proper planning and communication are essential for a smooth transition.
 - SOFR is just one of the alternative reference rates being considered globally to replace LIBOR. Other rates include the Sterling Overnight Index Average (SONIA) in the UK and the Euro Short-Term Rate (€STR) in the Eurozone.
-

Q.4902 Consider a bond whose current price is USD 150 with a cash flow in one year providing a present value of USD 50 and a cash flow in two years providing a present value of USD 100. Suppose further that a yield of 5% with semi-annual compounding applies on the bond. What is the value of modified duration?

A. 1.6667

B. 1.626

C. 1.3333

D. 1.3008

The correct answer is **B**.

We that know that,

$$\text{Modified duration} = \left[\frac{\text{Macaulay duration}}{\left(1 + \frac{y}{n}\right)} \right]$$

We, therefore, start by calculating the Macaulay duration:

$$\begin{aligned} \text{Macaulay duration} &= \sum_{i=1}^n t_i \left[\frac{\text{PV}(C_{t_i})}{P} \right] \\ &= \frac{50}{150} \times 1 + \frac{100}{150} \times 2 = 1.6667 \end{aligned}$$

Where $\text{PV}(C_{t_i})$ = present value of cash flows at time t_i and P is the price of the bond.

Thus,

$$\text{Modified duration} = \frac{1.6667}{\left(1 + \frac{0.05}{2}\right)} = 1.62604$$

Q.4904 Consider a portfolio that has a bond position worth USD 10,000. Suppose the position has a modified duration of 2 years and a convexity of 30. Assume that the term structure is flat. By how much does the value of the position change if interest rates increase by 15 basis points?

- A. The position decreases by \$30.
- B. The position increases by \$60.
- C. The position increases by \$30.
- D. The position does not change.

The correct answer is **A**.

$$\begin{aligned}\text{Change in bond's price} &= \text{Duration effect} + \text{Convexity effect} \\ &= [-\text{Duration} \times \text{Price} \times \text{Change in yield}] \\ &\quad + \left[\frac{1}{2} \times \text{Convexity} \times \text{Price} \times (\text{Change in yield})^2\right] \\ &= (-2 \times 10,000 \times 0.0015) + \left(\frac{1}{2} \times 30 \times 10,000 \times 0.0015^2\right) \\ &= -29.665 \approx -30\end{aligned}$$

This means that, with every increase in interest rates of 15 basis points, the bond's price will decrease by \$30

Q.4905 Suppose the zero-coupon interest rates (semi-annually compounded) for maturities of 0.5, 1.0, and 1.5 years are 2.5%, 3.0% and 3.5%, respectively. Consider a USD 1,000 face value, two-year bond that currently trades at USD 1,060 and pays coupons at a rate of 6% per year every six months. If the two-year zero-coupon interest rate is R, what is the value of R?

- A. 11.27%
- B. 2.88%
- C. 6.34%
- D. 1.08%

The correct answer is **B**.

R can be solved using the following equation:

$$\begin{aligned} \frac{30}{1 + \frac{0.025}{2}} + \frac{30}{(1 + \frac{0.030}{2})^2} + \frac{30}{(1 + \frac{0.035}{2})^3} + \frac{1030}{(1 + \frac{R}{2})^4} &= 1,060 \\ \Rightarrow \frac{1030}{(1 + \frac{R}{2})^4} &= 1,060 - 87.22 = 972.78 \\ \Rightarrow (1 + \frac{R}{2}) &= \left(\frac{1,030}{972.78}\right)^{\frac{1}{4}} = 1.0144 \end{aligned}$$

Thus,

$$R = 0.0144 \times 2 = 0.0288 = 2.88\%$$

Q.4908 In an FRA, an annual rate of 2% will be received, in exchange for a 6-month LIBOR to be paid on a principal of USD 2,000 for a six-month period starting in 15 months. If the six-month forward rate in 15 months is 2.5% per annum, what is the settlement on the FRA?

- A. USD 4.94 is paid out
- B. USD 12.35 is received
- C. USD 4.88 is paid out
- D. USD 4.94 is received

The correct answer is **A**.

The settlement in 15 months is given by,

$$\frac{(\text{Fixed rate} - \text{Forward rate}) \times \text{Term of the FRA} \times \text{Principal}}{1 + \frac{\text{forward rate}}{n}}$$

$$= \frac{(0.02 - 0.025) \times \frac{6}{12} \times 2,000}{1 + \frac{0.025}{2}} = -4.9383$$

Thus, USD 4.94 will be paid.

Q.4909 A three-year bond with a face value of USD 1,000 pays coupons of 8% per annum. The yield on the bond is 6% compounded annually. What is the Macaulay duration?

- A. 1.0
- B. 2.6312
- C. 2.7891
- D. 2.7206

The correct answer is **C**.

We first calculate the present value of cash flows as follows,

$$P = \frac{80}{1.06^1} + \frac{80}{1.06^2} + \frac{1080}{1.06^3} = 1,053.46$$

Thus, the Macaulay duration is given by,

$$\begin{aligned}\text{Macaulay Duration} &= \sum_{i=1}^n t_i \left[\frac{\text{PV}(C_t)}{\sum \text{PV}(C_t)} \right] \\ &= \frac{\frac{80}{1.06^1}}{1,053.46} \times 1 + \frac{\frac{80}{1.06^2}}{1,053.46} \times 2 + \frac{\frac{1080}{1.06^3}}{1,053.46} \times 3 \\ &= 2.7891\end{aligned}$$

Q.4910 A three-year bond with a face value of USD 1,000 pays coupons of 8% per annum. The yield on the bond is 6% compounded annually. What is the Modified duration?

A. 2.6312

B. 2.7206

C. 2.7891

D. 1.0

The correct answer is **A**.

We first calculate the present value of cash flows as follows,

$$P = \frac{80}{1.06^1} + \frac{80}{1.06^2} + \frac{1080}{1.06^3} = 1,053.46$$

The Macaulay duration is given by,

$$\begin{aligned}\text{Macaulay Duration} &= \sum_{i=1}^n t_i \left[\frac{PV(C_t)}{\sum PV(C_t)} \right] \\ &= \frac{\frac{80}{1.06^1}}{1,053.46} \times 1 + \frac{\frac{80}{1.06^2}}{1,053.46} \times 2 + \frac{\frac{1080}{1.06^3}}{1,053.46} \times 3 \\ &= 2.7891\end{aligned}$$

And thus, the modified duration is given by

$$\text{Modified Duration} = \frac{2.7891}{1.06} = 2.6312$$

Reading 43: Corporate Bonds

Q.899 Joe George has recently joined the corporate trust department of Maximal Investment Bank based in New Jersey. George is an expert in performing the duties of a trustee. These functions include authenticating the bonds at the time of their issue, keeping track of the bonds sold and bonds outstanding, making sure that the issuer does not exceed the authorized principal, and making sure the issuer is maintaining required ratios. Which of the following parties does Joe George, a trustee, represent?

- A. Bond issuer
- B. Bondholder
- C. Government
- D. Independent rating agency

The correct answer is **B**.

The trustee, in this case Joe George, represents the bondholder. The role of a trustee in a bond issuance is to act as a representative for the bondholders. This is because it can be resource-intensive for a bondholder to monitor the activities of the bond issuer and to ensure that the issuer is not violating the terms of the bond indenture. The trustee, therefore, steps in to perform these duties on behalf of the bondholders. These duties include authenticating the bonds at the time of their issue, keeping track of the bonds sold and those still outstanding, ensuring that the issuer does not exceed the authorized principal, and verifying that the issuer is maintaining the required ratios. By performing these duties, the trustee helps to protect the interests of the bondholders and ensure that the bond issuer is adhering to the terms of the bond indenture.

Choice A is incorrect. The bond issuer is not represented by the trustee. The issuer is the entity that sells the bonds to raise funds, and it's their responsibility to maintain certain ratios and not surpass the authorized principal. The trustee's role is to ensure that these responsibilities are met, but they do not represent the issuer.

Choice C is incorrect. The government does not directly involve in individual bond issues and hence does not need representation by a trustee in this context. While trustees must ensure compliance with relevant laws and regulations, they do not represent the government.

Choice D is incorrect. An independent rating agency evaluates a bond's credit risk but does not have a direct stake in its issuance or management, so it would be inappropriate for a trustee to represent them.

Things to Remember

- Trustee in a bond issuance acts as a representative for the bondholders to ensure the issuer complies with the terms of the bond indenture.
- Trustees authenticate bonds at issuance, track bonds sold and outstanding, monitor

authorized principal, and verify required ratios.

- Bond issuer is responsible for maintaining ratios and principal, not represented by the trustee.
 - Government does not directly involve in individual bond issues and does not need representation by a trustee.
 - Independent rating agencies evaluate credit risk but do not have a direct stake in bond issuance or management.
-

Q.900 Trustees are the corporate trust units of large banks that are representative of the interest of bondholders. From which of the following parties does the trustee receives fees for its services?

- A. Governments
- B. Bond issuers
- C. Bondholders
- D. They work independently

The correct answer is **B**.

Trustees are typically compensated by the bond issuers for their services. This is because the bond issuer is the entity that benefits from the services of the trustee. The trustee acts as a representative of the bondholders, ensuring that the bond issuer fulfills its obligations as outlined in the bond indenture. The trustee's role is to protect the interests of the bondholders, and they are compensated by the bond issuer for this service. It's important to note that while the trustee represents the interests of the bondholders, they do not receive fees from the bondholders themselves. The fees are paid by the bond issuer as part of the agreement when the bonds are issued.

Choice A is incorrect. Trustees do not typically receive fees from governments. While governments may issue bonds, the trustee's role is to represent the interests of bondholders and they are compensated by the bond issuers, not the government directly.

Choice C is incorrect. Although trustees represent the interests of bondholders, they do not receive their fees from them. The bond issuer pays for these services as part of their obligation to protect and manage the rights of bondholders.

Choice D is incorrect. Trustees do not work independently in terms of compensation for their services. They are paid by the entity issuing bonds, which means they cannot operate without a

relationship with a bond issuer.

Things to Remember

- Trustees play a crucial role in ensuring that the terms of the bond indenture are upheld and that the rights of bondholders are protected.
 - They act as intermediaries between the bond issuer and the bondholders, representing the interests of the bondholders in various matters.
 - Trustees also monitor the financial condition of the bond issuer to ensure that it remains capable of meeting its obligations to bondholders.
 - They may also handle the disbursement of funds to bondholders, such as interest payments or principal repayments.
 - Trustees are typically appointed at the time of bond issuance and their duties and responsibilities are outlined in the bond indenture.
-

Q.901 Christopher Ray is a junior research analyst in the fixed-income unit of a mid-sized investment bank in the U.S. The fixed-income unit categorizes the bonds based on the type of issuers. These categories or issuer types are public utilities, transportations, industrials, banks & finance companies, and Yankees. The analyst is given the task to categorize a bond issued by a German municipal government that has recently issued bonds in the U.S. to raise capital to finance a new kindergarten in the center of Frankfurt. In which category should this bond be classified?

- A. Public utilities
- B. Industrials
- C. Yankees
- D. Banks & Finance

The correct answer is **C**.

The term 'Yankees' in the context of bond classification refers to bonds issued in the U.S. by foreign entities. These entities can be sovereign governments, foreign banks, companies, or foreign government agencies. In this case, the bond issued by the German municipal government in the U.S. falls under the 'Yankees' category. This is because the issuer is a foreign government entity issuing bonds in the U.S. market. The purpose of the bond issuance, which is to finance a

kindergarten, does not influence the classification of the bond. The classification is solely based on the issuer type. Therefore, the bond issued by the German municipal government is correctly classified as a 'Yankee' bond.

Choice A is incorrect. Public utilities refer to bonds issued by entities that provide essential public services such as water, electricity, and gas. The bond in question is issued by a German municipal government for the construction of a kindergarten, which does not fall under the category of public utilities.

Choice B is incorrect. Industrials are bonds issued by industrial companies or corporations. In this case, the issuer is a German municipal government and not an industrial company or corporation.

Choice D is incorrect. Banks & Finance refers to bonds issued by financial institutions like banks and finance companies. Here, the issuer of the bond is a German municipal government which does not belong to this category.

Things to Remember

- Yankee bonds are issued in the U.S. market by foreign entities, including sovereign governments, foreign banks, companies, or government agencies.
 - Public utilities bonds are issued by entities that provide essential public services like water, electricity, and gas.
 - Industrial bonds are issued by industrial companies or corporations.
 - Bonds issued by financial institutions like banks and finance companies fall under the Banks & Finance category.
 - Classification of bonds is based on the type of issuer and not the purpose of the bond issuance.
-

Q.903 Which of the following should be used to calculate the coupon on a bond?

- A. The coupon or interest is calculated on the price of the bond.
- B. The coupon or interest is calculated on the amortized value of the bond.
- C. The coupon or interest is calculated on the par value of the bond.
- D. The coupon or interest is always fixed in dollar terms.

The correct answer is C.

The coupon or interest on a bond is calculated on the par value of the bond. The par value, also known as the face value, is the amount that the bond issuer originally receives from the bondholder and promises to repay upon the bond's maturity. The coupon rate is expressed as a percentage of this par value. For instance, if a bond has a par value of \$1,000 and a coupon rate of 5%, the annual coupon payments would be \$50. This is because the coupon rate is applied to the par value of the bond to determine the annual coupon payments. Therefore, the par value of the bond is the correct basis for calculating the coupon or interest.

Choice A is incorrect. The coupon or interest of a bond is not calculated on the price of the bond. The price of a bond can fluctuate due to market conditions, but this does not affect the calculation of the coupon or interest which is fixed at issuance based on par value.

Choice B is incorrect. While amortized value can be used in certain calculations related to bonds, it's not used for calculating coupons or interest payments. Amortized cost accounting involves gradually writing off an intangible asset over a period of time, which doesn't apply here.

Choice D is incorrect. Although it might seem that coupons are always fixed in dollar terms because they are often quoted as such, this statement isn't universally true for all types of bonds. For example, floating rate notes have variable coupons tied to reference rates like LIBOR or Euribor and hence their dollar amount isn't always fixed.

Things to Remember

- **Bond Coupon:** The coupon on a bond is the fixed annual interest payment that the issuer pays to the bondholder, usually expressed as a percentage of the bond's par value.
- **Par Value:** Also known as face value, it is the amount that the bond issuer promises to repay upon the bond's maturity. The coupon is calculated based on this par value.
- **Amortized Value:** This is the gradual reduction of a bond's discount or premium to its face value over time. It is not used for calculating coupon payments.
- **Floating Rate Notes:** These are bonds with variable interest rates that are adjusted

periodically based on a reference rate, such as LIBOR or Euribor.

Q.904 The bonds that are issued in the United States are also classified on the basis of their interest rates. Which of the following types of bonds are always issued at a discount price?

- A. Zero-coupon bonds.
- B. Straight-coupon bonds.
- C. Floating rate bonds.
- D. Premium bonds.

The correct answer is **A**.

Zero-coupon bonds are a type of bond that does not pay interest (a coupon) during its life. Instead, investors buy zero-coupon bonds at a deep discount from their face value, which is the amount a bond will be worth when it 'matures' or comes due. When a zero-coupon bond matures, the investor will receive one lump sum equal to the initial investment plus interest that has accrued. This makes zero-coupon bonds unique among most types of bonds on the market, which typically pay out interest semiannually. The discount in price effectively represents the 'interest' the bond pays to investors. Therefore, by definition, zero-coupon bonds are always issued at a discount price.

Choice B is incorrect. Straight-coupon bonds are not always issued at a discount price. They pay periodic interest payments and the face value at maturity, and can be issued at par, premium or discount depending on the market interest rates.

Choice C is incorrect. Floating rate bonds have variable interest rates that are tied to a benchmark such as LIBOR or treasury bills. These bonds are not necessarily issued at a discount because their coupon rate adjusts with changes in market rates.

Choice D is incorrect. Premium bonds are actually sold for more than their face value (at a premium), hence they cannot be the type of bond that is invariably issued at a discount price.

Things to Remember

- Zero-coupon bonds do not pay interest during their life, but are issued at a deep discount from their face value.
- Straight-coupon bonds pay periodic interest payments and the face value at maturity, and can be issued at par, premium, or discount.
- Floating rate bonds have variable interest rates tied to a benchmark and their coupon rate adjusts with market rates.
- Premium bonds are sold for more than their face value (at a premium).

Q.905 Linda Angola is a senior asset investment advisor at Bright Partners Co., an investment advisory firm based in Singapore. Angola is responsible for providing advisory services to a group of small-medium institutional clients. Jaguar Tires Co. is one of the largest clients of Angola and the management of the company intends to invest in a fixed income instrument that pays a fixed coupon at a specific date and also gets a share of the issuer's profit if the issuer earns profits above a certain threshold. Which of the following should Angola recommend to Jaguar Tires?

- A. Deferred-interest bonds.
- B. Zero-coupon bonds.
- C. Participating bonds.
- D. Straight bonds.

The correct answer is C.

Participating bonds are a type of fixed income instrument that not only pays a fixed coupon at a specific date but also allows the bondholder to share in the issuer's profit if the issuer's profits exceed a certain threshold. This type of bond is particularly attractive to investors who want to benefit from the potential upside of the issuer's performance while still receiving regular fixed income payments. The profit-sharing component of participating bonds is typically linked to the issuer's profitability or the performance of a specific asset or index. This makes participating bonds a unique and potentially lucrative investment option for investors like Jaguar Tires Co. who are looking for a combination of fixed income and potential upside participation.

Choice A is incorrect. Deferred-interest bonds do not meet the requirements of Jaguar Tires Co. These bonds defer interest payments to a future date, but they do not allow the bondholder to share in the issuer's profits.

Choice B is incorrect. Zero-coupon bonds are also not suitable for Jaguar Tires Co.'s needs. These bonds do not pay any interest until maturity and certainly do not provide an opportunity for profit sharing with the issuer.

Choice D is incorrect. Straight bonds, also known as plain-vanilla bonds, offer fixed interest payments and return of principal at maturity but they don't provide any provision for profit sharing with the issuer which is a requirement of Jaguar Tires Co.

Things to Remember

- Participating bonds offer a unique feature of profit-sharing with the issuer if profits exceed a certain threshold.
- Investors like Jaguar Tires Co. may find participating bonds attractive due to the combination of fixed income payments and potential upside participation.

- Deferred-interest bonds defer interest payments to a future date but do not involve profit-sharing with the issuer.
 - Zero-coupon bonds do not pay any interest until maturity and do not provide profit-sharing opportunities with the issuer.
 - Straight bonds, also known as plain-vanilla bonds, offer fixed interest payments and return of principal at maturity but do not include provisions for profit-sharing with the issuer.
-

Q.906 Muhammad Zubair is a retail bond investor that invests in various types of government and corporate bonds. On June 30, 2016, he purchased a 5-year zero-coupon bond for the price of \$855. The bond was issued on July 1, 2014, at the discount price of \$690 and the face value of the bond is \$1000. Based on the given information, identify which of the following options is consistent with the definition of original issue discount (OID)?

- A. In the given case, the original issue discount (OID) is the difference between \$690 and \$1000.
- B. In the given case, the original issue discount (OID) is the difference between \$690 and \$855.
- C. In the given case, the original issue discount (OID) is the difference between \$855 and \$1000.
- D. In the given case, the original issue discount (OID) is the difference between the current amortized value and \$1000.

The correct answer is **A**.

The original issue discount (OID) is a term used in the bond market to represent the difference between the face value of a bond and its original issue price. In the context of zero-coupon bonds, which do not pay periodic interest, the OID essentially represents the total return an investor will receive upon the bond's maturity. In this scenario, the bond was originally issued at a price of \$690, and it carries a face value of \$1000. Therefore, the OID is the difference between the face value and the original issue price, which is $\$1000 - \$690 = \$310$. This amount represents the total return that the original bondholder will receive if they hold the bond until maturity.

Choice B is incorrect. The original issue discount (OID) is not the difference between the purchase price of \$690 and the price at which Muhammad Zubair bought it (\$855). The OID is determined at the time of initial issuance, not when it was purchased by Zubair.

Choice C is incorrect. The original issue discount (OID) cannot be calculated as the difference between Zubair's purchase price (\$855) and face value (\$1000). Again, OID refers to the discount at which a bond was originally issued, not its subsequent purchase prices.

Choice D is incorrect. The original issue discount (OID) does not involve any current amortized value in its calculation. It solely depends on the initial issuance price and face value of a bond.

Things to Remember

- Original Issue Discount (OID) is the difference between the face value of a bond and its original issue price.
 - OID is commonly seen in zero-coupon bonds, where it represents the total return an investor will receive at maturity.
 - OID is determined at the time of initial issuance and is not affected by subsequent purchase prices.
 - OID is not related to the current amortized value of a bond.
-

Q.907 Sam Denis is a junior fixed-income analyst that is analyzing a number of corporate bonds to recommend to one of his clients. The bonds under analysis are classified by the type of issuers, type of risk, and expected return. Which of the following categories of bonds will have the lowest interest rates?

- A. Corporate Bonds.
- B. Debenture issues.
- C. Mortgage bonds.
- D. Straight-coupon discount bonds.

The correct answer is **C**.

Mortgage bonds have the lowest interest rate and the highest price among the four types of bonds. Because of the fact that mortgage bonds are secured by mortgage liens, the interest rate on these bonds is low and the prices are higher. Options A and D are incorrect because Corporate Bonds offer a higher yield relative to a government bond due to the higher risk of insolvency. Straight-coupon discount bonds are sold at discount prices. Thus, the interest rate on such bonds is high. Option B is also incorrect because debenture issues are unsecured bonds. Due to higher risk, the price of the bonds is lower, and the interest is higher.

Things to Remember

- Corporate Bonds offer a higher yield relative to a government bond due to the higher risk of insolvency.
 - Straight-coupon discount bonds are sold at discount prices, leading to higher interest rates.
 - Debenture issues are unsecured bonds, which results in lower prices and higher interest rates.
 - Mortgage bonds are secured by mortgage liens, leading to lower interest rates and higher prices.
-

Q.911 Ohio Automotive Inc. raised capital to finance its expansion into the SUVs market. A year ago, the firm issued a 4-year 6% semi-annual coupon bond. The bond has a special provision that allows the issuer to call its bond before the maturity of the bond. Which of the following options is consistent with the properties of a callable bond?

- A. It is beneficial for the issuer to call the bond in an increasing interest rate environment.
- B. It is beneficial for the issuer to call the bond in a decreasing interest rate environment.
- C. It is beneficial for the bondholder if the bond is called in a decreasing interest rate environment.
- D. It is beneficial for the bondholder if the bond is called in an increasing prices environment.

The correct answer is **B**.

A callable bond provides the issuer with the advantage of redeeming the bond before its maturity, especially in a decreasing interest rate environment. When interest rates decrease, bond prices increase. This allows the issuer to call back the previously issued bonds, which were issued at higher interest rates, and replace them with new bonds at the current lower interest rates. This strategy allows the issuer to save on interest payments, making it beneficial for them. Therefore, in a decreasing interest rate environment, it is advantageous for the issuer to call the bond.

Choice A is incorrect. It is not beneficial for the issuer to call the bond in an increasing interest rate environment. When interest rates increase, the cost of borrowing increases for issuers. Therefore, they would prefer to keep paying lower coupon payments on existing bonds rather than issuing new bonds at higher interest rates.

Choice C is incorrect. It is not beneficial for the bondholder if the bond is called in a decreasing interest rate environment. When a bond is called, it means that it's bought back by the issuer before its maturity date. In a decreasing interest rate environment, if a bond gets called, then investors will have to reinvest their money at lower prevailing market rates which may lead to less income from their investment.

Choice D is incorrect. While it might seem beneficial for a bondholder if the bond was called in an increasing prices environment because they would receive more money upfront; however, this scenario does not align with benefits of having callable bonds from issuer's perspective as calling back bonds when prices are high would mean higher costs for them.

Things to Remember

- A callable bond gives the issuer the right to redeem the bond before its maturity date.
- Callable bonds are typically issued with higher coupon rates to compensate investors for the risk of early redemption.
- Callable bonds are more advantageous for the issuer in a decreasing interest rate environment as they can refinance at lower rates.

- Investors holding callable bonds face reinvestment risk if the bonds are called in a declining interest rate environment.
 - Callable bonds are less attractive to investors compared to non-callable bonds due to the risk of early redemption.
-

Q.912 Hauser Corp., a German portable house construction firm, is raising \$500 million through 7-year 9% semi-annual coupon bonds. Classico Investment Company is interested in purchasing 33% of Hauser's total bond issue, but it has put forward a condition that requires the issuer to retire a portion of the principal of the debt each year until maturity rather than paying the whole capital at maturity. This condition is most likely associated with the:

- A. Fixed-price call provision.
- B. Make-whole call provision.
- C. Sinking fund provision.
- D. Tender offer provision.

The correct answer is **C**.

The sinking fund provision is the correct answer. This provision is a protective covenant in the bond indenture that requires the issuer to retire a certain portion of the bond issue each year. The issuer can achieve this by either buying back a certain number of bonds in the open market or using a lottery system to select which bonds to retire. The sinking fund provision is designed to reduce the risk to bondholders by ensuring that the issuer does not default on the entire bond issue at maturity. It also helps to reduce the potential impact of a default on the bondholders. The sinking fund provision is beneficial to the bondholders as it provides a form of repayment guarantee. However, it can be disadvantageous to the issuer, especially in a declining interest rate environment, as the issuer may have to retire the bonds at a premium. In the context of the question, Classico Investment Company's condition that requires Hauser Corp. to retire a portion of the principal of the debt each year until maturity is most likely associated with the sinking fund provision.

Choice A is incorrect. A fixed-price call provision allows the issuer to buy back a portion or all of the bonds before maturity at a predetermined price. This does not align with Classico's condition of retiring a portion of the principal annually.

Choice B is incorrect. A make-whole call provision allows the issuer to pay off remaining debt early, but it requires them to make bondholders whole by paying an amount that equates to the net present value of future coupon payments that will be missed due to early repayment. This does not meet Classico's requirement for annual retirement of principal.

Choice D is incorrect. A tender offer provision involves an offer by the issuing company to

purchase back its own securities from holders at a specified price during a short period time. It doesn't necessarily involve retiring part of principal annually until maturity as required by Classico.

Things to Remember

- **Bond Indenture:** A legal document that outlines the terms and conditions of a bond issue, including any protective covenants such as sinking fund provisions.
 - **Default Risk:** The risk that the issuer of a bond may not be able to make interest payments or repay the principal amount at maturity.
 - **Callable Bonds:** Bonds that give the issuer the option to redeem the bonds before maturity, usually at a predetermined price.
 - **Yield to Maturity (YTM):** The total return anticipated on a bond if it is held until it matures, taking into account the current market price, par value, coupon interest rate, and time to maturity.
 - **Interest Rate Environment:** The prevailing level of interest rates in the economy, which can impact the attractiveness of bond investments and the cost of borrowing for issuers.
-

Q.913 Matthias Schmidt is the Chief Financial Officer of Caribbean Shipping & Logistics Company. Three years ago, the company raised \$600 million through 5-years 5% coupon bonds to finance its two new vessels that will sail in the Arabian Sea. Due to abrupt growth in emerging economies like Pakistan, Bangladesh, and India, the company grew exponentially in the last three years. The firm's senior management informed the CFO that they should retire the debt before maturity as the firm now has enough funds to pay for further expansions. If the indenture of the bond did not include any mechanism for early retirement in its indenture, then determine which of the following mechanisms could be used.

- A. Fixed-price call provision.
- B. Sinking fund provision.
- C. Tender offer provision.
- D. Replacement fund provision.

The correct answer is **C**.

A tender offer provision is a mechanism that allows for the early retirement of debt, even if it is not included in the bond's indenture. In a tender offer, the issuer of the bond sends an offering circular to the bondholders of record. This circular presents the price that the issuer is willing to pay to buy back the bond, as well as the window of time during which bondholders can sell their bonds back to the issuer. This mechanism is particularly useful in situations where the issuer has sufficient funds to retire the debt before its maturity, as is the case with Caribbean Shipping & Logistics Company. By using a tender offer, the company can effectively manage its debt and potentially save on interest payments.

Choice A is incorrect. A fixed-price call provision allows the issuer to retire part or all of the bond issue at a predetermined price before maturity. However, in this case, it was mentioned that the bond's indenture did not include any mechanism for early retirement, which implies that there is no call provision.

Choice B is incorrect. A sinking fund provision requires the issuer to retire a portion of the bond issue each year by buying back bonds in the open market or through a lottery system. Again, since there was no such provision included in Caribbean Shipping & Logistics Company's bond indenture, this option cannot be used.

Choice D is incorrect. There isn't any standard financial instrument known as a "replacement fund provision." This seems to be an invented term and does not apply to this situation.

Things to Remember

- Early bond retirement: Companies may choose to retire debt before maturity if they have sufficient funds to do so, which can help save on interest payments and manage debt levels.
- Call provision: A fixed-price call provision allows the issuer to retire part or all of the bond issue at a predetermined price before maturity.
- Sinking fund provision: A sinking fund provision requires the issuer to retire a portion of the bond issue each year by buying back bonds in the open market or through a lottery system.
- Debt management: Effective debt management is crucial for companies to maintain financial health and flexibility in funding future expansions.

Q.916 Iron Partners Co. is a private equity firm that invests in distressed firms through various investment vehicles. Recently, the company acquired a mid-size paper manufacturing company through a leveraged buy-out (LBO). Initially, the equity firm acquired loans from commercial and investment banks to purchase the company. However, the company now issues bonds to pay off the debt of these banks. This activity of issuing debt to retire initial debt is called:

- A. Payment-in-kind (PIKs).
- B. Multi-term notes.
- C. Bridge financing.
- D. Rolling stock certificates.

The correct answer is **C**.

Bridge financing is a short-term financing option used by companies before they can secure permanent financing. It 'bridges' the gap between when a company's money is set to run out and when it can expect to receive an inflow of funds later. In this case, Iron Partners Co. used bridge financing to acquire the paper manufacturing company. Initially, they borrowed funds from commercial and investment banks to facilitate the leveraged buy-out. However, to repay these initial loans, they decided to issue bonds, a strategy known as bridge financing. This strategy is often used in leveraged buyouts and mergers and acquisitions to cover the period between when the deal is closed and when the target company can generate cash flow or the acquirer secures long-term financing.

Choice A is incorrect. Payment-in-kind (PIKs) refers to a type of financing where the interest or dividends are paid in a form other than cash, often in additional securities. This does not align with the scenario described where new debt is issued to retire initial debt.

Choice B is incorrect. Multi-term notes refer to a type of bond that has multiple maturity dates, which allows for periodic return of principal over the life of the bond. This does not match with the strategy used by Iron Partners Co., as they are issuing new bonds to repay their initial loans, not returning principal periodically.

Choice D is incorrect. Rolling stock certificates are financial instruments related to railroads and do not have any relevance in this context where a private equity firm issues bonds to repay its initial loans from banks.

Things to Remember

- Bridge financing is a common strategy used in leveraged buyouts and mergers and acquisitions to cover the period between the deal closing and securing long-term financing.
- Payment-in-kind (PIKs) involve paying interest or dividends in the form of additional securities rather than cash.
- Multi-term notes have multiple maturity dates, allowing for periodic return of principal over the bond's life.
- Rolling stock certificates are financial instruments related to railroads and are not

relevant in the context of private equity firms issuing bonds to repay initial loans.

Q.917 McMillan Electronics Company issued four types of debt to raise the capital of \$730 million. The debt includes \$400 million raised through secured bonds, \$110 million through subordinate bonds, \$40 million through unsecured loans, and \$180 million through debenture bonds. If the company defaults, the total value of the assets to be distributed is \$670 million. Determine the total claim that the debenture bondholders will receive.

- A. \$180 million
- B. \$160 million
- C. \$120 million
- D. \$0

The correct answer is **A**.

The debenture bondholders have the claim over the issuer's assets after the claim of secured bondholders is satisfied. Though debenture bonds do not have pledged assets or collaterals, the residual funds after satisfying the secured debt (and before satisfying the claims of creditors) are distributed among debenture bondholders.

Since the total value of the firm's asset is \$670 million:

The secured bondholders will receive \$400 million;

The debenture bondholders will receive \$180 million;

The banks will receive \$40 million; and

The subordinate bondholders will receive \$50 million.

Things to Remember

- Secured bonds have collateral backing them, which means they have a higher claim on the company's assets in case of default compared to other types of debt.
- Subordinate bonds have a lower claim on the company's assets compared to secured bonds and debenture bonds.
- Unsecured loans are not backed by any collateral and are considered riskier for lenders.
- Debenture bonds are unsecured bonds that are backed only by the creditworthiness and reputation of the issuer.
- Debenture bondholders have a claim on the company's assets after the claims of secured bondholders are satisfied.

Q.5352 A portfolio manager is considering purchasing a bond with a face value of \$1,000, a coupon rate of 5%, and a maturity of 5 years. Which of the following statements is most likely correct regarding the bond's price and yield if market interest rates increase?

- A. The bond's price will increase, and its yield will decrease.
- B. The bond's price will decrease, and its yield will increase.
- C. The bond's price and yield will both increase.
- D. The bond's price and yield will both decrease.

The correct answer is **B**.

When market interest rates rise, the price of a bond decreases, and its yield increases. This is because the fixed coupon rate of the bond becomes less attractive when compared to newly issued bonds that offer higher coupon rates. The yield of a bond is calculated by dividing the bond's coupon payments by its market price. Therefore, when bond prices increase, bond yields decrease, and vice versa. In this scenario, the increase in market interest rates would make the bond's fixed coupon rate less attractive, leading to a decrease in the bond's price and an increase in its yield.

Choice A is incorrect. An increase in market interest rates would not cause the bond's price to increase and its yield to decrease. In fact, the opposite is true. When market interest rates rise, the price of existing bonds falls because they become less attractive compared to new bonds that are issued with higher coupon rates (interest payments). Consequently, the yield on these existing bonds increases as their prices fall.

Choice C is incorrect. It's not accurate that both the bond's price and yield will increase if there is an increase in market interest rates. As explained above, when market interest rates rise, it leads to a decrease in bond prices while causing an increase in yields.

Choice D is incorrect. This choice suggests that both the bond's price and yield will decrease with an increase in market interest rates which contradicts basic principles of bond pricing and yields. As previously mentioned, when market interest rates rise, it results in a decrease in bond prices but causes an increase in yields.

Things to Remember

- Bond prices and yields have an inverse relationship -when market interest rates rise, bond prices fall and yields rise.
- Yield is a measure of the return on investment for a bond, taking into account the coupon payments and the bond's price.
- Duration is an important concept in bond investing, representing the sensitivity of a bond's price to changes in interest rates.

- Yield to maturity (YTM) is the total return anticipated on a bond if it is held until it matures.
 - Callable bonds give the issuer the right to redeem the bond before it matures, which can impact the bond's price and yield.
-

Q.6293 Meadow Trust Co., a corporate trustee, is engaged to oversee a new bond issuance by Alpha Corp., a mid-sized technology firm. The newly appointed CFO of Alpha Corp. is evaluating the responsibilities of Meadow Trust Co. as delineated in the bond indenture. Which of the following statements accurately describes the role of Meadow Trust Co.?

- A. Meadow Trust Co. is obligated to regularly audit Alpha Corp.'s financials to ensure compliance with the bond covenants.
- B. Meadow Trust Co. should proactively enforce actions beyond the bond indenture terms if it deems it necessary to protect bondholders' interests.
- C. Meadow Trust Co. must act in accordance with the bond indenture whenever bondholders collectively request it.
- D. Meadow Trust Co. is compensated by Alpha Corp. but is tasked with safeguarding the bondholders' interests as stipulated in the indenture.

The correct answer is **D**.

The role of a corporate trustee, such as Meadow Trust Co., is to act on behalf of the bondholders and safeguard their interests as specified in the bond indenture. Although the trustee is compensated by the issuer, in this case, Alpha Corp., their fiduciary duty is to the bondholders. This involves ensuring that the terms and covenants outlined in the indenture are adhered to and taking appropriate action if there are any breaches. The trustee's role is to provide an additional layer of protection for the bondholders, ensuring that the issuer complies with its obligations, thereby maintaining the integrity of the bond agreement. This structure helps balance the interests of both the issuer and the investors, fostering trust and confidence in the bond market.

A is incorrect. While the trustee may monitor the issuer's financial condition, their primary duty is not to audit financials regularly but to act as specified in the bond indenture.

B is incorrect. Trustees typically do not go beyond the terms of the indenture unless under extraordinary circumstances or legal obligations. Their actions are generally confined to the indenture.

C is incorrect. Trustees are obligated to act within the confines of the indenture, and not every bondholder request requires action unless it aligns with the indenture's terms.

Things to Remember

- The primary duty of a corporate trustee like Meadow Trust Co. is to act in the best interests of the bondholders, according to the terms specified in the bond indenture.
 - Corporate trustees are compensated by the bond issuer but maintain a fiduciary duty to the bondholders, ensuring compliance with bond covenants and addressing any breaches.
 - A bond indenture is a legally binding contract between the bond issuer and the bondholders, outlining the key terms, covenants, and responsibilities of all parties involved, including the trustee.
 - Trustees provide an additional layer of oversight, which helps protect bondholders by monitoring the issuer's adherence to the agreed-upon terms and taking necessary action in the event of non-compliance.
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Reading 44: Mortgages and Mortgage-Backed Securities

Q.918 Mortgage-backed loans played a significant role in the 2007-2009 financial crisis. After the crisis, the importance of securitization of these loans increased furthermore. In the United States, there are multiple entities that securitize mortgage loans. Which of the following types of loans is securitized through the Federal National Mortgage Association?

- A. Adjustable-rate mortgage loans.
- B. Non-agency loans.
- C. Jumbos.
- D. Agency loans.

The correct answer is **D**.

Agency loans, also known as conforming loans, are typically residential loans that are securitized through entities like the Federal National Mortgage Association (FNMA), Government National Mortgage Association (GNMA), and Federal Home Loan Mortgage Corporation (FHLMC). These loans conform to the guidelines set by these government-sponsored entities (GSEs). The GSEs buy these loans from lenders, package them into mortgage-backed securities (MBS), and guarantee the timely payment of principal and interest to the MBS investors. This process helps to provide liquidity to the mortgage market, enabling lenders to make more loans. The FNMA, commonly known as Fannie Mae, is one of the leading agencies in this process.

Choice A is incorrect. Adjustable-rate mortgage loans are not typically securitized through the Federal National Mortgage Association (FNMA). While FNMA does deal with adjustable-rate mortgages, it is not the primary type of loan they securitize.

Choice B is incorrect. Non-agency loans are those that are not backed by government agencies. The Federal National Mortgage Association, being a government-sponsored enterprise, primarily deals with agency loans and does not typically securitize non-agency loans.

Choice C is incorrect. Jumbos or jumbo loans exceed the conforming loan limits set by the Federal Housing Finance Agency (FHFA), and hence, they are not usually securitized through FNMA which primarily deals with conforming loans within the set limits.

Things to Remember

- Securitization is the process of pooling various types of debt, such as mortgages, auto loans, or credit card debt, and selling them as securities to investors.
- Government-sponsored entities (GSEs) like FNMA, GNMA, and FHLMC play a crucial role in the securitization of mortgage loans by providing liquidity to the market.
- Conforming loans are those that meet the guidelines set by GSEs and are typically

easier to securitize compared to non-conforming loans.

- Mortgage-backed securities (MBS) are financial products that are created by pooling together various mortgage loans and selling them to investors.
 - The 2007-2009 financial crisis was triggered in part by the collapse of the subprime mortgage market, which had a significant impact on the securitization of mortgage loans.
-

Q.919 Mohan Singh, a sales manager at a retail chain, has recently moved to Minnesota with his wife. He purchased a studio apartment with a mortgage loan of \$140,000 at 5% for 20 years. Which of the following is the most appropriate estimation of the monthly installments on this loan?

- A. \$1,004.61
- B. \$923.94
- C. \$763.84
- D. \$340.60

The correct answer is **B**.

The monthly installments of \$923.93 for the monthly periods of 20 years (12x20=240) at the rate of 5% results in the present value of the loan that is \$140,000.

The easiest way to solve this problem is with the help of the financial calculator:

$N=20 \times 12=240$; $I/Y=5/12=0.4167$; $PV=140,000$; $FV=0$; $CPT \Rightarrow PMT=-923.94$

This installment can also be calculated with the simple annuity formula as follows:

$$PV = \text{installment} \frac{[1 - (1 + r)^{-n}]}{r}$$

Since the annual rate of interest is 5%, the monthly rate can be approximated as $5\%/12 = 0.004167$ or 0.4167%

Thus, $140,000 = \text{installment} \frac{[1 - 1.004167^{-240}]}{0.004167}$

$$\text{Installment} = \frac{140,000}{\left\{ \frac{[1 - 1.004167^{-240}]}{0.004167} \right\}} = \$923.94$$

Q.920 Which of the following is the accurate explanation of a mortgage loan with a prepayment option?

- A. It allows the lender to demand the repayment before the maturity of the loan.
- B. It restricts the borrower from repaying the mortgage before the maturity date.
- C. It allows the borrower to demand the disbursement of the mortgage loan before the agreed disbursement date.

D. It allows the borrower to repay the mortgage loan before the maturity date of the loan.

The correct answer is **D**.

A mortgage loan with a prepayment option indeed allows the borrower to repay the mortgage loan before the maturity date of the loan. This option provides the borrower with the flexibility to manage their finances more effectively. If the borrower comes into a financial windfall or manages to save enough money, they can choose to repay the loan early, thereby saving on future interest payments. This option is particularly beneficial in a declining interest rate environment. If interest rates decline, the present value of future payments will be higher. Hence, the borrower can choose to retire the mortgage early, thereby saving on future interest payments. However, it's important to note that some lenders may charge a prepayment penalty to compensate for the loss of interest income. Therefore, borrowers should carefully read the loan agreement before deciding to prepay their mortgage.

Choice A is incorrect. This statement is not accurate because the prepayment option in a mortgage loan does not allow the lender to demand repayment before the maturity of the loan. Instead, it gives this right to the borrower.

Choice B is incorrect. This statement contradicts with what a prepayment option offers. The prepayment option does not restrict but allows borrowers to repay their mortgage loans before their maturity date.

Choice C is incorrect. The disbursement of a mortgage loan before its agreed date has nothing to do with a prepayment option. Prepayment refers specifically to repaying part or all of a debt obligation earlier than its due date, and not about demanding disbursement earlier.

Things to Remember

- Prepayment option in a mortgage loan allows the borrower to repay the loan before the maturity date, providing flexibility and potential interest savings.
- In a declining interest rate environment, prepaying a mortgage can be advantageous as it helps in saving on future interest payments.
- Some lenders may charge a prepayment penalty to compensate for the loss of interest income when borrowers choose to prepay their mortgage.
- It's essential for borrowers to carefully review the loan agreement to understand any prepayment terms and conditions before deciding to prepay their mortgage.

Q.921 Alison Garry and her partner have recently secured a mortgage to buy a condominium in a beach town in Costa Rica. The purchase price of the condo was \$40,000, and the mortgage has an interest rate of 3.25% for a period of 10 years. The mortgage also includes a prepayment option. In light of these circumstances, which of the following statements would be the most accurate regarding the appropriate use of the prepayment option?

- A. The use of the prepayment option is appropriate if the mortgage rate increases to 3.75%.
- B. The use of the prepayment option is appropriate if the mortgage rate remains at 3.25%.
- C. The use of the prepayment option is appropriate if the mortgage decreases to 3.0%.
- D. None of the above; prepayment is not a function of interest rates.

The correct answer is **C**.

A prepayment option mortgage loan allows the mortgage borrower to pay back the outstanding principal before its predefined maturity date. A mortgage borrower can benefit from the prepayment option if the interest rates decline. The prepayment option is valuable for the borrower when the mortgage rates decline. As the rates will decline, the present value of the remaining monthly payments will be greater than the principal outstanding. Hence, the borrower can gain by paying the principal outstanding in exchange for not having to make further mortgage payments.

Think of this along the lines of refinancing. If rates fall, you can pay off the existing mortgage and take on a new one that will come with lower coupon payments.

Things to Remember

- Prepayment option allows the borrower to pay off the outstanding principal before the maturity date of the loan.
- Prepayment option is beneficial for the borrower when interest rates decrease.
- Refinancing is a common strategy used by borrowers to take advantage of lower interest rates.
- Prepayment option can help borrowers save on interest costs over the life of the loan.

Q.922 Ahmed Saeed has recently graduated from the Frankfurt Finance School with a Bachelor's degree. He was invited by a small-size audit firm that provides audit services to small-medium companies and startups to take a test in order to join the firm as a junior risk analyst. In the test,

he was asked to identify the definition of securitization. Which of the following is the appropriate definition of securitization?

- A. Securitization is the process of securing the mortgage with a security or collateral, which the lender can use in case of default.
- B. Securitization is the process of setting a bankruptcy-remote entity with the sole purpose issuing bank loans to individual borrowers.
- C. Securitization is the process of setting a bankruptcy-remote entity with the sole purpose of acquiring asset-backed securities (ABSs).
- D. Securitization is the process of converting a group of nonmarketable assets, or expected future cash flows on the assets, into units of marketable securities.

The correct answer is **D**.

Securitization is indeed the process of converting a group of nonmarketable assets, or expected future cash flows on the assets, into units of marketable securities. This process is often carried out by financial institutions, such as banks, to transfer the risk associated with these assets to the market. By doing so, these institutions are able to generate funds for their business operations. The securitization process involves the creation of a special purpose vehicle (SPV) or special purpose entity (SPE), which purchases the assets from the originator. The SPV/SPE then issues securities that are backed by these assets, which are sold to investors. The cash flows generated by the underlying assets are used to service the securities. This process allows the originator to remove these assets from their balance sheet, thereby improving their financial position.

Choice A is incorrect. Securitization does not involve securing the mortgage with a security or collateral. This definition is more aligned with the concept of a secured loan, where the borrower pledges an asset as collateral for the loan.

Choice B is incorrect. While securitization does involve setting up a bankruptcy-remote entity, its purpose isn't to issue bank loans to individual borrowers. Rather, this entity's purpose is to hold assets and issue securities backed by these assets.

Choice C is incorrect. The statement incorrectly defines securitization as setting up an entity for acquiring asset-backed securities (ABSs). In reality, securitization involves creating ABSs from non-marketable assets or expected future cash flows on these assets.

Things to Remember

- Securitization helps financial institutions to transfer risk associated with assets to the market.
- Special Purpose Vehicles (SPVs) or Special Purpose Entities (SPEs) are created in the securitization process to purchase assets from the originator.

- The cash flows from the underlying assets are used to service the securities issued in securitization.
 - Securitization allows originators to improve their financial position by removing assets from their balance sheet.
 - Securitization is a key mechanism for liquidity management and risk transfer in financial markets.
-

Q.924 Billy Clark is an investment manager at the Sachsenhausen Investment Bank based in Dusseldorf. Clark manages a pool of mortgages and the assets constructed with the pool. If the pool prepaid 1.1% of its principal above its amortizing principal as the percentage of total outstanding principal in the month of February, then which of the following is the appropriate annualized constant prepayment rate he can come up with?

- A. 0.0302
- B. 0.1243
- C. 0.132
- D. 0.1402

The correct answer is **B**.

If the pool prepaid 1.1% of its principal above its amortizing principal as the percentage of total outstanding principal in the month, then its single monthly mortality rate or SMM is 1.1%.

$$\begin{aligned}\text{Constant prepayment rate or CPR} &= 1 - (1 - \text{SMM})^{12} \\ \text{CPR} &= 1 - (1 - 0.011)^{12} = 0.1243\end{aligned}$$

Q.926 Pooja Rao has recently joined Green Oceans Hedge Fund that invests in non-conventional investment assets and securities. Rao was instructed by the fund manager to evaluate four mortgage-backed securities of four different banks, and identify the riskiest MBS. Identify for Rao which of these MBSs has the highest risk.

- A. Prime fixed-rate MBS.
- B. Subprime fixed-rate MBS.

C. Prime adjustable-rate mortgage.

D. Subprime adjustable-rate mortgage.

The correct answer is **D**.

Subprime adjustable-rate mortgages (ARMs) are considered to be the riskiest type of mortgage-backed securities (MBSs). This is primarily due to the nature of the borrowers and the structure of the loans. Subprime mortgages are typically extended to borrowers who have poor creditworthiness, which is often reflected in higher loan-to-value (LTV) and debt-to-income (DTI) ratios. These borrowers are more likely to default on their loans, making the securities backed by these mortgages riskier. Additionally, the adjustable-rate structure of these mortgages adds another layer of risk. Unlike fixed-rate mortgages, ARMs have interest rates that can fluctuate over time. If interest rates rise, the borrowers' mortgage payments can increase significantly, further increasing the likelihood of default. Therefore, among the four options, subprime ARMs are the riskiest MBSs.

Choice A is incorrect. Prime fixed-rate MBSs are generally considered to be less risky than their subprime counterparts. This is because prime borrowers have a lower risk of defaulting on their loans, which reduces the overall risk of the MBS. Additionally, the fixed rate nature of these securities means that they are not subject to interest rate fluctuations, further reducing their risk profile.

Choice B is incorrect. Subprime fixed-rate MBSs carry more risk than prime fixed-rate MBSs due to the higher likelihood of default by subprime borrowers. However, they still have a lower level of risk compared to adjustable-rate mortgages (ARMs), as ARMs expose investors to additional interest rate risks.

Choice C is incorrect. While prime adjustable-rate mortgages do carry some additional interest rate risks compared with fixed-rate mortgages, they are still less risky than subprime ARMs due to the lower default rates associated with prime borrowers.

Things to Remember

- **Mortgage-Backed Securities (MBS):** MBS are investment products that are backed by a pool of mortgages. Investors in MBS receive payments based on the interest and principal payments made by borrowers on the underlying mortgages.
- **Fixed-Rate Mortgages:** Mortgages with a fixed interest rate that remains constant throughout the life of the loan.
- **Adjustable-Rate Mortgages (ARMs):** Mortgages with an interest rate that can change periodically based on changes in a specified index.
- **Loan-to-Value (LTV) Ratio:** A ratio used to assess the risk of a mortgage loan by

comparing the loan amount to the appraised value of the property.

- **Debt-to-Income (DTI) Ratio:** A ratio used to evaluate a borrower's ability to manage monthly payments and repay debts by comparing total monthly debt payments to gross monthly income.
 - **Default Risk:** The risk that a borrower will be unable to make required payments on a loan, leading to potential losses for investors in MBS.
 - **Interest Rate Risk:** The risk that changes in interest rates will affect the value of fixed-income securities, such as MBS.
-

Q.928 Boris Arkarov is a Russian real estate investor based in California. He has been investing in the real estate sector of California for the past 20 years and is famous for selling some luxurious villas to well-known Hollywood celebrities. In a magazine interview, Mr. Arkarov expressed that he wants to borrow to finance his personal residential estate in the suburbs of L.A that costs \$10.5 million. If the loan-to-value ratio is 78%, calculate the amount of monthly payment that Mr. Arkarov has to pay on a 20-year mortgage at the rate of 9%.

- A. \$94,471.22
- B. \$73,687.56
- C. \$45,750
- D. \$34,890.74

The correct answer is **B**.

1. Calculate the Loan Amount.

$$\begin{aligned}\text{Loan Amount} &= \text{Property Value} \times \text{Loan-to-Value Ratio} \\ &= \$10,500,000 \times 0.78 \\ &= \$8,190,000\end{aligned}$$

2. Calculate the Monthly Interest Rate.

$$\begin{aligned}\text{Monthly Interest Rate} &= \frac{\text{Annual Interest Rate}}{12} \\ &= \frac{0.09}{12} \\ &= 0.0075\end{aligned}$$

3. Calculate the Total Number of Payments.

$$\begin{aligned}\text{Total Payments} &= \text{Loan Term in Years} \times 12 \\ &= 20 \times 12 \\ &= 240\end{aligned}$$

4. Calculate the Monthly Payment.

$$\begin{aligned}P &= \frac{r \times PV}{1 - (1 + r)^{-n}} \\ &= \frac{0.0075 \times \$8,190,000}{1 - (1 + 0.0075)^{-240}} \\ &\approx \$73,687.56\end{aligned}$$

So, with a loan amount of \$8,190,000, a monthly interest rate of 0.0075, and 240 total payments, the monthly mortgage payment comes out to approximately \$73,687.56.

Things to Remember

- **Loan-to-Value Ratio (LTV):** The ratio of the loan amount to the appraised value or purchase price of the property.
- **Monthly Interest Rate:** The annual interest rate divided by 12 to get the monthly rate.
- **Loan Term:** The number of years over which the loan will be repaid.
- **Amortization:** The process of paying off a debt over time through regular payments.
- **Mortgage Payment Formula:** The formula to calculate the monthly mortgage payment based on the loan amount, interest rate, and loan term.

Q.933 Anna Henderson is a high net-worth individual investor with Galaxy Investments Inc. Anna has recently learned about the investments and returns of mortgage-backed securities. Jacob Glen, a dedicated investment manager, briefed Henderson that she does not need to concern about the contraction risk of mortgage pool as their investment products are designed to mitigate risk. Which of the following is the most appropriate explanation for the contraction risk?

- A. Contraction risk is the risk related to the increase in the expected life of a mortgage pool due to falling interest rates and higher prepayment rates.
- B. Contraction risk is the risk related to the increase in the expected life of a mortgage pool due to increasing interest rates and lower prepayment rates.
- C. Contraction risk is the risk related to the decrease in the expected life of a mortgage pool due to falling interest rates and higher prepayment rates.
- D. Contraction risk is the risk related to the decrease in the expected life of a mortgage pool due to increasing interest rates and lower prepayment rates.

The correct answer is **C**.

Contraction risk in the context of mortgage-backed securities refers to the risk associated with the decrease in the expected life of a mortgage pool. This decrease is typically triggered by higher prepayment rates, which are often a result of falling interest rates. When interest rates fall, borrowers are more likely to refinance their mortgages at these lower rates, leading to a higher rate of prepayments. This, in turn, reduces the expected life of the mortgage pool, hence the term 'contraction risk'. This risk is a significant concern for investors in mortgage-backed securities as it can impact the timing and amount of their expected cash flows.

Choice A is incorrect. This choice incorrectly states that contraction risk is related to the increase in the expected life of a mortgage pool due to falling interest rates and higher prepayment rates. In reality, falling interest rates and higher prepayment rates lead to a decrease in the expected life of a mortgage pool, which is known as contraction risk.

Choice B is incorrect. This choice inaccurately describes contraction risk as being associated with an increase in the expected life of a mortgage pool due to increasing interest rates and lower prepayment rates. However, these conditions actually result in extension risk, not contraction risk.

Choice D is incorrect. This option wrongly suggests that contraction risk involves a decrease in the expected life of a mortgage pool due to increasing interest rates and lower prepayment rates. These circumstances would actually cause extension risk rather than contraction risk.

Things to Remember

- Extension risk is the risk related to the increase in the expected life of a mortgage pool due to increasing interest rates and lower prepayment rates.
- Prepayment risk is the risk that borrowers will prepay their mortgages earlier than expected, impacting the cash flows of mortgage-backed securities.
- Mortgage-backed securities (MBS) are investment products that are backed by a pool of mortgages, where investors receive payments based on the interest and principal

payments from the underlying mortgages.

- Investors in MBS need to consider various risks such as interest rate risk, credit risk, and prepayment risk in addition to contraction risk.
-

Q.936 GrossHaus Investment Bank is one of the largest German investment banks that has more than 38% market share in financing residential and commercial properties. GrossHaus is not only involved in the mortgage business, but it also issues securities to its investors against mortgage pools. Robin Frazer has recently purchased a security from a bank that gives him a claim to the principal portion of a mortgage payment on a mortgage pool. Which of the following securities has Frazer has purchased from the bank?

- A. A planned amortization class tranche (PAC).
- B. A collateralized mortgage obligation (CMO).
- C. A strip.
- D. A mortgage back security (MBS).

The correct answer is **C**.

A strip is a type of security that allows an investor to separately purchase the principal portion and interest portion of the mortgage payments on a mortgage pool. In the context of the question, Robin Frazer has purchased a principal-only strip (PO) from GrossHaus Investment Bank. These PO strips are typically sold at a discount and increase in size over time as the principal component of the mortgage grows. This is in contrast to interest-only (IO) strips, which decrease in size over time as the principal due on the mortgage decreases. Therefore, the security that Frazer has purchased is a strip, making choice C the correct answer.

Choice A is incorrect. A planned amortization class tranche (PAC) is a type of mortgage-backed security that is designed to protect investors from prepayment risk. It does not entitle the investor to the principal portion of a mortgage payment on a mortgage pool, but rather provides scheduled payments over a period of time.

Choice B is incorrect. A collateralized mortgage obligation (CMO) is a type of security that pools together mortgages and then issues tranches to investors based on the risk profile and cash flow needs. While it does involve mortgages, it doesn't specifically entitle an investor to the principal portion of a single mortgage payment.

Choice D is incorrect. A Mortgage Backed Security (MBS) represents an ownership interest in mortgage loans made by financial institutions. It does not specifically provide entitlement to the principal portion alone as it includes both principal and interest payments from homeowners.

Things to Remember

- Principal-only (PO) strips allow investors to separately purchase the principal portion of mortgage payments.
 - Interest-only (IO) strips allow investors to separately purchase the interest portion of mortgage payments.
 - Collateralized Mortgage Obligations (CMOs) pool together mortgages and issue tranches based on risk profiles.
 - Prepayment risk is a concern for investors in mortgage-backed securities.
 - Mortgage-backed securities (MBS) represent ownership interests in mortgage loans.
-

Q.937 Since the mortgage borrowers have the option to prepay the underlying securities at any time, the valuation of the mortgage securities with this embedded options is not possible with the traditional valuation model. One of the mortgage securities valuation model uses probability distributions to value securities. In other words, it values the securities by allocating different probabilities to the multiple variables like future interest rate, shape of the yield curve, default rate, prepayment rate, etc. Which of the following models uses this approach?

- A. Binomial model approach.
- B. Best guess approach.
- C. Monte Carlo simulation.
- D. Black-Scholes Model.

The correct answer is **C**.

The Monte Carlo simulation is a computational algorithm that relies on repeated random sampling to obtain numerical results. In the context of mortgage-backed securities (MBS), the Monte Carlo simulation is used to value these complex financial instruments. The model does this by assigning different probabilities to various factors that can affect the value of the MBS. These factors include future interest rates, the shape of the yield curve, default rates, prepayment rates, recovery rates, and interest rate volatility. By considering these variables and their associated probabilities, the Monte Carlo simulation generates a range of possible outcomes. The average of these outcomes is then taken as the estimated value of the MBS. This approach allows for a more nuanced and accurate valuation of MBS, taking into account the embedded prepayment option and the various factors that can influence the value of these securities.

Choice A is incorrect. The Binomial model approach is a method used for the valuation of

options and other financial instruments. It constructs a recombining binomial lattice (tree) for the short rate, from which one can calculate the price of fixed income securities. However, it does not assign probabilities to various factors such as future interest rates, yield curve shape, default rates, prepayment rates etc., which are crucial in valuing mortgage-backed securities.

Choice B is incorrect. The Best guess approach relies on subjective judgment rather than probabilistic modeling to estimate values and risks associated with financial instruments. This approach lacks the systematic and comprehensive analysis provided by probabilistic models like Monte Carlo simulation in assessing various factors affecting MBS valuation.

Choice D is incorrect. The Black-Scholes Model is used primarily for pricing European options and does not incorporate multiple risk factors or their probabilities as required in MBS valuation. It assumes markets are efficient with no transaction costs or taxes; these assumptions may not hold true in real-world scenarios involving mortgage-backed securities.

Things to Remember

- Mortgage-backed securities (MBS) are financial instruments that represent an ownership interest in a pool of mortgage loans.
- Prepayment risk is a key consideration when valuing MBS, as borrowers can choose to prepay their mortgages early, impacting the cash flows to MBS investors.
- Monte Carlo simulation is a powerful tool for valuing complex securities by considering multiple variables and their associated probabilities.
- Factors such as future interest rates, yield curve shape, default rates, prepayment rates, recovery rates, and interest rate volatility all play a role in MBS valuation.
- The Binomial model approach is commonly used for valuing options and constructing binomial lattices, but it does not assign probabilities to various factors like Monte Carlo simulation.
- The Best guess approach relies on subjective judgment rather than probabilistic modeling, which may not provide a comprehensive analysis for valuing MBS.
- The Black-Scholes Model is primarily used for pricing European options and may not be suitable for valuing MBS due to its assumptions about market efficiency.

Q.3456 Suppose that an investor owns a pass-through security with an initial principal of \$500 million. The remaining mortgage balance at the beginning of a certain month is \$400 million. Assuming that the SMM is 0.4125% and the scheduled principal payment is \$5 million, the estimated *prepayment for the month* is:

- A. \$1.65 million
- B. \$1.25 million
- C. \$2.04 million
- D. \$1.63 million

The correct answer is **D**.

Note: We only take into account the remaining mortgage balance, not the initial amount of principal.

$$\begin{aligned}\text{Prepayment for month } i(\text{in } \$) &= \text{SMM}(\text{beginning balance} - \text{scheduled principal repayment}) \\ &= 0.4125\%(\$400\text{m} - \$5\text{m}) = \$1,629,375\end{aligned}$$

Q.3457 A fund holds & a 10 million nominal of the GNMA 5.5% 30-year bond with a face value of 100. It enters into a one-month dollar roll with a repo dealer bank in which it sells the security at a price of 100-08 and buys it back at a forward price of par. Assuming that the security experiences a 2% paydown (scheduled principal plus prepayments) during the term of the trade, estimate the value of the drop.

- A. \$225,000
- B. \$22,500,000
- C. \$800,000
- D. \$200,500

The correct answer is **A**.

Drop is the price difference between the front month, i.e., month of sale, and the back month, i.e., the month of purchase.

The bonds are sold for 100-08, hence proceeds = $\$10m(100 + 8/32)/100 = \$10,025,000$

Purchasing the security at par in the back month, bearing in mind that the security has experienced a 2% paydown, will cost:

$$\$10m \times (1 - 2\%) \times (100/100) = \$9,800,000$$

$$\text{Drop} = \$10,025,000 - \$9,800,000 = \$225,000$$

Q.3458 Consider a pool of mortgages that were issued exactly 16 months ago at an effective interest rate of 6% p.a(they are beginning the 17th month). What is the CPR, and what is the SMM assuming 150 PSA?

- A. CPR = 0.2141%; SMM = 0.01786%
- B. CPR = 4.8%; SMM = 0.409%
- C. CPR = 5.1%; SMM = 0.4353%
- D. CPR = 3.4%; SMM = 0.2878%

The correct answer is **B**.

Assuming 100 PSA

$$\text{CPR}(\text{month } t \mid t \leq 30) = 6\% \times t/30$$

$$\text{CPR}(\text{month } 17) = 6\% \times 16/30 = 3.2\%$$

$$150 \text{ PSA implies that } \text{CPR}(\text{at the beginning of month } 17) = 1.5 \times 3.2\% = 4.8\%$$

$$\text{SMM} = 1 - (1 - \text{CPR})^{1/12} = 1 - (1 - 0.048)^{1/12} = 0.409\%$$

Note: The Public Securities Association model prepayment benchmark is one of the models used to estimate the monthly rate of prepayment. It is based on the assumption that rather than remaining constant, the monthly repayment rate gradually increases as a mortgage pool ages. The model assumes that:

(I) $\text{CPR} = 0.2\%$ for the first month after origination, increasing by 0.2% every month up to 30 months

(II) $\text{CPR} = 6\%$ for months 30 to 360

Things to Remember

- Conditional Prepayment Rate (CPR) is the annualized percentage of the principal of a pool of mortgages that is expected to prepay in a year, given a certain set of assumptions.
- Single Monthly Mortality (SMM) is the percentage of the remaining principal balance of a pool of mortgages that is expected to prepay in a month.
- PSA (Public Securities Association) is a prepayment benchmark used to measure prepayment speeds for mortgage-backed securities.
- The PSA model assumes a gradual increase in prepayment rates over time, starting at 0.2% and increasing by 0.2% each month up to 30 months, then remaining constant at 6% from month 30 to 360.
- Prepayment models are essential in valuing mortgage-backed securities and assessing the risks associated with prepayment behavior.

Q.3459 TBA prices of the Freddie Mac 5% for June 10 and July settlements are \$104.20 and \$103.00, respectively. The accrued interest to be added to each of these prices is \$0.160. The expected total principal paydown (scheduled principal plus prepayments) is 2% of the outstanding balance and the prevailing short-term rate is 1.5%. Also, assume that the actual/360-day convention is applied. An investor wishes to roll a balance of \$10 million. Determine the value of the roll.

A. \$436,000

- B. \$13,045
- C. \$109,680
- D. \$97,698

The correct answer is **D**.

Proceeds from selling the June 10 TBA are: $\$10\text{m} \times (104.20 + 0.160)/100 = \$10,436,000$
Investing these proceeds to July 10 at 1.5% interest earns interest of:
 $\$10,436,000 \times 30/360 \times 1.5\% = \$13,045$

Purchasing the July 10 TBA, which has experienced a 2% paydown, will cost:
 $\$10\text{m} \times (1 - 2\%) \times (103.0 + 0.160)/100 = \$10,109,680$

Net proceeds from the roll therefore are: $\$10,436,000 + \$13,045 - \$10,109,680 = \$339,365$

If the investor does not roll, the net proceeds are the coupon plus principal paydown:
 $\$10\text{m} \times (5\%/12 + 2\%) = \$241,667$

Value of the roll = net proceeds from the roll - net proceeds without roll: =
 $\$339,365 - \$241,667 = \$97,698$

Things to Remember

- TBA (To Be Announced) prices are used in the mortgage-backed securities market to represent a generic MBS for future delivery.
- Accrued interest is the interest that has accumulated on a bond since the last interest payment date.
- Principal paydown refers to the reduction in the outstanding balance of a loan or security due to scheduled principal payments or prepayments.
- The prevailing short-term rate is the current interest rate in the short-term market, which can impact investment decisions.
- The actual/360-day convention is a day-count convention where the actual number of days in a month and year are used for interest calculations.

Q.3460 A mortgage-backed portfolio includes four mortgage investments as follows:

- Mortgage 1: \$140,000 in current value, 5% interest rate, 5 years remaining duration
- Mortgage 2: \$100,000 in current value, 4% interest rate, 6 years remaining duration
- Mortgage 3: \$50,000 in current value, 6% interest rate, 3 years remaining duration
- Mortgage 4: \$60,000 in current value, 3% interest rate, 2 years remaining duration

What is the weighted average coupon of the portfolio?

- A. 4.5%
- B. 5.1%
- C. 4.9%
- D. 4.0%

The correct answer is **A**.

The weighted average coupon (WAC) is the weighted-average interest rate of mortgages that underlie a mortgage-backed security (MBS) at the time the securities were issued. It represents the **average interest rate** of a pool of mortgages with varying interest rates.

Total value of portfolio = \$140,000 + \$100,000 + \$50,000 + \$60,000 = \$350,000

We then compute the percentage value of each mortgage:

- Mortgage 1 %value = $\$140,000 / \$350,000 = 40\%$
- Mortgage 2 %value = $\$100,000 / \$350,000 = 28.6\%$
- Mortgage 3 %value = $\$50,000 / \$350,000 = 14.3\%$
- Mortgage 4 %value = $\$60,000 / \$350,000 = 17.1\%$

The percentage values of each mortgage are then multiplied by their respective interest rates:

- $40\% * 5\% = 2\%$
- $28.6\% * 4\% = 1.1\%$
- $14.3\% * 6\% = 0.9\%$
- $17.1\% * 3\% = 0.5\%$

The resulting figures are then totaled to produce a WAC of approx. 4.5%.

Things to Remember

- Weighted Average Coupon (WAC) is a crucial metric in analyzing mortgage-backed securities (MBS).
 - WAC is calculated by taking the weighted average of the interest rates of the underlying mortgages in a portfolio.
 - It helps investors assess the overall interest rate risk and potential returns of the MBS.
 - Understanding the WAC is essential for risk management and investment decision-making in the fixed income market.
 - Investors should also consider factors like prepayment risk, credit risk, and market conditions when evaluating MBS investments.
-

Q.3461 A mortgage-backed portfolio includes four mortgage investments as follows:

- Mortgage 1: \$150,000 in current value, 5% interest rate, 5 years remaining duration
- Mortgage 2: \$100,000 in current value, 6% interest rate, 6 years remaining duration
- Mortgage 3: \$50,000 in current value, 4% interest rate, 3 years remaining duration
- Mortgage 4: \$80,000 in current value, 7% interest rate, 2 years remaining duration

What is the weighted average maturity (n years) of the portfolio?

- A. 4.4
- B. 5.1
- C. 4.9
- D. 4.0

The correct answer is **A**.

Weighted average maturity (WAM) is the weighted average amount of time until the maturities

on mortgages in an MBS.

Total value of portfolio = \$150,000 + \$100,000 + \$50,000 + \$80,000 = \$380,000

We then compute the percentage value of each mortgage:

- Mortgage 1 %value = $\$150,000 / \$380,000 = 39.5\%$
- Mortgage 2 %value = $\$100,000 / \$380,000 = 26.3\%$
- Mortgage 3 %value = $\$50,000 / \$380,000 = 13.2\%$
- Mortgage 4 %value = $\$80,000 / \$380,000 = 21\%$

The percentage values of each mortgage are then multiplied by the remaining duration until maturity:

- $39.5\% * 5 \text{ years} = 1.975 \text{ years}$
- $26.3\% * 6 \text{ years} = 1.578 \text{ years}$
- $13.2\% * 3 \text{ years} = 0.396 \text{ years}$
- $21\% * 2 \text{ years} = 0.42 \text{ years}$

The resulting figures are then totaled to produce a WAM of approx. 4.4 years

Things to Remember

- Weighted average maturity (WAM) is a measure used in fixed income analysis to calculate the average time until a portfolio's securities mature, weighted by the proportion of the portfolio each security represents.
- WAM is an important metric for investors in mortgage-backed securities (MBS) as it helps in assessing the interest rate risk associated with the portfolio.
- Duration is another key concept in fixed income analysis, representing the sensitivity of a bond's price to changes in interest rates.
- Weighted average duration (WAD) is another metric that combines the concept of duration with the weighting of each security in the portfolio.
- Investors often use WAM and WAD to manage their exposure to interest rate risk and

make informed investment decisions.

Q.3463 Consider the following residential mortgage:

- Loan amount: \$500,000
- Annual rate of interest: 5.5% (fixed)
- Term: 10 years
- Start date: 01/01/2019

If payments are made monthly, what is the amortized amount for the 10-year loan?

- A. \$5,520.00
- B. \$5,426.21
- C. \$5,225.50
- D. 4,834.55

The correct answer is **B**.

A fixed-rate mortgage is a mortgage loan that has a fixed interest rate for the entire term of the loan. Equal payments are made over the life of the mortgage.

$$PMT = \frac{\text{Principal}}{\left(\frac{1-(1+r)^{-n}}{r}\right)}$$

Where:

PMT = amortized amount for each month

r = monthly interest rate (annual rate/12)

n = total number of months

$r = 0.055/12 = 0.004583$, $n = 10 * 12 = 120$

$$PMT = \frac{500,000}{\left(\frac{1-(1+0.004583)^{-120}}{0.004583}\right)} = \$5,426.21$$

On a financial calculator,

$$n = 120;$$

$$I/Y = 0.4583 (5.5/12);$$

$$PV = -500,000;$$

$$FV = 0;$$

$$CPT \Rightarrow PMT = \$5,426.21$$

Q.3464 A 30-year \$150,000 mortgage has a fixed mortgage rate of 6 percent compounded monthly. What portion of the first payment is the principal?

A. \$150

B. \$149.33

C. \$500

D. \$750

The correct answer is **B**.

A fixed-rate mortgage is a mortgage loan that has a fixed interest rate for the entire term of the loan. Equal payments are made over the life of the mortgage.

$$PMT = \frac{\text{Principal}}{\left(\frac{1-(1+r)^{-n}}{r}\right)}$$

Where:

PMT = amortized amount for each month

r = monthly interest rate (annual rate/12)

n = total number of months

$$r = 0.06/12 = 0.005, n = 30 * 12 = 360$$

$$PMT = \frac{150,000}{\left(\frac{1-(1+0.005)^{-360}}{0.005}\right)} = \$899.33$$

On a financial calculator,

$$N = 360;$$

$$I/Y = 0.5 \% (6/12\%);$$

$$PV = -150,000;$$

$$FV = 0;$$

$$CPT \Rightarrow PMT = \$899.33$$

The interest portion for a month is equal to the mortgage balance at the beginning of the month (\$150,000 in this case) multiplied by the interest rate per month.

$$\text{Interest portion} = 0.005 * 150,000 = \$750$$

The remaining portion of the payment, $\$899.33 - \$750 = \$149.33$, goes to reduce the principal balance.

Q.3465 Consider a 15-year \$500,000 mortgage with a 6 percent interest rate. After 10 years, the borrower (the mortgage issuer) pays off the outstanding principal. How much will the lender receive?

A. \$180,000

B. \$220,000

C. \$6,313

D. \$218,245

The correct answer is **D**.

We first need to know the monthly payment, PMT.

$$PMT = \frac{\text{Principal}}{\left(\frac{1 - (1 + r)^{-n}}{r} \right)}$$

Where:

PMT = amortized amount for each month

r = monthly interest rate (annual rate/12)

n = total number of months

$$r = 0.06/12 = 0.005, n = 15 * 12 = 180$$

$$PMT = \frac{500,000}{\left(\frac{1-(1+0.005)^{-180}}{0.005}\right)} = \$4,219.28$$

On a financial calculator,

$$n = 180;$$

I/Y = 0.5 (6/12); (when using the financial calculator, we input the rates in percentage form. In this case, 0.005 = 0.5%)

$$PV = -500,000;$$

$$FV = 0;$$

$$CPT \Rightarrow PMT = \$4,219.28$$

From this point, there are two ways to go: First. We could construct an amortization table for the mortgage, but this would consume a lot of time. But there's a simpler approach. After 10 years, we can treat this mortgage as though it were a 5-year mortgage with payments of \$4,219.28 and an interest rate of 6 percent. We can then solve for the outstanding principal balance using the same formula.

$$4,219.28 = \frac{\text{Outstanding Principal}}{\left(\frac{1-(1+0.005)^{-60}}{0.005}\right)}$$

Solving for the outstanding principal gets us \$218,245

Q.3467 Consider a 15-year \$500,000 mortgage with a rate of 6 percent. Ten years into the mortgage, rates have fallen to 5 percent. What would be the monthly saving to a homeowner from refinancing the outstanding mortgage balance at the lower rate? (Round your answer to the nearest unit.)

- A. \$265
- B. \$101
- C. \$0
- D. \$111

The correct answer is **B**.

We first need to know the monthly payment, PMT.

$$PMT = \frac{\text{Principal}}{\left(\frac{1-(1+r)^{-n}}{r}\right)}$$

Where:

PMT = amortized amount for each month

r = monthly interest rate (annual rate/12)

n = total number of months

$$r = 0.06/12 = 0.005, n = 15 * 12 = 180$$

$$PMT = \frac{500,000}{\left(\frac{1-(1+0.005)^{-180}}{0.005}\right)} = \$4,219.28$$

On a financial calculator,

n = 180;

I/Y = 0.5 (6/12); ; (not 0.005, because it's already in percentage format)

PV = -500,000;

FV = 0;

CPT $\Rightarrow$ PMT = \$4,219.28

From this point, there are two ways to go: First. We could construct an amortization table for the mortgage, but this would consume a lot of time. But there's a simpler approach. After 10 years, we can treat this mortgage as though it were a 5-year mortgage with payments of \$4,219.28 and an interest rate of 6 percent. We can then solve for the mortgage balance using the same formula.

$$4,219.28 = \frac{\text{Principal}}{\left(\frac{1-(1+0.005)^{-60}}{0.005}\right)}$$

Solving for the principal gets us \$218,245

For comparability, we calculate the new payments assuming a loan of \$218,245, a 5-year life, and a rate of 5 percent.

$$PMT = \frac{218,245}{\left(\frac{1-(1+0.004167)^{-60}}{0.004167}\right)} = \$4,118.59$$

Thus, the saving is approx.

\$101 per month ($\$4,219.28 - \$4,118.59 = \100.69).

Note: it would be misleading to compare the payments on the old loan to a new, 15-year loan.

Q.3585 The Public Securities Association (PSA) prepayment benchmark assumes that the monthly prepayment rate for a mortgage pool:

- A. remains constant for the first 30 months and then increases by 0.2% for months 30 to 360
- B. increases by 0.2% every month up to 30 months
- C. decreases by 0.2% every month up to 30 months
- D. remains constant for the first 30 months and then decreases by 0.2% for months 30 to 360

The correct answer is **B**.

The Public Securities Association (PSA) prepayment benchmark assumes that the monthly prepayment rate for a mortgage pool increases as it ages (becomes seasoned). The PSA is expressed as a monthly series of Conditional Prepayment Rates (CPRs). The model assumes that:

- CPR = 0.2% for the first month after origination, increasing by 0.2% every month up to 30 months; and
- CPR = 6% for months 30 to 360

A mortgage pool whose prepayment speed (experience) is in line with the assumptions of the PSA model is said to be 100% PSA. Similarly, a pool whose prepayment experience is two times the CPR under the PSA model is said to be 200% PSA (or 200 PSA).

Choice A is incorrect. The PSA prepayment benchmark does not assume that the monthly prepayment rate remains constant for the first 30 months and then increases by 0.2% for months 30 to 360. Instead, it assumes that the rate increases by 0.2% every month up to the first 30 months.

Choice C is incorrect. The PSA model does not assume a decrease in the monthly prepayment rate over time, but rather an increase of 0.2% each month until it reaches a certain point (usually around month 30).

Choice D is incorrect. Similar to choice A, this option incorrectly assumes that the monthly prepayment rate remains constant for an initial period and then decreases afterwards, which contradicts with how PSA model works.

Things to Remember

- Conditional Prepayment Rate (CPR): The percentage of the remaining principal balance that is prepaid in a single period, typically a month.
 - Public Securities Association (PSA): A prepayment benchmark used in the mortgage-backed securities market to standardize prepayment assumptions.
 - Seasoning: Refers to the age of a mortgage pool, with prepayment rates typically increasing as the pool ages.
 - 100% PSA: Indicates that the prepayment speed of a mortgage pool aligns with the assumptions of the PSA model.
 - 200% PSA: Indicates that the prepayment speed of a mortgage pool is twice the CPR under the PSA model.
-

Q.3586 During periods of falling interest rates, the refinancing of mortgage loans will:

- A. Accelerate prepayments and reduce the average life of the MBS
- B. Decelerate prepayments and increase the average life of the MBS
- C. Accelerate prepayments and increase the average life of the MBS
- D. Decelerate prepayments, but and decreases the average life of the MBS

The correct answer is **A**.

During periods of falling interest rates, homeowners are more likely to refinance their mortgage loans. Refinancing allows homeowners to replace their existing mortgage with a new one that has a lower interest rate. This process accelerates prepayments because homeowners are paying off their old mortgage faster than originally planned. Prepayments are the early return of principal on a mortgage security. When prepayments accelerate, the average life of the mortgage-backed security (MBS) reduces. The average life of an MBS is the weighted-average time to the return of a dollar of principal, measured in years. When homeowners prepay their mortgages, the principal is returned sooner, reducing the average life of the MBS.

Choice B is incorrect. Falling interest rates typically encourage borrowers to refinance their mortgage loans, which accelerates prepayments. This in turn reduces the average life of the MBS, not increase it as this option suggests.

Choice C is incorrect. While it's true that falling interest rates accelerate prepayments due to increased refinancing activity, this actually reduces the average life of the MBS rather than increasing it. The faster prepayment speed means that investors receive their principal back sooner than expected, thus shortening the average life of the security.

Choice D is incorrect. This choice incorrectly suggests that falling interest rates decelerate prepayments and decrease the average life of MBS. In reality, lower interest rates lead to an acceleration in prepayments due to increased refinancing activity by borrowers and consequently reduce (not decrease) the average life of MBS.

Things to Remember

- Prepayments: Early return of principal on a mortgage security
 - Weighted-average life of an MBS: The average time to the return of a dollar of principal, measured in years
 - Refinancing: Process of replacing an existing mortgage with a new one that has a lower interest rate
 - Interest rates: Falling interest rates typically lead to increased refinancing activity by homeowners
 - Accelerated prepayments: Occur when homeowners pay off their mortgages faster than originally planned
-

Q.3587 Which of the following are considered as weights while determining the weighted average maturity of a mortgage pass-through security?

- A. Time to maturity as a percentage of the total life of the security
- B. Remaining principal of the i^{th} mortgage to the total principal
- C. Absolute value of coupon payments due
- D. Remaining number of months to maturity for each mortgage loan

The correct answer is **B**.

The weighted average maturity (WAM) of a mortgage pass-through security is calculated by assigning weights to the remaining life of each mortgage in the pool. The weight assigned to each mortgage, denoted as w_i , is determined by the ratio of the remaining principal of the i^{th} mortgage (P_i) to the total remaining principal of all mortgages in the pool. This is mathematically

represented as: $w_i = P_i / \sum_{i=1}^n P_i$. This method of weighting ensures that mortgages with larger remaining principals have a greater impact on the WAM, reflecting the fact that these mortgages will take longer to be fully repaid and thus have a greater effect on the average maturity of the security.

Choice A is incorrect. The time to maturity as a percentage of the total life of the security is not used as weights in calculating WAM. Instead, it's the remaining principal balance of each mortgage that is used.

Choice C is incorrect. The absolute value of coupon payments due does not serve as weights for calculating WAM. This would be more relevant in determining cash flows rather than the average time it will take for mortgages to be repaid.

Choice D is incorrect. While the remaining number of months to maturity for each mortgage loan does play a role in determining WAM, it's not used as weights in its calculation. Rather, these are multiplied by their respective weights (i.e., remaining principal balances) to arrive at WAM.

Things to Remember

- Weighted Average Maturity (WAM) is a measure used in the analysis of mortgage-backed securities to determine the average time until the principal is expected to be repaid.
 - WAM helps investors assess the interest rate risk and prepayment risk associated with mortgage pass-through securities.
 - Prepayment risk refers to the risk that borrowers will repay their mortgages earlier than expected, impacting the cash flows to investors.
 - Factors such as interest rates, economic conditions, and borrower behavior can influence prepayment risk in mortgage-backed securities.
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Reading 45: Interest Rate Futures

Q.686 An investment manager at Galaxy Investments Co. is analyzing the interest earned on the corporate bond of Aero Supermarts, a chain of grocery stores. The bond pays semiannual interest of 13% on March 1st and September 1st. Using the appropriate day count convention, determine the interest earned on the bond between September 1st and February 12th.

- A. 6.5%
- B. 5.92%
- C. 5.82%
- D. 5.87%

The correct answer is **C**.

For measuring the interest earned on municipal bonds and corporate bonds, we use the 30 days per month convention or 30/360 convention.

Days between September 1st and February 12th (30 days convention) = 162 days

Days between September 1st and March 1st (30 days convention) = 181 days

Therefore, the interest earned is:

$$6.5\% * \frac{162}{181} = 5.82\%$$

Q.687 Fahim Zakaria, a fund manager based in Qatar, manages a sovereign fund for the Qatari government. The fund has more than \$12.6 billion in assets under management. The fund invests only in the shares of blue-chip firms and the sovereign bonds/bills of different countries. If the manager wants to include a 164-days U.S. Treasury bill which is quoted as 9, then what is the cash price of the bill?

- A. \$96
- B. \$95.9
- C. \$93.7
- D. \$91

The correct answer is **B**.

The cash price of the bill can be calculated with the following formula:

$$\begin{aligned}\text{Discount} &= \left(\frac{\text{No. of days to maturity}}{360 \text{ days}} \right) * \text{Quoted price} \\ \text{Discount} &= \left(\frac{164}{360} \right) * 9 = \$4.1\end{aligned}$$

$$\text{Cash price} = \text{Face Value} - \text{Discount} = 100 - 4.1 = \$95.9$$

Note that,

$$Q = \frac{360}{n}(100 - C)$$

and we can thus make C the subject,

$$C = 100 - \frac{n}{360}Q$$

So, following the last formula, $\text{Discount} = \frac{n}{360}Q$

Where

C=Cash price

Q= Quoted price

Q.690 An analyst wishes to estimate the dirty price of a \$100,000 face value Treasury bond with an annual coupon of 8% paid semiannually on December 10th and on June 10th. The bond was purchased on April 8, 2017, when the quoted price was 93-8. If the Treasury bond's maturity date is December 10th, 2020, what is the bond's dirty price?

- A. \$93,250
- B. \$94,865
- C. \$95,125
- D. \$95,865

The correct answer is **D**.

We will calculate the dirty price of the bond using the following steps:

Let's assume the face value of the bond is \$100, and an 8% coupon bond will pay \$4 on June 10th and December 10th.

Since the bond was purchased on April 8th, 2017, the last coupon was paid on December 10th, 2016.

The number of actual days from December 10th to April 8th is 119 days.

The number of actual days between December 10th and June 10th is 182 days.

$$\text{Accrued interest} = 119 / 182 * \$4 = \$2.615$$

The quoted price of 93-8 is equal to $\$93 + 8/32$, as the Treasury bond prices are quoted in dollars and thirty-seconds of a dollar.

$$\text{Dirty price} = \text{Quoted price} + \text{Accrued interest} = \$93.25 + \$2.615 = \$95.865$$

or \$95,865 with the face value of \$100,000

Things to Remember

- Dirty price of a bond includes both the clean price (quoted price) and the accrued interest
- Accrued interest is the interest that has accumulated since the last coupon payment
- Treasury bond prices are quoted in dollars and thirty-seconds of a dollar
- Calculation of accrued interest involves the actual number of days since the last coupon payment
- Understanding the concept of face value, coupon rate, and coupon payment dates in bond valuation

Q.691 Futures contracts on Treasury bonds are well-known investment instruments for large investment banks and sovereign funds. Such futures include a large variety of Treasury bonds with maturities ranging from 15 to 25 years. Since the deliverable bonds under the futures contracts have different market values, the exchanges have taken a measure for adjusting the price received by the short position holder in the futures contracts according to the specific Treasury bond or note delivered.

Which of the following is that measure?

- A. Dirty price
- B. Convexity adjustment
- C. Conversion factor
- D. Clean price

The correct answer is **C**.

The measure that exchanges have implemented to adjust the prices of Treasury bond futures is the conversion factor. This factor is used to calculate the price received by the short party of the bond. The quote price applicable for the bond delivered is the product of the conversion factor and the futures contract's most recent settlement price. The formula to calculate the cash received by the short party is: (Most recent settlement price * Conversion factor) + Accrued interest. This ensures that the short party receives a fair price for the bond, regardless of its market value.

Choice A is incorrect. The dirty price of a bond includes the accrued interest in addition to its market price. It does not adjust the price received by the holder of the short position in futures contracts based on the specific Treasury bond or note delivered.

Choice B is incorrect. Convexity adjustment refers to a correction made to account for changes in interest rates and their impact on bonds with different durations and maturities. While it's an important concept in fixed income securities, it doesn't directly relate to adjusting prices for varying deliverable bonds under futures contracts.

Choice D is incorrect. The clean price of a bond excludes any accrued interest, representing only its market value. Similar to dirty price, it does not serve as a measure for adjusting prices based on specific deliverable Treasury bonds or notes under futures contracts.

Things to Remember

- Conversion factor is a key concept in the valuation of Treasury bond futures contracts, ensuring fair pricing for the short position holder.
- Accrued interest is an important component in calculating the final price received by

the short party in futures contracts.

- Dirty price includes accrued interest, while clean price excludes it, affecting the overall valuation of bonds.
 - Convexity adjustment is crucial in assessing the impact of interest rate changes on bond prices, but it is not directly related to adjusting prices for specific deliverable bonds in futures contracts.
 - Understanding the differences between dirty price, clean price, and conversion factor is essential for effective risk management in fixed income investments.
-

Q.692 Amy Jackson, a fixed investment manager at Flaxes Investment Company headquartered in Toronto, has a short position in 15-year Treasury bond contracts with a \$100,000 face value for each contract. The last quoted price of the contract is 91-28, while the accrued interest on the bond is \$3.29 (for \$100 face value). If the conversion factor for the deliverable bond under the contract is 1.471, then which of the following is true?

- A. The cash received by the short-position party is \$138,438.
- B. The cash paid by the short-position party is \$138,438.
- C. The cash paid by the short-position party is \$139,987.
- D. The cash received by the short-position party is \$139,987.

The correct answer is **A**.

Amy has a short position in the contract. Therefore,

$$\begin{aligned}\text{Cash received by the short-position party} &= (\text{Most recent settlement price} * \text{Conversion factor}) + \text{Accrued interest} \\ &= (91 + \frac{28}{32}) * 1.471 + \$3.29 = \$138,438\end{aligned}$$

or \$138,438 for a \$100,000 face value contract

Q.693 An investor with a short position is about to deliver a bond and has four bonds to choose from which as listed in the following table. The last settlement price (quoted futures price) is \$96.25. Determine which bond is the cheapest-to-deliver.

Bond	Quoted bond price	Conversion factor
1	99	1.02
2	122	1.22
3	107	1.1
4	112	1.15

- A. Bond 1
- B. Bond 2
- C. Bond 3
- D. Bond 4

The correct answer is **A**.

The CTD bond minimizes the following:

$$\text{quoted bond price} - (\text{QFP} \times \text{CF})$$

Where:

QFP = quoted futures price (most recent settlement price)

CF = conversion factor for the bond delivered

The expression above calculates the cost of delivering the bond.

Thus, we can compute the cost of delivery of each bond as follows:

Bond 1: $99 - (96.25 \times 1.02) = \0.83

Bond 2: $122 - (96.25 \times 1.22) = \4.58

Bond 3: $107 - (96.25 \times 1.1) = \1.13

Bond 4: $112 - (96.25 \times 1.15) = \1.31

Bond 1 is the cheapest to deliver with a cost of delivery of \$0.83.

Q.695 Since there is a large universe of Treasury bonds and futures contracts on those bonds, there is a large number of Treasury bonds available to be delivered at any point in a month.

However, due to the imperfection of conversion factors used by exchanges, at times it is cheaper to deliver one bond as compared to another bond. Which of the following options truly defines the cheapest-to-deliver option?

- A. The cheapest-to-deliver option allows the long position holder of the futures contract on a Treasury bond to choose which is the cheapest bond to receive.
- B. The cheapest-to-deliver option allows the long position holder of the futures contract on a Treasury bond to choose which is the cheapest bond to deliver.
- C. The cheapest-to-deliver option allows the short position holder of the futures contract on a Treasury bond to choose which is the cheapest bond to receive.
- D. The cheapest-to-deliver option allows the short position holder of the futures contract on a Treasury bond to choose which is the cheapest bond to deliver.

The correct answer is **D**.

The cheapest-to-deliver option allows the short position holder of the futures contract on a Treasury bond to choose which is the cheapest bond to deliver. In futures contracts, the party holding the short position is obligated to deliver the underlying asset, in this case, a Treasury bond, at the contract's expiration. The 'cheapest-to-deliver' option refers to the most cost-effective bond that the short position holder can deliver to fulfill their obligation under the contract. This is primarily due to the imperfect nature of conversion factors used by exchanges, which can sometimes make it more economical to deliver one bond over another. Therefore, the short position holder has the discretion to choose the bond that is cheapest for them to deliver, thereby minimizing their costs.

Choice A is incorrect. The long position holder of the futures contract on a Treasury bond does not have the option to choose which bond to receive. This choice lies with the short position holder who delivers the bond.

Choice B is incorrect. Again, it's a misunderstanding of roles in futures contracts. The long position holder does not deliver bonds; they are on the receiving end of this transaction.

Choice C is incorrect. While it correctly identifies that the short position holder has an option, it incorrectly states that they choose which bond to receive. In reality, they decide which bond to deliver based on cost-effectiveness.

Things to Remember

- Treasury bonds are long-term debt securities issued by the US Department of the Treasury to finance the government's operations and pay off debts.
- Futures contracts are standardized agreements to buy or sell a specified asset at a predetermined price at a specified time in the future.

- Cheapest-to-deliver option is important in futures contracts as it allows the short position holder to minimize costs by choosing the most cost-effective bond to deliver.
- Conversion factors are used in the calculation of the delivery amount of the underlying asset in a futures contract.
- Short position holders in futures contracts are obligated to deliver the underlying asset, while long position holders are obligated to receive the asset.

Q.696 Henry Louis is a derivative investment manager at the Global First Investment Bank in Singapore. He manages a portfolio of fixed income assets and interest rate futures. He currently has a short position in a futures contract on GILTS (U.K. equivalent to U.S. Treasury securities). As the delivery month is approaching, the manager has to choose the cheapest-to-deliver bond from the four available bonds. If the last settlement price is 95-16, which of the following bond is the cheapest to deliver?

Bond	Quoted bond price	Conversion factor
A	\$99	1.011
B	\$97	1.001
C	\$103	1.069
D	\$107	1.072

- A. Bond A
- B. Bond B
- C. Bond C
- D. Bond D

The correct answer is **C**.

The cheapest-to-deliver bond among the four available bonds is Bond C. To find out cheapest-to-deliver bond, we will use the following formula:

$$\text{Cost of delivery} = \text{Quoted bond price} - (\text{Last settlement price} \times \text{Conversion factor})$$

Bond	Quoted bond price	Conversion factor	Cost of delivery
A	\$99	1.011	$\$99 - (95.5 \times 1.011) = \2.45
B	\$97	1.001	$\$97 - (95.5 \times 1.001) = \1.40
C	\$103	1.069	$\$103 - (95.5 \times 1.069) = \0.91
D	\$107	1.072	$\$107 - (95.5 \times 1.072) = \4.62

Q.698 According to recent data, the most frequently traded futures contract in the United States is the 3-month Eurodollar futures contract that is traded on the Chicago Mercantile Exchange (CME). Which of the following is the appropriate and complete definition of a Eurodollar futures contract?

- A. It is the foreign currencies futures contract on euros/dollars. The underlying interest rate of the contract is the 3-month forward LIBOR.
- B. It is the foreign currencies futures contract on euros/dollars. The underlying interest rate of the contract is the 3-month forward LIBOR and the 3-month U.S. risk-free rate.
- C. It is the interest rate futures contract on Eurodollars or on the U.S. dollars deposited outside of the U.S. The underlying interest rate of the contract is the 3-month U.S. risk-free rate.
- D. It is the interest rate futures contract on Eurodollars or on the U.S. dollars deposited outside of U.S. The underlying interest rate of the contract is the 3-month forward LIBOR.

The correct answer is **D**.

A Eurodollar futures contract is an interest rate futures contract on Eurodollars, which are U.S. dollars deposited outside of the U.S. The underlying interest rate of the contract is the 3-month forward London Interbank Offered Rate (LIBOR). The face value of the contract is \$1 million, and the price change in the futures contract is a minimum of \$25, which is equal to the change of one tick or one basis point. This contract allows investors to hedge against interest rate risk, and its popularity can be attributed to its high liquidity and the transparency of the LIBOR.

Choice A is incorrect. While it correctly identifies the Eurodollar futures contract as a foreign currencies futures contract, it incorrectly states that the underlying interest rate of the contract is the 3-month forward LIBOR. The Eurodollar futures contract is not based on euros/dollars but on U.S. dollars deposited outside of U.S.

Choice B is incorrect. This choice also misidentifies the Eurodollar futures contract as a foreign currencies futures contract and further complicates this misunderstanding by adding that both the 3-month forward LIBOR and 3-month U.S. risk-free rate are underlying interest rates for this type of contract, which is not accurate.

Choice C is incorrect. Although it correctly identifies that a Eurodollar futures contracts are based on U.S. dollars deposited outside of U.S., it incorrectly states that its underlying interest rate is the 3-month U.S. risk-free rate when in fact, it's based on the 3-month forward LIBOR.

Things to Remember

- Eurodollar futures contract is one of the most actively traded futures contracts in the world.

- It is traded on the Chicago Mercantile Exchange (CME).
 - The contract size for Eurodollar futures is \$1 million.
 - Price changes in Eurodollar futures are in minimum increments of \$25, which is equivalent to one tick or one basis point.
 - Investors use Eurodollar futures to hedge against interest rate risk.
 - The popularity of Eurodollar futures is due to their high liquidity and the transparency of the LIBOR.
-

Q.699 Alina Escobar is a junior derivatives analyst at the derivatives investment unit of a financial institution. The company holds a mid-day meeting where managers discuss investment strategies according to recent trends in the market. Escobar's manager asked her to estimate the price of a March Eurodollar futures contract that is quoted as 94.25. Estimate the effective dollar price that the firm will have to pay if the firm ultimately decides to invest in the March Eurodollar futures contract.

- A. \$942,500
- B. \$985,625
- C. \$991,750
- D. \$1,050,870

The correct answer is **B**.

The (%) price of EDF contracts is quoted as $100 - R$, where R is the annualized LIBOR rate. This is also known as the index price. For example, an EDF with a quoted price of 94.25 implies that R , the annualized LIBOR rate at expiry, is expected to be 5.75%.

However, the curriculum requires us to concentrate on the 3-month EDF.

It follows that that to get the actual or effective dollar price of the contract given the index price, we have to divide the yield by four so as to reflect the three-month rate.

So if the quoted (index) price is 94.25, here's how we get the effective price:

Step 1: Convert the annual rate into a three-month rate

Annual rate = 5.75%, so three-month rate = $5.75\%/4 = 1.4375\%$

Thus, the effective percentage price is $100 - 1.4375 = 98.5625$

Step 2: Convert into an effective dollar price

Each EDF contract has a face amount of \$1 million.

Thus, effective price = $98.5625/100 * \$1,000,000 = \$985,625$

Note: the quoted price is initially in percentages, but the effective price is in dollars.

Note also that, any Eurodollar futures contract consists of a eurodollar time deposit, whose principal is \$1 million with a maturity period of three months.

This information need not necessarily be provided in the question, so there is a need to memorize this.

Things to Remember

- Eurodollar futures contracts are based on the 3-month USD LIBOR rate.
- The quoted price of Eurodollar futures contracts is expressed as 100 minus the expected annualized LIBOR rate at expiry.
- To calculate the effective dollar price, the annualized rate needs to be converted to a 3-month rate by dividing it by four.
- Each Eurodollar futures contract has a face value of \$1 million.
- The effective dollar price is calculated by multiplying the effective percentage price by the face value of the contract.

Q.701 As the Eurodollar futures contracts are marked to market on a daily basis, the Eurodollar futures contract can result in differences between actual forward rates and those implied by futures contracts. Generally, in the longer maturities Eurodollar futures contracts, the implied forward rates (futures interest rates) are greater than the actual forward rates. Which of the following is useful in reducing the mentioned above difference between the implied forward rates and the actual forward rates?

- A. Convexity adjustment.
- B. Conversion factor.
- C. Duration adjustment.
- D. Dirty price.

The correct answer is **A**.

Convexity adjustment is a method used by analysts and managers to reduce the difference between the implied forward rates by the futures and the actual forward rates. This discrepancy arises due to the daily settlement feature of Eurodollar futures contracts. In contracts with longer maturities, the implied forward rates are often greater than the actual forward rates. The convexity adjustment method helps to reduce this difference, making it a useful tool in the management of Eurodollar futures contracts.

Choice B is incorrect. The conversion factor is a tool used in bond futures contracts to account for differences in the characteristics of different bonds, such as coupon rates and time to maturity. It does not have any direct relevance to the discrepancy between implied forward rates and actual forward rates in Eurodollar futures contracts.

Choice C is incorrect. Duration adjustment refers to a strategy used by portfolio managers to adjust the sensitivity of their bond portfolios to changes in interest rates. While it can help manage interest rate risk, it does not directly address the issue of discrepancies between implied and actual forward rates in Eurodollar futures contracts.

Choice D is incorrect. The dirty price of a bond includes accrued interest while clean price excludes it. This concept applies primarily to bonds and has no direct impact on minimizing the difference between implied forward rates and actual forward rates in Eurodollar futures contracts.

Things to Remember

- Eurodollar futures contracts are used to hedge or speculate on future interest rate movements.
- Implied forward rates are calculated based on the prices of Eurodollar futures contracts.
- Actual forward rates are the market's expectation of future interest rates.
- Convexity adjustment helps in reducing the difference between implied and actual forward rates in Eurodollar futures contracts.
- Duration adjustment is a risk management strategy related to bond portfolios.
- Conversion factor is used in bond futures contracts to adjust for bond characteristics.

Q.702 Xiaoping Yu is an investment manager at Shanghai Derivatives Investors Lounge, an investment company that solely invests in derivatives. At the beginning of the current fiscal year,

Yu and her team constructed a hedge with interest rates futures contracts by taking long positions in futures contracts. Yu believes that the interest rates will start to increase in the foreseeable future. Which of the following actions should Yu take in order to protect her position?

- A. As the prices of interest rate futures contract will increase, Yu should take a long position in interest futures contracts.
- B. As the prices of interest rate futures contract will decrease, Yu should take a long position in interest futures contracts.
- C. As the prices of interest rate futures contract will increase, Yu should take a short position in interest futures contracts.
- D. As the prices of interest rate futures contract will decrease, Yu should take a short position in interest futures contracts.

The correct answer is **D**.

As interest rates rise, the prices of interest rate futures contracts fall. This is due to the inverse relationship between interest rates and the price of interest rate futures contracts. An investor who has taken a long position in futures contracts stands to lose money if interest rates increase. Therefore, to hedge their portfolio against this risk, an investor should take a short position in interest rate futures contracts. This strategy allows the investor to profit from the falling prices of the futures contracts, thereby offsetting the losses from their long position. This is a common risk management strategy used in derivative trading to protect against adverse price movements.

Choice A is incorrect. This choice suggests that Yu should take a long position in interest futures contracts as the prices will increase. However, this is not correct because when interest rates rise, the prices of interest rate futures contracts typically decrease. Therefore, taking a long position would not be an appropriate strategy to safeguard her position.

Choice B is incorrect. While it correctly states that the prices of interest rate futures contract will decrease with rising interest rates, it incorrectly suggests taking a long position in these contracts. A long position would mean buying more contracts expecting their price to increase which contradicts with the given scenario where we expect prices to fall due to rising interest rates.

Choice C is incorrect. This choice incorrectly assumes that the prices of interest rate futures contract will increase when in fact they are expected to decrease with rising interest rates as per our scenario. Hence, taking a short position based on this assumption would not be beneficial for Yu.

Things to Remember

- Interest rate futures contracts are financial derivatives that derive their value from changes in interest rates.

- Long positions in interest rate futures contracts benefit from falling interest rates, while short positions benefit from rising interest rates.
 - Hedging with futures contracts involves taking an offsetting position to protect against adverse price movements in the underlying asset.
 - Investors use futures contracts to manage risk, speculate on price movements, and hedge their portfolios.
 - Understanding the relationship between interest rates and futures prices is crucial for effective risk management in derivative trading.
-

Q.703 Anil Kumar has recently joined Axe Investment Bank as a junior analyst through a global analyst recruitment program. In order to impress the management of the asset management unit of the bank, Kumar decides to create a combined position in interest rate futures contracts that does not change in value with small changes in yield. Which of the following can help him create such a position?

- A. Convexity-hedging.
- B. Duration-based hedging.
- C. Hedging with FRAs.
- D. DV01 hedging.

The correct answer is **B**.

Duration-based hedging is a strategy that involves adjusting the duration of a portfolio to hedge against interest rate risk. Duration is a measure of the sensitivity of the price of a bond or a bond portfolio to a change in interest rates. In this context, Anil Kumar can use duration-based hedging to create a combined position in interest rate futures contracts that does not change in value with small changes in yield. This is achieved by creating a position that has zero duration. When the position has zero duration, it means that the value of the position does not change due to small changes in yield. This is exactly what Anil Kumar wants to achieve. Therefore, duration-based hedging is the correct strategy for him to use.

Choice A is incorrect. Convexity-hedging is a strategy that aims to protect against large yield changes, not minor fluctuations. It involves managing the curvature of the price-yield relationship of a bond portfolio, which becomes significant when interest rates change dramatically.

Choice C is incorrect. Hedging with FRAs (Forward Rate Agreements) primarily protects

against changes in short-term interest rates and does not necessarily remain unaffected by minor yield fluctuations. FRAs are more suitable for hedging specific future interest rate exposures rather than general yield movements.

Choice D is incorrect. DV01 hedging involves adjusting a portfolio to make its dollar value change by one basis point for each basis point change in yield, which means it would be affected by even minor yield fluctuations, contrary to what Anil Kumar wants.

Things to Remember

- Duration is a key concept in managing interest rate risk and measures the sensitivity of the price of a bond or a bond portfolio to changes in interest rates.
 - Convexity is another measure that helps in understanding the curvature of the price-yield relationship of a bond or a bond portfolio.
 - FRAs (Forward Rate Agreements) are derivative contracts that help in managing interest rate risk by locking in a future interest rate.
 - DV01 (dollar value of 01) is a measure that quantifies the change in the dollar value of a bond or a bond portfolio for a one basis point change in yield.
 - Hedging strategies are used to mitigate risks such as interest rate risk, currency risk, and commodity price risk in financial markets.
-

Q.3555 A 3.125% government bond is priced for settlement on April 12, 2016. The bond makes quarterly coupon payments on June 30th, September 30th, December 31st, and March 31st. What is the bond's accrued interest per 100 of par value?

- A. 0.672
- B. 0.103
- C. 0.4121
- D. 0.1713

The correct answer is **B**.

Government/Treasury bonds make use of the actual/actual day count convention. In other words, we compare the actual number of days that have elapsed since the last coupon payment date and the actual number of days between the coupon dates.

Here, the coupon dates of interest are 31 March and 30th June; we have 91 days (30 days on April, 31 in May, and 30 in June)

The number of days that have elapsed since 31st March is 12.

Thus, the accrued interest = $12/91 * 3.125/4 = 0.1030\%$

Per \$100 face value, that's \$0.1030

Note: 3.125% is an annual rate, that's why we have to divide by 4 to get the quarterly (3-month) rate

Note: Another approximate approach to the calculations would be:

Accrued interest = $(12/365) * (0.03125) = 0.1027$

Things to Remember

- Government/Treasury bonds typically use the actual/actual day count convention for calculating accrued interest
 - Accrued interest is calculated based on the actual number of days since the last coupon payment date and the actual number of days between coupon dates
 - For quarterly coupon payments, the annual coupon rate needs to be divided by 4 to get the quarterly rate
 - Accrued interest is typically calculated as (actual days since last coupon / total days in coupon period) * (coupon rate)
-

Q.4916 Suppose that the continuously compounded forward rate for the period between year 2 and year 4 is 2.5%. Suppose further that the continuously compounded interest rate of the two-year zero-coupon is 5.5%. What is the continuously compounded 4-year zero-coupon interest rate?

- A. 8%
- B. 4%
- C. 2.625%
- D. 3.375%

The correct answer is **B**.

The 4-year zero-coupon interest rate is calculated by making R_2 the subject of the formula in the forward rate formula as follows:

$$\begin{aligned} R_2 &= \frac{F(T_2 - T_1) + R_1 T_1}{T_2} \\ &= \frac{0.025(4 - 2) + 0.055 \times 2}{4} = 0.04 = 4\% \end{aligned}$$

Q.4918 Suppose a firm has a \$20,000,000 portfolio of Treasury bonds with a portfolio duration of 5. Suppose further that the cheapest to deliver bond has a duration of 3 and that the six-month treasury bond futures price is \$200,000. What is the number of futures contracts to fully hedge the portfolio?

- A. 60 contracts should be shorted.
- B. 166 contracts should be shorted.
- C. 166 contracts should be bought.
- D. 500 contracts should be shorted.

The correct answer is **B**.

The number of futures contracts to fully hedge the portfolio is given by

$$N = -\frac{P \times DP}{FC \times DF}$$

Where:

P = forward value of the fixed-income portfolio being hedged

DP = duration of the portfolio at the maturity date of the hedge

FC = futures contract price

DF = duration of the asset underlying the futures

Thus,

$$N = \frac{\$20,000,000 \times 5}{\$200,000 \times 3} = -166.6667 \approx -166$$

Thus, 166 contracts should be shorted

The negative sign implies that the number of contracts taken up must be the opposite of the original position. If the investor is **short** a portfolio, for example, they must **long** N contracts to produce a position with zero duration.

Q.4919 A Treasury bond pays coupons at the rate of 12% per year on March 1 and September 1. What is the accrued interest between March 1 and April 30 per USD 100 of face value?

- A. 1.0004
- B. 0.9783
- C. 1.9672
- D. 1.9565

The correct answer is **D**.

12% coupon rate per year implies a coupon of \$6 on each of these dates. If we want to compute the accrued interest as of April 30, we will have to determine the actual number of days between April 30 and the last coupon date, i.e., March 1. We have 60 days (= 30 + 30, in March and April, respectively).

The reference period, March 1 to September 1, has 184 actual days.

Thus,

$$\text{Accrued interest} = \frac{60}{184} \times 6 = 1.9565$$

Q.4920 A corporate bond pays coupons at the rate of 12% per year on March 1 and September 1. What is the accrued interest between March 1 and April 30 per USD 100 of face value?

- A. 0.9833
- B. 0.9779
- C. 1.0
- D. 1.9667

The correct answer is **D**.

In the case of a corporate bond, we use the 30/360-day count convention. 12% coupon rate per year implies a coupon of \$6 on each of these dates.

Based on the 30/360 convention, the number of days between March 1 and April 30

That's 59 days (= 29 + 30, in March and April, respectively).

The reference period, March 1 to September 1, has 180 days(= 29 + 30+ 30+ 30+ 30+ 30+ 1, in March, April, May, June, July, August, and September, respectively).

Thus,

$$\text{Accrued interest} = \frac{59}{180} \times 6 = 1.9667$$

Q.4921 Suppose a Treasury bill lasts for 230 days and has a quoted price of 5.5 What is the cash price of the Treasury bill?

A. 91.2717

B. 96.4861

C. 96.5342

D. 91.3913

The correct answer is **B**.

We know that,

$$P = \frac{360}{n}(100 - Y)$$

Where P is the quoted price, and Y is the corresponding cash price per USD 100 of face value.

We can make Y the subject of the formula so that:

$$\begin{aligned} Y &= 100 - \frac{Pn}{360} \\ &= 100 - \frac{5.5 \times 230}{360} = 96.4861 \end{aligned}$$

Q.4922 A Treasury bond pays coupons at the rate of 10% per year on June 1 and December 1. What is the accrued interest between June 1 and July 31 per USD 100 of face value (rounded to two decimal places)?

- A. \$1.64
- B. \$1.5
- C. \$1.62
- D. \$1.66

The correct answer is **A**.

The number of days between June 1 and December 1 is 183 (29, 31, 31, 30, 31, 30, 1 in June, July, August, September, October, November, and December, respectively).

The number of days between June 1 and July 31 is 60 (29, 31 in June and July, respectively). The accrued interest is therefore:

$$\frac{60}{183} \times 5 = 1.639$$

Q.4923 Suppose that the bond that will be cheapest to deliver in a Treasury bond futures contract pays coupons of 6% per annum on March 1 and September 1 and will be delivered on July 1. Suppose further that the bond's quoted price on June 1 is 120.00. If all interest rates are 5 % continuously compounded, what is the clean futures price on July 1?

A. 121.5

B. 121.33

C. 120.01

D. 123.51

The correct answer is C.

We have 92 days(=30+30+31+1 in March, April, May, and June respectively) between March 1 and June 1 and 184 days(=30+30+31+30+31+31 +1 in March, April, May, June, July, August, and September, respectively)

We know that,

$$\begin{aligned}\text{Dirty price} &= \text{Quoted price} + \text{Accrued interest} \\ &= 120 + \frac{92}{184} \times 3 = 121.50\end{aligned}$$

There are no coupon payments for the 30-day period between June 1 and July 1.

The dirty futures price is therefore

$$121.50e^{\frac{30}{365} \times 0.05} = 122.00$$

We have 122 days(=30+30+31+30+1 March, April, May, June and in July, respectively) between March 1 and July 1.

The accrued interest on July 1 is, therefore,

$$3 \times \frac{122}{184} = 1.9891$$

The clean futures price is, therefore: $122.00 - 1.9891 = 120.0109$

Q.4924 Suppose that the bond that will be cheapest to deliver in a Treasury bond futures contract pays annual coupons of 6% per annum on March 1 and September 1 and will be delivered on July 1. Suppose further that the bond's quoted price on June 1 is 120.00, and its conversion factor is 1.2424. If all interest rates are 5 % continuously compounded, what is the estimated futures price on July 1?

- A. 98.78
- B. 96.6
- C. 97.66
- D. 122.0

The correct answer is **B**.

We have 92 days(=30+30+31+1 in March, April, May, and June respectively) between March 1 and June 1 and 184 days(=30+30+31+30+31+31 +1 in March, April, May, June, July, August, and September, respectively) between March 1 and September 1

We know that,

$$\begin{aligned}\text{Dirty price} &= \text{Quoted price} + \text{Accrued interest} \\ &= 120 + \frac{92}{184} \times 3 = 121.50\end{aligned}$$

There are no coupon payments for the 30-day period between June 1 and July 1.

The dirty futures price is therefore

$$121.50e^{\frac{30}{365} \times 0.05} = 122.00$$

We have 122 days(=30+31+30+31 and 1 in March, April, May, June, and July, respectively) between March 1 and July 1.

The accrued interest on July 1 is, therefore,

$$3 \times \frac{122}{184} = 1.9891$$

The clean futures price is, therefore: $122.00 - 1.9891 = 120.0109$

Thus, the futures price can be estimated by

$$\frac{\text{Clean futures price}}{\text{Conversion factor}} = \frac{120.0109}{1.2424} = 96.5960 \approx 96.60$$

Q.4926 Suppose that a nine-month interest is expected to be paid on a USD 30,000,000 bond. Suppose further that three-month Eurodollar futures contracts are used to hedge the nine-month interest and that the nine-month period starts at the maturity of the futures contract that will be used. How many three-month Eurodollar futures contracts are necessary to hedge the nine-month interest? (Ignore the differences between Eurodollar futures and FRAs)

- A. 30
- B. 45
- C. 90
- D. 60

The correct answer is **C**.

The change in the value of the instrument for a 1-basis point parallel shift in the interest rate is

$$\text{USD } 30,000,000 \times \frac{9}{12} \times 0.0001 = \text{USD } 2250$$

We know that the interest rate per three months changes by 0.0025%, which is equivalent to USD 25 on a principal of USD 1 million. In other words, a Eurodollar contract is designed in such a way that one basis point(0.01) move in the futures price leads to a profit or loss of \$25.

Thus, the number of three-month Eurodollar futures contracts necessary to hedge the nine-month interest is

$$\frac{2250}{25} = 90$$

Q.4927 Suppose that a bond portfolio of USD 1,000,000 has a duration of 5. Suppose further that the current Treasury bond futures price is USD 104 and that the cheapest to deliver bond has a duration of 5 at maturity. What is the number of contracts that should be traded to fully hedge the portfolio?

- A. 52
- B. 96
- C. 10
- D. 100

The correct answer is **C**.

Note that, futures contract involves the delivery of bonds with a face value of USD 100,000.

Thus, the value of one futures contract is USD 104,000

The number of contracts that should be traded (with a negative number indicating a short position) for a hedge should be

$$-\frac{E_V}{E_F}$$

E_F is the increase in value of one futures contract for a 1-basis point downward parallel shift in the zero curve, and

E_V is the increase in value of a trader's position for a 1-basis point downward parallel shift in the zero curve.

Thus, the increase in value of a trader's position for a 1-basis point downward parallel shift in the zero curve is given by:

$$E_V = 1,000,000 \times 5 \times 0.0001 = 500$$

Similarly, the increase in value of one futures contract for a 1-basis point downward parallel shift in the zero curve is given by:

$$E_F = 104,000 \times 5 \times 0.0001 = 52$$

Thus, the number of contracts required is

$$= -\frac{500}{52} = -9.615 \approx -10$$

Thus, 10 contracts should be shorted.

Q.4929 Suppose that the bond that will be cheapest to deliver in a Treasury bond futures contract pays annual coupons of 6% per annum on March 1 and September 1 and will be delivered on June 1. Suppose further that the bond's clean futures price is 121.4848 on June 1, and its conversion factor is 1.2424. If all interest rates are 5% continuously compounded, what is the dirty price on June 1?

- A. 121.99
- B. 119.46
- C. 123.0
- D. 123.51

The correct answer is C.

We have 184 days(=30+30+31+30+31+31 +1 in March, April, May, June, July, August, and September, respectively)

We have 92 days between March 1 and June 1(= 30 + 30 + 31 + 1 in March, April, May, and June, respectively)

The accrued interest on June 1 is, therefore,

$$3 \times \frac{92}{184} = 1.5$$

We know that,

$$\text{Clean Futures Price} = \text{Dirty Futures Price} - \text{Accrued Interest}$$

Thus,

$$\text{Dirty Futures Price} = \text{Clean Futures Price} + \text{Accrued Interest} = 121.4848 + 1.5 = 122.9848 \approx 123$$

Q.4934 What undesirable trading behavior involves using market developments after 4 pm to cancel or carry out a trade?

- A. Late Trading.
- B. Market Timing.

C. Directed Brokerage.

D. Front Running.

The correct answer is **A**.

Late trading refers to the practice of placing orders to buy or sell securities after the close of a trading day's official trading hours (typically 4 pm in the U.S.), but still having these orders executed at the closing price of the same trading day. This is considered undesirable and illegal because it allows traders to take advantage of post-close market developments and information that are not yet reflected in the closing price, thereby creating an unfair trading environment. Late trading is particularly problematic in the context of mutual funds, where it can lead to dilution of the fund's assets and harm to long-term investors. It is considered a serious offense and is punishable by law.

Choice B is incorrect. Market Timing refers to the strategy of making buy or sell decisions of financial assets (often stocks) by attempting to predict future market price movements. This practice does not involve exploiting post-4 pm market developments for trade execution or cancellation.

Choice C is incorrect. Directed Brokerage involves a mutual fund directing its portfolio transactions to a particular broker in return for that broker selling more shares of the fund. It does not involve taking advantage of post-market developments.

Choice D is incorrect. Front Running refers to the unethical practice where a broker executes orders on a security for its own account while taking advantage of advance knowledge of pending orders from its customers; it doesn't relate to actions taken based on information after 4 pm.

Things to Remember

- Late trading is considered illegal and unethical in the financial markets.
- Regulators have strict rules and penalties in place to prevent and punish late trading activities.
- Market timing, while not the same as late trading, can also have negative impacts on market efficiency and investor returns.
- Directed brokerage can create conflicts of interest and raise concerns about fair treatment of investors.
- Front running is another form of unethical behavior that can harm investors and erode trust in the financial markets.

Reading 46: Swaps

Q.705 Which of the following options is a correct differentiating feature between swaps and forward contracts?

- A. Forward contracts trade on over-the-counter markets, while swaps trade on exchanges.
- B. Forward contract holders have an obligation, while swap holders have the right to buy or sell the underlying security in the future.
- C. Forward contracts can be customized, but swaps are standardized.
- D. There is only one exchange of cash flow at a future date in a forward contract, whereas there are many exchanges of cash flows on multiple future dates in a swap.

The correct answer is **D**.

A forward contract and a swap are both financial derivatives used for hedging and speculation in financial markets. However, they differ significantly in their structure and operation. A forward contract is a private agreement between two parties to buy or sell an asset at a specified future date for a price agreed upon today. The key feature of a forward contract is that there is only one exchange of cash flows that occurs at a future date. This exchange is the culmination of the contract, where the agreed-upon asset is delivered, and the payment is made. On the other hand, a swap is a derivative contract through which two parties exchange financial instruments. These instruments can be almost anything, but most swaps involve cash flows based on a notional principal amount agreed upon by both parties. Unlike forward contracts, swaps involve multiple exchanges of cash flows on various future dates. These cash flows are often calculated over the notional principal amount using different interest rates. Therefore, choice D accurately differentiates between swaps and forward contracts.

Choice A is incorrect. Both forward contracts and swaps are traded over-the-counter (OTC), not on exchanges. These instruments are typically customized to the needs of the parties involved, which makes them unsuitable for exchange trading.

Choice B is incorrect. Both forward contract holders and swap holders have obligations, not rights, to buy or sell the underlying security in the future. The difference lies in their structure: a forward contract involves a single future transaction, while a swap involves multiple transactions over time.

Choice C is incorrect. Both forward contracts and swaps can be customized according to the needs of the parties involved. This flexibility is one of their key features that make them popular in financial markets.

Things to Remember

- Forward contracts and swaps are both types of derivative contracts used for hedging and speculation in financial markets.

- Both forward contracts and swaps are traded over-the-counter (OTC) and not on exchanges.
 - Forward contracts involve a single exchange of cash flows at a future date, while swaps involve multiple exchanges of cash flows on various future dates.
 - Forward contracts are private agreements between two parties to buy or sell an asset at a specified future date, while swaps involve the exchange of financial instruments based on a notional principal amount.
 - Both forward contracts and swaps can be customized to meet the specific needs of the parties involved.
-

Q.707 Leipzig Auto Company is one of the largest auto interior maker firms in Germany. On April 1st, 2017, the company entered into a 3-year swaps contract with the Allied-Swiss Bank to pay quarterly cash flows equal to the fixed interest rate of 5.7% on a notional principal of €30 million. In return, Allied-Swiss Bank agreed to pay the quarterly cash flow equal to the 3-months LIBOR on the same notional principal. After reviewing the terms of the contract, determine Allied-Swiss Bank's position in the swap contract.

- A. Fixed-rate-payer
- B. Libor-receiver
- C. Floating-rate-payer
- D. Floating-rate-receiver

The correct answer is **C**.

In a swap contract, the parties involved agree to exchange cash flows, which are calculated on a notional principal amount. The cash flows are determined by different interest rates - one party pays a fixed rate, and the other pays a floating rate. In this case, Allied-Swiss Bank has agreed to pay the floating rate (3-month LIBOR), while receiving the fixed rate (5.7%) from Leipzig Auto Company. Therefore, Allied-Swiss Bank is the floating-rate-payer in this swap contract. This position allows the bank to hedge against the risk of falling interest rates. If the LIBOR decreases, the bank's payments will also decrease, while it continues to receive the fixed rate from Leipzig Auto Company.

Choice A is incorrect. Leipzig Auto Company, not Allied-Swiss Bank, is the fixed-rate-payer in this swap contract. They are obligated to make quarterly payments at a fixed interest rate of 5.7% on a notional principal of €30 million.

Choice B is incorrect. This choice incorrectly identifies Allied-Swiss Bank as the receiver of the LIBOR-based payments. In fact, it's Leipzig Auto Company that will receive these floating-rate payments from Allied-Swiss Bank.

Choice D is incorrect. While it might seem logical to assume that since Allied-Swiss Bank isn't receiving fixed-rate payments they must be receiving floating-rate ones, this isn't correct either. The bank's role in this swap contract is to pay out based on the 3-month LIBOR rate, making them a floating-rate-payer rather than receiver.

Things to Remember

- **Swap Contracts:** In a swap contract, two parties agree to exchange cash flows based on different interest rates on a notional principal amount.
 - **Fixed-Rate-Payer:** The party in a swap contract that agrees to pay a fixed interest rate is known as the fixed-rate-payer.
 - **Floating-Rate-Payer:** The party in a swap contract that agrees to pay a floating interest rate (such as LIBOR) is known as the floating-rate-payer.
 - **LIBOR (London Interbank Offered Rate):** LIBOR is a benchmark interest rate at which major global banks lend to one another in the international interbank market for short-term loans.
 - **Interest Rate Risk:** Swap contracts are often used to hedge against interest rate risk, where one party can protect themselves from fluctuations in interest rates by entering into a swap agreement.
-

Q.708 Swaps are customizable derivative contracts between two parties that trade in the over-the-counter (OTC) markets around the world. Financial intermediaries and companies have used swaps for multiple purposes. The most popular and basic swap agreement, which is used worldwide, is called the plain vanilla swap. Which of the following is the underlying variable of plain vanilla swaps?

- A. Currency exchange rates
- B. Interest rates
- C. Volatility
- D. Equities

The correct answer is **B**.

The underlying variable of plain vanilla swaps is interest rates. Plain vanilla swaps, also known as interest rate swaps, are the most common type of swap agreement used worldwide. In these swaps, one party agrees to pay cash flows equal to a fixed interest rate set at a predetermined time, in exchange for a floating interest rate. This allows the parties involved to hedge against interest rate risk. The interest rate is the key variable that determines the cash flows in these swaps, making it the underlying variable of plain vanilla swaps.

Choice A is incorrect. While currency exchange rates are indeed an underlying variable in certain types of swaps, specifically currency swaps, they are not the underlying variable in plain vanilla swaps. Plain vanilla swaps typically involve the exchange of a fixed interest rate for a floating interest rate.

Choice C is incorrect. Volatility is not the underlying variable in plain vanilla swaps. It might be an important factor to consider when pricing or valuing derivatives such as options, but it does not directly underlie a plain vanilla swap agreement.

Choice D is incorrect. Equities can be an underlying asset for some types of derivative contracts like equity swaps or equity options, but they are not associated with plain vanilla interest rate swap agreements which primarily deal with exchanging cash flows based on different interest rates.

Things to Remember

- Swaps are derivative contracts that allow parties to exchange cash flows or other financial instruments based on certain variables.
- Plain vanilla swaps are the most basic type of swap agreement, typically involving the exchange of fixed and floating interest rates.
- Interest rates are the underlying variable in plain vanilla swaps, determining the cash flows exchanged between parties.
- Currency exchange rates are a key variable in currency swaps, a different type of swap agreement.
- Volatility is important in pricing options and other derivatives, but it is not the underlying variable in plain vanilla swaps.
- Equities can be underlying assets in equity swaps or options, but they are not associated with plain vanilla interest rate swaps.

Q.709 Kevin Bernard, the head of the derivatives trading department at Savers Bank, entered into a 3-year swap agreement on September 30, 2015, with Moon Traders. In the agreement, Savers Bank agreed to pay Moon Traders an interest rate of 5%, paid semiannually on the principal of \$100 million. In return, Moon Traders agreed to pay Savers the LIBOR rate on the same principal. If the LIBOR prevailing on March 30, 2016, is 4.95% paid semiannually, which of the following statements is true?

- A. Savers Bank will pay \$25,000 to Moon Traders on March 30, 2016.
- B. Savers Bank will receive \$25,000 from Moon Traders on March 30, 2016.
- C. Savers Bank will pay \$25,000 to Moon Traders on September 30, 2016.
- D. Savers Bank will receive \$25,000 from Moon Traders on September 30, 2016.

The correct answer is C.

Savers Bank will pay \$25,000 to Moon Traders on September 30, 2016.

On the other hand, Moon Traders will pay \$2.475 million to Savers Bank.

Savers Bank's cash outflow = $\frac{5\%}{2} * 100 \text{ million} = -\2.5 million

Savers Bank's cash inflow = $\frac{4.95\%}{2} * 100 \text{ million} = \2.475 million

Net cash flow on September 30, 2016 = 2,475,000 – 2,500,000 = –25,000

One crucial element of plain vanilla swaps is that the exchange of funds related to a particular Libor (floating rate) takes place one period (six months in the case of our example) after the Libor rate is observed. In other words, at the beginning of each period, both payments for the end of the period - fixed and floating - are known. This allows both parties to budget for their future payments.

Things to Remember

- Interest Rate Swap: A financial derivative agreement between two parties to exchange interest rate cash flows, typically one party pays a fixed interest rate while the other pays a floating interest rate based on a notional principal amount.
- LIBOR (London Interbank Offered Rate): The benchmark interest rate at which major global banks lend to one another in the international interbank market for short-term loans.
- Notional Principal: The hypothetical amount of principal that is used to calculate interest payments in an interest rate swap, but is not actually exchanged.

- Cash Flow: The movement of money into or out of a business, project, or financial product, including both inflows (receipts) and outflows (payments).
 - Net Cash Flow: The difference between cash inflows and cash outflows in a given period, which indicates the overall financial position of an entity.
-

Q.710 Muhammad Ali is a credit risk manager at Green Flag Investment Company. Recently, Green Flag Investment Company borrowed \$50 million from another investment bank at the rate of the 6-month LIBOR plus 50 basis points. Ali worries that the LIBOR can significantly increase due to the current economic situation of the country, which can increase the investment company's liability. If Ali intends to change the floating rate liability into a fixed rate liability, which of the following positions can transform the floating rate liability into a fixed rate liability?

- A. A fixed-rate payer position in a swap.
- B. A short position in interest futures.
- C. A long position in a forward rate agreement.
- D. A short position in call options.

The correct answer is **A**.

A fixed-rate payer position in a swap is the correct choice because it allows the transformation of a floating rate liability into a fixed rate liability. In a swap contract, two parties agree to exchange one stream of cash flows against another stream. In this case, Ali would enter into a swap contract where he agrees to pay a fixed rate and receive a floating rate. The floating rate he receives from the swap contract would offset the floating rate he has to pay on the initial loan. Therefore, his net payment would be the fixed rate from the swap contract, effectively transforming the floating rate liability into a fixed rate liability. This strategy is often used by investors and companies to hedge against the risk of interest rate fluctuations.

Choice B is incorrect. Taking a short position in interest rate futures would not transform the floating rate liability into a fixed one. Instead, it would be used to hedge against falling interest rates, which is not Ali's concern here. He is worried about rising rates.

Choice C is incorrect. A long position in a forward rate agreement (FRA) allows the holder to lock in an interest rate for borrowing or lending money in the future. However, this does not convert a floating-rate liability into a fixed-rate one as it only hedges against future changes in interest rates and does not affect existing liabilities.

Choice D is incorrect. A short position on call options gives the holder the right to sell an asset at an agreed price before a certain date but does not provide any transformation from floating to fixed-rate liabilities.

Things to Remember

- **Interest Rate Swaps:** In an interest rate swap, two parties exchange interest rate cash flows, with one party paying a fixed rate and the other paying a floating rate. This is commonly used to manage interest rate risk.
 - **LIBOR (London Interbank Offered Rate):** LIBOR is a benchmark interest rate at which major global banks lend to one another in the international interbank market for short-term loans.
 - **Interest Rate Futures:** These are standardized contracts to buy or sell a specified amount of a financial instrument at a predetermined price on a specified future date.
 - **Forward Rate Agreements (FRAs):** FRAs are over-the-counter contracts that allow parties to lock in an interest rate for a future period, providing protection against interest rate movements.
 - **Call Options:** Call options give the holder the right, but not the obligation, to buy an asset at a specified price within a specified time period.
-

Q.711 Assume that you are a swap dealer and have just acted as a counterparty in an interest rate swap. The notional principal for the swap was \$7.5 million and you are now obligated to make five annual payments of 8 percent interest. The floating rate that you will receive is 8.2 percent, and the floating payments to you are annual as well.

If the floating rate remains unchanged for the first two years and then falls by 1.5 percent for the remainder of the contract, what will be your net payments for the five years?

- A. \$62,000
- B. \$30,500
- C. \$203,500
- D. \$262,500

The correct answer is **D**.

You will receive a total of \$30,000 for the first two years [$\$7,500,000 * (0.082 - 0.080) * 2$].

The new floating rate afterward that you will receive is $8.2\% - 1.5\% = 6.7\%$.

You will pay a total of \$292,500 for the last three years [$\$7,500,000 * (0.067 - 0.08) * 3$ years].

Thus, your net payment over the five years will be \$262,500 ($\$30,000 - \$292,500 = -\$262,500$).

Things to Remember

- Interest Rate Swap: A financial derivative contract in which two parties agree to exchange interest rate cash flows, based on a specified notional amount, over a set period of time.
- Notional Principal: The amount on which the exchange of interest payments in a swap is based, but it is not actually exchanged.
- Floating Rate: An interest rate that is allowed to move up and down with the market or along with an index.
- Fixed Rate: An interest rate that remains constant throughout the life of the swap.
- Net Payments: The difference between the total payments received and the total payments made over the life of the swap.

Q.712 Hygiene Foods Inc. is one of the largest fast-moving consumer goods (FMCG) company in

Malaysia. Last year, it entered into a 3-years swap agreement to pay semiannual cash flows equal to the 6-months LIBOR plus 20 basis point on the notional principal of \$400 million. Which of the following parties is most likely the counterparty of the given swap agreement?

- A. The exchange
- B. A clearinghouse
- C. Financial intermediaries
- D. The central bank

The correct answer is **C**.

Financial intermediaries are typically the counterparties in swap agreements. Swap agreements bear resemblance to forward contracts in terms of their customizability, lack of regulation, and over-the-counter (OTC) nature. However, the key difference lies in the involvement of financial intermediaries such as banks or brokerage houses in swap agreements. These intermediaries earn a spread of 3 to 5 basis points in the transaction between the two parties. In exchange for this spread, they guarantee the execution of the agreement even if one party defaults. In the context of the swap agreement entered into by Hygiene Foods Inc., a financial intermediary is most likely the counterparty, ensuring the agreement's fulfillment even in the event of a default.

Choice A is incorrect. An exchange is not likely to be the counterparty in a swap agreement. Exchanges are platforms where securities, commodities, derivatives and other financial instruments are traded. They do not participate as counterparties in trades.

Choice B is incorrect. A clearinghouse acts as an intermediary between buyers and sellers in financial markets, ensuring that transactions are conducted smoothly and efficiently. However, they do not typically act as counterparties in swap agreements.

Choice D is incorrect. The central bank does not engage directly with individual companies for such transactions like swaps; instead it implements monetary policy and provides regulatory oversight for the financial system.

Things to Remember

- Swap agreements involve the exchange of cash flows based on different interest rates or other financial variables.
- Financial intermediaries, such as banks or brokerage houses, play a crucial role in facilitating swap agreements by acting as counterparties.
- Counterparties in swap agreements provide liquidity, ensure transaction execution, and manage risks associated with the agreement.
- Swap agreements are customizable and are traded over-the-counter (OTC), meaning

they are not standardized like exchange-traded instruments.

- Central banks are responsible for monetary policy and regulatory oversight, but they do not typically engage directly in swap agreements with individual companies.
-

Q.714 Faheem Salami has recently joined a large investment bank that acts as a financial intermediary in a number of swaps agreements. The bank also acts as the market maker when the counterparties to swaps are unavailable. Salami's boss asked him to calculate the swap rate of the 6-month interest rate swap when the 6-month LIBOR is 4.3%.

Salami also knows that the 6-month risk-free rate is 3.9%, and the bid and offer rates for the swap are 4.02 and 4.08, respectively.

Which of the following rates is the accurate swap rate for the specific swap agreement?

- A. 3.90%
- B. 4.05%
- C. 4.10%
- D. 4.30%

The correct answer is **B**.

A swap rate is the average of the bid-and-offer rates (4.02 and 4.08) of the swap agreement. Financial intermediaries post these bid and offer rates when making the market for these swaps. Due to the versatility of swap agreements, it is difficult to find two counterparties with uniform terms. Therefore, financial intermediaries take the position of the counterparty in all swaps agreement.

Things to Remember

- **Swap Rate:** The swap rate is the fixed interest rate that two parties agree to exchange in a swap agreement. It is the rate at which the cash flows are exchanged in the swap.
- **LIBOR (London Interbank Offered Rate):** LIBOR is the average interest rate at which major global banks borrow from one another. It is a key benchmark for short-term interest rates.
- **Risk-Free Rate:** The risk-free rate is the theoretical rate of return of an investment with zero risk of financial loss. It is often used as a benchmark in pricing financial

instruments.

- **Market Maker:** A market maker is a financial intermediary that stands ready to buy and sell financial instruments at quoted prices. They provide liquidity to the market by facilitating trading.
 - **Bid and Offer Rates:** Bid rate is the price at which a market maker is willing to buy a financial instrument, while the offer rate is the price at which they are willing to sell. The bid-offer spread represents their profit margin.
-

Q.715 Otto Cornell is the head of the finance department of Easy Home Appliances Inc. The firm intends to enter into a swap agreement to convert its outstanding floating-rate liability into a fixed rate liability. Therefore, the firms decided to enter into a fixed-rate payer position in a 2-year swap rate agreement to pay the quarterly cash flow equal to a 6% fixed rate to the Great Spanish Bank (GSB). In return, GSB agreed to pay quarterly cash flow equal to 3-month LIBOR to the firm. Since, it is the first transaction of this nature, the head of the financial department does not know who is required to facilitate the preparation of the confirmation of swap or the master agreement. Which of the following is most likely to facilitate the confirmation?

- A. Great Spanish Bank.
- B. Financial intermediaries.
- C. The International Swaps and Derivatives Association.
- D. The Securities and Exchange Commission (SEC).

The correct answer is **C**.

The International Swaps and Derivatives Association or ISDA has been facilitating the preparation of the confirmation or the master agreement for swap agreements. The confirmation of the swap agreement consists of the definition of clauses and terminologies, consequences in case of default, notional principals, day counts convention, rates, etc.

A is incorrect. While the Great Spanish Bank is a counterparty in the swap agreement, it is not responsible for facilitating the preparation of the confirmation or the master agreement.

B is incorrect. Financial intermediaries may assist in executing the swap but do not typically handle the preparation of the detailed legal documentation, which is standardized by ISDA.

D is incorrect. The SEC is a regulatory body that oversees securities markets in the United States, but it does not facilitate the preparation of swap agreements or their confirmations.

Things to Remember

- Swap agreements are financial contracts where two parties agree to exchange cash flows based on a specified notional amount.
 - ISDA, the International Swaps and Derivatives Association, plays a crucial role in standardizing and facilitating the preparation of swap agreements and their confirmations.
 - Confirmation of a swap agreement includes important details such as payment terms, interest rates, notional amounts, and other key provisions.
 - Financial intermediaries may assist in executing swaps but are not typically responsible for preparing the legal documentation.
 - The SEC, Securities and Exchange Commission, is a regulatory body overseeing securities markets and is not involved in the preparation of swap agreements.
-

Q.718 Cherry Inc. and Sang Wang are the two market leaders in the tablet PC market. Cherry Inc. is an American company while Sang Wang is headquartered in Japan. Both companies are considering taking on debt in either USD or Yen. The following table shows the borrowing rates for both companies.

	USD	Yen
Cherry Inc.	3.5%	9%
Sang Wang	5.5%	10.4%

Considering the comparative advantage argument, estimate the total gain both companies can have if they enter into a currency swaps contract.

- A. 4.7%
- B. 2.2%
- C. 1.4%
- D. 0.6%

The correct answer is **D**.

The table suggests that the interest rate in the Japanese Yen is higher than the U.S dollar. As per the comparative argument, Cherry has an absolute advantage in both USD and Yen.

The total gain as per the comparative advantage is equal to the difference of the difference:

Difference in USD = $(5.5\% - 3.5\%) = 2.0\%$

Difference in Yen = $10.4\% - 9\% = 1.4\%$

Total gain for both companies = $(5.5\% - 3.5\%) - (10.4\% - 9\%) = 0.6\%$

Q.719 Black Corporation and UK Fabrics have entered into a 3-year currency swap agreement with periodic annual payments, where Black agreed to pay 5% in British pound (GBP) on the principal amount of GBP 100 million to UK Fabrics. In addition, UK Fabrics agreed to pay 7% to Black on the principal of \$120 million. The currency exchange rate at the initiation of the swap was USD 1.26 per GBP. If the interest rate in the United States and Great Britain are flat at 5.5% and 6.8%, respectively, but the dollar has appreciated in value against the GBP, then determine which of the following is true.

- A. The value of the swap to Black Corporation will increase.
- B. The value of the swap to UK Fabrics will increase.
- C. The value of the swap to both the companies will increase.
- D. The value of the swap will be unaffected by subsequent changes in exchange rates.

The correct answer is **A**.

The value of the swap to Black Corporation will increase. This is because Black Corporation has to pay interest in GBP, and with the appreciation of the US dollar against the GBP, it will need fewer dollars to purchase the necessary pounds. This effectively reduces the cost of the swap for Black Corporation, thereby increasing its value. Furthermore, the increase in the value of the dollar also increases the risk of default for UK Fabrics, as it will need more pounds to purchase the necessary dollars to meet its interest payment obligations. This further increases the value of the swap to Black Corporation.

Choice B is incorrect. The value of the swap to UK Fabrics will not increase if the US dollar appreciates against the GBP. This is because UK Fabrics has agreed to pay a 7% interest rate in USD, and if the USD appreciates, it would mean that they would have to pay more in GBP terms. Therefore, an appreciation of the USD against the GBP would decrease rather than increase the value of this swap agreement for UK Fabrics.

Choice C is incorrect. The value of a currency swap does not necessarily increase for both parties when exchange rates fluctuate. In this case, Black Corporation benefits from an appreciation of USD as it receives payments in this currency while paying out in GBP. Conversely, UK Fabrics suffers from an appreciation of USD as it pays out in this currency while receiving payments in GBP.

Choice D is incorrect. The value of a currency swap can be affected by subsequent changes in exchange rates because these changes alter how much each party must pay or receive at each payment date and at maturity when principal amounts are exchanged back again.

Things to Remember

- Currency swap agreements involve the exchange of cash flows in different currencies based on a specified principal amount.
- Interest rate differentials between the two currencies involved can impact the value of the swap to each party.
- Exchange rate movements can affect the cost and value of the swap to the parties involved.
- Risk of default increases for the party paying in a currency that appreciates, as they need more of their own currency to make payments.
- Understanding the impact of currency movements and interest rate differentials is crucial in analyzing the value of currency swap agreements.

Q.721 Green Grass Co. intends to enter into a 5-year fixed for floating interest rate swap with MNG Bank. Green agrees to pay annual cash flow equal to a fixed interest rate of 5% on the principal of €140 million to the bank in exchange for receiving the annual cash flow equal to 1-year LIBOR plus 50 basis points on the same notional principal from the bank. However, Green Grass does not want to exchange the notional principal at the inception of the swap. Instead, it wants to decrease the principal in a predetermined manner. Which of the following swaps is most suitable for this transaction?

- A. Basis swap
- B. Amortizing swap
- C. Deferred swap
- D. Step up swap

The correct answer is **B**.

An amortizing swap is a type of interest rate swap where the notional principal amount is reduced over time, typically in line with the decrease in the outstanding amount of a loan. In this case, Green Grass Co. does not want to exchange the notional principal at the inception of the swap, but instead, it wants to decrease the principal in a predetermined manner. This is a characteristic feature of an amortizing swap. The principal amount in an amortizing swap decreases over time, which can be beneficial for a company like Green Grass Co. that wants to gradually reduce its exposure to interest rate risk. The reduction in the principal amount can be structured to match the amortization schedule of the underlying loan, making it a suitable choice for this transaction.

Choice A is incorrect. A basis swap involves exchanging one type of floating rate payments for another type of floating rate payments. It does not involve a reduction in the principal amount over time, which is what Green Grass Co. wants.

Choice C is incorrect. A deferred swap refers to a situation where the start date of the swap agreement is postponed to a future date, but it does not involve reducing the principal amount over time.

Choice D is incorrect. In a step-up swap, the interest rate increases over time according to a predetermined schedule, but it does not involve reducing the principal amount over time as desired by Green Grass Co.

Things to Remember

- Interest Rate Swap: A financial derivative contract where two parties agree to exchange interest rate cash flows, based on a notional principal amount, to hedge against interest rate risk.
- Notional Principal: The hypothetical amount of principal that is used to calculate

interest payments in an interest rate swap, but is not actually exchanged between the parties.

- **LIBOR (London Interbank Offered Rate):** The benchmark interest rate at which major global banks lend to one another in the international interbank market for short-term loans.
 - **Basis Points:** One basis point is equal to one-hundredth of a percentage point (0.01%).
 - **Interest Rate Risk:** The risk that changes in interest rates will adversely affect the value of an investment or a company's financial position.
-

Q.722 Heidelberg Brewery wants to enter into a 3-year swap agreement with Everest Investment Co. Heidelberg intends to pay semiannual cash flows equal to the 10-year swap rate on the principal of €100 million to the Everest Investment in exchange for receiving semiannual cash flows from the investment company equal to the 6-month LIBOR on the same notional principal. Which of the following swaps is most suitable for this transaction?

- A. LIBOR-to-floating swap
- B. Step-up swap
- C. Constant maturity swap
- D. LIBOR in arrears swap

The correct answer is **C**.

A constant maturity swap (CMS) is a derivative with a payoff that is linked to a reference rate, typically a swap rate such as the LIBOR. In a CMS, one party pays a series of fixed payments while receiving a series of floating payments indexed to a swap rate (constant maturity). The swap rate is reset periodically, typically every six months, and is usually tied to the rate of a specific maturity of swap. In this case, Heidelberg Brewery is planning to pay semiannual cash flows equal to the 10-year swap rate and receive semiannual cash flows equal to the 6-month LIBOR. This is exactly how a CMS works, making it the most suitable swap for this transaction.

Choice A is incorrect. A LIBOR-to-floating swap would involve exchanging a fixed interest rate for a floating interest rate based on the LIBOR. However, in this case, both sides of the swap are floating rates: one side is the 10-year swap rate and the other side is the 6-month LIBOR.

Choice B is incorrect. A step-up swap involves an agreement where one party makes payments that increase over time while receiving a fixed or floating rate in return. This scenario does not describe such an arrangement as there's no mention of increasing payments over time.

Choice D is incorrect. In a LIBOR in arrears swap, one party pays a fixed or floating rate and receives the LIBOR determined at the end of each period rather than at its beginning (i.e., "in arrears"). This situation doesn't match with Heidelberg Brewery's proposed arrangement as they expect to receive semiannual payments equivalent to 6-month LIBOR without any delay or "arrear" condition.

Things to Remember

- Constant Maturity Swap (CMS) is a type of interest rate swap where one party pays a fixed rate and receives a floating rate based on a specific maturity of swap rate.
 - LIBOR (London Interbank Offered Rate) is a benchmark interest rate at which major global banks lend to one another in the international interbank market for short-term loans.
 - LIBOR in arrears swap involves receiving the LIBOR rate determined at the end of each period rather than at the beginning.
 - Step-up swap involves increasing payments over time in the swap agreement.
-

Q.723 Henry Coelho, the Chief Financial Officer of Imperial Hotels & Resorts, intends to enter into a 4-year interest rate swap with a financial institution. Imperial agrees to pay quarterly cash flow equal to the 3-month LIBOR rate on the notional principal of \$200 million to the financial institution in exchange for receiving the annual cash flow equal to 550 basis points on the same notional principal from the financial institution. However, Coelho does not want to enter into a swap at the moment, but he wants to purchase an instrument that allows him to enter into the swap agreement on specified terms at a predetermined date. Which of the following instruments is most suitable for Coelho?

- A. Extendable Swap
- B. Callable Swap
- C. Swaption
- D. Constant Maturity Swap

The correct answer is **C**.

A swaption is an option that gives the holder the right to enter into a swap agreement with a predetermined fixed rate in exchange for a floating rate at a future time. This instrument is most suitable for Coelho's situation as it allows him to enter into the swap agreement on specified

terms at a predetermined date. Swaptions are essentially options on forward swap agreements. They provide the holder with the right, but not the obligation, to enter into a swap agreement at a future date. This gives the holder flexibility and protection against adverse movements in interest rates. Swaptions can be used for a variety of purposes, including hedging interest rate risk, speculating on future interest rate movements, and enhancing portfolio yield. They are commonly used by corporations, financial institutions, and investment funds.

Choice A is incorrect. An extendable swap allows the holder to lengthen the duration of an existing swap at a predetermined date, which does not align with Coelho's need to enter into a new swap agreement at a future date.

Choice B is incorrect. A callable swap gives one party the right but not the obligation to terminate the contract before its maturity, which again does not meet Coelho's requirement of entering into a new agreement in future.

Choice D is incorrect. A constant maturity swap involves swapping fixed interest payments for floating rate payments linked to an interest rate index with a continuously changing (or "constant") maturity, such as 1-year LIBOR or 5-year Treasury notes. This type of instrument would not provide Coelho with the optionality he seeks.

Things to Remember

- Swaptions are commonly used to hedge interest rate risk, speculate on future interest rate movements, and enhance portfolio yield.
- Swaptions provide the holder with the right, but not the obligation, to enter into a swap agreement at a future date.
- An extendable swap allows the holder to lengthen the duration of an existing swap at a predetermined date.
- A callable swap gives one party the right but not the obligation to terminate the contract before its maturity.
- A constant maturity swap involves swapping fixed interest payments for floating rate payments linked to an interest rate index with a continuously changing maturity.

Q.3556 What is the difference between a fixed-for-floating swap and a forward contract?

A. The payment date would be unlikely to match in a fixed-for-floating swap while the exact expiration date is known as a forward contract.

B. All the fixed-rate payments in a swap are equal, while in a forward contract, only one fixed payment is made on the settlement date.

C. The floating-rate payments in a swap are known at the start of the contract while future payments in a forward contract unknown at the contract initiation.

D. A fixed-for-floating swap involves exchanging principal amounts while a forward contract does not involve principal exchanges.

The correct answer is **B**.

A fixed-for-floating swap and a forward contract are both financial derivatives used for hedging and speculation. However, they differ significantly in their structure and payment terms. In a fixed-for-floating swap, there are multiple settlement periods, and at each of these periods, equal fixed-rate payments are made. This is a key characteristic of swaps - they involve a series of cash flows over time, and in the case of a fixed-for-floating swap, these cash flows are fixed. On the other hand, a forward contract has only one settlement period at the end of the contract. At this time, a single payment is made, which was agreed upon at the initiation of the contract. Therefore, while swaps involve multiple, equal fixed-rate payments, forward contracts involve a single payment made at the end of the contract.

Choice A is incorrect. The payment dates in a fixed-for-floating swap are predetermined and known to both parties at the start of the contract, similar to a forward contract where the expiration date is also known at the outset. Therefore, it's not accurate to say that payment dates would be unlikely to match in a fixed-for-floating swap.

Choice C is incorrect. In a fixed-for-floating swap, floating-rate payments are not known at the start of the contract as they depend on future interest rates or other variables that can change over time. Similarly, future payments in a forward contract are also unknown at initiation because they depend on future market prices.

Choice D is incorrect. A fixed-for-floating swap typically involves the exchange of interest payments based on a notional principal amount, but the principal itself is not exchanged. In a forward contract, the agreement is to buy or sell an asset at a future date for a specified price, also without exchanging principal amounts.

Things to Remember

- Fixed-for-floating swaps involve multiple settlement periods with equal fixed-rate payments, while forward contracts have only one settlement period with a single fixed payment.
- Swaps involve a series of cash flows over time, while forward contracts involve a single payment at the end of the contract.
- Payment dates in fixed-for-floating swaps are predetermined and known at the start of the contract, similar to the expiration date in a forward contract.
- Floating-rate payments in a swap are not known at the start of the contract as they

depend on future interest rates, similar to future payments in a forward contract.

- In a fixed-for-floating swap, the principal amounts are not exchanged, only the interest payments are exchanged based on a notional principal amount.
-

Q.3557 Tiara Enterprises (TIEN) has just announced its plans to establish a facility in New York, USA, to meet the increased demand for its products. TIEN plans to fund the expansion with debt and in order to hedge the risk of borrowing, TIEN has entered into a plain vanilla interest rate swap with a notional principal of \$50 million. TIEN would make semiannual payments at the rate of 12% with the counterparty making floating rate payments at the Euribor rate.

Assuming a 360-day year, if the Euribor was 13.5% on the last settlement date and is 11.0% on the current settlement date, what is the amount that TIEN would receive on the last settlement date?

- A. \$250,000
- B. \$625,000
- C. \$465,000
- D. \$375,000

The correct answer is **D**.

TIEN's payment: $(\$50 \text{ million})(180/360)(12\%) = \$3,000,000$

Counterparty's payment: $(\$50 \text{ million})(180/360)(0.135) = \$3,375,000$

Therefore, TIEN would receive a net amount of \$375,000.

Things to Remember

- Interest Rate Swap: A financial derivative agreement between two parties to exchange interest rate cash flows, typically fixed rate payments for floating rate payments.
- Notional Principal: The amount on which the exchange of interest payments in an interest rate swap is based, but not actually exchanged.
- Plain Vanilla Swap: The most basic type of interest rate swap, involving a fixed rate and a floating rate.
- Euribor Rate: The Euro Interbank Offered Rate, the average interest rate at which Eurozone banks offer to lend unsecured funds to other banks in the euro wholesale money market.
- Settlement Date: The date on which payments are made by the parties involved in the interest rate swap.

Q.3558 Company X seeks a 4-year fixed-rate US dollar funding while Company Y seeks a 4-year fixed-rate Japanese yen funding. Company X's direct borrowing all-in-cost is 10.50% in dollars and 8% in Japanese yen. Company Y's direct borrowing all-in-cost is 9.30% in dollars and 9% in Japanese yen. What is the maximum gain for all parties involved through this swap?

- A. 2.2%
- B. 1%
- C. 1.2%
- D. 0.2%

The correct answer is **A**.

	X	Y	Difference
Dollar	10.5%	9.3%	1.2%
Yen	8%	9%	-1%

X has a comparative advantage in Yen borrowing;

Y has a comparative advantage in Dollar borrowing;

When a comparative advantage exists, the implication is that the parties involved can reduce their borrowing costs by entering into a swap agreement. The net borrowing savings (maximum gain) by entering into a swap is the difference between the differences;

$$\begin{aligned}\Delta\text{Dollar} - \Delta\text{Yen} \\ = 1.2\% - -1\% = 2.2\%\end{aligned}$$

Q.3560 Understanding the implications of various financial transactions is crucial in banking and investment. With that in mind, if Core Bank sells a swaption, it has:

- A. The obligation to enter into a swap if the swaption is exercised
- B. The right, but not the obligation to enter into a swap if the swaption is exercised
- C. The obligation to make a payment to the counterparty if the swaption is exercised
- D. The right, but not the obligation to make a payment to the counterparty if the swaption is exercised

The correct answer is **A**.

When Core Bank sells a swaption, it essentially takes on the obligation to enter into a swap if the

swaption is exercised. A swaption, being a type of option, gives the buyer the right, but not the obligation, to enter into a swap agreement. The seller of the swaption, in this case, Core Bank, is obligated to honor the terms of the swap if the buyer decides to exercise the swaption. This is similar to the obligation of an option seller who must fulfill the terms of the option contract if the buyer exercises their right. In the context of a swaption, the swap agreement could involve exchanging fixed interest payments for floating interest payments, or vice versa, depending on the terms of the swaption.

Choice B is incorrect. Core Bank, as the seller of the swaption, does not have a right but an obligation. If the swaption is exercised by the holder, Core Bank must enter into a swap agreement. It does not have any choice in this matter.

Choice C is incorrect. While it's true that entering into a swap might involve making payments to the counterparty depending on how interest rates move, selling a swaption specifically results in an obligation to enter into a swap if it's exercised by the holder, not merely making a payment.

Choice D is incorrect. Similar to Choice B, this option incorrectly suggests that Core Bank has some sort of right or choice in this scenario. As the seller of the swaption, Core Bank would be obligated to enter into a swap if it's exercised; they do not have any rights or options here.

Things to Remember

- A swaption is a type of financial option that grants the holder the right, but not the obligation, to enter into an underlying swap agreement.
 - Swaptions can be used to hedge against interest rate risk or to speculate on interest rate movements.
 - When a party sells a swaption, they take on the obligation to enter into the swap if the swaption is exercised by the buyer.
 - Swaps are derivative contracts that involve the exchange of cash flows based on different financial instruments or interest rates.
 - Understanding the terms and implications of financial options like swaptions is essential for managing risk in banking and investment activities.
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